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**More-Than-People's Communes: Veterinary Workers,
Nonhuman Animals, and One Health in Mao-Era China**

presented by **Jongsik Christian Yi**

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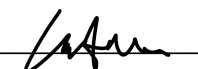
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超越人民公社：毛泽东时代中国的兽医工作者、非人类动物与
“同一健康”

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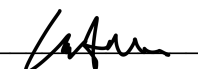
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More-Than-People’s Communes:
Veterinary Workers, Nonhuman Animals, and One Health in Mao-Era China

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Jongsik Christian Yi
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毛泽东时代中国的兽医工作者、非人类动物与“同一健康”

呈交的学位论文
by
Jongsik Christian Yi
to
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为部分满足要求
学位
哲学博士
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More-Than-People’s Communes:

Veterinary Workers, Nonhuman Animals, and One Health in Mao-Era China

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兽医工作者、非人类动物与毛泽东时代的“同一健康”

Abstract

Long before COVID-19, global health experts and citizens perceived China as a past and future epicenter of pandemic disease. The high density of its human and nonhuman populations, the prevalence of wet markets, and the relentless destruction of its ecology have been linked to the authoritarianism of its government and the apparent political submissiveness of its people. This dissertation historicizes this perception by examining the communal veterinary system during China’s Maoist period (1949–1976). I show the bottom-up efforts to preserve the vitality of humans, domestic animals, and the environment through the work and knowledge of local “people’s communes.” At the center of these initiatives were large numbers of local veterinary workers. Veterinarians, animal disease prevention workers, animal caretakers, and breeders undergirded the economic, intellectual, and moral order in which farm animals and veterinary expertise were regarded as common goods. I argue that prior to the post-1978 capitalist reforms in China, there were “more-than-people’s communes,” or places where diverse humans cared for, healed, and exploited nonhuman beings to weather the revolution’s radicalism and unpredictability.

This dissertation complicates the understandings of rural China, Maoist mass science, and human-animal relations. First and foremost, I ask a historiographical question: Despite famine, exploitation, and violence in rural China, how did local communities persevere and survive? Engaging with historians and social scientists who have studied the subsistence ethics, agency, and

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本论文使人们对中国农村、毛泽东时代的群众科学以及人与动物关系的理解复杂化。首先，我提出了一个史学问题： 尽管中国农村存在饥荒、剥削和暴力， 但当地社区是如何坚持和生存下来的？ 我与研究生存伦理、能动性和

forms of resistance of local individuals and communities, I claim that to fully appreciate such communal capacity for subsistence and well-being, we must recognize the contribution made by nonhuman animals. Local veterinary workers were at the forefront of enabling and sustaining the animal contributions to communal survival. They were by no means elite experts, but rather products of the ideal of mass science, or an anti-technocratic, popular democratic, and even “decolonial” vision of science. While bringing the cases of veterinary workers to the scholarly conversation about mass science, this dissertation reshapes the discussion about mass science by insisting that the ideal within grassroots veterinary spheres was enacted through not only human action but also by animal initiative. Foregrounding human-animal relations within more-than people’s communes, this dissertation does not insist that “animal agency” was equal to or even more significant than that of local state cadres, veterinarians, or peasants. Instead, it juxtaposes human and nonhuman actors whenever possible. In so doing, I do not erase the presence of diverse farm animals in the communes and I seek to denaturalize anthropocentric and eugenic aspects of the communal life.

我主张，为了充分认识这种社区的生存和福祉能力，我们必须承认非人类动物所做的贡献。当地兽医工作者站在最前沿，促成并维持了动物对社区生存的贡献。他们绝非精英专家，而是大众科学理想的产物，或者说是一种反技术官僚、人民民主甚至“去殖民化”的科学愿景。本论文将兽医工作者的案例引入关于大众科学的学术讨论，通过强调基层兽医领域中的理想不仅通过人类行动，而且通过动物的积极性来实施，从而重塑了关于大众科学的讨论。本论文在“超人民公社”中突出人与动物的关系，并不坚持“动物能动性”与地方国家干部、兽医或农民的能动性相等或甚至更重要。相反，它尽可能地并置了人类和非人类行为者。通过这样做，我没有抹去公社中各种农场动物的存在，我试图使公社生活中以人类为中心和优生学的方面去自然化。

Contents

Figures and Tables
Acknowledgements

Introduction

Maoist Communes and One Health
More-Than-People’s Communes
Veterinary Workers and Mass Science
Chapter Outline
Archives and Other Sources

Chapter One: Infectious Animal Diseases

Introduction
Collectives and Epizootics
Lockdown, Quarantine, and Disinfection
Mass Animal Vaccination and Mass Science
Producing Vaccines On site
The Choice between Treatment and Culling
Conclusion

Chapter Two: Science for the Commune

Introduction
The Earlier Encounters, 1949–1956
From Folk to Collective Veterinarians, 1956–1962
The Scientization of Folk Veterinary Knowledge
Making TCVM a Socialist Science
The Codification of TCVM, 1956–1962

目录

图表致谢

引言

毛泽东时代的公社与“同一健康”
超越人民公社
兽医工作者与大众科学
章节大纲
档案及其他来源

Chapter One: Infectious Animal Diseases

引言
集体与流行病封锁、隔离与消毒大规模动物疫苗接种与大规模科学生产疫苗现场治疗与扑杀的选择结论

第二章：公社的科学

引言早期遭遇，1949-1956年从民间到集体兽医，1956-1962年民间兽医知识的科学化使中兽医成为社会主义科学中兽医的法典化，1956-1962年

Chapter Three: Grassroots Experts

Introduction

Stratified Veterinary Education, 1958–1966

Enduring Disparities, 1960–1966

Upending Hierarchies, 1966–1976

The Veterinarian’s Way under High Maoism

Conclusion

Chapter Four: The Ideology of Care

Introduction

Collectivization, Animals, and Caretakers in the Mid-1950s

The Great Leap Forward and Swine Production, 1958–1962

The Ideology of Animal Care in the 1960s

Outliers in More-than-People’s Communes

Animal Care and Feminism

Conclusion

Chapter Five: Animal Eugenics

Introduction

Terms, Technologies, and Assumptions

Institutional Change

Black and White Pigs

Cattle Must Plow a Field

Gendered Local Knowledge of Equine Reproduction

Conclusion

第三章：基层专家

引言

S分层兽医教育，1958–1966

持续的差异，1960–1966

颠覆等级制度，1966–1976

高毛泽东时代兽医之道 结论

第四章：关怀的意识形态

引言

20世纪50年代中期的集体化、动物与饲养员 大跃进与生猪生产，1958-1962年 20世纪60年代动物护理的意识形态 超出人民公社的异类 动物护理与女权主义 结论

第五章：动物优生学

引言 术语、技术和假设 制度变革 黑白猪 牛必须耕地 马匹繁殖的性别化地方知识 结论

Epilogue

- A Village Veterinarian in the Post-Mao era
- Recognizing without Celebrating More-Than-People’s Communes

Bibliography

结语

- 后毛泽东时代的一位乡村兽医
- 承认而非赞美“超人公社”

参考书目

Figures and Tables

Figures

- Figure 1. Bovine and Swine Acupunctural Points.
- Figure 2. A 1967 Medical Receipt from Shangtuhai Commune Veterinary Station, Wuchuan County, Inner Mongolia.
- Figure 3. A Commune Veterinarian Treating Lambs in Mandoula Commune, Baotou, Inner Mongolia.
- Figure 4. The 1970s Certificates of Swine Vaccination.
- Figure 5. A Vaccine Producer Collecting Samples of Spleen and Lymph Nodes from a Rabbit.
- Figure 6. The Inaugural General Meeting of the ITCVM of CAAS on July 1, 1958
- Figure 7. Acupunctural Needles for Horses, Cattle, and Pigs.
- Figure 8. Meng Shiqing’s Map of Ovine Acupunctural Points.
- Figure 9. Adolescent Caretakers and Animals in Changning Commune, Yuci County, Shanxi Province, 1960.
- Figure 10. Xu Xuelin and Zhang Jinmei Standing by a Pregnant Sow in the Pigsty.
- Figure 11. Local Breeders in Yanheng Cattle Breeding Station, Sichuan.
- Figure 12. Visual Explanation on How to Conduct Artificial Insemination.
- Figure 13. The Roadmap to “Self-Reproduction within the Local Herds.”

Tables

- Table 1. Numbers and Incomes of Local Veterinarians in Gansu Province, 1964.
- Table 2. The Total Number of Pigs Raised in China, 1955–1977.
- Table 3. The Total Number of Large Livestock (Bovines, Equines, and Camels) Raised in China, 1955–1977.
- Table 4. The Knowledge about the Reproduction of Major Livestock.

图表

图

- 图 1. 牛和猪的针灸穴位。
- 图 2. 1967 年内蒙古乌川县上土海公社兽医站的医疗收据 ty,
- 图 3. 内蒙古包头市满都拉公社的一名公社兽医正在治疗羔羊。
- 图 4. 20 世纪 70 年代猪疫苗接种证书。
- 图 5. 疫苗生产商从兔子身上采集脾脏和淋巴结样本。
- 图 6. 中国农业科学院中兽医研究所于 1958 年 7 月 1 日召开的首次全体会议
- 图 7. 马、牛、猪针灸针。
- 图 8. 孟世清的绵羊针灸穴位图。
- 图 9. 1960 年山西省榆次县长宁公社的青少年饲养员和动物。
- 图 10. 徐学林和张金梅站在猪圈里一头怀孕的母猪旁。
- 图 11. 四川盐亭牛繁育站的当地育种员。
- 图 12. 人工授精操作的图解。
- 图 13. “本地畜群自繁自育”路线图。

表格

- 表 1. 1964 年甘肃省乡村兽医人数及收入。
- Table 2. The Total Number of Pigs Raised in China, 1955–1977.
- 表 3. 1955-1977 年中国饲养的大型牲畜（牛、马和骆驼）总数。
- Table 4. The Knowledge about the Reproduction of Major Livestock.

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东南亚科学与医学史在思想上令人震撼。我对后殖民/去殖民研究与科学技术研究之间交叉点的看法，在很大程度上得益于沃里克（Warwick）的真知灼见。即使他回到澳大利亚，悉尼和波士顿之间14小时的时差也从未成为问题。

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如果没有2019年和2021年我在中国进行的两轮实地考察期间所获得的帮助和支持，这个项目不可能完成。在北京大学，张大庆、张力、唐文佩、陈琦、苏晶晶、John Alekna、张萌、李玉荣、姜珊、庞静怡、陈雪阳、刘瑞塔、陈俊彦、金亚楠、娄玲、高子扬、莫晓聪、甘霖、崔云怡和李一璇让我把北大称为我的家。清华大学的陆懿丽、王成伟、李欣然、乔宇、戴碧云和臧龙凯，以及南开大学的于新中、陈思言和陈拓，都对我的工作表现出真正的兴趣，邀请我做讲座，或者请我吃美味的中国菜。在广西，我的

dear East China Normal University *shixiong* Li Yufeng helped me find and arrange interviews with former local veterinarians and caretakers such as “Mr. Chen,” “Mrs. Ling,” and “Mrs. Hong.” It was a great pleasure to give a guest lecture at Nanning Normal University and meet Yufeng’s students. In Langfang, Hebei, Wei *shushu* and Wang *shushu* willingly arranged my interview with a former commune veterinarian “Mr. Wen.” And my fieldwork would not have been nearly as productive without the mentorship and friendship that I enjoyed with admirable Chinese scholars and colleagues since 2014 including Dong Guoqiang, Yang Kuisong, Shen Zhihua, Han Gang, Zhang Jishun, Li Gongzhong, Xu Jin, Jiang Huajie, He Zhiming, Deng Guang, Zhao Songjie, and Yang Fang.

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华东师范大学的李玉峰师兄帮我找到了“陈先生”“凌女士”和“洪女士”等当地前兽医和护理人员，并安排了对他们的采访。我很荣幸能在南宁师范大学做客座讲座，并与玉峰的学生见面。在河北廊坊，魏叔叔和王叔叔欣然安排了我对前公社兽医“温先生”的采访。如果没有我自2014年以来与董国强、杨奎松、沈志华、韩钢、张济顺、李功忠、徐进、蒋华杰、何志明、邓光、赵松杰和杨芳等令人钦佩的中国学者和同事的指导和友谊，我的田野调查就不会如此富有成效。

我的博士研究得到了富布赖特外国学生项目、美国学术团体理事会、美国历史协会、亚洲研究协会、亨利·鲁斯基金会、哈佛大学费正清中国研究中心、哈佛大学亚洲中心、哈佛大学韦瑟海德国际事务中心、蒋经国基金会、埃舍里克-叶氏家族基金会以及哈佛大学科学史系的支持和资助。

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Mukharji, Megan Raby, Matt Rothwell, Aaron Sachs, Dagmar Schäfer, Sigrid Schmalzer, Grace Shen, Wayne Soon, Kristin Stapleton, William Summers, Eric Tagliacozzo, Zuoyue Wang, and Taomo Zhou. Sharing the historicities and concerns of living in and beyond contemporary South Korea, teachers at my alma mater Korea University and other institutions in Seoul, Daejeon, and Jeonju have been my trailblazers. Among them include Insun Yu, Seonmin Kim, Hun Lee, Hanung Kim, Jun Hyung Chae, Hyong Jin Yoon, Woon-Ok Yeom, Yong-Jin Hong, Donghyuk Kim, Whanbhum Song, Soon-il Chong, Chihyung Jeon, Hyungsub Choi, Tae-ho Kim, Manyong Moon, Bohm Soon Park, Jongtae Lim, Doogab Yi, Scott Knowles, Jung Lee, and especially Sang-Soo Park who first introduced the importance and pleasure of studying China to me.

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Mukharji, Megan Raby, Matt Rothwell, Aaron Sachs, Dagmar Schäfer, Sigrid Schmalzer, Grace Shen, Wayne Soon, Kristin Stapleton, William Summers, Eric Tagliacozzo, Zuoyue Wang 和 Taomo Zhou。我的母校韩国大学以及首尔、大田和全州其他机构的老师们，与我分享了生活在当代韩国及以外地区的历史和担忧，他们是我的开拓者。其中包括 Insun Yu、Seonmin Kim、Hun Lee、Hanung Kim、Jun Hyung Chae、Hyong Jin Yoon、Woon-Ok Yeom、Yong-Jin Hong、Donghyuk Kim、Whanbhum Song、Soon-il Chong、Chihyung Jeon、Hyungsub Choi、Tae-ho Kim、Manyong Moon、Bohm Soon Park、Jongtae Lim、Doogab Yi、Scott Knowles、Jung Lee，尤其是 Sang-Soo Park，他首次向我介绍了研究中国的重要性和乐趣。

我是一个渴望友谊的人。我很幸运在马萨诸塞州剑桥市拥有好朋友和同学。我会想念哈佛大学科学中心的“鱼缸”、469室和252室，在那里我与Erik Baker、Lucian Bessmer、Frank Blibo、Brad Bolman、Alyssa Botelho、Emma Broder、Anna Christensen、Hannah Conway、Taylor Coplen、Iman Darwish、Max Ehrenfreund、Erin Freedman、Jonathan Galka、Nayanika Ghosh、Aaron Gluck-Thaler、Shireen Hamza、Kit Heintzman、Jordan Howell、Kiran Kumbhar、Michelle LaBonte、Colleen Lanier-Christensen、Oliver Lazarus、Vi Le、Gustave Lester、Angélica Márquez-Osuna、Jamie Marsella、Francis Newman、Tiffany Nichols、Udodiri Okwandu、Meg Perret、Kat Poje、Sara Press、Chris Rudeen、Tasha Schoenstein、Caleb Shelburne、Beatrice Steinert、Salina Suri、Alexis Turner、Aaron Van Neste、Gili Vidan、Molly Walker、Tina Wei和Che Yeun一起呼吸。如果我对亚洲的文化和历史知识有所了解，那都来自于我与Kanghun Ahn、Aurélien Bellucci、Wenjiao Cai、Jannis Jizhou Chen、Mark Chen、Ann Da In Choi、Jeonghun Choi、Aniket De、Yuting Dong、Ruodi Duan、Peter Dziedzic、Peng Hai、Sujin Elisa Han、John Hayashi、Kangni Huang、Dingru Huang、Hyeok Hweon Kang、Juhee Kang的对话，

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Last but not least, I send my sincere love to my family. My parents Hyun-hye Yang and Gyu-tae Yi have been fully sympathetic to every stage of doctoral studies as they went through it themselves many years ago in Tokyo. I know they are proud of me. I want them to know that I am proud of them too. Visiting my in-laws in Wonju is the best way to refresh myself. My parents-in-

Sung, Donghyun Woo, Taemin Woo, Dora Yao, Dongkyu Yeom, Jennifer Yip, Sangwoon Yoo, Yaming You 和 Jinghong Zhang, 让我意识到在崎岖的学术道路上我并不孤单。

<style id='1'>Many friends in the Boston/Cambridge area, Seoul, Beijing, Shanghai, and Chengdu who are not interested in local veterinary workers in China have been the guardians of my work-life balance. Those who are in this sense essential in completing my doctoral studies are Jinwon Chung, Dae-Young Lee, Jubee Sohn, Jaewook Lee, Seongmin Kim, Jesung Lee, Yong Hwan Kim, Daekyum Kim, Ashley Kim, Ohjae Gowen, Seokweon Jeon, Dorothy Ahn, Haneui Bae, Kimberly Ohn, Gangsim Eom, Johannes Makar, Vivien Chung, Yunhyae Kim, Sun Young Park, Inhwan Oh, Mona Mun, Donghwan Kim, Jeongmoon Hahm, Yona Yoeun Chung, Seongwon Kim, Ellie Jungmin Han, Hanseul Kim, Deogwoon Anna Kim, JungEun Goo, Nak-seung Patrick Hyun, Heedoo Yang, Haedo Cho, Ukjin Kwon, GiHun Choi, Cui Huishan, Cui Huizhen, Suk Young Lee, Gahyun Lee, Chae Un Yim, Ha Young Roh, Byeongjun Kim, Dong Ah Choi, Woong Rae Son, KyungHee Lee, Kyeongwan Kang, Hena Lee, Yein Kim, Jiseon Choe, Seokwon Gong, Hyunkyum Kim, Ji Ye Myung, Seoyoung Oh, Daeun Choo, Sunhee Kyung, Heejin Mok, Wani Yu, Leo Kim (the father of Si-yoon—congrats!), Myoungjae Kim, Hyowon Chae, Na Hyun Kim, Lin Luo, Yu Youhan, and Yan Cui. I also must not leave out my generous editors, Jena Gaines and Andy Daily. Jena and Andy not only have helped me make this dissertation more readable, but also rooted for me as I ran the marathon of writing.</style>我在波士顿/剑桥地区、首尔、北京、上海和成都的许多朋友，他们对中国的当地兽医工作者不感兴趣，但他们一直是我工作与生活平衡的守护者。在某种意义上，对完成我的博士学业至关重要的人包括：Jinwon Chung、Dae-Young Lee、Jubee Sohn、Jaewook Lee、Seongmin Kim、Jesung Lee、Yong Hwan Kim、Daekyum Kim、Ashley Kim、Ohjae Gowen、Seokweon Jeon、Dorothy Ahn、Haneui Bae、Kimberly Ohn、Gangsim Eom、Johannes Makar、Vivien Chung、Yunhyae Kim、Sun Young Park、Inhwan Oh、Mona Mun、Donghwan Kim、Jeongmoon Hahm、Yona Yoeun Chung、Seongwon Kim、Ellie Jungmin Han、Hanseul Kim、Deogwoon Anna Kim、JungEun Goo、Nak-seung Patrick Hyun、Heedoo Yang、Haedo Cho、Ukjin Kwon、GiHun Choi、Cui Huishan、Cui Huizhen、Suk Young Lee、Gahyun Lee、Chae Un Yim、Ha Young Roh、Byeongjun Kim、Dong Ah Choi、Woong Rae Son、KyungHee Lee、Kyeongwan Kang、Hena Lee、Yein Kim、Jiseon Choe、Seokwon Gong、Hyunkyum Kim、Ji Ye Myung、Seoyoung Oh、Daeun Choo、Sunhee Kyung、Heejin Mok、Wani Yu、Leo Kim（Si-yoon 的父亲——恭喜！）、Myoungjae Kim、Hyowon Chae、Na Hyun Kim、Lin Luo、Yu Youhan 和 Yan Cui。我还要感谢我的慷慨编辑 Jena Gaines 和 Andy Daily。Jena 和 Andy 不仅帮助我使这篇论文更具可读性，还在我撰写论文的马拉松过程中为我加油鼓劲。

最后但同样重要的是，我向我的家人致以真挚的爱。我的父母 Hyun-hye Yang 和 Gyu-tae Yi 对我博士学习的每个阶段都给予了充分的理解，因为他们多年前在东京也经历过。我知道他们为我感到骄傲。我想让他们知道，我也为他们感到骄傲。去原州拜访我的岳父母是最好的放松方式。我的岳父母——

law, Younghye Park and Wonbae Kim, sister-in-law, Jiwon Kim, and brother-in-law, Jonghyuk Kim, have kindly helped me discover my hidden talent for agriculture on our family farm. I look forward to harvesting peaches, peanuts, walnuts, chilis, and the like with them for many years to come. My paternal grandmother Kyeong-sim Martha Kim and deceased grandfather Papal Knight Sang-rae Francesco Yi always encouraged me to live up to my baptismal name, “Christian.” I interpret this as a message to live a life full of love, tolerance, and compassion. My maternal grandmother Jong-ja Kim and deceased grandfather Hoe-ryeon Paul Yang would be pleased to hear that their eldest grandson has done his job in America where they had hoped to travel together. My uncle and aunts, Gyusoo Yi, Yeounsuk Lee, deceased Hyunkyung Yang, Hyunjoo Yang, and Hyun Yang, have been deeply taking care of me. My canine siblings Green and Dasom might not be impressed by their brother having earned a doctoral degree. They are impressed with me just as I am, like all companion animals are. I wish I had a language to let Green and Dasom know that their companionship has shaped my intellectual journey. My wife, Yewon, has allowed me to get into intellectual adventures literally across the ocean. I will always remember that anything that I achieve, I owe to her support and patience. Thank you so much, Yewon, for having been by my side. I shall pay back your love with mine.

法律, Younghye Park 和 Wonbae Kim, 我的嫂子 Jiwon Kim 和我的姐夫 Jonghyuk Kim, 他们都曾好心地帮助我在我们的家庭农场中发现我隐藏的农业天赋。我期待在未来的许多年里与他们一起收获桃子、花生、核桃、辣椒等等。我的祖母 Kyeong-sim Martha Kim 和已故的祖父 Papal Knight Sang-rae Francesco Yi 总是鼓励我活出我的洗礼名 “Christian” 的意义。我将此解读为一种充满爱、宽容和同情的生活信息。我的外祖母 Jong-ja Kim 和已故的外祖父 Hoe-ryeon Paul Yang 如果知道他们的大外孙在美国完成了他的工作, 他们一定会很高兴, 因为他们曾希望一起去美国旅行。我的叔叔和阿姨们, Gyusoo Yi、Yeounsuk Lee、已故的 Hyunkyung Yang、Hyunjoo Yang 和 Hyun Yang, 一直都在深深地关爱着我。我的狗狗兄弟姐妹 Green 和 Dasom 可能不会因为它们的兄弟获得了博士学位而感到印象深刻。它们对我的印象就像我一样, 就像所有伴侣动物一样。我希望我有一种语言能让 Green 和 Dasom 知道它们的陪伴塑造了我的智力旅程。我的妻子 Yewon 允许我进行跨越海洋的智力冒险。我将永远记住, 我所取得的任何成就都归功于她的支持和耐心。非常感谢你, Yewon, 一直在我身边。我将用我的爱回报你的爱。

Introduction

This dissertation is a product of our era of ecological imbalance. My earlier intellectual journey was preoccupied with the current planetary crisis which has many names, ranging from global warming to the Anthropocene to the sixth mass extinction. Whatever we call it, as a reader of Donna

Haraway, Bruno Latour, Anna Tsing, Immanuel Wallerstein, Walter Mignolo, and John Bellamy

Foster, among others, I recognize that it is rooted in a quincentenary *longue durée* that has witnessed the rise of global capitalism and colonial legacies.¹ In addition, my training in the disciplines of the history of science and science and technology studies (STS) equipped me to depart from notions of

science and technology as universal, neutral solutions to social problems.² In the vague hope of

redirecting the broader historical trajectory, I was eager to look for inspiration from the rural,

socialist, and non-Western worlds—something that seemed *different*. Collectivized agricultural

communities in Maoist China thereby became my focal point. Devouring primary and secondary

material that seemed relevant, a bit of romanticization, or in Gayatri Spivak’s words, “strategic

essentialism,” was unmistakable in my earlier approach to rural China.³

¹ Donna Haraway, *Staying with the Trouble: Making Kin in the Chtulucene* (Durham: Duke University Press, 2016); Bruno Latour, *Facing Gaia: Eight Lectures on the New Climatic Regime* (Cambridge: Polity, 2017); Anna Tsing, *The Mushroom at the End of the World: On the Possibility of Life in Capitalist Ruins* (Princeton: Princeton University Press, 2015); Immanuel Wallerstein, *World-systems Analysis: An Introduction* (Durham: Duke University Press, 2004); Walter Mignolo, *The Darker Side of Western Modernity: Global Futures, Decolonial Options* (Durham: Duke University Press, 2011); John Bellamy Foster, *The Ecological Rift: Capitalism’s War on the Earth* (New York: Monthly Review Press, 2010).

² Warwick Anderson, “Postcolonial Technoscience,” *Social Studies of Science* 32.5 (October 2002): 643–658; Lorraine Daston and Peter Galison, *Objectivity* (New York: Zone Books, 2007); Naomi Oreskes and Erik Conway, *Merchants of Doubt: How a Handful of Scientists Obscured the Truth on Issues from Tobacco Smoke to Global Warming* (New York: Bloomsbury, 2011); Warwick Anderson, “Remembering the Spread of Western Science,” *Historical Records of Australian Science* 29.2 (July 2018): 73–81.

³ Elisabeth Eide, “Strategic Essentialism,” *The Wiley Blackwell Encyclopedia of Gender and Sexuality Studies* (Singapore: John Wiley & Sons, 2016): 1–2. <https://doi.org/10.1002/9781118663219.wbegss554>.

引言

本论文是生态失衡时代的产物。我早期的学术生涯一直关注当前的全球危机，这场危机有许多名称，从全球变暖到人类世再到第六次大规模灭绝。无论我们称之为“什么”，作为唐娜·哈拉维、布鲁诺·拉图尔、安娜·青、伊曼纽尔·沃勒斯坦、沃尔特·米尼奥洛和约翰·贝拉米·福斯特等人的读者，我认识到它根植于长达五百年的漫长时期，见证了全球资本主义和殖民遗产的兴起。¹此外，我在科学史以及科学技术研究（STS）学科方面的训练使我能够摆脱将科学技术视为解决社会问题的普遍、中立方案的概念。²为了模糊地希望改变更广泛的历史轨迹，我渴望从农村、社会主义和非西方世界中寻找灵感——一些看起来与众不同的东西。因此，毛泽东时代中国的集体化农业社区成为我的焦点。在阅读似乎相关的原始和次要材料时，我早期研究中国农村的方法中，一点浪漫主义，或者用伽耶特里·斯皮瓦克的话说，“策略性本质主义”，是显而易见的。³

¹ 唐娜·哈拉维，《与麻烦共存：在克苏鲁世中结亲》（达勒姆：杜克大学出版社，2016）；布鲁诺·拉图尔，《面对盖亚：新气候体制的八次讲座》（剑桥：政体出版社，2017）；安娜·秦，《世界尽头的蘑菇：资本主义废墟中生命的可能》（普林斯顿：普林斯顿大学出版社，2015）；伊曼纽尔·沃勒斯坦，《世界体系分析：导论》（达勒姆：杜克大学出版社，2004）；沃尔特·米尼奥洛，《西方现代性的阴暗面：全球未来，去殖民选择》（达勒姆：杜克大学出版社，2011）；约翰·贝拉米·福斯特，《生态裂痕：资本主义对地球战争》（纽约：每月评论出版社，2010）。

² Warwick Anderson, “Postcolonial Technoscience,” *Social Studies of Science* 32.5 (October 2002): 643–658; Lorraine Daston and Peter Galison, *Objectivity* (New York: Zone Books, 2007); Naomi Oreskes and Erik Conway, *Merchants of Doubt: How a Handful of Scientists Obscured the Truth on Issues from Tobacco Smoke to Global Warming* (New York: Bloomsbury, 2011); Warwick Anderson, “Remembering the Spread of Western Science,” *Historical Records of Australian Science* 29.2 (July 2018): 73–81.

³ 伊丽莎白·艾德，“策略本质主义”，《威利·布莱克威尔性别与性研究百科全书》（新加坡：约翰·威利父子出版社，2016）：1-2. <https://doi.org/10.1002/9781118663219.wbegss554>。

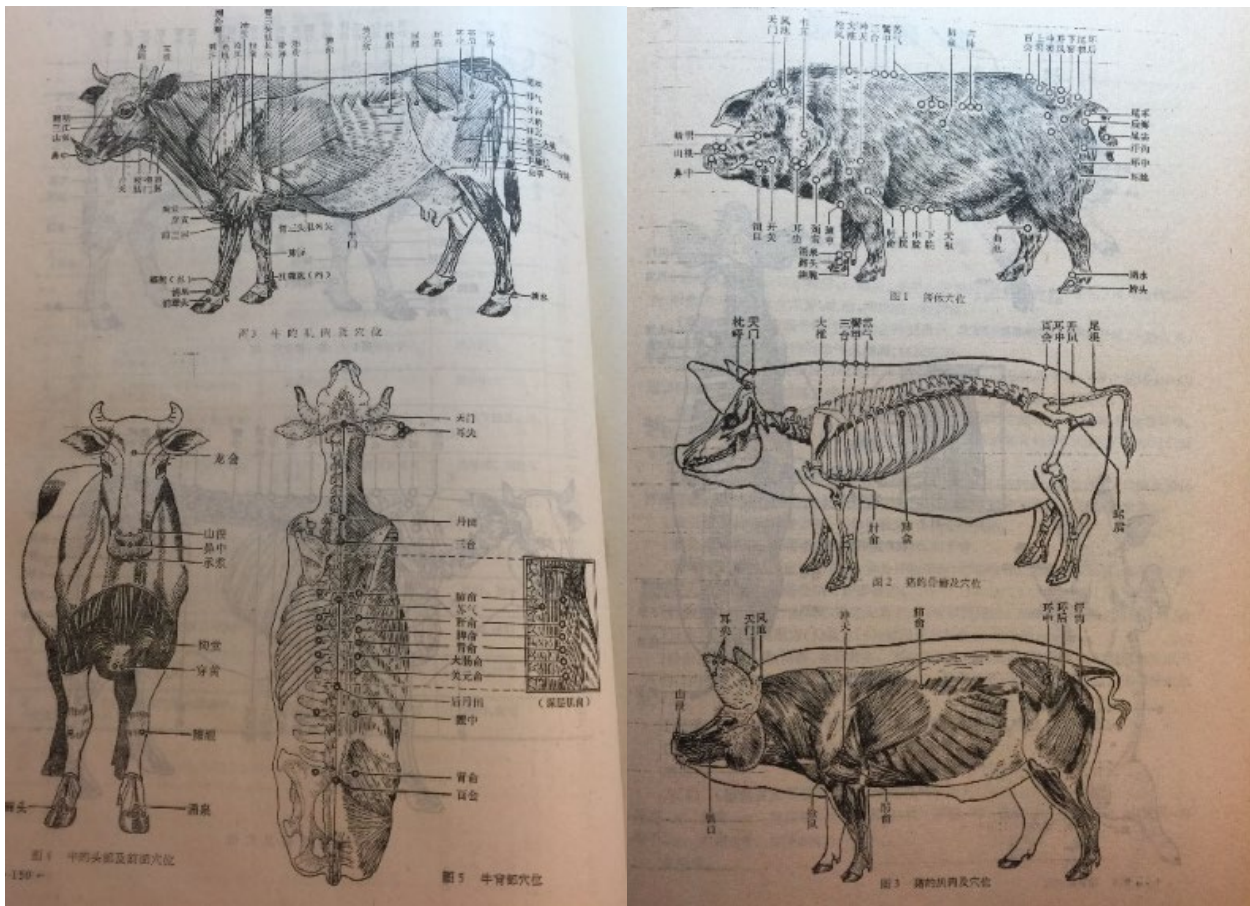


Figure 1. Bovine and swine acupunctural points. Source: Quanguo zhongshouyi jingyan xuanbian bianshenzu 全国中兽医经验选编编审组, *Quanguo zhongshouyi jingyan xuanbian* 全国中兽医经验选编 (Beijing: Kexue chubanshe, 1977), 53 and 150.

I first came across the 1977 book, *Nationwide Selection of Traditional Chinese Veterinary Experience*, in early 2019 at the Harvard-Yenching Library.⁴ Born in Japan and raised in South Korea, I was well-aware of modern Chinese medicine, or Traditional Chinese Medicine (TCM), and its

⁴ Quanguo zhongshouyi jingyan xuanbian bianshenzu 全国中兽医经验选编编审组, *Quanguo zhongshouyi jingyan xuanbian* 全国中兽医经验选编 (Beijing: Kexue chubanshe, 1977).

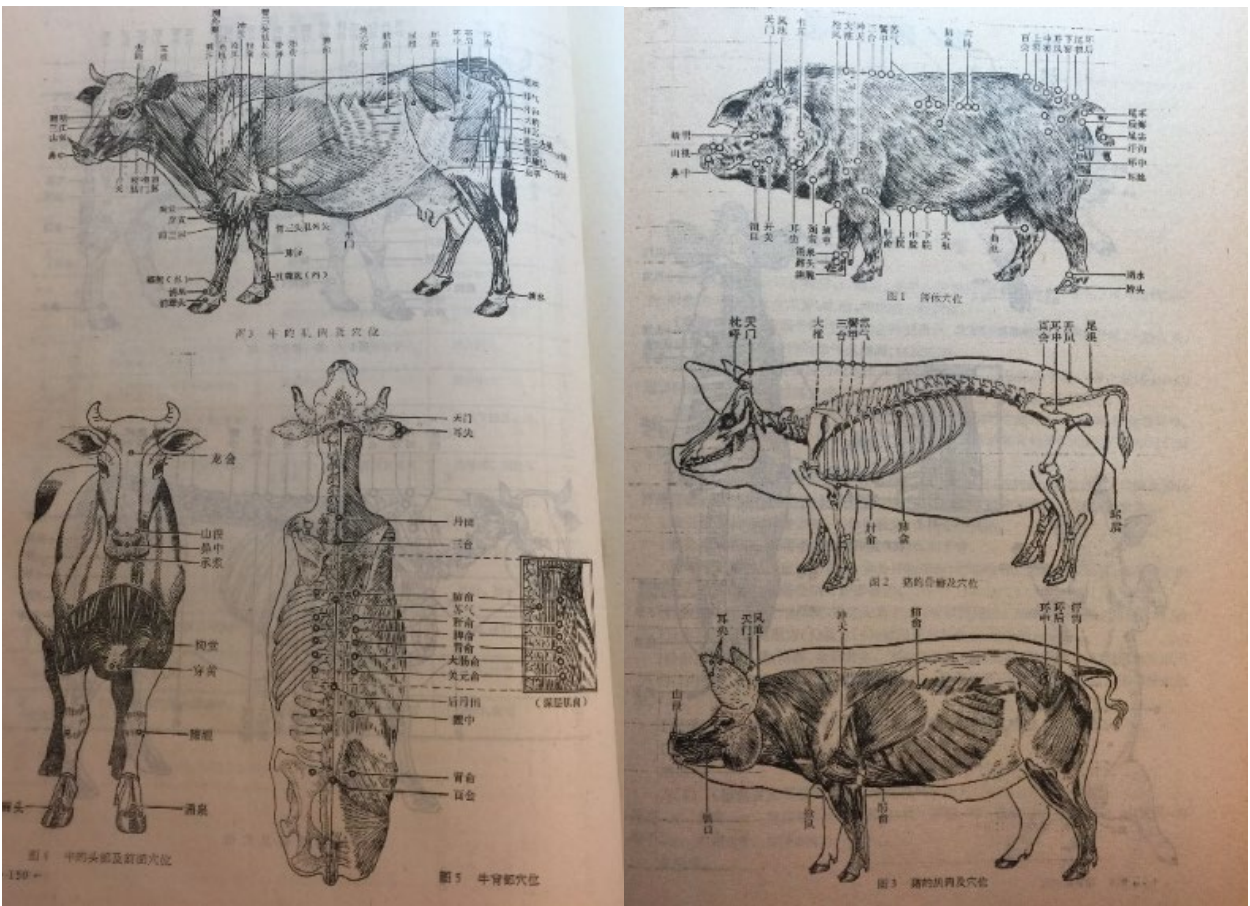


图 1。牛和猪的针灸穴位。来源：全国中兽医经验选编编审组，《全国中兽医经验选编》（北京：科学出版社，1977），第53页和第150页。

2019年初，我在哈佛燕京图书馆首次接触到1977年的《全国中兽医经验选编》一书。⁴我出生于日本，在韩国长大，对现代中医，即传统中医（TCM）及其

⁴ 全国中兽医经验选编编审组，《全国中兽医经验选编》（北京：科学出版社，1977）。

analogues in other East Asian countries. But I had never heard about its veterinary equivalent until the fortuitous encounter with this book. As soon as I opened it, I was fascinated by the elaborate illustrations of the distribution of animal acupunctural points (Figure 1). Based on my satisfying experience in traditional Korean medical clinics, I was not surprised by the fact that Chinese people tried to treat animal diseases with acupuncture and medicinal herbs. What surprised me was how ignorant I had been about animals, their doctors, and the labor and knowledge to keep nonhuman commune members healthy. I soon realized that I was not alone. The rich scholarship on twentieth-century rural China had few insights into the ways in which humans and farm animals lived together.⁵ Immediately identifying some keywords—care and healing; multispecies companionship and interdependence; and exploitation and killing—I felt in my bones that my dissertation would be about this world of communal veterinary workers and nonhuman animals.

Maoist Communes and One Health

This dissertation is also the product of COVID-19 era. I was able to travel to China in the summer of 2019 and the first half of 2021. I collected tens of thousands of archival documents and other forms of primary sources. I expected that these materials would open a door to shed light on human-animal relations in the communes and allow me to present an appealing alternative to so-called Modernity. Between these two fieldwork trips, however, the pandemic began. Global health experts and citizens across the globe posited that the virus SARS-CoV-2 first emerged in the

⁵ One of the few exceptions is Peter Braden’s Ph.D. dissertation on the bovine experiences of the Chinese Civil War and Revolution. Peter W. Braden, “Serve the People: Bovine Experiences in China’s Civil War and Revolution, 1935–1961,” Ph.D. diss., University of California at San Diego, 2020. On nonhuman animals in pre-1911 Chinese history, see Roel Sterckx, Martina Siebert, and Dagmar Schäfer, eds., *Animals Through Chinese History: Earliest Times to 1911* (Cambridge: Cambridge University Press, 2018).

在其他东亚国家也有类似的情况。但我从未听说过它的兽医等效物，直到偶然遇到这本书。我一打开它，就被动物针灸穴位分布的精美插图所吸引（图1）。根据我在韩国传统医疗诊所的满意经验，我对中国人试图用针灸和草药治疗动物疾病的事实并不感到惊讶。令我惊讶的是，我对动物、它们的医生以及保持非人类公社成员健康的劳动和知识是多么无知。我很快意识到我不是一个人。关于20世纪中国农村的丰富学术研究很少深入探讨人类和农场动物如何共同生活。⁵ 我立即识别出一些关键词——关怀与治愈；多物种的陪伴与相互依存；以及剥削与杀戮——我深切地感觉到我的论文将是关于这个由公共兽医工作者和非人类动物组成的世界。

毛泽东时代的公社与“同一健康”

这篇论文也是新冠疫情时代的产物。我得以在2019年夏天和2021年上半年前往中国。我收集了数万份档案文件和其他形式的原始资料。我曾期望这些材料能为阐明公社中的人与动物关系打开一扇门，并让我能提出一个有吸引力的、不同于所谓“现代性”的替代方案。然而，在这两次田野调查之间，疫情爆发了。全球健康专家和世界各地的公民都认为SARS-CoV-2病毒最初出现在

⁵ 少数例外之一是彼得·布雷登（Peter Braden）关于中国内战和革命中牛的经历的博士论文。彼得·W·布雷登（Peter W. Braden），《为人民服务：1935-1961年中国内战和革命中的牛的经历》（Serve the People: Bovine Experiences in China’s Civil War and Revolution, 1935–1961），博士论文，加州大学圣地亚哥分校，2020年。关于1911年以前中国历史上的非人类动物，参见罗尔·斯特克克斯（Roel Sterckx）、马丁娜·西伯特（Martina Siebert）和达格玛·舍费尔（Dagmar Schäfer）主编的《中国历史上的动物：最早时期至1911年》（Animals Through Chinese History: Earliest Times to 1911）（剑桥：剑桥大学出版社，2018年）。

complex of seafood markets in the city of Wuhan, Hubei Province. More specialists soon followed with many other hypotheses and research results about the animal origin of the virus and theories that bats or pangolins were the possible medium of transspecies transmission.⁶ It did not take long to see that if I were to discuss China and animals, I would not be able to avoid relating them to the current pandemic more specifically.

Some scientists and scholars argue that COVID-19 is the “disease of the Anthropocene” and for good reason. Population growth and the intensification of industrialization on a global scale have enabled ever closer contact with wild animals and their environments, which provides more opportunities for diseases to pass between species. The lack of care for nonhuman beings on the planet has only accelerated this tendency.⁷ The ongoing COVID-19 pandemic is only one singular example of the results that the long-standing, larger crisis of ecological imbalance can bring about. Therefore, my initial association of the Anthropocenic questions with the topic of communal veterinary practices in Maoist China still seemed valid. But I needed to add an additional layer to frame my project more critically: One Health as a framework of reference for comprehensively understanding and effectively responding to transspecies pandemics.

The definition provided by the US Centers for Disease Control and Prevention (CDC) describes One Health as “an approach that recognizes that the health of people is closely connected

⁶ For example, see WHO Headquarters, “WHO-convened Global Study of the Origins of SARS-CoV-2” (March 30, 2021), accessed April 2, 2022. <https://www.who.int/emergencies/diseases/novel-coronavirus-2019/origins-of-the-virus>.

⁷ Renee Salas, James Shultz, and Caren Solomon, “The Climate Crisis and Covid-19: A Major Threat to the Pandemic Response,” *The New England Journal of Medicine* 383.11 (September 2020): 70; Cristina O’Callaghan-Gordo and Josep M. Antó, “COVID-19: The disease of the Anthropocene,” *Environmental Research* 187 (August 2020): 109683; Thomas Heyd, “Covid-19 and climate change in the times of the Anthropocene,” *The Anthropocene Review* 8.1 (April 2021): 21–36; Eva Horn, “Tipping Points: The Anthropocene and COVID-19,” in Gerard Delanty, ed., *Pandemics, Politics, and Society Critical Perspectives on the Covid-19 Crisis* (Berlin: De Gruyter, 2021), 123–138; Gala Argent, “Human-Animal Relationships and Welfare in the Anthropocene: Pandemics, Climate Change, and Other Disasters,” *Journal of Applied Animal Welfare Science* (March 2022), DOI: 10.1080/10888705.2022.2042299.

位于湖北省武汉市的海鲜市场。很快，更多专家提出了许多其他假设和研究结果，涉及病毒的动物起源以及蝙蝠或穿山甲可能是跨物种传播媒介的理论。⁶ 没过多久我就发现，如果我要讨论中国和动物，就无法避免将它们更具体地与当前的疫情联系起来。

一些科学家和学者认为，COVID-19是“人类世的疾病”，这是有充分理由的。全球范围内的人口增长和工业化加剧使得人类与野生动物及其环境的接触日益密切，这为疾病在物种间传播提供了更多机会。对地球上非人类生物的漠视只会加速这种趋势。⁷ 持续的COVID-19大流行只是生态失衡这一长期、更大危机可能带来结果的一个单一例子。因此，我最初将人类世问题与毛泽东时代中国的公共兽医实践主题联系起来仍然是有效的。但我需要增加一个额外的层面来更批判性地构建我的项目：将“同一个健康”作为全面理解和有效应对跨物种大流行的参考框架。

美国疾病控制与预防中心 (CDC) 提供的定义将“同一健康”描述为“一种认识到人类健康密切相关的方针”

⁶ 例如，参见世卫组织总部，“世卫组织召集的 SARS-CoV-2 溯源全球研究”（2021 年 3 月 30 日），2022 年 4 月 2 日访问。<https://www.who.int/emergencies/diseases/novel-coronavirus-2019/origins-of-the-virus>。

⁷ Renee Salas、James Shultz 和 Caren Solomon，“气候危机与 Covid-19：对大流行应对的重大威胁”，《新英格兰医学杂志》383.11（2020 年 9 月）：70；Cristina O’ Callaghan-Gordo 和 Josep M. Antó，“COVID-19：人类世的疾病”，《环境研究》187（2020 年 8 月）：109683；Thomas Heyd，“Covid-19 与人类世的气候变化”，《人类世评论》8.1（2021 年 4 月）：21–36；Eva Horn，“临界点：人类世与 COVID-19”，载于 Gerard Delanty（编），《大流行病、政治与社会：Covid-19 危机批判性视角》（柏林：De Gruyter，2021），123–138；Gala Argent，“人类世的人类与动物关系及福利：大流行病、气候变化及其他灾害”，《应用动物福利科学杂志》（2022 年 3 月），DOI: 10.1080/10888705.2022.2042299。

to the health of animals and our shared environment.”⁸ One Health, like many other ideas and ideals, did not come into being out of the blue. The history of the connections and disconnections between human and animal health in Europe and North America since the early modern period has influenced the emergence of the terms “One Medicine” or “One Health, One World” in the second half of the twentieth century.⁹ More historians of medicine and veterinary medicine have become interested in tracing how the particular historical trajectory of the West gradually led to the notion of One Health.¹⁰

However, it is unclear how China might be situated in the Western-centric history of One Health, even though health specialists have perceived China as an epicenter of transspecies pandemics long before COVID-19.¹¹ The anthropologist Lyle Fearnley examines how the idea of China as a pandemic epicenter was constructed at the turn of the twenty-first century. China carried out reforms and opened its borders to international scientific exchange after the late 1970s. This post-socialist breakthrough enabled Western virologists and spatial ecologists access to Southern China, pinpointing the region as the epicenter of highly pathogenic avian influenza (HPAI)

⁸ The US Centers for Disease Control and Prevention, “One Health Basics,” accessed April 2, 2022, <https://www.cdc.gov/onehealth/basics/index.html>.

⁹ J. Zinsstag, E. Schelling, D. Waltner-Toews, and M. Tanner, “From “One Medicine” to “One Health” and Systemic Approaches to Health and Well-being,” *Preventive Veterinary Medicine* 101.3 (September 2011): 148–156; Ronald M. Atlas and Stanley Maloy, eds., *One Health: People, Animals, and the Environment* (Washington, DC: ASM Press, 2014).

¹⁰ For example, Joanna Swabe, *Animals, Disease and Human Society: Human-Animal Relations and the Rise of Veterinary Medicine* (London: Routledge, 1999); Abigail Woods, Michael Bresalier, Angela Cassidy, and Rachel Mason Dentinger, *Animals and the Shaping of Modern Medicine: One Health and Its Histories* (London: Palgrave MacMillan, 2018).

¹¹ On the public health and more human-specific pandemics in modern China, see: Ruth Rogaski, *Hygienic Modernity: Meanings of Health and Disease in Treaty-Port China* (Berkeley: University of California Press, 2004); Miriam Gross, *Farewell to the God of Plague: Chairman Mao’s Campaign to Deworm China* (Berkeley: University of California Press, 2016); Mary Augusta Brazelton, *Mass Vaccination: Citizens’ Bodies and State Power in Modern China* (Ithaca: Cornell University Press, 2019); Xun Zhou, *The People’s Health: Health Intervention and Delivery in Mao’s China, 1949–1983* (Montreal: McGill-Queen’s University Press, 2020); Xiaoping Fang, *China and the Cholera Pandemic: Restructuring Society Under Mao* (Pittsburgh: University of Pittsburgh Press, 2021).

对动物健康和我们一起环境的关注。”⁸ “同一健康”与其他许多思想和理念一样，并非凭空出现。自近代早期以来，欧洲和北美人类健康与动物健康之间联系与脱节的历史，影响了20世纪下半叶“同一医学”或“同一健康，同一世界”等术语的出现。⁹越来越多的医学史和兽医学史学家开始关注西方特定的历史轨迹如何逐渐促成了“同一健康”的概念。¹⁰

然而，尽管健康专家早在COVID-19之前就已将中国视为跨物种大流行的中心，但中国在以西方为中心的“同一健康”历史中如何定位尚不明确。¹¹ 人类学家莱尔·费恩利（Lyle Fearnley）研究了21世纪之交中国作为大流行病中心的观念是如何形成的。20世纪70年代末，中国进行了改革并向国际科学交流开放了边境。这种后社会主义的突破使西方病毒学家和空间生态学家得以进入中国南方，并将该地区确定为高致病性禽流感（HPAI）的中心。

⁸ 美国疾病控制与预防中心，“一体化健康基础知识”，2022年4月2日访问，<https://www.cdc.gov/onehealth/basics/index.html>。

⁹ J. Zinsstag, E. Schelling, D. Waltner-Toews 和 M. Tanner，“从‘一体化医学’到‘一体化健康’以及健康和福祉的系统方法”，《预防兽医学》101.3（2011年9月）：148-156；Ronald M. Atlas 和 Stanley Maloy，编著，《一体化健康：人类、动物与环境》（华盛顿特区：ASM 出版社，2014年）。

¹⁰ For example, Joanna Swabe, *Animals, Disease and Human Society: Human-Animal Relations and the Rise of Veterinary Medicine* (London: Routledge, 1999); Abigail Woods, Michael Bresalier, Angela Cassidy, and Rachel Mason Dentinger, *Animals and the Shaping of Modern Medicine: One Health and Its Histories* (London: Palgrave MacMillan, 2018).

¹¹ On the public health and more human-specific pandemics in modern China, see: Ruth Rogaski, *Hygienic Modernity: Meanings of Health and Disease in Treaty-Port China* (Berkeley: University of California Press, 2004); Miriam Gross, *Farewell to the God of Plague: Chairman Mao’s Campaign to Deworm China* (Berkeley: University of California Press, 2016); Mary Augusta Brazelton, *Mass Vaccination: Citizens’ Bodies and State Power in Modern China* (Ithaca: Cornell University Press, 2019); Xun Zhou, *The People’s Health: Health Intervention and Delivery in Mao’s China, 1949–1983* (Montreal: McGill-Queen’s University Press, 2020); Xiaoping Fang, *China and the Cholera Pandemic: Restructuring Society Under Mao* (Pittsburgh: University of Pittsburgh Press, 2021).

pandemics. However, Fearnley’s trailblazing monograph barely touches on the prehistory: How did Chinese scientists, physicians, veterinarians, and peasants who lived and worked in close proximity to nonhuman animals in the preceding Maoist period understand and cope with the question of animal health and its impact on human well-being?¹²

The vacuum in the scholarly understanding of the historical relationship between China and One Health seems to have been filled by moral condemnations of the country. Facts and values about the People’s Republic of China (PRC), based both on well-grounded observations and Cold War-minded biases, are entangled. The high density of its human and nonhuman populations, the prevalence of wet markets, the relentless destruction of its ecology, and the increased use of animal-based drugs in TCM therapeutics have been linked to the authoritarianism of its government and the political submissiveness of its people.¹³ These factors not only continue to strengthen the idea of China as the pandemic epicenter, but also fuel anti-Asian and anti-Chinese racism.¹⁴

All these views of China and its relation to pandemics and global health are not necessarily wrong. However, they do need to be historicized. In an effort to intervene in these conversations, my dissertation complicates such observations by examining the communal veterinary system in the Maoist period (1949–1976). I show that there were bottom-up efforts to preserve the vitality of

¹² Lyle Fearnley, *Virulent Zones: Animal Disease and Global Health at China’s Pandemic Epicenter* (Durham: Duke University Press, 2020); For ethnographic research on the health and economic values of poultry in contemporary Vietnam, see Natalie Porter, *Viral Economies: Bird Flu Experiments in Vietnam* (Chicago: University of Chicago Press, 2019).

¹³ Liz Chee examines the historical development of the incorporation of animal parts and tissue to both state-sanctioned and more popular forms of Chinese medicine, or “faunal medicalization.” Liz P. Y. Chee, *Mao’s Bestiary: Medical Animals and Modern China* (Durham: Duke University Press, 2021).

¹⁴ Xiao Tan, Rennie Lee, and Leah Ruppanner, “Profiling Racial Prejudice During COVID-19: Who Exhibits Anti-Asian Sentiment in Australia and the United States?” *The Australian Journal of Social Issues* 56.4 (December 2021): 464–484; Rachell Sanchez-Rivera, “The Legacies of ‘Race’ Science, Anti-Chinese Racism, and COVID-19 in Mexico,” *Bulletin of Latin American Research* 39 (December 2020): 35–38.

大流行病。然而，费恩利开创性的专著几乎没有触及史前史：在之前的毛泽东时代，与非人类动物近距离生活和工作的中国科学家、医生、兽医和农民是如何理解和应对动物健康及其对人类福祉影响的问题的？¹²

学术界对中国与“同一健康”之间历史关系的理解空白似乎已被对中国的道德谴责所填补。关于中华人民共和国（PRC）的事实和价值观，既基于有充分根据的观察，也基于冷战思维的偏见，两者纠缠不清。其人类和非人类人口的高密度、活体动物市场的普遍存在、生态的无情破坏以及中医药疗法中动物源药物使用量的增加，都与政府的威权主义和人民的政治顺从性联系在一起。¹³ 这些因素不仅继续强化了中国是疫情中心的观念，也助长了反亚洲和反华种族主义。¹⁴

所有这些关于中国及其与大流行病和全球健康关系的观点不一定是错误的。然而，它们确实需要被历史化。为了介入这些对话，我的论文通过考察毛泽东时期（1949-1976）的公社兽医系统，使这些观察变得复杂。我表明，当时存在自下而上的努力来保护

¹² Lyle Fearnley, *Virulent Zones: Animal Disease and Global Health at China’s Pandemic Epicenter* (Durham: Duke University Press, 2020); 关于当代越南家禽健康和经济价值的民族志研究，参见 Natalie Porter, *Viral Economies: Bird Flu Experiments in Vietnam* (Chicago: University of Chicago Press, 2019)。

¹³ Liz Chee 考察了将动物器官和组织纳入国家认可的 a 以及更受欢迎的中医形式，即“动物医学化”的历史发展。Liz P. Y. Chee, *Mao’s Bestiary: Medical Animals and Modern China*（达勒姆：杜克大学出版社，2021 年）。

¹⁴ Xiao Tan, Rennie Lee, and Leah Ruppanner, “Profiling Racial Prejudice During COVID-19: Who Exhibits Anti-Asian Sentiment in Australia and the United States?” *The Australian Journal of Social Issues* 56.4 (December 2021): 464–484; Rachell Sanchez-Rivera, “The Legacies of ‘Race’ Science, Anti-Chinese Racism, and COVID-19 in Mexico,” *Bulletin of Latin American Research* 39 (December 2020): 35–38.

humans, domestic animals, and the environment through the work and knowledge of local “people’s communes.” At the center of these initiatives were local veterinary workers. Thousands of veterinarians, animal caretakers, and breeders struggled to undergird the economic, intellectual, and moral order in which farm animals and veterinary expertise were regarded as common goods to be shared and cared for communally. Although the knowledge and labor of these veterinary workers in the communes were not justified or promoted in terms of microbiological or ecological languages, their deep commitment to the communal well-being of diverse human and nonhuman beings was in line with the contemporary ideals of One Health. The veterinary practices in the Chinese communes can help us ponder how we might live with our fellow constituents, in all their forms, on Earth as an ecological commune.

More-Than-People’s Communes

At the heart of the dissertation project is my effort to reconceptualize collectivized communities in Maoist rural China into “more-than-people’s communes” (多物种的人民公社)—a play on their official name—by focusing on local veterinary workers and farm animals. I define a more-than-people’s commune as a place where different kinds of local inhabitants and healers lived together with nonhuman animals to maintain the subsistence capacity of the community. Without sustaining more-than-people’s communes on their own, it would have been impossible for individual humans and nonhumans at the grassroots level to weather the revolution’s radicalism and unpredictability.

Although it may sound timely and beautiful, it also may sound too romanticized and idealistic. The well-established literature in the fields of modern Chinese history and politics has critically examined various characteristics of agricultural collectivism and rural societies under Mao.

通过当地“人民公社”的工作和知识，将人类、家畜和环境联系起来。这些举措的核心是当地的兽医工作者。成千上万的兽医、动物护理员和饲养员努力支撑经济、知识和道德秩序，其中农场动物和兽医专业知识被视为共同财产，应共同分享和照护。尽管这些公社中兽医工作者的知识和劳动并未以微生物学或生态学语言来证明或推广，但他们对不同人类和非人类生物的共同福祉的深切承诺与当代“同一健康”的理想相符。中国公社中的兽医实践可以帮助我们思考，我们如何才能与地球上所有形式的同伴，作为一个生态公社共同生活。

More-Than-People’s Communes

本论文项目的核心是我努力将毛泽东时代中国农村的集体化社区重新概念化为“多物种的人民公社”（这是对其官方名称的戏仿），重点关注当地兽医工作者和农场动物。我将“多物种的人民公社”定义为一个地方，在那里，不同种类的当地居民和治疗者与非人类动物共同生活，以维持社区的生存能力。如果没有独立维持“多物种的人民公社”，基层的人类和非人类个体将不可能经受住革命的激进和不可预测性。

尽管这听起来很及时和美好，但也可能听起来过于浪漫和理想化。现代中国历史和政治领域的成熟文献批判性地审视了毛泽东时代农业集体主义和农村社会的各种特征。

Rural China often represented backwardness, feudalism, and superstition in contrast to cities that have symbolized progress, the Enlightenment, and science.¹⁵ The communes have also been regarded as sites of inefficiency under the planned economy and non-private ownership.¹⁶ They were also the helpless target of state exploitation. Collectivization was, first and foremost, a result of the state’s desire to control and extract agrarian surpluses for the sake of primitive accumulation and rapid industrialization.¹⁷ These problematic elements culminated in uncontrolled outbreaks of state and mass violence especially during land reform, the anti-Rightist movement, the Great Leap Famine, and the Cultural Revolution.¹⁸ Many scholars seem to believe that there is little room for any rosy reinterpretations of Maoist communes.¹⁹

On top of the historiographic focus on exploitation, extraction, and violence in rural China, another question equally worth asking is: Despite all the adversities, how did local collectives persevere and survive? Driven by this problematic, groups of social scientists and historians have studied the subsistence ethics, agency, and forms of resistance of local individuals and

¹⁵ On the discourses of the urban-rural dichotomy, see Jeremy Brown, *City Versus Countryside in Mao’s China: Negotiating the Divide* (Cambridge: Cambridge University Press, 2012); David Faure and Tao Tao Liu, eds., *Town and Country in China: Identity and Perception* (London: Palgrave Macmillan, 2002).

¹⁶ Frank Dikötter, *Mao’s Great Famine: The History of China’s Most Devastating Catastrophe, 1958-1962* (New York: Walker & Co., 2010).

¹⁷ Schurmann Franz, *Ideology and Organization in Communist China* (Berkeley: University of California Press, 1966); Ho-fung Hung, *The China Boom: Why China Will Not Rule the World* (New York: Columbia University Press, 2015); Tiejun Wen 温铁军, *Ten Crises: The Political Economy of China’s Development, 1949-2020* (Singapore: Palgrave Macmillan, 2021).

¹⁸ Yang Su, *Collective Killings in Rural China during the Cultural Revolution* (Cambridge: Cambridge University Press, 2011); Xun Zhou, ed, *The Great Famine in China, 1958-1962: A Documentary History* (New Haven: Yale University Press, 2012).

¹⁹ Joshua Eisenman, however, complicates the conventional understanding of the Mao-era rural communes by arguing that the communes laid the crucial foundation for the economic growth in the post-Mao period. Joshua Eisenman, *Red China’s Green Revolution: Technological Innovation, Institutional Change, and Economic Development under the Commune* (New York: Columbia University Press, 2018).

在中国农村，落后、封建和迷信常常是其代名词，与此形成对比的是，城市则象征着进步、启蒙和科学。¹⁵ 人民公社也被认为是计划经济和非私有制下效率低下的场所。¹⁶ 它们也是国家剥削的无奈目标。集体化首先是国家为了原始积累和快速工业化而控制和榨取农业剩余的产物。¹⁷ 这些问题最终导致了国家和大规模暴力的失控爆发，尤其是在土地改革、反右运动、大跃进饥荒和文化大革命期间。¹⁸ 许多学者似乎认为，对毛泽东时代公社的任何美好重新解读都没有多少空间。¹⁹

除了史学上对中国农村剥削、榨取和暴力的关注之外，另一个同样值得探讨的问题是：尽管面临种种逆境，地方集体是如何坚持和生存下来的？受此问题的驱动，社会科学家和历史学家群体研究了地方个体和

¹⁵ On the discourses of the urban-rural dichotomy, see Jeremy Brown, *City Versus Countryside in Mao’s China: Negotiating the Divide* (Cambridge: Cambridge University Press, 2012); David Faure and Tao Tao Liu, eds., *Town and Country in China: Identity and Perception* (London: Palgrave Macmillan, 2002).

¹⁶ Frank Dikötter, *Mao’s Great Famine: The History of China’s Most Devastating Catastrophe, 1958-1962* (New York: Walker & Co., 2010).

¹⁷ Schurmann Franz, *Ideology and Organization in Communist China* (Berkeley: University of California Press, 1966); Ho-fung Hung, *The China Boom: Why China Will Not Rule the World* (New York: Columbia University Press, 2015); Tiejun Wen 温铁军, *Ten Crises: The Political Economy of China’s Development, 1949-2020* (Singapore: Palgrave Macmillan, 2021).

¹⁸ 杨苏，《文化大革命期间中国农村的集体杀戮》（剑桥：剑桥大学出版社，2011）；周逊，编，《中国大饥荒，1958-1962：文献史》（纽黑文：耶鲁大学出版社，2012）。

¹⁹ 然而，约书亚·艾森曼（Joshua Eisenman）认为，公社为毛泽东时代后期的经济增长奠定了关键基础，从而使人们对毛泽东时代农村公社的传统理解复杂化。约书亚·艾森曼，《红色中国的绿色革命：公社下的技术创新、制度变革和经济发展》（纽约：哥伦比亚大学出版社，2018年）。

communities.²⁰ For example, in his 1989 monograph the anthropologist Shu-min Huang describes how a dedicated village cadre and his community in Fujian Province resiliently grappled with the serial upheavals of the Maoist period.²¹ In the 1990s and 2000s, prominent China specialists such as Anita Chan, Richard Madsen, Jonathan Unger, and Edward Friedman, Paul Pickowicz, and Mark Selden led the blooming of village studies, shifting the historiographic weight from the Communist state to rural communities at the grassroots.²² More recently, historians Gail Hershatter, Jacob Eyferth, and Sigrid Schmalzer further emphasized how women, papermakers, and youth in rural China proactively and tenaciously carried on under given conditions.²³

Taken together, this scholarship suggests that socialist systems in China did not always work in top-down, pre-determined ways. Rather, they were something more unstable and experimental, and rested on the support and participation of resilient nonelites. Scholars also have disclosed the diverse forms of conflicts between the interventionist state and resistant social groups. In this vein, they show that everyday life under Maoist communes was more complex than the still-dominant

²⁰ In this sense, I find James Scott’s pioneering book still inspirational. James C. Scott, *The Moral Economy of the Peasant: Rebellion and Subsistence in Southeast Asia* (New Haven: Yale University Press, 1976).

²¹ Shu-min Huang, *The Spiral Road: Change in a Chinese Village Through the Eyes of a Communist Party Leader* (Boulder: Westview Press, 1989).

²² Anita Chan, Richard Madsen, and Jonathan Unger, *Chen Village: The Recent History of a Peasant Community in Mao’s China* (Berkeley: University of California Press, 1984); Edward Friedman, Paul G. Pickowicz, and Mark Selden, *Chinese Village, Socialist State* (New Haven: Yale University Press, 1991); Edward Friedman, Paul G. Pickowicz, and Mark Selden, *Revolution, Resistance, and Reform in Village China* (New Haven: Yale University Press, 2005).

²³ Jacob Eyferth, *Eating Rice from Bamboo Roots: The Social History of a Community of Handicraft Papermakers in Rural Sichuan, 1920-2000* (Cambridge, Mass.: Harvard University Asia Center, 2009); Gail Hershatter, *The Gender of Memory: Rural Women and China’s Collective Past* (Berkeley: University of California Press, 2011); Sigrid Schmalzer, *Red Revolution, Green Revolution: Scientific Farming in Socialist China* (Chicago: University of Chicago Press, 2016). Also see Jacob Eyferth, “State Socialism and the Rural Household: How Women’s Handloom Weaving (and Pig-Raising, Firewood-Gathering, Food-Scavenging) Subsidized Chinese Accumulation,” *International Review of Social History* (February 2022): 1–19, <https://doi.org/10.1017/S0020859021000717>. Also see Ralph Thaxton, *Catastrophe and Contention in Rural China: Mao’s Great Leap Famine and the Origins of Righteous Resistance in Da Fo Village* (Cambridge: Cambridge University Press, 2008); Huaiyin Li, *Village China Under Socialism and Reform: A Micro-History, 1948–2008* (Stanford: Stanford University Press, 2009).

社区。²⁰ 例如，人类学家黄树民在其1989年的专著中描述了福建省一位敬业的村干部和他的社区如何坚韧地应对毛泽东时代的一系列剧变。²¹ 在20世纪90年代和21世纪初，安妮塔·陈（Anita Chan）、理查德·马德森（Richard Madsen）、乔纳森·昂格尔（Jonathan Unger）以及爱德华·弗里德曼（Edward Friedman）、保罗·皮科维奇（Paul Pickowicz）和马克·塞尔登（Mark Selden）等著名中国问题专家引领了乡村研究的蓬勃发展，将史学重心从共产主义国家转向了基层农村社区。²² 最近，历史学家盖尔·赫沙特（Gail Hershatter）、雅各布·艾弗斯（Jacob Eyferth）和西格丽德·施马尔策（Sigrid Schmalzer）进一步强调了中国农村的妇女、造纸工人和青年如何在既定条件下积极而顽强地生活。²³

总而言之，这项研究表明中国的社会主义制度并非总是以自上而下、预先确定的方式运作。相反，它们更不稳定、更具实验性，并依赖于坚韧的非精英阶层的支持和参与。学者们还揭示了干预主义国家与抵抗性社会群体之间冲突的多种形式。在这方面，他们表明毛泽东时代公社的日常生活比仍然占主导地位的观点更为复杂

²⁰ 从这个意义上说，我发现詹姆斯·斯科特（James Scott）的开创性著作仍然具有启发性。詹姆斯·C·斯科特（James C. Scott），《农民的道德经济：东南亚的叛乱与生存》（纽黑文：耶鲁大学出版社，1976年）。

²¹ 黄树民，《螺旋之路：一个中国村庄通过一位共产党员领导人眼中的变迁》（博尔德：Westview出版社，1989）。

²² Anita Chan, Richard Madsen, and Jonathan Unger, *Chen Village: The Recent History of a Peasant Community in Mao’s China* (Berkeley: University of California Press, 1984); Edward Friedman, Paul G. Pickowicz, and Mark Selden, *Chinese Village, Socialist State* (New Haven: Yale University Press, 1991); Edward Friedman, Paul G. Pickowicz, and Mark Selden, *Revolution, Resistance, and Reform in Village China* (New Haven: Yale University Press, 2005).

²³ Jacob Eyferth, *Eating Rice from Bamboo Roots: The Social History of a Community of Handicraft Papermakers in Rural Sichuan, 1920-2000* (Cambridge, Mass.: Harvard University Asia Center, 2009); Gail Hershatter, *The Gender of Memory: Rural Women and China’s Collective Past* (Berkeley: University of California Press, 2011); Sigrid Schmalzer, *Red Revolution, Green Revolution: Scientific Farming in Socialist China* (Chicago: University of Chicago Press, 2016). Also see Jacob Eyferth, “State Socialism and the Rural Household: How Women’s Handloom Weaving (and Pig-Raising, Firewood-Gathering, Food-Scavenging) Subsidized Chinese Accumulation,” *International Review of Social History* (February 2022): 1–19, <https://doi.org/10.1017/S0020859021000717>. Also see Ralph Thaxton, *Catastrophe and Contention in Rural China: Mao’s Great Leap Famine and the Origins of Righteous Resistance in Da Fo Village* (Cambridge: Cambridge University Press, 2008); Huaiyin Li, *Village China Under Socialism and Reform: A Micro-History, 1948–2008* (Stanford: Stanford University Press, 2009).

image of helpless destitution under authoritarian state power. These diverse approaches to rural China in the Maoist era in the interdisciplinary field of Chinese studies eventually led me to pay attention to both multifarious contradictions and their deep everydayness in rural communes.

Building on the scholarship on rural China under Mao, I argue that the agency, subsistence capacity, and conflicts of local communes were more than human. Here I am indebted to the wisdom of the rising field of animal history that argues that beneath the social developments and watershed events in human history were unrecognized nonhuman contributions. For instance, Harriet Ritvo, Timothy Mitchell, Rohan Deb Roy, and James Hevia have analyzed how exotic animals in English zoos, mosquitos in the colonies, and draft animals on battlefronts were at the center of making and maintaining the hegemony of British Empire.²⁴ Andrew Robichaud, Jessica Wang, and Chris Pearson challenge dominant narratives of urban development in New York, London, and Paris by articulating the historical roles of livestock and pets.²⁵ Tiago Saraiva traces how pigs were used by the Nazi regime to extend its reach into the remote countryside and construct the ideology of autarky and Aryanism.²⁶ Like other historians of animals,²⁷ I bring the

²⁴ Harriet Ritvo, *The Animal Estate: The English and Other Creatures in the Victorian Age* (Cambridge, Mass.: Harvard University Press, 1987); Timothy Mitchell, *Rule of Experts: Egypt, Techno-Politics, Modernity* (Berkeley: University of California Press, 2022); Rohan Deb Roy, *Malarial Subjects: Empire, Medicine and Nonhumans in British India, 1820-1909* (Cambridge: Cambridge University Press, 2017); James Hevia, *Animal Labor and Colonial Warfare* (Chicago: University of Chicago Press, 2018).

²⁵ Andrew Robichaud, *Animal City: The Domestication of America* (Cambridge, Mass.: Harvard University Press, 2019); Jessica Wang, *Mad Dogs and Other New Yorkers: Rabies, Medicine, and Society in an American Metropolis, 1840-1920* (Baltimore: Johns Hopkins University Press, 2019); Chris Pearson, *Dogopolis: How Dogs and Humans Made Modern New York, London, and Paris* (Chicago: University of Chicago Press, 2021).

²⁶ Tiago Saraiva, *Fascist Pigs: Technoscientific Organisms and the History of Fascism* (Cambridge, Mass.: MIT Press, 2016).

²⁷ Also see Gregg Mitman, *Reel Nature: America's Romance with Wildlife on Film* (Cambridge, Mass.: Harvard University Press, 1999); Nigel Rothfels, *Savages and Beasts: The Birth of the Modern Zoo* (Baltimore: Johns Hopkins University Press, 2002); Thomas Fleischman, *Communist Pigs: An Animal History of East Germany's Rise and Fall* (Seattle: University of Washington Press, 2020).

在专制国家权力下无助的贫困景象。这些对毛泽东时代中国农村的多元化研究方法，最终使我开始关注农村公社中多方面的矛盾及其深刻的日常性。

在毛泽东时代中国农村研究的基础上，我认为地方公社的能动性、生存能力和冲突不仅仅是人类的。在此，我受益于新兴的动物史领域的智慧，该领域认为，人类历史上的社会发展和分水岭事件背后，存在着未被承认的非人类贡献。例如，哈里特·里特沃（Harriet Ritvo）、蒂莫西·米切尔（Timothy Mitchell）、罗汉·德布·罗伊（Rohan Deb Roy）和詹姆斯·赫维亚（James Hevia）分析了英国动物园中的异域动物、殖民地的蚊子以及战场上的役畜如何成为建立和维持大英帝国霸权的核心。²⁴ 安德鲁·罗比肖（Andrew Robichaud）、杰西卡·王（Jessica Wang）和克里斯·皮尔逊（Chris Pearson）通过阐明牲畜和宠物在历史上的作用，挑战了纽约、伦敦和巴黎城市发展的主流叙事。²⁵ 蒂亚戈·萨拉瓦（Tiago Saraiva）追溯了纳粹政权如何利用猪将其影响力扩展到偏远农村，并构建自给自足和雅利安主义的意识形态。²⁶ 像其他动物史学家一样，²⁷ 我将

²⁴ Harriet Ritvo, *The Animal Estate: The English and Other Creatures in the Victorian Age*（马萨诸塞州剑桥：哈佛大学出版社，1987）；Timothy Mitchell, *Rule of Experts: Egypt, Techno-Politics, Modernity*（伯克利：加州大学出版社，2022）；Rohan Deb Roy, *Malarial Subjects: Empire, Medicine and Nonhumans in British India, 1820-1909*（剑桥：剑桥大学出版社，2017）；James Hevia, *Animal Labor and Colonial Warfare*（芝加哥：芝加哥大学出版社，2018）。

²⁵ Andrew Robichaud, *Animal City: The Domestication of America*（马萨诸塞州剑桥：哈佛大学出版社，2019）；Jessica Wang, *Mad Dogs and Other New Yorkers: Rabies, Medicine, and Society in an American Metropolis, 1840-1920*（巴尔的摩：约翰霍普金斯大学出版社，2019）；Chris Pearson, *Dogopolis: How Dogs and Humans Made Modern New York, London, and Paris*（芝加哥：芝加哥大学出版社，2021）。

²⁶ 蒂亚戈·萨拉瓦 (Tiago Saraiva)，《法西斯猪：技术科学有机体与法西斯主义史》(*Fascist Pigs: Technoscientific Organisms and the History of Fascism*) (马萨诸塞州剑桥：麻省理工学院出版社，2016)。

²⁷ 另见格雷格·米特曼 (Gregg Mitman)，《真实自然：美国与电影中的野生动物的浪漫史》(*Reel Nature: America's Romance with Wildlife on Film*) (马萨诸塞州剑桥：哈佛大学出版社，1999)；奈杰尔·罗斯费尔斯 (Nigel Rothfels)，《野蛮人与野兽：现代动物园的诞生》(*Savages and Beasts: The Birth of the Modern Zoo*) (巴尔的摩：约翰霍普金斯大学出版社，2002)；托马斯·弗莱希曼 (Thomas Fleischman)，《共产主义猪：东德兴衰的动物史》(*Communist Pigs: An Animal History of East Germany's Rise and Fall*) (西雅图：华盛顿大学出版社，2020)。

relations between communal veterinary workers and farm animals to discussions of the agrarian life in the Maoist period.

Engaging in the literature on animal history and investigating human-animal relations within more-than-people’s communes, I do not insist that “animal agency” was equal to or even more significant than that of local state cadres, veterinarians, or peasants.²⁸ Instead, how this dissertation treats animals is to juxtapose human and nonhuman actors whenever possible. In so doing, I do not erase the presence of diverse farm animals in the communes. I do not belittle animals’ ability to work for humans, or to interrupt and reshape human actors’ preexisting ways of thinking, acting, and organizing the sociopolitical.

In other words, I present “more-than-people’s communes” not necessarily as a utopian benchmark in the era of the Anthropocene, but as an experimental site of trial and error that the indeterminant revolution brought about and in which both humans and nonhumans participated. This dissertation will show that such “more-than-people’s communes” indeed existed.

Veterinary Workers and Mass Science

Who then lived in and sustained more-than-people’s communes? I draw special attention to the groups of veterinary workers (兽医人员 or 畜牧兽医工作者). Among them were local

²⁸ For more discussions on animal agency, see Jason Hribal, “Animals, Agency, and Class: Writing the History of Animals from Below,” *Human Ecology Review* 14.1 (July 2007): 101–112; Susan Nance, *Entertaining Elephants: Animal Agency and the Business of the American Circus* (Baltimore: Johns Hopkins University Press, 2013); Chris Pearson, “Dogs, History, and Agency,” *History and Theory* 52.4 (December 2013): 128–145; David Gary Shaw, “The Torturer’s Horse: Agency and Animals in History,” *History and Theory* 52.4 (December 2013): 146-167.

从公社兽医与农场动物之间的关系，到毛泽东时代农业生活中的讨论。

在研究动物历史文献和调查“超人公社”中人与动物关系时，我并不认为“动物能动性”与地方国家干部、兽医或农民的能动性相等，甚至更重要。²⁸相反，本论文对待动物的方式是尽可能地将人类和非人类行为者并置。这样做，我并没有抹去公社中各种农场动物的存在。我也没有贬低动物为人类工作的能力，或者它们打断和重塑人类行为者原有思维、行动和组织社会政治的能力。

换句话说，我提出“超人公社”不一定作为人类世时代的一个乌托邦基准，而是作为不确定革命所带来的一个试验性场所，人类和非人类都参与其中。本论文将表明，这种“超人公社”确实存在。

兽医工作者与大众科学

那么，谁生活在并维持着“超人类”的公社呢？我特别关注兽医工作者（兽医人员或畜牧兽医工作者）群体。其中有当地的

²⁸ 更多关于动物能动性的讨论，请参见 Jason Hribal, “Animals, Agency, and Class: Writing the History of Animals from Below” , *Human Ecology Review* 14.1 (2007年7月): 101–112; Susan Nance, *Entertaining Elephants: Animal Agency and the Business of the American Circus* (Baltimore: Johns Hopkins University Press, 2013); Chris Pearson, “Dogs, History, and Agency” , *History and Theory* 52.4 (2013年12月): 128–145; David Gary Shaw, “The Torturer’s Horse: Agency and Animals in History” , *History and Theory* 52.4 (2013年12月): 146-167。

veterinarians, veterinary cadres (兽医干部), animal epidemic prevention workers (兽疫防疫员), collective animal caretakers (饲养员), and animal breeders (配种员).

What I call “local veterinarians” in this dissertation were a heterogenous group of animal healers who came to work at commune-level veterinary stations (公社级兽医站, Figure 2) or were in charge of veterinary services exclusively for sub-communal collectives, namely production brigades and production teams. Local veterinarians can be categorized based on their educational backgrounds and forms of employment into “folk veterinarians” (民间兽医), “collective veterinarians” (集体兽医 or 社内兽医), and “barefoot veterinarians” (赤脚兽医).

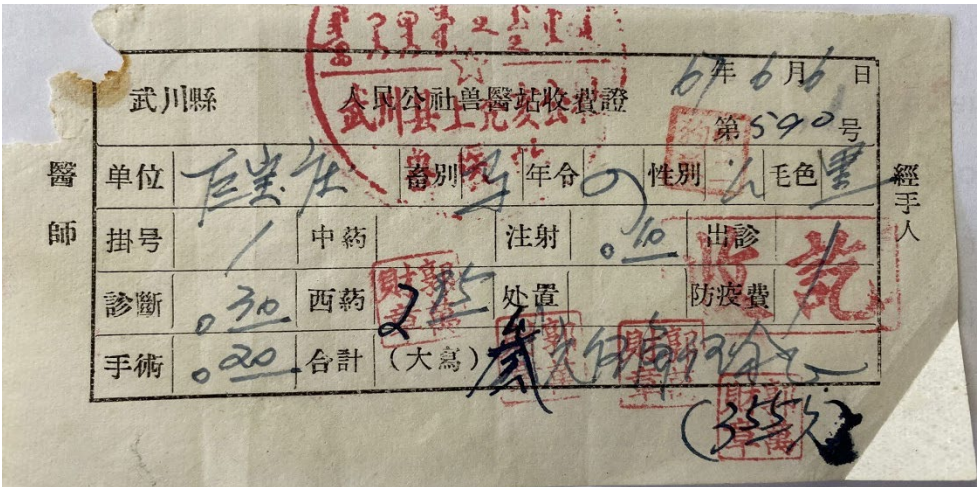


Figure 2. A 1967 medical receipt from Shangtuhai Commune Veterinary Station, Wuchuan County, Inner Mongolia. The animal patient was a nine-year-old black stallion from Jubaozhuang Production Team. The commune veterinarian prescribed injection, surgery, and “Western medicines” to the patient. Source: Author’s Collection.

Folk veterinarians refer to animal healers, either full-time or half-peasant-half-veterinarian, who actively treated livestock with vernacular remedies before the Communist Party seized power in 1949. Most of them learned veterinary knowledge from family or village elders and were once

兽医、兽医干部、兽疫防疫员、饲养员和配种员。

本论文中我所说的“地方兽医”是一群异质的动物治疗师，他们要么在公社级兽医站（图2）工作，要么专门负责社以下集体（即生产大队和生产队）的兽医服务。地方兽医可根据其教育背景和就业形式分为“民间兽医”、“集体兽医”（或社内兽医）和“赤脚兽医”。

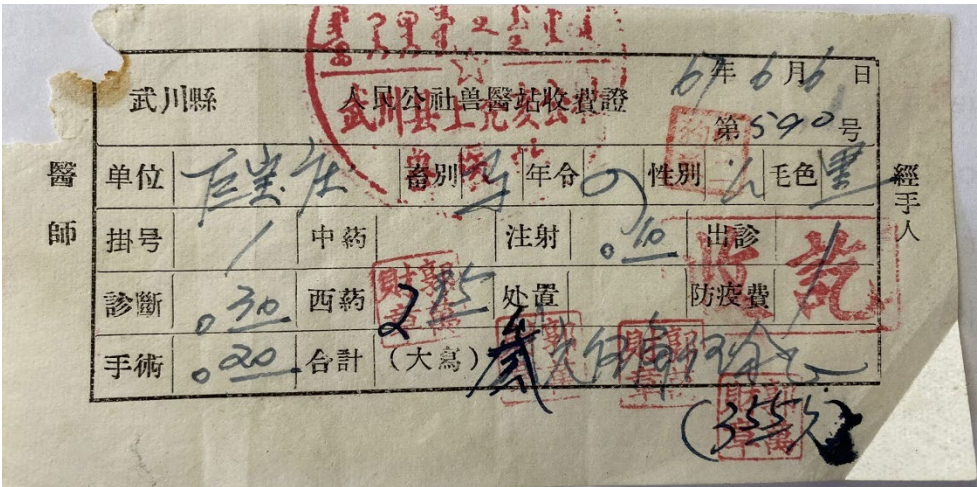


图2。1967年内蒙古五原县上土亥公社兽医站的医疗收据。患病动物是聚宝庄生产队的一匹九岁黑种马。公社兽医给这匹马开了注射、手术和“西药”。来源：作者收藏。

民间兽医指的是全职或半农半兽医的动物治疗师，他们在1949年共产党掌权之前，积极用民间疗法治疗牲畜。他们中的大多数人从家人或村里的长辈那里学习兽医知识，曾经是

private practitioners. As agricultural collectivization proceeded in the 1950s, the Communist state gradually organized and transformed folk veterinarians into collective veterinarians, or full-time animal doctors in a commune or brigade (Figure 3). New generations of collective veterinarians were also educated through apprenticeship under the tutelage of former folk veterinarians as well as various short-term trainings run by provincial, prefectural, or county governments. Barefoot veterinarians were a particular product of the Cultural Revolution. Their background, knowledge, and practices were not dramatically different from their senior and fellow collective veterinarians. But, similar to “barefoot doctors” in the realm of public health, their presence was more politically charged than other types of animal healers as they symbolized the justification of high Maoism in the Cultural Revolution era.



Figure 3. A commune veterinarian treating lambs in Mandoula Commune, Baotou, Inner Mongolia. Source: Author’s Collection.

私人执业者。随着20世纪50年代农业集体化的推进，共产主义国家逐步将民间兽医组织并改造为集体兽医，或公社或大队的全职动物医生（图3）。新一代集体兽医也通过在原民间兽医的指导下学徒以及省、地、县政府举办的各种短期培训而获得教育。赤脚兽医是文化大革命的特殊产物。他们的背景、知识和实践与他们的前辈和同伴集体兽医没有显著差异。但是，与公共卫生领域的“赤脚医生”类似，他们的存在比其他类型的动物治疗师更具政治色彩，因为他们象征着文化大革命时期高毛泽东思想的正当性。



图3。内蒙古包头市满都拉公社的一名公社兽医正在治疗羔羊 资料来源：作者收藏。 lia.

Veterinary cadres were middle or low-level state agents who were in charge of livestock farming and veterinary affairs. They were not themselves veterinarians, but they could diagnose and treat common, mild ailments and were trained in how to handle vaccines, how to give animals an injection, possible immune reactions, and how to care for inoculated animals. More importantly, veterinary cadres dealt with the “political” tasks related to animal disease control such as making a specific action plan of lockdown, quarantine, and disinfection; explaining the rationales of such plans to local peasants; and distributing necessary information and related propaganda materials to every corner of the collective.

Animal epidemic prevention workers were young peasants who were passionate about participating in the state’s anti-animal disease campaigns. As they actively assisted veterinary cadres and veterinarians, they were seen by the state as the wellspring of future veterinary expertise. Animal caretakers were those who were responsible for taking care of collectively owned livestock. Caretakers were usually the first people who noticed when something was wrong with animals and called in veterinarians. They themselves were capable of treating common, mild diseases and ailments. They were also at the forefront of everyday prevention efforts such as keeping individual animals clean and conducting regular disinfections of collective farms. Last but not least, animal breeders were those who worked to increase the quantity and quality of livestock. They selected “superior” males out of local herds; mated them with female animals; carried out artificial insemination; and desexed or culled undesirable animals.

These veterinary workers, with few exceptions, emerged from the peasantry. Most of them were not trained in higher educational institutions. As such, the credibility of their expertise was often questioned not only by contemporary actors but also by historians. How could we understand the nature of the expertise of these local veterinary workers then? The suspicion about peasant

兽医干部是中低层国家代理人，负责畜牧业和兽医事务。他们本身不是兽医，但可以诊断和治疗常见的轻微疾病，并接受过疫苗处理、动物注射、可能的免疫反应以及如何护理接种动物的培训。更重要的是，兽医干部处理与动物疾病控制相关的“政治”任务，例如制定具体的封锁、隔离和消毒行动计划；向当地农民解释这些计划的理由；以及向集体的每个角落分发必要的信息和相关宣传材料。

动物防疫员是热衷于参与国家动物疫病防治运动的年轻农民。由于他们积极协助兽医干部和兽医，他们被国家视为未来兽医专业知识的源泉。动物饲养员是负责照料集体所有牲畜的人。饲养员通常是第一个发现动物出现问题并召集兽医的人。他们自己能够治疗常见的轻微疾病。他们也站在日常预防工作的第一线，例如保持个体动物清洁和对集体农场进行定期消毒。最后但同样重要的是，动物育种员是致力于提高牲畜数量和质量的人。他们从当地畜群中选择“优质”雄性；将它们与雌性动物交配；进行人工授精；并对不理想的动物进行去势或淘汰。

这些兽医工作者，除少数例外，都出身于农民。他们中的大多数人没有在高等教育机构接受过培训。因此，他们的专业知识的可信度不仅受到当时人的质疑，也受到历史学家的质疑。那么，我们该如何理解这些地方兽医工作者的专业知识的性质呢？对农民的怀疑

knowledge in Maoist China seems to have its root in historiography. Historians of science in the Soviet Union and other socialist countries saw the prioritization of amateur, indigenous knowledge over professional, “scientific” knowledge in the name of proletarian revolution as a sign of the most fundamental error of socialist science: Politics and ideology overwhelm science.²⁹

More recently, however, there has been a growing interest in the ideal of “mass science” in Maoist China among historians of science who sympathize with postcolonial critiques of science.³⁰ Mass science was a vision of science inclusive of the masses of ordinary working people. It opposed scientific elitism and ivory tower professionalization. The experiences of revolutionary workers and peasants on the shop floor or in the field were recognized as the fount of knowledge. If certain intellectual and scientific ideas seemed isolated from working people’s reality, they were easily deemed “bourgeois,” “capitalistic,” or “idealist” as opposed to “proletarian,” “socialist,” and “materialist.” While some scholars attribute the destruction of professional science and higher education during the Maoist period, especially during the Cultural Revolution, to this controversial

²⁹ On this historiography on socialist science, see Zhores Medvedev, *The Rise and Fall of T. D. Lysenko* (New York: Columbia University Press, 1965); David Joravsky, *The Lysenko Affair* (Cambridge, Mass.: Harvard University Press, 1970); Valery N. Soyfer, *Lysenko and the Tragedy of Soviet Science* (New Brunswick: Rutgers University Press, 1994); Nikolai Krementsov, *Stalinist Science* (Princeton: Princeton University Press, 1997); Laurence Schneider, *Biology and Revolution in Twentieth-century China* (Lanham: Rowman and Littlefield, 2003); Danian Hu, *China and Albert Einstein: The Reception of the Physicist and His Theory in China, 1917–1979* (Cambridge, Mass.: Harvard University Press, 2005). As for more complicated accounts of the relationships between socialist ideology and science, see Gary Werskey, *The Visible College: A Collective Biography of British Scientists and Socialists of the 1930s* (London: Allen and Unwin, 1978); Loren R. Graham, *What Have We Learned About Science and Technology from the Russian Experience?* (Stanford: Stanford University Press, 1998); Gary Werskey, “The Marxist Critique of Capitalist Science: A History of Three Movements?” *Science and Culture* 16.4 (December 2007): 397–461.

³⁰ On postcolonial STS and history of science, see footnote 2. Also see Itty Abraham, “The Contradictory Spaces of Postcolonial Techno-Science,” *Economic and Political Weekly* 41.3 (January 2006): 210–217; John Law and Wen-yuan Lin, “Provincializing STS: Postcoloniality, Symmetry, and Method,” *East Asian Science, Technology, and Society* 11.2 (June 2017): 211–227.

毛泽东时代中国的知识似乎根植于史学。苏联和其他社会主义国家的科学史学家认为，以无产阶级革命的名义，将业余的、本土的知识置于专业的、“科学的”知识之上，是社会主义科学最根本的错误：政治和意识形态压倒了科学。²⁹

然而，最近，对后殖民科学批判持同情态度的科学史学家对毛泽东时代中国“群众科学”的理想越来越感兴趣。³⁰群众科学是一种包容普通劳动人民的科学愿景。它反对科学精英主义和象牙塔式的专业化。革命工人、农民在车间或田间的经验被认为是知识的源泉。如果某些知识和科学思想似乎脱离了劳动人民的现实，它们很容易被认为是“资产阶级的”、“资本主义的”或“唯心主义的”，而不是“无产阶级的”、“社会主义的”和“唯物主义的”。虽然一些学者将毛泽东时代，特别是文化大革命期间，专业科学和高等教育的破坏归因于这一有争议的观点

²⁹ On this historiography on socialist science, see Zhores Medvedev, *The Rise and Fall of T. D. Lysenko* (New York: Columbia University Press, 1965); David Joravsky, *The Lysenko Affair* (Cambridge, Mass.: Harvard University Press, 1970); Valery N. Soyfer, *Lysenko and the Tragedy of Soviet Science* (New Brunswick: Rutgers University Press, 1994); Nikolai Krementsov, *Stalinist Science* (Princeton: Princeton University Press, 1997); Laurence Schneider, *Biology and Revolution in Twentieth-century China* (Lanham: Rowman and Littlefield, 2003); Danian Hu, *China and Albert Einstein: The Reception of the Physicist and His Theory in China, 1917–1979* (Cambridge, Mass.: Harvard University Press, 2005). As for more complicated accounts of the relationships between socialist ideology and science, see Gary Werskey, *The Visible College: A Collective Biography of British Scientists and Socialists of the 1930s* (London: Allen and Unwin, 1978); Loren R. Graham, *What Have We Learned About Science and Technology from the Russian Experience?* (Stanford: Stanford University Press, 1998); Gary Werskey, “The Marxist Critique of Capitalist Science: A History of Three Movements?” *Science and Culture* 16.4 (December 2007): 397–461.

³⁰ On postcolonial STS and history of science, see footnote 2. Also see Itty Abraham, “The Contradictory Spaces of Postcolonial Techno-Science,” *Economic and Political Weekly* 41.3 (January 2006): 210–217; John Law and Wen-yuan Lin, “Provincializing STS: Postcoloniality, Symmetry, and Method,” *East Asian Science, Technology, and Society* 11.2 (June 2017): 211–227.

notion, others frame mass science as an anti-technocratic, popular democratic, and even “decolonial” vision of science.³¹

Xiaoping Fang, for example, analyzes how the Chinese Communist Party’s (CCP) commitment to the ideal of mass science gave many local peasants the opportunity to participate in short-term training organized by local governments and thus become “barefoot doctors.” These grassroots experts provided villagers with first aid and basic healthcare when biomedically trained doctors were rare and helped local peasants familiarize themselves with Western medicine.³² Sigrid Schmalzer scrutinizes how mass science worked in the field of agricultural sciences. Under the Maoist ideological affinity for local knowledge, ordinary peasants were entitled to produce and practice “scientific farming” through diverse opportunities such as mass scientific experiment movements and four-level agricultural extension systems.³³ This unique socio-intellectual context, enabled by the pursuit of mass science, was widely relevant to local veterinary activities. Communal veterinary workers, too, were products of mass science.

This dissertation reshapes this conversation by foregrounding nonhuman animals. I argue that mass science was enacted within grassroots veterinary spheres through not only human action but also animal initiative. The revolutionary state’s commitment to the ideal of mass science did

³¹ Schmalzer, *Red Revolution, Green Revolution*; Gross, *Farewell to the God of Plague*; Rui Kunze and Marc Andre Matten, *Knowledge Production in Mao-era China: Learning from the Masses* (Lanham: Lexington Books, 2021); Fa-ti Fan and Shun-Ling Chen, “Citizen, Science, and Citizen Science,” *East Asian Science, Technology, and Society* 13.2 (June 2019): 181–193; Danian Hu, “Despite or Due to the Cultural Revolution: The Development of Chinese Science, Technology, and Medicine in the 1960s and 1970s,” *Endeavor* 43.3 (September 2017): 78–84.

³² Xiaoping Fang, *Barefoot Doctors and Western Medicine in China* (Rochester: University of Rochester Press, 2012). On barefoot doctors, also see Daqing Zhang and Paul Unschuld, “China’s Barefoot Doctor: Past, Present, and Future,” *The Lancet* 372. 9653 (29 November–5 December 2008): 1865–1867; Miriam Gross, “Between Party, People, and Profession: The Many Faces of the ‘Doctor’ during the Cultural Revolution,” *Medical History* 62.3 (July 2018): 333–359.

³³ Schmalzer, *Red Revolution, Green Revolution*. For more case studies regarding mass science, see Kunze and Matten, *Knowledge Production in Mao-era China*.

概念，另一些人则将大众科学视为一种反技术官僚、人民民主，甚至是“去殖民化”的科学愿景。³¹

例如，方小平分析了中国共产党（CCP）对大众科学理想的承诺如何使许多当地农民有机会参加地方政府组织的短期培训，从而成为“赤脚医生”。这些基层专家在生物医学培训的医生稀缺时为村民提供急救和基本医疗保健，并帮助当地农民熟悉西医。³² Sigrid Schmalzer 仔细研究了大众科学在农业科学领域的作用。在毛泽东思想对地方知识的亲和力下，普通农民有权通过大众科学实验运动和四级农业推广系统等多种机会生产和实践“科学耕作”。³³ 这种独特的社会智力背景，通过追求大众科学得以实现，与当地兽医活动广泛相关。社区兽医工作者也是大众科学的产物。

本论文通过突出非人类动物来重塑这一对话。我认为，群众科学不仅通过人类行动，而且通过动物的积极性，在基层兽医领域得以实施。革命国家对群众科学理想的承诺做到了

³¹ Schmalzer, 《红色革命，绿色革命》；Gross, 《瘟神告别》；Rui Kunze 和 Marc Andre Matten, 《毛泽东时代中国的知识生产：向群众学习》（Lanham: Lexington Books, 2021）；Fa-ti Fan 和 Shun-Ling Chen, “公民、科学和公民科学”，《东亚科学、技术与社会》13.2（2019 年 6 月）：181-193；Danian Hu, “尽管或由于文化大革命：中国科学、技术和医学在 20 世纪 60 年代和 70 年代的发展”，《奋进》43.3（2017 年 9 月）：78-84。

³² Xiaoping Fang, *Barefoot Doctors and Western Medicine in China* (Rochester: University of Rochester Press, 2012). On barefoot doctors, also see Daqing Zhang and Paul Unschuld, “China’s Barefoot Doctor: Past, Present, and Future,” *The Lancet* 372. 9653 (29 November–5 December 2008): 1865–1867; Miriam Gross, “Between Party, People, and Profession: The Many Faces of the ‘Doctor’ during the Cultural Revolution,” *Medical History* 62.3 (July 2018): 333–359.

³³ Schmalzer, 《红色革命，绿色革命》。有关大众科学的更多案例研究，请参见 Kunze 和 Matten, 《毛泽东时代中国的知识生产》。

enable many local peasants to become collective veterinarians, caretakers, breeders, and so on. But that was not enough. They more than often failed to gain the trust of fellow peasants. Facing this problem of trust, not a few of them, knowing that they were the symbols of Maoism, tried to seek a political solution.³⁴ By ideologically attacking and ousting veterinary elites or senior healers, they hoped to take their positions. However, larger numbers of local veterinary workers chose to move closer to their animal patients and immerse themselves in clinical work. They tried to consolidate their expertise and gain the peasantry’s trust by accumulating clinical experience with ill animals whose bodies and behaviors ultimately determined the success and failure of treatments. In this way, animals co-produced the knowledge and professional authority of local veterinary workers.

Chapter Outline

This dissertation is thematically divided into five chapters. While articulating the historical actors’ knowledges and practices, each chapter presents a particular veterinary issue in the more-than-people’s communes as a means to examine the dilemmas of the Chinese Revolution. Put together, the five chapters demonstrate that more-than-people’s communes in Maoist China involved unique, experimental combinations of individualism and communalism; private and collective economies; exploitation and care; elitism and socio-intellectual egalitarianism; and human-centric instrumentalism and a sense of multispecies coexistence.

³⁴ The problem of trust is inextricable from the construction of scientific knowledge. Simon Schaffer and Steven Shapin, *Leviathan and the Air-Pump: Hobbes, Boyle, and the Experimental Life* (Princeton: Princeton University Press, 2011 [1985]); Theodore M. Porter, *Trust in Numbers: The Pursuit of Objectivity in Science and Public Life* (Princeton: Princeton University Press, 2020 [1995]).

使许多当地农民成为集体兽医、饲养员、育种员等等。但这还不够。他们往往未能赢得其他农民的信任。面对信任问题，不少人深知自己是毛泽东思想的象征，试图寻求政治解决方案。³⁴ 通过思想上攻击和驱逐兽医精英或资深治疗师，他们希望取代他们的位置。然而，更多的当地兽医选择更接近他们的动物病人，沉浸在临床工作中。他们试图通过积累对病畜的临床经验来巩固他们的专业知识并赢得农民的信任，因为病畜的身体和行为最终决定了治疗的成功与否。通过这种方式，动物共同生产了当地兽医的知识和专业权威。

章节大纲

本论文在主题上分为五章。每一章在阐述历史参与者的知识和实践的同时，都以人畜混居公社中的一个特定兽医问题为切入点，以审视中国革命的困境。总而言之，这五章表明，毛泽东时代的人畜混居公社涉及个人主义与集体主义、私人经济与集体经济、剥削与关怀、精英主义与社会智力平等主义、以人类为中心的工具主义与多物种共存意识的独特实验性结合。

³⁴ 信任问题与科学知识的构建密不可分。西蒙·沙弗和史蒂文·沙平，《利维坦与气泵：霍布斯、波义耳与实验生活》（普林斯顿：普林斯顿大学出版社，2011 [1985]）；西奥多·M·波特，《信任数字：科学与公共生活中的客观性追求》（普林斯顿：普林斯顿大学出版社，2020 [1995]）。

Chapter One situates epizootic and zoonotic diseases alongside the question of state-society relations in Maoist China. This chapter explores how diverse local veterinary workers responded in concert to swine fever, foot and mouth disease, anthrax, and brucellosis, among other diseases. It questions the state-centric approach to the historical operations of multispecies biopower by foregrounding bottom-up communal initiatives. It was local veterinary cadres, folk animal healers, caretakers, and epidemic prevention workers who spearheaded a series of patterns to fight infectious animal diseases—lockdowns, quarantines, disinfection, mass animal vaccination, mass education, and prioritizing the treatment of infected animals over mass culling. I also argue that the Chinese state, or more precisely central and provincial governments, did not properly intervene into the life of nonhumans in collectivized communities and the subsistence of rural residents closely connected to those animals.

If the first chapter is an introduction that showcases how veterinary workers in more-than-people's communes worked as a team to protect livestock and livelihoods from animal diseases, the following four chapters emphasize each group of veterinary workers: local veterinarians (Chapters Two and Three), caretakers (Chapter Four), and breeders (Chapter Five). The central question that runs through Chapter Two is: what kind of knowledge did local veterinarians have to possess in order to sustain more-than-people's communes? It delves into the construction and codification of what came to be known as Traditional Chinese Veterinary Medicine (TCVM). It examines how China's folk animal healers, formerly private practitioners, were transformed into collective veterinarians in the 1950s. I argue that this process of collectivizing local veterinary personnel and their knowledge was a prerequisite for the codification of TCVM. In so doing, I show that socialist collectivism could facilitate knowledge production. With this case study on folk veterinary medicine, this chapter complicates the idea that social and political factors in socialist countries, unlike their

第一章将动物流行病和人畜共患病与毛泽东时代中国的国家与社会关系问题联系起来。本章探讨了各地兽医工作者如何协同应对猪瘟、口蹄疫、炭疽病和布鲁氏菌病等疾病。它通过突出自下而上的社区举措，质疑了以国家为中心的跨物种生物权力历史运作方法。正是地方兽医干部、民间动物治疗师、护理人员和防疫工作者率先采取了一系列模式来对抗传染性动物疾病——封锁、隔离、消毒、大规模动物疫苗接种、大众教育以及优先治疗受感染动物而非大规模扑杀。我还认为，中国政府，或者更准确地说，中央和省级政府，没有适当地干预集体化社区中非人类的生命以及与这些动物密切相关的农村居民的生计。

如果说第一章是引言，展示了“超人”公社的兽医工作者如何作为一个团队来保护牲畜和生计免受动物疾病的侵害，那么接下来的四章则分别侧重于每一类兽医工作者：地方兽医（第二章和第三章）、护理人员（第四章）和饲养员（第五章）。贯穿第二章的核心问题是：地方兽医必须具备什么样的知识才能维持“超人”公社？它深入探讨了后来被称为中兽医（TCVM）的构建和编纂。它考察了中国民间动物治疗师（以前是私人执业者）在20世纪50年代如何转变为集体兽医。我认为，这种地方兽医人员及其知识的集体化过程是中兽医编纂的先决条件。通过这样做，我表明社会主义集体主义可以促进知识生产。通过这个关于民间兽医学的案例研究，本章使社会主义国家中的社会和政治因素与它们的

liberal democratic counterparts, had a negative impact on the construction of science and that Maoism was inherently hostile to science.

Moving from the question of expertise to that of experts in more-than-people’s communes, Chapter Three examines how grassroots veterinarians struggled to stabilize their social status and establish their professional authority during the high Maoist period (1958–1976). Navigating the complicated realities in which elitism, anti-intellectualism, meritocracy, and egalitarianism were intertwined, collective veterinarians’ endeavors reflected a larger story of Maoist China’s approach to socio-intellectual inequalities. This chapter shows that although inspired by Maoist egalitarianism, mass scientific veterinary education in the first 17 years of the PRC (1949–1966) was meritocratic and sometimes even elitist. The Cultural Revolution (1966–1976) in the local veterinary realm was an anti-intellectual backlash against the socio-intellectual hierarchy and disparities that the preceding period brought about. Nonetheless, even under high Maoism, to be recognized as a decent veterinarian in one’s community, the healer must have proven his or her clinical competence and earned the trust of the other villagers. In the end, both elite and barefoot veterinarians appropriated Maoist voluntarist and practice-centric languages to frame their immersion in clinical work as a way to burnish their healing expertise, shunning both meritocracy and anti-intellectualism.

Building on the nature of the expertise and the meanings of being an expert, Chapter Four sheds light on the everyday labor of animal care that was indispensable to sustaining the nonhuman capacity of communes. How did the state and collectives make some rural residents pursue such an onerous yet unappreciated vocation without relying on the market mechanism or giving livestock ownership to individuals? The chapter examines the workings of Maoist ideology in animal care by critically analyzing propagandistic materials about “model” animal caretakers. It illuminates that animal care was an integral part of the state’s imperative to establish collectivism as well as the

与自由民主国家相比，对科学建设产生了负面影响，毛泽东思想本质上是敌视科学的。

第三章从专业知识问题转向人民公社中专家的问题，探讨了在毛泽东时代（1958-1976）基层兽医如何努力稳定其社会地位并确立其专业权威。在精英主义、反智主义、英才教育和平等主义交织的复杂现实中，集体兽医的努力反映了毛泽东时代中国处理社会智力不平等问题的更大图景。本章表明，尽管受到毛泽东平等主义的启发，但中华人民共和国成立后的前17年（1949-1966）的大众科学兽医教育是精英主义的，有时甚至是精英化的。地方兽医领域的文化大革命（1966-1976）是对前一时期造成的社会智力等级制度和差异的反智主义反弹。然而，即使在毛泽东时代，要在社区中被认可为一名合格的兽医，治疗师必须证明其临床能力并赢得其他村民的信任。最终，精英兽医和赤脚兽医都运用毛泽东的唯意志论和以实践为中心的语言，将他们沉浸在临床工作中视为磨练其治疗专业知识的一种方式，从而避开了精英主义和反智主义。

第四章以专业知识的性质和作为专家的意义为基础，阐述了对维持公社非人类能力不可或缺的日常动物护理工作。国家和集体是如何让一些农村居民在不依赖市场机制或不将牲畜所有权赋予个人的情况下，从事这种艰巨却不受重视的职业的？本章通过批判性地分析有关“模范”动物饲养员的宣传材料，审视了毛泽东思想在动物护理中的作用。它阐明了动物护理是国家建立集体主义以及

imposition of moral transformations on rural residents. This chapter also draws attention to the frequent criticism of “bad” caretakers and insists that the need to discuss these undesirable workers can be taken as circumstantial evidence of outliers. It reveals the tensions and discrepancies between the state ideal of a multispecies community of care and peasants’ resistant forms of living side by side with animals.

If people’s communes were to be more-than-people’s communes, a certain level of quantity and quality had to be maintained in animal populations. Animal breeders were at the forefront of this important endeavor and the protagonists of the last chapter. I demonstrate that breeders in the state-owned breeding farms and their counterparts in collectives had different views on what constituted “superior” livestock, the treatment of “inferior” animals, and how the value produced by animals needed to be distributed. Although their assumptions and measures were different, I problematize how both the state and local collectives attempted to control nonhuman populations for their own purposes. I conclude that animal breeding itself was therefore eugenic and anthropocentric. No matter how unappealing the knowledge and practices of the breeders were, it was an essential part of the ways in which more-than-people’s communes maintained their multispecies well-being.

Archives and Other Sources

To support these findings and arguments, I refer to primary documents from the Beijing Municipal Archives, Gansu Provincial Archives, Inner Mongolia Bayannuur Municipal Archives, Guangxi Zhuang Autonomous Region Archives, Guangxi Yulin Municipal Archives, and Guangxi Bobai County Archives. Archival sources about rural communities on the outskirts of the capital give us a

对农村居民施加道德改造。本章还提请注意对“坏”看护人的频繁批评，并坚持认为讨论这些不受欢迎的工人可以作为异常情况的旁证。它揭示了国家多物种关怀共同体理想与农民与动物共存的抵抗形式之间的紧张关系和差异。

如果人民公社要成为超人民公社，就必须保持一定数量和质量的动物种群。动物饲养员是这项重要工作的先锋，也是上一章的主角。我证明，国有养殖场的饲养员和集体养殖场的饲养员对什么是“优等”牲畜、“劣等”动物的处理以及动物产生的价值如何分配有不同的看法。尽管他们的假设和措施不同，但我探讨了国家和地方集体如何都试图为了自己的目的控制非人类种群。我得出结论，动物育种本身因此是优生学和人类中心主义的。无论饲养员的知识和实践多么不吸引人，它都是超人民公社维持其多物种福祉方式的重要组成部分。

档案及其他来源

为支持这些发现和论点，我参考了北京市档案馆、甘肃省档案馆、内蒙古巴彦淖尔市档案馆、广西壮族自治区档案馆、广西玉林市档案馆和广西博白县档案馆的原始文献。关于首都郊区农村社区的档案资料为我们提供了

typical sense of how they responded to animal diseases when the central government’s presence was highly conspicuous. Cases from Guangxi, the southwestern most provincial-level administrative unit and farthest from the capital, forms a comparative pair with sources from Beijing, allowing us to weigh the dynamics between the central government and local communities. Gansu Province and Inner Mongolia were central in the development of veterinary medicine in modern China as their geography includes both staple farming and pastoral areas. Sources from Gansu and Inner Mongolia therefore inform us on the situation in places where veterinary expertise and experts were regarded to be more crucial than elsewhere. These archival materials from the different regions enabled me to construct representative, inclusive narratives on more-than-people’s communes that account for possible regional differences.

This dissertation is also based on memoirs, clinical notes, experiment reports, promotional materials regarding “model” veterinary workers, and self-criticism records of “bad” or “poor” workers. It includes both published and unpublished sources. Given the obvious propagandistic nature of some of these documents, we should ask to what extent their prescriptive and embellished accounts reflect historical reality. Some historians of modern China may insist that to understand rural China in the Maoist period it is not convincing to rely on propagandistic materials. But historians like Sigrid Schmalzer emphasize that “how we read our sources is more important than which sources we keep and which we throw away.”³⁵ As such, I highlight the fact that this dissertation does refer to didactic and propagandistic documents. I have tried my best to maintain a critical distance to these sources rather than taking them at face value. In the same vein, I specify,

³⁵ Sigrid Schmalzer, “Beyond Bias: Critical Analysis and Layered Reading of Mao-Era Sources,” *Positions: Asia Cultures Critique* 29.4 (November 2021): 760.

在中央政府高度显眼的情况下，他们如何应对动物疾病的典型意义。广西（中国西南最远的省级行政单位，离首都最远）的案例与北京的资料形成对比，使我们能够权衡中央政府与地方社区之间的动态。甘肃省和内蒙古在现代中国兽医学的发展中居于核心地位，因为它们的地理位置包括主要农耕区和牧区。因此，来自甘肃和内蒙古的资料使我们了解了兽医专业知识和专家被认为比其他地方更重要的地方的情况。来自不同地区的这些档案材料使我能够构建具有代表性的、包容性的关于“不止于人民公社”的叙述，其中考虑了可能的区域差异。

本论文还基于回忆录、临床笔记、实验报告、关于“模范”兽医工作者的宣传材料以及“不良”或“差劲”工作者的自我批评记录。它包括已发表和未发表的资料。鉴于其中一些文件明显的宣传性质，我们应该思考它们的规范性和修饰性描述在多大程度上反映了历史现实。一些研究现代中国的历史学家可能坚持认为，要理解毛泽东时期的中国农村，仅仅依靠宣传材料是不可信的。但像西格丽德·施马尔泽这样的历史学家强调，“我们如何解读我们的资料比我们保留哪些资料和丢弃哪些资料更重要。”³⁵ 因此，我强调本论文确实参考了说教性和宣传性文件。我已尽力与这些资料保持批判性距离，而不是照单全收。同样，我明确指出，

³⁵ Sigrid Schmalzer, “超越偏见：毛泽东时代资料的批判性分析和分层解读”，*Positions: Asia Cultures Critique* 29.4（2021年11月）：760。

whenever possible, the contexts in which certain unpublished documents were produced so that readers can be aware of their propagandistic features.

Although this dissertation is firmly based on written sources, I also conducted oral historical interviews with two former commune veterinarians and two former caretakers in Langfang, Hebei and Bobai, Guangxi. I have used these oral historical data in a limited fashion and only when they seem to complement the other documents.

尽可能地了解某些未发表文件产生的背景，以便读者能够意识到它们的宣传性质。

尽管本论文坚定地以书面资料为基础，但我也对河北廊坊和广西博白的两位前公社兽医和两位前饲养员进行了口述历史访谈。我有限地使用了这些口述历史数据，并且仅在它们似乎能补充其他文件时才使用。

Chapter One:
Infectious Animal Diseases

Introduction

To prevent infectious diseases from spreading and to protect the life and property of its constituents, the state should intervene in and control the everyday life of individuals. How this statement has been adopted by contemporaries at given times and in different places is one of the central questions of the field of the history of medicine. Historians of medicine have focused extensively on diverse infectious diseases to document how different forms of centralized governments around the globe came to converge on what Foucault describes as “biopower”³⁶ in tandem with different analytical focuses such as colonialism, race, gender, and so on.³⁷ Twentieth-century China was not exceptional. Historians Ruth Rogaski, Sean Hsiang-lin Lei, Miriam Gross, Mary Augusta Brazelton, and Xiaoping Fang, among others, have shown that the modern Chinese state developed bacteriology and epidemiology on its soil, monopolized peoples’ health information, and, whenever necessary, mobilized coercive and violent means to eliminate epidemics.³⁸

³⁶ Michel Foucault, *The History of Sexuality Vol. 1: An Introduction*, trans. Robert Hurley (New York: Pantheon Books, 1978), 135–145.

³⁷ Charles Rosenberg, *The Cholera Years: The United States in 1832, 1849, and 1866* (Chicago: University of Chicago Press, 1962); Allan Brandt, *No Magic Bullet: A Social History of Venereal Disease in the United States since 1880* (Oxford: Oxford University Press, 1985); David Arnold, *Colonizing the Body: State Medicine and Epidemic Disease in Nineteenth-Century India* (Berkeley: University of California Press, 1993); Nayan Shah, *Contagious Divides: Epidemics and Race in San Francisco’s Chinatown* (Berkeley: University of California Press, 2001); Warwick Anderson, *Colonial Pathologies: American Tropical Medicine, Race, and Hygiene in the Philippines* (Durham: Duke University Press, 2006); Jim Downs, *Sick from Freedom: African-American Illness and Suffering During the Civil War and Reconstruction* (Oxford: Oxford University Press, 2012); Nancy N. Chen and Lesley A. Sharp, eds., *Bioinsecurity and Vulnerability* (Santa Fe: School for Advanced Research Press, 2014).

³⁸ Ruth Rogaski, *Hygienic Modernity: Meanings of Health and Disease in Treaty-Port China* (Berkeley: University of California Press, 2004); Sean Hsiang-lin Lei, *Neither Donkey Nor Horse: Medicine in the Struggle Over China’s Modernity* (Chicago: University of Chicago Press, 2014); Miriam Gross, *Farewell to the God of Plague: Chairman Mao’s Campaign to Deworm China* (Berkeley: University of California Press, 2016); Mary Augusta Brazelton, *Mass Vaccination: Citizens’ Bodies and State Power*

第一章：
动物传染病

引言

为了防止传染病蔓延，保护民众的生命和财产，国家应该干预和控制个人的日常生活。这一说法在特定时期和不同地点是如何被同时代人采纳的，是医学史领域的核心问题之一。医学史学家广泛关注各种传染病，以记录全球各地不同形式的中央政府如何趋同于福柯所描述的“生命权力”³⁶，同时结合殖民主义、种族、性别等不同的分析重点。³⁷ 20世纪的中国也不例外。历史学家Ruth Rogaski、Sean Hsiang-lin Lei、Miriam Gross、Mary Augusta Brazelton和Xiaoping Fang等人已经表明，现代中国国家在其本土发展了细菌学和流行病学，垄断了人民的健康信息，并在必要时动用强制和暴力手段来消除流行病。³⁸

³⁶ 米歇尔·福柯，《性史：第一卷：导论》，罗伯特·赫尔利译（纽约：万神殿图书，1978），第135-145页。

³⁷ Charles Rosenberg, *The Cholera Years: The United States in 1832, 1849, and 1866* (Chicago: University of Chicago Press, 1962); Allan Brandt, *No Magic Bullet: A Social History of Venereal Disease in the United States since 1880* (Oxford: Oxford University Press, 1985); David Arnold, *Colonizing the Body: State Medicine and Epidemic Disease in Nineteenth-Century India* (Berkeley: University of California Press, 1993); Nayan Shah, *Contagious Divides: Epidemics and Race in San Francisco’s Chinatown* (Berkeley: University of California Press, 2001); Warwick Anderson, *Colonial Pathologies: American Tropical Medicine, Race, and Hygiene in the Philippines* (Durham: Duke University Press, 2006); Jim Downs, *Sick from Freedom: African-American Illness and Suffering During the Civil War and Reconstruction* (Oxford: Oxford University Press, 2012); Nancy N. Chen and Lesley A. Sharp, eds., *Bioinsecurity and Vulnerability* (Santa Fe: School for Advanced Research Press, 2014).

³⁸ 露丝·罗加斯基，《卫生现代性：通商口岸中国的健康与疾病的意义》（伯克利：加州大学出版社，2004）；雷祥麟，《非驴非马：中国现代化斗争中的医学》（芝加哥：芝加哥大学出版社，2014）；米里亚姆·格罗斯，《告别瘟神：毛主席的中国除虫运动》（伯克利：加州大学出版社，2016）；玛丽·奥古斯塔·布拉泽尔顿，《大规模疫苗接种：公民身体与国家权力

More recently, this concept of biopower has expanded from humans to encompass nonhumans. Increased awareness of the connectedness between human and animal health, as well as growing interest in animal rights and historical animal-human relations, have led to the idea that the development and exercise of biopower by many empires and nation-states were indeed multispecies. Animals, livestock products, and zoonotic pathogens needed to come under the biopolitical purview of the modern state. According to scholars of veterinary medicine and animal disease, how the operations of multispecies biopower were justified varies depending on: various political economies of agriculture; different understandings of disease; available technologies and materials; the degree to which the veterinary realm is professionalized; and conflicting notions over what deserves healing and what is killable.³⁹

Despite historical variability and contingency, there seems, however, to be a common assumption that central governments’ efforts to strengthen their power over life via bureaucratic means is the most important determinant of how biopower came into being. The story of multispecies biopower, therefore, has mostly taken the form of a top-down expansion of the state apparatus and of rational, if not scientific, health and population management. With few exceptions, the interventions of historians of epidemic disease revolve around the state’s priorities, policies, and actions, whether appreciating its timely successes or criticizing its certain short-sightedness,

in Modern China (Ithaca: Cornell University Press, 2019); Xiaoping Fang, *China and the Cholera Pandemic: Restructuring Society Under Mao* (Pittsburgh: University of Pittsburgh Press, 2021).

³⁹ Abigail Woods, *A Manufactured Plague: The History of Foot and Mouth Disease in Britain* (London: Earthscan, 2004); Priscilla Wald, *Contagious: Cultures, Carriers, and the Outbreak Narrative* (Durham: Duke University Press, 2007); Karen Brown and Daniel Gilfoyle, eds., *Healing the Herds: Disease, Livestock Economies, and the Globalization of Veterinary Medicine* (Athens: Ohio University Press, 2010); Karen Brown, *Mad Dogs and Meerkats: A History of Resurgent Rabies in Southern Africa* (Athens: Ohio University Press, 2011); Neel Ahuja, *Bioinsecurities: Disease Interventions, Empire, and the Government of Species* (Durham: Duke University Press, 2016); Rohan deb Roy, *Malarial Subjects: Empire, Medicine and Nonhumans in British India, 1820-1909* (Cambridge: Cambridge University Press, 2017); Natalie Porter, *Viral Economies: Bird Flu Experiments in Vietnam* (Chicago: University of Chicago Press, 2019).

最近，这种生物权力（biopower）的概念已从人类扩展到非人类。人们对人类和动物健康之间相互联系的认识不断提高，以及对动物权利和历史上海人与动物关系的兴趣日益增长，促使人们认为许多帝国和民族国家发展和行使的生物权力确实是多物种的。动物、牲畜产品和人畜共患病原体需要纳入现代国家的生物政治管辖范围。根据兽医学和动物疾病学者的说法，多物种生物权力的运作方式因以下因素而异：农业的各种政治经济；对疾病的不同理解；可用的技术和材料；兽医领域的专业化程度；以及关于什么值得治愈和什么可以被杀死的相互冲突的观念。³⁹

尽管存在历史变异性和偶然性，但似乎有一个共同的假设，即中央政府通过官僚手段加强其生命权力的努力是生物权力形成的最重要决定因素。因此，多物种生物权力的故事大多采取了自上而下的国家机器扩张以及理性（如果不是科学的）健康和人口管理的形式。除了少数例外，流行病史学家的干预都围绕着国家的优先事项、政策和行动，无论是赞赏其及时的成功，还是批评其某些短视之处，

in Modern China（伊萨卡：康奈尔大学出版社，2019）；方小平，*China and the Cholera Pandemic: Restructuring Society Under Mao*（匹兹堡：匹兹堡大学出版社，2021）。

³⁹ Abigail Woods, *A Manufactured Plague: The History of Foot and Mouth Disease in Britain*（伦敦：Earthscan，2004）；Priscilla Wald, *Contagious: Cultures, Carriers, and the Outbreak Narrative*（达勒姆：杜克大学出版社，2007）；Karen Brown 和 Daniel Gilfoyle，编，*Healing the Herds: Disease, Livestock Economies, and the Globalization of Veterinary Medicine*（雅典：俄亥俄大学出版社，2010）；Karen Brown, *Mad Dogs and Meerkats: A History of Resurgent Rabies in Southern Africa*（雅典：俄亥俄大学出版社，2011）；Neel Ahuja, *Bioinsecurities: Disease Interventions, Empire, and the Government of Species*（达勒姆：杜克大学出版社，2016）；Rohan deb Roy, *Malarial Subjects: Empire, Medicine and Nonhumans in British India, 1820-1909*（剑桥：剑桥大学出版社，2017）；Natalie Porter, *Viral Economies: Bird Flu Experiments in Vietnam*（芝加哥：芝加哥大学出版社，2019）。

illegitimacy, or brutality. Especially in the cases of epidemics in Maoist China (1949–1976), the aggressive state has been a focal point. In the scholarly narratives, fragmented individuals, regardless of whether they were portrayed as helpless or potentially resistant, as well as nonhumans, beneficent or harmful, ended up being subject to authoritarian state surveillance and violence.⁴⁰

In this chapter, I problematize the understandable yet overdetermined centrality of the state in understanding the historical unfolding of multispecies biopower in Maoist China as it obscures non-state historical actors’ perspectives and agencies. I shed light on how rural communities responded to major epizootic and zoonotic diseases. I particularly pay attention to the labor and knowledge of grassroots veterinary cadres, folk animal doctors, and epidemic prevention workers who worked in people’s communes and lower collectives, namely production brigades and production teams, under the condition of agricultural collectivization since 1953. I argue that it was these local veterinary workers, and not necessarily the central or provincial governments, who spearheaded a series of patterns in fighting infectious animal diseases— lockdowns, quarantines, disinfection, mass vaccination, mass education, and prioritizing the treatment of infected animals over mass culling. These techniques were only later recognized by the state as the “Comprehensive Prevention and Treatment” (综合性防治, hereafter CPT). By carrying out this CPT scheme, local veterinary workers cared about and sustained the vitality of their own communities where diverse human and nonhuman beings had to depend on one another to survive the revolution’s radicalism and uncertainty. In other words, I foreground bottom-up communal initiatives, which were concurrent with top-down health bureaucratization and medical scientific popularization that have

⁴⁰ Gross, *Farewell to the God of Plague*; Brazelton, *Mass Vaccination*, 123–143; Fang, *China and the Cholera Pandemic*.

非法、不正当或残暴。特别是在毛泽东时代（1949-1976）的流行病案例中，侵略性国家一直是焦点。在学术叙事中，无论是被描绘成无助的还是潜在的反抗者，以及有益或有害的非人类，最终都受到专制国家的监视和暴力。⁴⁰

在本章中，我将对国家在理解毛泽东时代中国多物种生物权力历史演变过程中所扮演的中心角色提出质疑，尽管这种中心地位可以理解但却被过度强调，因为它掩盖了非国家历史行动者的视角和能动性。我将阐明农村社区如何应对主要的动物流行病和人畜共患病。我特别关注基层兽医干部、民间兽医和防疫工作者的劳动和知识，他们在1953年农业集体化以来的人民公社和较低级集体，即生产大队和生产队中工作。我认为，正是这些地方兽医工作者，而不是中央或省级政府，率先开创了一系列防治动物传染病的模式——封锁、隔离、消毒、大规模疫苗接种、大众教育以及优先治疗受感染动物而非大规模扑杀。这些技术后来才被国家认可为“综合性防治”（以下简称CPT）。通过实施CPT方案，地方兽医工作者关心并维持了他们自己社区的活力，在那里，不同的人类和非人类生物必须相互依存才能在革命的激进主义和不确定性中生存下来。换句话说，我强调自下而上的社区倡议，这些倡议与自上而下的卫生官僚化和医学科学普及化同时进行，而这些

⁴⁰ Gross, 《瘟疫之神的告别》； Brazelton, 《大规模疫苗接种》, 123–143； Fang, 《中国与霍乱大流行》。

already been examined elsewhere, in the process of protecting living things from contagious diseases.

Collectives and Epizootics

The CCP started agrarian collectivization in 1953. The goal was to maximize the state’s capacity to control and extract an agrarian surplus for the sake of primitive accumulation to support rapid industrialization. The Communists first transformed the village, having approximatively 30 families and 145 people, into the most basic collective unit, called “mutual aid team,” “elementary cooperatives,” or “production team.” Now each household in the village could no longer claim private ownership, and the collective became a unit of production and distribution. The scale of collectives rose throughout the decade. On average, seven villages (production teams) formed an “advanced cooperatives” or a “production brigade” that consisted of about 220 households and 980 people. After 1958, the largest cooperatives, “people’s communes,” appeared and had an average population of 15,000. This three-level collective structure incorporating the grassroots rural society belonged to a county and was officially differentiated from the lowest state apparatus, the county government. Communes and lower collectives outlived Mao Zedong’s death in 1976 and were maintained until roughly 1985.⁴¹

Not only human peasants but nonhuman animals went through collectivization. Draft animals—oxen, donkeys, horses, and mules—and meat animals—pigs, sheep, chickens, ducks, geese, and rabbits—were expropriated as cooperatives’ public property. Every collective now

⁴¹ Franz Schurmann, *Ideology and Organization in Communist China* (Berkeley: University of California Press, 1966), 455; Andrew G. Walder, *China Under Mao: A Revolution Derailed* (Cambridge, Mass.: Harvard University Press, 2015), 56.

已在别处进行过研究，旨在保护生物免受传染病侵害。

集体与疫病

中国共产党于1953年开始实行农业集体化。其目标是最大限度地提高国家控制和提取农业剩余的能力，以实现原始积累，支持快速工业化。共产党首先将约有30户家庭和145人的村庄，转变为最基本的集体单位，称为“互助组”、“初级社”或“生产队”。现在，村里的每户家庭都不能再声称拥有私人所有权，集体成为生产和分配的单位。集体规模在整个十年中不断扩大。平均而言，七个村庄（生产队）组成一个“高级社”或“生产大队”，约有220户家庭和980人。1958年后，出现了最大的合作社——“人民公社”，平均人口为15,000人。这种将基层农村社会纳入其中的三级集体结构隶属于一个县，并与最低一级的国家机构——县政府正式区分开来。公社和下级集体在毛泽东1976年去世后仍然存在，并一直维持到大约1985年。⁴¹

不仅人类农民，非人类动物也经历了集体化。役畜——牛、驴、马和骡子——以及肉畜——猪、羊、鸡、鸭、鹅和兔子——被征用为合作社的公共财产。现在每个集体

⁴¹ Franz Schurmann, 《共产主义中国的意识形态与组织》（伯克利：加州大学出版社，1966），第455页；Andrew G. Walder, 《毛泽东领导下的中国：一场脱轨的革命》（马萨诸塞州剑桥：哈佛大学出版社，2015），第56页。

established its animal farms and selected some peasants as designated caretakers, who were tasked with taking care of the animals in those sheds. There were three to seven caretakers in a production team, and a caretaker usually looked after 20-30 pigs and five bovines or equines.⁴² There were fewer veterinarians than caretakers. Production teams usually had no veterinarians at all and, when one was necessary, had to call in brigade veterinarians. Each brigade was to have at least one full-time veterinarian. Every commune had a communal veterinary station staffed with five or more veterinarians who were responsible for the health of around 15,000 farm animals.⁴³ Although it cannot be denied that collectivization, especially at its inception and during the Great Leap Famine (1959–1961), negatively affected animal wellbeing,⁴⁴ local veterinary workers were able to gradually stabilize the system of communal management of livestock throughout the 1960s, as the latter chapters show.

The biggest threat to collective animals were epizootic diseases, or animal infectious disease. If China can be divided into agricultural and pastoral regions, the former includes most of China proper where the Han Chinese have lived, whereas the latter refers to some provinces in the northwest and northeast and Inner Mongolia which have been home to diverse ethnic minorities. In the agricultural regions, poultry, pigs, oxen, water buffalos, and donkeys were widespread, and therefore local veterinary workers mainly dealt with avian flu, anthrax, foot-and-mouth disease (FMD), swine fever, and swine pneumonia. In the grasslands and deserts, horses, camels, yaks, sheep, and goats were more common, and protecting these animals from anthrax, FMD, brucellosis,

⁴² Author’s interview with former caretakers Mrs. Hong and Mrs. Ling in Bobai County, Guangxi, June 2021.

⁴³ Author’s interview with former commune veterinarian Mr. Wen in Langfang, Hebei, May 2021.

⁴⁴ Peter W. Braden, “Serve the People: Bovine Experiences in China’s Civil War and Revolution, 1935–1961,” PhD diss., University of California at San Diego, 2020.

建立了畜牧场，并挑选了一些农民作为指定饲养员，负责照看棚圈里的牲畜。一个生产队有三到七名饲养员，一名饲养员通常照看20-30头猪和五头牛或马。⁴²兽医比饲养员少。生产队通常根本没有兽医，当需要时，必须请大队兽医。每个大队至少要有一名专职兽医。每个公社都设有一个公社兽医站，配备五名或更多兽医，负责约15,000头牲畜的健康。⁴³尽管不可否认，集体化，尤其是在其初期和大跃进饥荒（1959-1961）期间，对动物福利产生了负面影响，⁴⁴但地方兽医工作者 在整个1960年代能够逐步稳定牲畜的集体管理体系，正如后面章节所示。

对集体饲养的动物最大的威胁是流行病，即动物传染病。如果将中国划分为农耕区和畜牧区，前者包括汉族居住的中国大部分本土地区，而后者则指西北和东北的一些省份以及内蒙古，这些地区是不同少数民族的家园。在农耕区，家禽、猪、牛、水牛和驴子广泛分布，因此地方兽医工作者主要处理禽流感、炭疽、口蹄疫（FMD）、猪瘟和猪肺炎。在草原和沙漠地区，马、骆驼、牦牛、绵羊和山羊更为常见，保护这些动物免受炭疽、口蹄疫、布鲁氏菌病、

⁴² 作者于2021年6月在广西博白县采访了前饲养员洪女士和凌女士。

⁴³ 作者于2021年5月在河北廊坊采访了前公社兽医温先生。

⁴⁴ Peter W. Braden, “为人民服务：1935-1961年中国内战和革命中的牛类经验”， 博士论文，U加州大学圣地亚哥分校，2020年。

ovine parasitic diseases, sheep pox, and glanders were the highest priority. Grassroots veterinary workers paid particular attention to anthrax, brucellosis, FMD, and swine fever, among others. The first two could also infect human beings and the last two were most devastating economically.⁴⁵

As early as April 1959, at the height of the formation of people’s communes during the Great Leap Forward, the central government was aware that the concentration of livestock could make them more vulnerable to a variety of epizootics as it would “increases the chance of infection of diseases from one animal to another.”⁴⁶ Consequently, it emphasized that each communal and sub-communal collective had to proactively prevent and respond to animal diseases within its jurisdiction in the earliest stages of epidemic. The diversity of the environment and animal species across the country’s vast rural area also made this grassroots collective-based approach appear more sensible.⁴⁷ The conventional framework that associates medical and veterinary activities only with the centralizing state’s increasing biopower hinders us from grasping local communities’ resilience and competence in the face of formidable diseases.

Veterinary cadres and epizootic prevention workers, along with collective veterinarians and caretakers, were the main forces of the grassroots initiative against epizootics. Veterinary cadres (兽医干部 or 畜牧兽医干部) were middle or low-level CCP members who were in charge of livestock farming and veterinary affairs. They themselves were not veterinarians, but they could diagnose and

⁴⁵ Guangxi sheng nongyeting 广西省农业厅, “Guangxi sheng 1954 nian jiachu baoyu jihua dagang” 广西省 1954 年家畜保育计划大纲 (1954), Guangxi Zhuang Autonomous Region Archives (hereafter GARA), X163-001-0394-0083; “1958–1962 nian shouyi fangzhi jianyi shuzi” 1958–1962 年兽疫防治建议数字 (1966), Gansu Provincial Archives (hereafter GPA), 222-001-0410-0008.

⁴⁶ Nongyebu xumu shouyiju 农业部畜牧兽医局, *Renmin gongshe shouyi gongzuo shouce* 人民公社兽医工作手册 (Beijing: Nongye chubanshe, 1959), 3.

⁴⁷ Nongyebu xumu shouyiju, *Renmin gongshe shouyi gongzuo shouce*, 3 and 19.

绵羊 寄生虫病、羊痘和鼻疽是重中之重。基层兽医工作者特别关注炭疽、布鲁氏菌病、口蹄疫和猪瘟等。前两种疾病也会感染人类，后两种疾病在经济上最具破坏性。⁴⁵

早在1959年4月，在大跃进期间人民公社组建的高峰期，中央政府就意识到牲畜的集中可能使它们更容易受到各种流行病的影响，因为它会“增加疾病从一种动物传播到另一种动物的机会”。⁴⁶ 因此，中央政府强调，每个公社和次级公社集体都必须在其管辖范围内，在疫情爆发的最早阶段，积极预防和应对动物疾病。全国广大农村地区环境和动物物种的多样性也使得这种以基层集体为基础的方法显得更加明智。⁴⁷ 将医疗和兽医活动仅与中央集权国家日益增长的生物权力联系起来的传统框架，阻碍了我们理解地方社区在面对可怕疾病时的韧性和能力。

兽医干部和疫病防治人员，以及集体兽医和饲养员，是基层防治疫病工作的主要力量。兽医干部（兽医干部或畜牧兽医干部）是负责畜牧业和兽医事务的中低层中共党员。他们自己不是兽医，但他们可以诊断和

⁴⁵ 广西省农业厅，《广西省1954年家畜保育计划大纲》（1954），广西壮族自治区档案馆（以下简称GARA），X163-001-0394-0083；《1958–1962年兽疫防治建议数字》（1966），甘肃省档案馆（以下简称GPA），222-001-0410-0008。

⁴⁶ 农业部畜牧兽医局，《人民公社兽医工作手册》（北京：农业出版社，1959），第3页。

⁴⁷ 农业部畜牧兽医局，《人民公社兽医工作手册》，第3页和第19页。

treat common, mild ailments and were trained in how to handle vaccines, how to give animals an injection, what were the possible immune reactions, and how to care for inoculated animals. More importantly, veterinary cadres dealt with the “political” tasks related to animal disease control such as making a specific action plan of lockdown, quarantine, and disinfection; explaining the rationales of such plan to local peasants; and distributing necessary information and related propaganda materials to every corner of the collective.⁴⁸

These veterinary cadres did not necessarily represent the top-down workings of the state. Scholars who work on socialist China’s grassroots society have questioned the conventional view that local individuals and communities were completely overwhelmed by the authoritarian state. Debates about the nature of local Communist cadres, for example, provides one example of this historiographical move. Joseph Esherick argues that it is difficult to conceptualize rank-and-file Communists in the countryside as “an organizational weapon of obedient apparatchiks commanded by the Party Center.”⁴⁹ Rather, they were strategic “straddlers” between roles as state agents and representatives of local communities’ interests, sometimes complying with and sometimes adroitly derailing and undermining directives handed down by the upper levels of government.⁵⁰ Grassroots cadres in the veterinary realm were no different in this regard. They were, on the one hand,

⁴⁸ Pingle xian nonglin jishu zhidaozhan 平乐县农林技术指导站, “Guanyu 1953 nian xumu shouyi gongzuo zongjie” 关于 1953 年畜牧兽医工作总结 (December 24, 1953), GARA, X163-001-0250-0092; “Dali fangzhi zhu chuanranbing de chubu yijian” 大力防治猪传染病的初步意见 (1954), GARA, X163-001-0391-0093.

⁴⁹ Joseph W. Esherick, “Ten Theses on the Chinese Revolution,” *Modern China* 21.1 (January 1995): 61–64. Philip C. C. Huang similarly presents the “community operation with state assistance” to explain the nature of the lower-level governance in rural China. See Li-An Zhou, “Understanding China: A Dialogue with Philip Huang,” *Modern China* 45.4 (July 2019): 398–400.

⁵⁰ Benjamin L. Read, *Roots of the State: Neighborhood Organization and Social Network in Beijing and Taipei* (Stanford: Stanford University Press, 2012); Sang-Soo Park, “Neighborhood Space in 1950s Beijing: Urban Governance in the Early PRC,” Harvard-Yenching Institute Working Paper Series, 2015. <https://www.harvard-yenching.org/research/hyi-working-paper-series-park-sang-soo/>. Also, see Shu-min Huang, *The Spiral Road: Change in a Chinese Village Through The Eyes Of a Communist Party Leader* (Boulder: Westview Press, 1989).

治疗常见的轻微疾病，并接受了关于如何处理疫苗、如何给动物注射、可能的免疫反应以及如何护理接种疫苗的动物的培训。更重要的是，兽医干部处理与动物疾病控制相关的“政治”任务，例如制定封锁、隔离和消毒的具体行动计划；向当地农民解释此类计划的理由；以及向集体的每个角落分发必要的信息和相关宣传材料。⁴⁸

这些兽医干部不一定代表国家的自上而下运作。研究社会主义中国基层社会的学者对传统观点提出了质疑，即地方个人和社区完全被专制国家所压倒。例如，关于地方共产党员性质的辩论就提供了这种史学转变的一个例子。约瑟夫·埃舍里克认为，很难将农村的普通共产党员概念化为“听命于党中央的顺从的机关人员的组织武器”。⁴⁹相反，他们是国家代理人和地方社区利益代表之间的战略性“跨界者”，有时遵守，有时巧妙地阻挠和破坏上级政府下达的指令。⁵⁰兽医领域的基层干部在这方面也H A RV A RD U N IV ERS ITY不例外。他们一方面，

⁴⁸ Pingle xian nonglin jishu zhidaozhan 平乐县农林技术指导站, “Guanyu 1953 nian xumu shouyi gongzuo zongjie” 关于 1953 年畜牧兽医工作总结 (December 24, 1953), GARA, X163-001-0250-0092; “Dali fangzhi zhu chuanranbing de chubu yijian” 大力防治猪传染病的初步意见 (1954), GARA, X163-001-0391-0093.

⁴⁹ Joseph W. Esherick, “Ten Theses on the Chinese Revolution,” *Modern China* 21.1 (1995年1月): 61–64. Philip C. C. Huang 同样提出了“国家协助下的社区运作”来解释中国农村基层治理的性质。参见Li-An Zhou, “Understanding China: A Dialogue with Philip Huang”, *Modern China* 45.4 (2019年7月): 398–400.

⁵⁰ Benjamin L. Read, *Roots of the State: Neighborhood Organization and Social Network in Beijing and Taipei* (斯坦福: 斯坦福大学出版社, 2012); Sang-Soo Park, “Neighborhood Space in 1950s Beijing: Urban Governance in the Early PRC,” 哈佛燕京学社工作论文系列, 2015. <https://www.harvard-yenching.org/research/hyi-working-paper-series-park-sang-soo/>. 另请参见Shu-min Huang, *The Spiral Road: Change in a Chinese Village Through The Eyes Of a Communist Party Leader* (博尔德: Westview出版社, 1989)。

executors of veterinary policies and, on the other, actors who were keen to preserve their communities’ animal workforce for the sake of themselves and their fellow villagers.

Local cadres were initially reliant on a large number of traditional veterinary practitioners from the era before collectivization. This diverse group of folk animal healers were called “Chinese veterinarians” (中兽医) as opposed to “Western veterinarians” (西兽医), which broadly referred to Western-trained veterinarians or practitioners of Western veterinary medicine. While they were less familiar with vaccines and serums than acupuncture and moxibustion, they soon became vaccinators after short training sessions and local cadres were eager to mobilize them in the anti-epizootic campaigns.⁵¹ After the mid-1950s when the grassroots collective veterinary system was established, there were no longer private folk veterinarians. Former folk veterinarians were now absorbed into commune- or brigade-level veterinary institutions. And, thanks to these grassroots experts, communes became able to reproduce the next generations of veterinarians who generally were called collective veterinarians or “barefoot animal doctors,” particularly during the Cultural Revolution (1966–1976).⁵²

Grassroots cadres and veterinarians were supported by the even greater number of epizootic prevention workers (防疫员) and caretakers (饲养员). As collectivization intensified, each production team was required to educate at least one passionate young peasant to produce a prevention worker. In this way, by 1960 there were roughly 20,000 prevention workers in each

⁵¹ Zhao Qingsen 赵庆森, “Kaizhan jiachu fangyi baohu shengkou jiankang” 开展家畜防疫保护牲畜健康, *Renmin Ribao* 人民日报 (August 18, 1949).

⁵² “Xingan xian zhongshouyi dai tudi zanxing banfa” 新干县中兽医带徒弟暂行办法, *Minjian shouyi tongxun* 民间兽医通讯 3 (January 1958): 5–8; “Zhangye zhuanqu shi ruhe zhuangdale minjian shouyi duiwu de” 张掖专区是如何壮大了民间兽医队伍的 (1958), GPA, 227-001-0410-0016; Qianxian xumu shouyi gongzuozhan 乾县畜牧兽医工作站, *Chijiao shouyi shouce* 赤脚兽医手册 (Qianxian: Qianxian hongse yinshuachang, 1971).

一方面是兽医政策的执行者，另一方面是渴望为自己和村民保留社区动物劳动力的行动者。

地方干部最初依赖于集体化之前的大量传统兽医。这群形形色色的民间动物治疗师被称为“中兽医”（中兽医），与“西兽医”（西兽医）相对，后者泛指受西方训练的兽医或西方兽医学的从业者。虽然他们对疫苗和血清的熟悉程度不如针灸和艾灸，但经过短期培训后，他们很快就成为了疫苗接种者，地方干部也急于在抗疫运动中动员他们。⁵¹ 20世纪50年代中期，基层集体兽医制度建立后，不再有私人兽医。以前的民间兽医被吸收到公社或大队一级的兽医机构中。而且，多亏了这些基层专家，公社得以培养出下一代兽医，他们通常被称为集体兽医或“赤脚兽医”，尤其是在文化大革命（1966-1976）期间。⁵²

基层干部和兽医得到了数量更多的防疫员和饲养员的支持。随着集体化的深入，每个生产队都被要求培养至少一名有热情的年轻农民成为防疫员。通过这种方式，到1960年，每个县大约有20,000名防疫员。

⁵¹ 赵庆森，“开展家畜防疫保护牲畜健康”，人民日报（1949年8月18日）。

⁵² “新干县中兽医带徒弟暂行办法”，民间兽医通讯 3（1958年1月）：5-8；“张掖专区是如何壮大了民间兽医队伍的”（1958），GPA, 227-001-0410-0016；乾县畜牧兽医工作站，赤脚兽医手册（乾县：乾县红色印刷厂，1971）。

province.⁵³ These prevention workers would assist cadres in enforcing animal disease control (Figure 4). Although caretakers were not expected to play a particular role in the midst of epidemic, they were at the forefront of everyday prevention efforts such as keeping individual animals clean and healthy and conducting regular disinfections of collective farms.⁵⁴ All these different veterinary workers worked in concert with each other to prevent or minimize losses from infectious animal diseases throughout the Maoist era.

省。⁵³ 这些防疫人员将协助干部执行动物疫病控制（图4）。虽然不指望饲养员在疫情爆发时发挥特殊作用，但他们站在日常预防工作的第一线，例如保持个体动物清洁健康，并对集体农场进行定期消毒。⁵⁴ 在整个毛泽东时代，所有这些不同的兽医工作者相互协作，以预防或最大限度地减少传染性动物疾病造成的损失。



Figure 4. The 1970s certificates of swine vaccination. The main vaccinators were prevention workers, while the form of the certificate was designed and verified by the county veterinary station. Source: Author’s Collection.

⁵³ “Baozheng yangzhu shengchan gaosudu fazhan, gedi jiti yangchang dazhua fangyi zhushu” 保证养猪生产高速度发展, 各地集体猪场大抓防疫注射, *Renmin Ribao* 人民日报 (April 10, 1960).

⁵⁴ “Guanyu jiben xiaomie zhuwen de yijian” 关于基本消灭猪瘟的意见 (1964), GPA, 222-001-0346-0005.



图4。20世纪70年代的生猪疫苗接种证书。主要接种人员是防疫人员，而证书的形式由县兽医站设计和验证。来源：作者收藏。

⁵³ “保证养猪生产高速度发展, 各地集体猪场大抓防疫注射”, 《人民日报》(1960年4月10日)。

⁵⁴ “关于基本消灭猪瘟的意见” (1964), GPA, 222-001-0346-0005。

As the following sections will show, local veterinary workers went through trials and errors and came up with a series of protocols to respond to epizootics throughout the 1950s. They typically included lockdown, quarantine, disinfection, vaccination, treatment, and mass education. In the late 1950s, the central government retrospectively recognized these bottom-up actions and christened them as the “Comprehensive Prevention and Treatment.” Upper-level Communist leaders especially legitimated the CPT in the name of Mao’s famous call for the “mass line,” as its grassroots-centered characteristics facilitated the participation of a wide range of local constituents in public duties for the revolution.⁵⁵ This nationwide paradigm of the CPT remained in place in the 1960s and 1970s without major changes and with few regional variants.

Lockdown, Quarantine, and Disinfection

The most basic step of the CPT was to impose lockdowns and quarantines in infected areas and to disinfect animals, sheds, roads, and vehicles. The new Communist state’s early experiences of putting these measures into effect were geographically concentrated in the pastoral regions of the northwest where nomadic people had relied on animals for generations. As early as March 1951, the Gansu Provincial Government reported to Beijing that while confronting an outbreak of FMD in the province, they established watchhouses and inspection stations on major thoroughfares to control and disinfect the traffic. They also shut off large waterways. Infected animals were put into

⁵⁵ “Quanguo zhongxi shouyi yanjiu gongzuohuiyi queding” 全国中西兽医研究工作会议确定, *Renmin Ribao* 人民日报 (February 11, 1959); Gansu sheng renmin weiyuanhui 甘肃省人民委员会, “Guanyu xiaomie shengchu koutiyi gongzuo zongjie baogao” 关于消灭牲畜口蹄疫病工作总结 (April 26, 1965), GPA, 128-002-0718-0002.

如下文所示，在整个 20 世纪 50 年代，当地兽医工作者经历了反复试验，并制定了一系列应对动物疫病的方案。这些方案通常包括封锁、隔离、消毒、疫苗接种、治疗和大众教育。20 世纪 50 年代后期，中央政府追溯性地认可了这些自下而上的行动，并将其命名为“综合防治”。上层共产党领导人特别以毛泽东著名的“群众路线”号召的名义使综合防治合法化，因为其以基层为中心的特点促进了广大地方民众参与革命的公共职责。⁵⁵ 这种全国性的综合防治模式在 20 世纪 60 年代和 70 年代保持不变，没有重大变化，区域差异也很小。

封锁、隔离和消毒

综合防治最基本的步骤是在疫区实行封锁和隔离，并对动物、兽棚、道路和车辆进行消毒。新生的共产主义国家早期实施这些措施的经验主要集中在西北牧区，那里的人们世代都依靠动物为生。早在 1951 年 3 月，甘肃省政府就向北京报告说，在应对该省口蹄疫爆发时，他们在主要交通要道设立了检查站和检疫站，以控制和消毒交通。他们还关闭了主要水路。受感染的动物被放入

⁵⁵ “全国中西兽医研究工作会议确定”，《人民日报》（1959年2月11日）；甘肃省人民委员会，“关于消灭牲畜口蹄疫病工作总结”（1965年4月26日），GPA, 128-002-0718-0002。

quarantine and looked after separately. If the diseases had already spread within a particular herd, the entire herd was separated from the healthy herds and kept in a designated grazing district.⁵⁶

Some characteristics of this early experience shaped what would become the CPT. The 1951 report, for example, did not take into consideration the mass culling of infected animals. It was at odds with the dominant Western practice of widely terminating compromised animals for the efficient and rapid elimination of epizootics.⁵⁷ I will come back to this issue of culling in a later section. Here, the more important point is that the approach to animal diseases in the early PRC years was based not on sheer bureaucratic capacity, but on local people’s participation. According to the report, the provincial government’s preventative efforts needed to be “truly combined with [those of] the masses of people.” Also, “traditional animal healers, exemplary model workers, female peasants, shepherds, and so forth could play a central role in epizootic prevention.” Therefore, local leaders “absolutely tried to have them join” the temporary “small group for livestock protection.” It is not surprising that a nascent government which lacked human and material resources had no choice but to depend on local non-state actors.⁵⁸

Reliance on grassroots participation, however, did not disappear as state power stabilized throughout the first decade of the PRC. Rather, as the ambitious collectivization and consequential concentration of farm animals made the herds more vulnerable to epizootics, high-ranking leaders conceived that letting local communities collectively and flexibly respond to animal epidemics would

⁵⁶ Xibei junzheng weiyuanhui 西北军政委员会, “Jieshao 51 nian koutiyi fangzhi jingyan” 介绍 51 年口蹄疫防治经验 (April 8, 1952), GPA, 222-002-0068-0001.

⁵⁷ For example, see Woods, *A Manufactured Plague*.

⁵⁸ Xibei junzheng weiyuanhui, “Jieshao 51 nian koutiyi fangzhi jingyan.”

隔离并分开照料。如果疾病已在特定畜群中蔓延，则将整个畜群与健康畜群分开，并饲养在指定的放牧区。⁵⁶

这种早期经验的一些特点塑造了后来的CPT。例如，1951年的报告没有考虑大规模扑杀受感染的动物。这与西方普遍采用的为有效快速消除流行病而广泛扑杀受感染动物的做法背道而驰。⁵⁷我将在后面的章节中回到扑杀这个问题。这里，更重要的一点是，中华人民共和国早期对动物疾病的处理不是基于纯粹的官僚能力，而是基于当地人民的参与。根据报告，省政府的预防工作需要“真正与[群众]相结合”。此外，“传统兽医、模范工人、女农民、牧羊人等可以在流行病预防中发挥核心作用”。因此，地方领导“绝对努力让他们加入”临时的“牲畜保护小组”。一个缺乏人力和物力资源的新生政府别无选择，只能依靠地方非国家行为者，这并不奇怪。⁵⁸

然而，随着国家权力在中华人民共和国成立后的第一个十年中趋于稳定，对基层参与的依赖并未消失。相反，由于雄心勃勃的集体化和随之而来的牲畜集中使得畜群更容易受到动物疫病的影响，高级领导人认为，让地方社区集体灵活地应对动物疫情将

⁵⁶ Xibei junzheng weiyuanhui 西北军政委员会, “Jieshao 51 nian koutiyi fangzhi jingyan” 介绍 51 年口蹄疫防治经验 (April 8, 1952), GPA, 222-002-0068-0001.

⁵⁷ 例如，参见伍兹，《人造瘟疫》。

⁵⁸ 西北军政委员会，“介绍51年口蹄疫防治经验。”

be more efficient than centralized methods.⁵⁹ Against this backdrop, a Gansu provincial report on FMD in 1956 criticized the leadership of Linxia County for overestimating their own administrative capacity and not asking folk veterinarians for help. The county leaders were so complacent and lackadaisical that they thought that they could contain the spread of the disease with limited support from a few teenage boys who were asked to assist in the inspection stations.⁶⁰

The bottom-up initiative in the response to epizootics in the mid- and late-1950s led to growing sophistication in related countermeasures. Folk veterinarians and peasants made clear the point that the hay and filth from an infected animal’s shed had to be incinerated.⁶¹ They suggested that not only the movement of living ill animals but also that of any kinds of animal products from infected areas must be rigidly checked. Also, as each collective became the main unit in the fight against epizootic diseases, they came to pay attention to developing a collaborative prevention system “among counties, townships, and collectives.” Before long, these collective-led developments would be officially recognized by the central government as the CPT program.⁶²

It is also worth articulating collectives’ discussions on when lockdowns and quarantines should be lifted. To be certain, both collectives and county- or upper-level governments agreed that lockdown and quarantine “should not be eased too early.” At least when it comes to FMD, there seemed to be a consensus as early as 1956 that the end of an outbreak meant the status that “the majority of infected cattle regain their health, *and* that infection no longer increases (emphasized by

⁵⁹ Nongyebu xumu shouyiju, *Renmin gongshe shouyi gongzuo shouce*.

⁶⁰ Gansu sheng xumuting 甘肃省畜牧厅, “Guanyu koutiyi fangzhi gongzuo di er jieduan de qingkuang baogao” 关于口蹄疫防治工作第二阶段的情况报告 (October 22, 1956), GPA, 190-002-0157-0009.

⁶¹ “Dali fangzhi zhu chuanranbing de chubu yijian.”

⁶² Gansu sheng xumuting, “Guanyu koutiyi fangzhi gongzuo di er jieduan de qingkuang baogao.”

比集中式方法更有效率。⁵⁹ 在此背景下，1956年甘肃省一份关于口蹄疫的报告批评临夏县领导高估了自己的行政能力，没有向民间兽医寻求帮助。县领导们自满而散漫，他们认为只需少数被要求协助检查站工作的少年提供有限支持，就能控制疾病的传播。⁶⁰

20世纪50年代中后期，应对动物疫病的自下而上举措促使相关防控措施日益完善。民间兽医和农民明确指出，必须焚烧受感染动物棚舍中的干草和污物。⁶¹ 他们建议，不仅要严格检查病畜的活体移动，还要严格检查来自疫区的任何动物产品。此外，随着每个集体成为抗击动物疫病的主要单位，他们开始关注发展“县、乡、集体之间”的协作预防体系。不久，这些集体主导的发展将被中央政府正式认可为CPT计划。⁶²

同样值得阐明的是，集体就何时解除封锁和隔离进行了讨论。可以肯定的是，集体和县级或更高级别的政府都同意，封锁和隔离“不应放松过早”。至少就口蹄疫而言，早在1956年似乎就已达成共识，即疫情的结束意味着“大多数受感染的牛恢复健康，并且感染不再增加”的状态（由

⁵⁹ 农业部畜牧兽医局，《人民公社兽医工作手册》。

⁶⁰ 甘肃省畜牧厅，《关于口蹄疫防治工作第二阶段的情况报告》（1956年10月22日），GPA, 190-002-0157-0009。

⁶¹ “大力防治猪传染病的初步意见。”

⁶² 甘肃省畜牧厅，“关于口蹄疫防治工作第二阶段的情况报告。”

the author).” Gansu provincial leaders more carefully legislated that only fourteen days after achieving such status, collectives could lift lockdown with the approval of county governments.⁶³

Given that the endpoint of lockdown and quarantine was determined according to the double criteria of the recovery of herds and the disappearance of contagion, this consensus reflected that Chinese veterinary workers stressed preserving the animal power of a community as much as eliminating all pathogens within the jurisdiction.⁶⁴

In the early 1960s, the collective-based CPT was further institutionalized. When FMD spread from the northeast and Inner Mongolia and started to engulf Beijing in late 1963 and early 1964, the division of labor between upper-level governments and grassroots collectives emerged. Although Beijing Municipal Government and Hebei Provincial Government guided the overall response to the outbreak, their direct actions were largely confined to major railway stations, intercity bus terminals, and municipal distribution channels for food and goods. People’s communes and lower collectives on the outskirts of the capital took the lead in protecting their own communities. The large number of commune veterinarians, prevention workers, caretakers, and even militias stood guard at inspection stations along country roads and disinfected collective animal farms.⁶⁵

⁶³ Gansu sheng xumuting, “Guanyu koutiyi fangzhi gongzuo di er jieduan de qingkuang baogao.”

⁶⁴ Gansu sheng xumuting, “Guanyu koutiyi fangzhi gongzuo di er jieduan de qingkuang baogao.”

⁶⁵ Beijing shi xumu shouyi xuehui jiachu chuanranbing zu 北京市畜牧兽医学学会家畜传染病组, “Duiyu Beijing shi fangzhi koutiyi de jinji jianyi” 对于北京市防制口蹄疫的紧急建议 (January 24, 1964), Beijing Municipal Archives (hereafter BMA), 010-001-00446-00099; Beijing shi nonglinju 北京市农林局, Beijing shi gongxiaoshe 北京市供销社, Beijing shi fushipin shangyeju 北京市副食品商业局, Beijing shi shangye 北京市商业局, Beijing shi waimaoju 北京市外贸局, “Guanyu jiaqiang jianyi yanfang jiachu koutiyi chuanru de jinji tongzhi” 关于加强检疫严防家畜口蹄疫传入的紧急通知 (December 19, 1963), BMA, 119-001-00795-00106.

作者注)。甘肃省领导立法规定，集体在达到该状态十四天后，经县政府批准方可解除封锁。⁶³鉴于封锁和隔离的终点是根据畜群恢复和传染病消失的双重标准确定的，这一共识反映出中国兽医工作者既强调保护社区的畜力，也强调消除辖区内所有病原体。⁶⁴

20世纪60年代初，以集体为基础的CPT进一步制度化。1963年末至1964年初，当口蹄疫从东北和内蒙古蔓延并开始席卷北京时，上级政府和基层集体之间的分工出现了。尽管北京市政府和河北省政府指导了疫情的总体应对，但它们的直接行动主要局限于主要火车站、城际汽车站以及食品和商品的市政分销渠道。首都郊区的人民公社和下级集体率先保护自己的社区。大量的公社兽医、防疫人员、护理人员，甚至民兵，都在乡村公路沿线的检查站站岗，并对集体养殖场进行消毒。⁶⁵

⁶³ 甘肃省畜牧厅， “关于口蹄疫防治工作第二阶段的情况报告。”

⁶⁴ 甘肃省畜牧厅， “关于口蹄疫防治工作第二阶段的情况报告。”

⁶⁵ 北京市畜牧兽医学学会家畜传染病组， “对于北京市防制口蹄疫的紧急建议”（1964年1月24日），北京市档案馆（以下简称BMA）， 010-001-00446-00099；北京市农林局，北京市供销社，北京市副食品商业局，北京市商业局，北京市外贸局， “关于加强检疫严防家畜口蹄疫传入的紧急通知”（1963年12月19日）， BMA， 119-001-00795-00106。

The bottom-up initiative in deploying lockdown and quarantine often gave local collectives and the upper-level authorities a misleading and unrealistic idea that each collective, as a unit, should be self-sufficient in responding to epizootic diseases. Although no one disagreed that harmonious inter-communal collaboration in prevention was desirable, it was very common that a collective blamed another for disseminating pathogens by failing to carry out its responsibilities. When a suburban community in Beijing grappled with swine fever in 1964, for example, its residents pinpointed “a bicycle rider who came from outside to sell pigs” as the origin of the contagion. These and similar grassroots disputes led the central government to discourage inter-collective animal trade and to come up with the slogan, “self-breeding and self-raising” (自繁自养).⁶⁶ Now provincial and county-level governments expected communes and lower collectives to train some peasants as breeders so that grassroots communities could reproduce and maintain farm animals on their own with trading with the outside. “Self-breeding and self-raising” also meant that collectives needed to be self-sufficient not only in maximizing the number of collective livestock, but also in minimizing the loss of animals when epizootic diseases struck. And this demand for communal self-sufficiency clearly became the core part of the CPT, especially during the last decade of the Maoist era.⁶⁷

Unlike the common idea that the centralized authoritarian state in China controlled and choked off individuals by increasing its biopower, the grassroots veterinary workers who undergirded the CPT program were challenged by the troublesome ideal of self-sufficiency. In many cases, the goal was hardly achieved. After major epizootics, it turned out that collectives needed help

⁶⁶ “Beijing shi xumu shouyi xuehui juxing zhuwen fangzhi zuotanhui jianbao” 北京市畜牧兽医学会举行猪瘟防治座谈会简报 (July 28, 1964), BMA, 010-001-00187-00055.

⁶⁷ Zhonggong zhongyang 中共中央 and Guowuyuan 国务院, “Guanyu fazhan dashengchu de jixiang guiding” 关于发展大牲畜的几项规定 (November 22, 1962), in *Xumu shouyi gongzuo cankao ziliao* 畜牧兽医工作参考资料, ed. Gansu sheng xumuting 甘肃省畜牧厅 (Lanzhou: Gansu sheng xumuting, 1964), 1–3.

自下而上地推行封锁和隔离，常常让地方集体和上级政府产生一种误导性和不切实际的想法，即每个集体作为一个单位，在应对动物疫病时都应自给自足。尽管没有人不同意预防工作中和谐的社区间协作是可取的，但集体之间互相指责对方未能履行职责而传播病原体的情况却很常见。例如，1964年北京郊区的一个社区在应对猪瘟时，其居民将“一个从外面来卖猪的骑自行车的人”认定为传染源。这些以及类似的基层纠纷导致中央政府不鼓励集体间的牲畜交易，并提出了“自繁自养”的口号（自繁自养）。⁶⁶ 现在，省县级政府期望公社和下级集体培训一些农民成为饲养员，以便基层社区可以自行繁殖和饲养农场动物，而无需与外界进行交易。“自繁自养”也意味着集体不仅要在最大化集体牲畜数量方面自给自足，而且要在动物疫病爆发时将动物损失降到最低。这种对社区自给自足的要求显然成为CPT的核心部分，尤其是在毛泽东时代的最后十年。⁶⁷

与普遍认为的中国中央集权专制国家通过增强其生物权力来控制 and 扼杀个人的观点不同，支撑CPT项目的基层兽医工作者受到了自给自足这一棘手理想的挑战。在许多情况下，这一目标很难实现。在重大流行病之后，事实证明集体需要帮助

⁶⁶ “北京市畜牧兽医学会举行猪瘟防治座谈会简报”（1964年7月28日），BMA, 010-001-00187-00055。

⁶⁷ 中共中央和国务院，“关于发展大牲畜的几项规定”（1962年11月22日），载甘肃省畜牧厅编畜牧兽医工作参考资料（兰州：甘肃省畜牧厅，1964），1-3。

from the outside to recoup the loss of livestock. Some pig-raising collectives in Guangxi “relied on the [outside] market to buy new pigs” to refill their collective farms hit by another severe swine fever outbreak in 1972. Unfortunately, the newly acquired pigs brought about a new spike of infections. What the upper-level governments, which were ideologically rich yet materially deficient, could do was to criticize particular collectives for violating the principle of self-breeding and self-raising.⁶⁸ Under the paradigm of the CPT, it was grassroots veterinary cadres and workers who, for better or worse, bore the heavy burden of protecting nonhuman collective members by hook or by crook.

Mass Animal Vaccination and Mass Science

Lockdown, quarantine, and disinfection were the first actions that local veterinary cadres and workers took when animal infectious diseases broke out. But however meticulously these measures were carried out, there were always casualties, for the most part livestock but also, sometimes, human. For instance, when anthrax swept Jianling Brigade, Guangxi in 1962, 15 oxen and 18 people contracted the disease; ten oxen and three peasants died. Even three pigs and a dog died from licking up the infected oxen’s guts and bone marrow.⁶⁹ For a community whose population was less than 1,000 people, this was not insignificant damage. As such, prevention was highly emphasized as

⁶⁸ Yulin diqu nonglinju 玉林地区农林局, “Guanyu shengzhu yibing wenti de baogao” 关于生猪疫病问题的报告 (October 16, 1972), Guangxi Yulin Municipal Archives (hereafter GYMA), 064-001-012-09.

⁶⁹ Yulin zhuan yuan gongshu 玉林专员公署, “Guanyu Pingnan xian Wujiang deng xiang fasheng niu tanjubing ganran ren chu bingwang de tongbao” 关于平南县乌江等乡发生牛炭疽病感染人畜病亡的通报 (December 15, 1962), GYMA, 030-009-013-04.

从外部弥补牲畜损失。广西一些养猪集体“依靠[外部] 市场购买新猪”，以补充其在1972年遭受另一次严重猪瘟爆发的集体农场。不幸的是，新购买的猪带来了新的感染高峰。上级政府在思想上富足但在物质上匮乏，他们所能做的就是批评某些集体违反了自繁自养的原则。⁶⁸在CPT范式下，正是基层兽医干部和工人，无论好坏，都承担着不择手段保护非人类集体成员的重担。

大规模动物疫苗接种和大众科学

封锁、隔离和消毒是地方兽医干部和工人在动物传染病爆发时采取的首要行动。但无论这些措施多么细致地执行，总会有伤亡，大部分是牲畜，有时也包括人类。例如，1962年炭疽病席卷广西健陵大队时，有15头牛和18人感染了这种疾病；10头牛和3名农民死亡。甚至有3头猪和1条狗因舔食受感染牛的内脏和骨髓而死亡。⁶⁹对于一个人口不足1000人的社区来说，这不是微不足道的损失。因此，预防被高度强调为

⁶⁸ 玉林地区农林局, 《关于生猪疫病问题的报告》（1972年10月16日），广西玉林市档案馆（以下简称GYMA），064-001-012-09。

⁶⁹ 玉林专员公署, 《关于平南县乌江等乡发生牛炭疽病感染人畜病亡的通报》（1962年12月15日），GYMA, 030-009-013-04。

part of the CPT program, which came to produce another famous slogan: “rather prevention than treatment (防重于治).” And for prevention, there was no better option than mass vaccination.⁷⁰

Mass animal vaccination required mass education. It was different than bringing local villagers to inspection stations and asking them to control and disinfect the traffic of people and animals over a certain period of time. If local collectives were to administer vaccines, someone present needed to have some practical knowledge, such as how to discern the epizootic disease in question from other disorders; how to store vaccines appropriately; what is the correct dosage; how to bind animals while giving a shot; what are possible immune reactions; and so on. Ideally, someone in the collectives should know who Louis Pasteur was and how vaccinations work. Therefore, the story of the CPT, as it included mass animal vaccination, was that of the broad circulation of knowledge across different social stratum. In this regard, it was a component in realizing the Maoist ideal of mass science.⁷¹

Local veterinary cadres played an integral role in this process of “gradually fostering the scientific prevention practice of the masses” (逐渐养成群众科学的防疫习惯). In fact, they themselves were beneficiaries of mass science. In 1951–53, the first cohorts of local Communist veterinary cadres were produced through short-term training organized by provincial or county-level governments. They were not regarded as experts in healing animals, but they were intensively

⁷⁰ Cai Ziwei 蔡子伟, “Jiji cujin xumuye shengchan dayuejin” 积极促进畜牧业生产大跃进, *Renmin Ribao* 人民日报 (June 4, 1958).

⁷¹ Schmalzer, *Red Revolution, Green Revolution*; Zhonghua renmin gongheguo nongyebu 中华人民共和国农业部, “Guanyu zhengdun he chongshi xumu shouyizhan de zhishi” 关于整顿和充实畜牧兽医站的指示 (January 3, 1963), in *Xumu shouyi gongzuo cankao ziliao* 畜牧兽医工作参考资料, ed. Gansu sheng xumuting 甘肃省畜牧厅 (Lanzhou: Gansu sheng xumuting, 1964), 43–45.

CPT 项目的一部分，该项目后来提出了另一个著名口号：“防重于治（防重于治）。”对于预防，没有比大规模疫苗接种更好的选择。⁷⁰

大规模动物疫苗接种需要大规模教育。这与将当地村民带到检查站，要求他们在特定期限内控制和消毒人畜交通不同。如果当地集体要接种疫苗，在场的人需要具备一些实践知识，例如如何区分所讨论的动物疫病与其他疾病；如何妥善储存疫苗；正确剂量是多少；注射时如何捆绑动物；可能的免疫反应是什么；等等。理想情况下，集体中的某个人应该知道路易·巴斯德是谁以及疫苗接种是如何进行的。因此，CPT 的故事，因为它包括大规模动物疫苗接种，是知识在不同社会阶层广泛传播的故事。在这方面，它是实现毛泽东群众科学理想的一个组成部分。⁷¹

地方兽医干部在“逐渐养成群众科学的防疫习惯”这一过程中发挥了不可或缺的作用。事实上，他们自己就是群众科学的受益者。1951年至1953年，第一批地方共产主义兽医干部通过省或县级政府组织的短期培训培养出来。他们不被视为动物治疗专家，但他们被深入地

⁷⁰ 蔡子伟，“积极促进畜牧业生产大跃进”，人民日报（1958年6月4日）。

⁷¹ Schmalzer, *Red Revolution, Green Revolution*；中华人民共和国农业部，“关于整顿和充实畜牧兽医站的指示”（1963年1月3日），载甘肃省畜牧厅编畜牧兽医工作参考资料（兰州：甘肃省畜牧厅，1964），第43-45页。

introduced to the “common senses regarding scientific disease prevention and therapeutics that the masses could easily get” and the handling of “preventative materials.”⁷²

In the 1950s and after, while utilizing their knowledge and leading epizootic prevention within their collectives, veterinary cadres were keen on recruiting “activists” (积极分子), passionate villagers who showed strong support for and loyalty to the Communist Party. They encouraged the activists to participate in the regular vaccinating of livestock and gave them a chance to learn through practice (边做边学). The cadres also organized on-the-ground technical lectures for the activists or recommended them for opportunities for provincial or county-level veterinary workshops. Local veterinary cadres in these ways helped some ordinary peasants become “the most reliable grassroots workforce” (最可靠的基层力量). These activists were the “wellspring of future [veterinary] cadres,” and as such one or two activists in each elementary cooperative or production team came to be officially titled “epizootic prevention workers.”⁷³

Let us take a closer look at how veterinary cadres and activists-turned-prevention workers worked together by examining a specific case from Beijing in 1952. When swine fever hit the capital in the early spring of that year, veterinary cadres mobilized 342 youths from 47 nearby villages and organized 6-day intensive vaccination technical sessions. During this training period, they were able to inject swine fever vaccines in 3,745 pigs, 42.5% of the total 8,337 swine population in the participating villages. To vaccinate the remaining pigs, the class was split into work teams of 4 to 12

⁷² “Dali fangzhi zhu chuanranbing de chubu yijian.”

⁷³ “Dali fangzhi zhu chuanranbing de chubu yijian.”

向大众普及“关于科学疾病预防和治疗的常识，大众可以轻松掌握”以及“预防材料”的处理。⁷²

在20世纪50年代及以后，兽医干部在利用他们的知识并在其集体内部领导动物疫病预防的同时，积极招募“积极分子”（积极分子），这些热情的村民对共产党表现出强烈的支持和忠诚。他们鼓励积极分子参与牲畜的定期疫苗接种，并给他们通过实践学习的机会（边做边学）。干部们还为积极分子组织了实地技术讲座，或推荐他们参加省级或县级兽医培训班。地方兽医干部通过这些方式帮助一些普通农民成为“最可靠的基层力量”（最可靠的基层力量）。这些积极分子是“未来[兽医]干部”的“源泉”，因此每个初级合作社或生产队都有一两名积极分子被正式命名为“动物疫病防治员”。⁷³

让我们通过一个1952年北京的具体案例，来仔细看看兽医干部和转变为防疫工作者的积极分子是如何协同工作的。当年早春，猪瘟袭击了首都，兽医干部动员了来自附近47个村庄的342名青年，组织了为期6天的猪瘟疫苗接种技术强化培训班。在培训期间，他们为3745头猪注射了猪瘟疫苗，占参与村庄8337头猪总数的42.5%。为了给剩余的猪接种疫苗，培训班被分成4到12人的工作小组

⁷² “大力防治猪传染病的初步意见。”

⁷³ “大力防治猪传染病的初步意见。”

people and sent down to unvisited villages. While giving animals a shot, they put informative posters on billboards and explained the benefits of vaccination door to door.⁷⁴

Some peasants, however, were still worried that the vaccine might harm their precious animals. Villagers on the outskirts of Beijing even openly demonstrated their mistrust in the technical competence of the teams, saying that “I will give you the vaccination fee if my pigs will be alive in six months.”⁷⁵ Guangxi peasants, too, were skeptical about vaccinators. They complained that after getting a vaccine, “most oxen got emaciated and lethargic, limped, and had diarrhea.” They therefore found vaccination to be disturbing agricultural work. Once a few of the vaccinated oxen died, they explicitly refused to get their livestock vaccinated.⁷⁶ Only after a great number of animals contracted diseases and the spread looked uncontrollable did the peasants call in veterinary workers and “direly asked for healing the infected animals.”⁷⁷ To have the ideal of mass science as well as the CPT program fully work in the grassroots veterinary realm, the vaccinators—local veterinary cadres and prevention workers—must have gained their fellow peasants’ trust.

The cadres and prevention workers, however, lacked degrees from elite institutions or any other social capital to address this issue of trust. It is worth noting that from the perspective of local peasants, the veterinary cadres, not to mention the peasants-turned-prevention workers, were by no means anonymous state agents arriving from above. Most were neighbors from the same or nearby

⁷⁴ “Zhuwen liuxing zhili qingkuang” 猪瘟流行治理情况 (April 24, 1952), BMA, 009-002-00296-073.

⁷⁵ “Zhuwen liuxing zhili qingkuang.”

⁷⁶ Fuzhong xian renmin zhengfu 富钟县人民政府, “Fuzhong xian 1953 nian chunji shouyi fangyi gongzuo zongjie” 富钟县 1953 年春季兽医防疫工作总结 (April 30, 1953), GARA, X163-001-0256-0098.

⁷⁷ Wuzhou shi renmin zhengfu 梧州市人民政府, “Guanyu Wuzhou shi 1954 nian xumu shouyi gongzuo de baogao” 关于梧州市 1954 年畜牧兽医工作的报告 (1954), GARA, X163-001-0395-0056.

人员，并派往未曾到访的村庄。在给动物注射疫苗的同时，他们在广告牌上张贴宣传海报，并挨家挨户解释疫苗的好处。⁷⁴

然而，一些农民仍然担心疫苗可能会伤害他们宝贵的动物。北京郊区的村民甚至公开表达了他们对团队技术能力的怀疑，声称“如果我的猪能在六个月内活下来，我就给你疫苗接种费。”⁷⁵ 广西的农民也对疫苗接种人员持怀疑态度。他们抱怨说，接种疫苗后，“大多数牛都变得消瘦、嗜睡、跛行，并出现腹泻。”因此，他们认为疫苗接种扰乱了农业生产。一旦有几头接种过疫苗的牛死亡，他们就明确拒绝给牲畜接种疫苗。⁷⁶ 只有当大量动物感染疾病且疫情看起来无法控制时，农民才请来兽医，并“迫切要求治疗受感染的动物。”⁷⁷ 为了使大众科学的理想以及CPT计划在基层兽医领域充分发挥作用，疫苗接种人员——当地兽医干部和防疫人员——必须赢得同乡农民的信任。

然而，干部和防疫人员缺乏名牌院校的学历或任何其他社会资本来解决信任问题。值得注意的是，在当地农民看来，兽医干部，更不用说农民出身的防疫人员，绝不是从天而降的匿名国家代理人。他们大多是来自同村或附近村庄的邻居，

⁷⁴ “猪瘟流行治理情况”（1952年4月24日），BMA，009-002-00296-073。

⁷⁵ “猪瘟流行治理情况。”

⁷⁶ 富钟县人民政府，“富钟县1953年春季兽医防疫工作总结”（1953年4月30日），GARA，X163-001-0256-0098。

⁷⁷ 梧州市人民政府，“关于梧州市 1954 年畜牧兽医工作的报告”（1954），GARA，X163-001-0395-0056。

villages. Furthermore, collectivization and the merging of several villages into a certain scale of collective meant these local actors shared a broader sense of community. Against this backdrop, veterinary cadres and workers tried to form face-to-face ties with peasants and convince them to support vaccination.⁷⁸

For instance, in December 1953, Pingle County in Guangxi, where 37,318 households, 39,723 cattle, and a larger number of pigs and chickens lived, was facing the outbreak of several animal diseases including anthrax, brucellosis, and swine erysipelas. Each collective within the county had its own group of veterinary cadres and activist-assistants to vaccinate farm animals. These veterinary workers “explained in detail the benefits of shots” as well as described anticipated immune reactions to the village elders whom they were familiar with for long. And, about 10 days after giving animals a shot, they made a “follow-up visit” (复查访问) to collective farms. A peasant there told the cadres that “on the next day after the vaccination, some pigs did not eat and the spot where they got the shot swelled up. If you had not explained it to us beforehand, I would have been very worried.”⁷⁹

Similarly, veterinary workers in Fuzhong County, Guangxi took advantage of their personal closeness to villagers who worked side by side with farm animals. From February 28 to March 31, 1953, a team of three veterinary cadres, a folk veterinarian, and an unknown number of prevention workers in charge of vaccinating livestock in their own cooperatives administered 583 rinderpest vaccines, 35 anthrax vaccines, 1,467 brucellosis vaccines, and 4,139 avian Newcastle disease vaccines. In the process, they encountered some “skeptics” and “bystanders” hanging around the

⁷⁸ Fuzhong xian renmin zhengfu, “Fuzhong xian 1953 nian chunji shouyi fangyi gongzuo zongjie.”

⁷⁹ Pingle xian nonglin jishu zhidaozhan, “Guanyu 1953 nian xumu shouyi gongzuo zongjie.”

村庄。此外，集体化以及将几个村庄合并成一定规模的集体，意味着这些地方行动者拥有更广泛的社区意识。在此背景下，兽医干部和工作人员试图与农民建立面对面的联系，并说服他们支持疫苗接种。⁷⁸

例如，1953年12月，广西平乐县有37318户家庭、39723头牛以及大量的猪和鸡，正面临炭疽病、布鲁氏菌病和猪丹毒等多种动物疾病的爆发。该县的每个集体都有自己的兽医干部和积极分子助理团队，负责为农场动物接种疫苗。这些兽医工作人员“详细解释了注射的好处”，并向他们长期熟悉的村里老人描述了预期的免疫反应。大约在给动物注射后10天，他们对集体农场进行了“复查访问”（复查访问）。一位农民告诉干部：“接种疫苗的第二天，有些猪不吃东西，注射部位肿胀。如果你们没有事先向我们解释，我就会非常担心。”⁷⁹

同样，广西富钟县的兽医人员利用他们与村民的私人亲近关系，与村民并肩工作。从1953年2月28日到3月31日，一个由三名兽医干部、一名民间兽医和数量不详的负责为合作社牲畜接种疫苗的防疫人员组成的团队，共接种了583剂牛瘟疫苗、35剂炭疽疫苗、1467剂布鲁氏菌病疫苗和4139剂禽新城疫疫苗。在此过程中，他们遇到了一些“怀疑论者”和“旁观者”围观

⁷⁸ Fuzhong xian renmin zhengfu, “Fuzhong xian 1953 nian chunji shouyi fangyi gongzuo zongjie.”

⁷⁹ 平乐县农林技术指导站， “关于1953年畜牧兽医工作总结。”

vaccination sites. The team made sure to have them see for themselves that the vaccinated animals were okay and to elaborate the mechanism of vaccination and post-vaccinal issues.⁸⁰ To grassroots peasants, the trust in epizootic vaccines was little differentiated from their trust in the vaccinators. Hence, the direct human relationship between peasants and veterinary workers within communities could increase the general credibility of vaccination as well as the CPT program.

A decade later in 1963, as local experiences were continuously accumulated, the central Ministry of Agriculture reaffirmed the method of communal mass animal vaccination. Taking swine fever, for instance, the Ministry claimed that more than 10 years of preventative work verified the effectiveness of both vaccines and local vaccinators. China had already secured a valid vaccine technology, namely the rabbit-attenuated swine fever vaccine, which was easy and cheap to produce, quick to ensure immunity (four days after the vaccination), and long-lasting (effective for a year and a half). Central leaders also stipulated that “each commune should train its full-time prevention workers, and each brigade needs to have a prevention worker.” And, when epizootic and zoonotic diseases spread, each county would guide its communes and lower collectives to “combine the capabilities of veterinary cadres, veterinarians, and prevention workers” and organize a CPT taskforce. In doing so, grassroots communities could “take the three-in-one methodology incorporating local leaders, technicians, and the masses into the unfolding of the prevention and treatment work.”⁸¹ Collective-based mass science and trust in this approach were the foundation of Maoist China’s fight against contagious animal diseases.

⁸⁰ Fuzhong xian renmin zhengfu, “Fuzhong xian 1953 nian chunji shouyi fangyi gongzuo zongjie.”

⁸¹ Zhonghua renmin gongheguo nongyebu 中华人民共和国农业部, “Guanyu zai disange wunian jihua qinei jibenshang xiaomic zhuwen de zhishi” 关于在第三个五年计划期内基本上消灭猪瘟的指示 (July 22, 1963), in *Xumu shouyi gongzuo cankao ziliao* 畜牧兽医工作参考资料, ed. Gansu sheng xumuting 甘肃省畜牧厅 (Lanzhou: Gansu sheng xumuting, 1964), 51–52.

疫苗接种点。团队确保让他们亲眼看到接种疫苗的动物安然无恙，并详细阐述了疫苗接种的机制和接种后的问题。⁸⁰ 对于基层农民来说，他们对动物疫苗的信任与对疫苗接种人员的信任没有太大区别。因此，农民与社区内兽医工作者之间直接的人际关系可以提高疫苗接种和CPT计划的总体可信度。

十年后，即1963年，随着地方经验的不断积累，中央农业部重申了社区性大规模动物疫苗接种的方法。以猪瘟为例，农业部声称，十多年的预防工作验证了疫苗和地方疫苗接种人员的有效性。中国已经获得了一种有效的疫苗技术，即兔化猪瘟疫苗，这种疫苗生产简单、成本低廉、免疫速度快（接种后四天）、持续时间长（一年半有效）。中央领导还规定，“每个公社都应培训其专职防疫人员，每个大队都需要有一名防疫人员。”并且，当动物疫病和人畜共患病蔓延时，各县将指导其公社和下级集体“结合兽医干部、兽医和防疫人员的能力”，组织CPT工作队。通过这样做，基层社区可以“将地方领导、技术人员和群众三结合的方法融入到防治工作的开展中。”⁸¹ 基于集体的群众科学和对这种方法的信任是毛泽东时代中国对抗动物传染病的基础。

⁸⁰ 抚钟县人民政府，“抚钟县1953年春季兽医防疫工作总结。”

⁸¹ 中华人民共和国农业部，“关于在第三个五年计划期内基本上消灭猪瘟的指示”（1963年7月22日），载甘肃省畜牧厅编《畜牧兽医工作参考资料》（兰州：甘肃省畜牧厅，1964），51-52页。

Producing Vaccines On site

In the 1960s and 70s, as the CPT program based on self-sufficient collectives became the standard in response to animal diseases, the centrality of the grassroots was more and more taken for granted.

In 1963, the central government proclaimed the principal of “the state to finance, collectives to labor (国家出本, 社队出工).” It was supposed to mean that the central or provincial governments would provide communes and lower collectives with vaccines, syringes, disinfectants, and so on, while each collective should be in charge of organizing the actual work and bearing the cost of labor. As we will see the case study of the on-site production of the rabbit-attenuated swine fever vaccine shortly, the material supply that the state promised did not always work out. The labor of grassroots veterinary workers was calculated by “work points” (工分), and such points would translate into a certain salary or food rations. All these expenses should be paid from the budget of collectives, not the state. Certainly, what we often vaguely call “the state” in Maoist China wanted to avoid responsibility for the health of communal animals, which reveals the mismatch with the conventional image of the intervening, overcontrolling Big Brother state.⁸²

This tendency even led to the idea that local collectives should be able to produce vaccines, especially swine fever vaccines, on their own without asking the upper-level state apparatus for help. The Ministry of Agriculture determined that when it comes to the rabbit-attenuated swine fever vaccine, “local procurement” was the rule.⁸³ More specifically, it required each county to select one

⁸² Zhonghua renmin gongheguo nongyebu, “Guanyu zai disange wunian jihua qinei jibenshang xiaomie zhuwen de zhishi,” 52.

⁸³ Zhonghua renmin gongheguo nongyebu, “Guanyu zai disange wunian jihua qinei jibenshang xiaomie zhuwen de zhishi,” 53.

疫苗就地生产

在20世纪60年代和70年代，随着以自给自足的集体为基础的CPT项目成为应对动物疾病的标准，基层的重要性越来越被认为是理所当然的。1963年，中央政府颁布了“国家出本，社队出工”的原则。国家出本，社队出工”。这本应意味着中央或省级政府将向公社和下级集体提供疫苗、注射器、消毒剂等，而每个集体应负责组织实际工作并承担劳务费用。正如我们即将看到的兔化猪瘟疫苗就地生产的案例研究，国家承诺的物资供应并非总能实现。基层兽医人员的劳动按“工分”计算（工分），这些工分将转化为一定的工资或口粮。所有这些费用都应由集体预算支付，而不是国家。当然，我们在毛泽东时代的中国经常模糊地称之为“国家”的实体，希望避免对集体动物健康的责任，这揭示了与干预、过度控制的老大哥国家的传统形象不符。⁸²

这种趋势甚至催生了一种观点，即地方集体应该能够自行生产疫苗，特别是猪瘟疫苗，而无需向更高层级的国家机构寻求帮助。农业部规定，在兔化弱毒猪瘟疫苗方面，“地方采购”是惯例。⁸³ 更具体地说，它要求每个县选择一个

⁸² 中华人民共和国农业部， 《关于在第三个五年计划期内基本上消灭猪瘟的指示》， 第52页。

⁸³ 中华人民共和国农业部， 《关于在第三个五年计划期内基本上消灭猪瘟的指示》， 第53页。

or two communes under its jurisdiction to establish “vaccine manufacturing posts (制苗点).” Veterinary cadres on site would lead these institutions, which were expected to supply all the vaccines that were needed in the entire county. Each vaccine station should systematically raise enough fertile female rabbits to “guarantee the sources of the vaccine production.” The Ministry also warned local leaders that having more than two vaccine manufacturing posts in their county should be discouraged. Although the rabbit-attenuated swine fever vaccine “is easy to produce, it is also easy to contaminate when it is not properly handled, which could have a huge impact.” Now local communes, following the planning of the county to which they were affiliated, needed to be self-sufficient even in the supply of swine fever vaccine.⁸⁴

The vaccine was indeed producible on site as long as local manufacturing posts could acquire viral antigens and the needed number of rabbits. Grassroots veterinary cadres injected the classical swine fever virus into healthy rabbits. The virus would not kill rabbits under normal conditions and would cause only several types of thermal reactions. The “typical thermal reaction” refers to having 24-48 hours of a latent period before running a fever. Rabbits would have 1-2 degrees Celsius higher temperature than normal for more than 18 hours. The “mild thermal reaction” shows 24–72 latent hours and more than 12 hours of a 0.5–1 degree Celsius higher temperature. The “questionable thermal reaction” has irregularly short or long latent and high temperature periods, which signals that the rabbit’s immune system may have responded to other

⁸⁴ Zhonghua renmin gongheguo nongyebu 中华人民共和国农业部, “Guanyu xiaomie zhuwen jianmiezhan de tongzhi” 关于消灭猪瘟歼灭战的通知 (April 28, 1964), in *Shouyi gongzuo wenjian xuanbian* 兽医工作文件选编, ed. Shaanxi sheng xumu shouyi zongzhan 陕西省畜牧兽医总站 (N.p., 1975), 7–9.

或其管辖下的两个公社建立“制苗点”。现场兽医干部将领导这些机构，这些机构预计将供应全县所需的所有疫苗。每个疫苗站应系统地饲养足够多的可繁殖母兔，以“保证疫苗生产的来源”。农业部还警告地方领导，不鼓励在其县内设立两个以上的疫苗生产点。尽管兔化猪瘟疫苗“易于生产，但如果处理不当也容易受到污染，这可能会产生巨大影响”。现在，地方公社根据其所属县的规划，甚至在猪瘟疫苗供应方面也需要自给自足。⁸⁴

只要当地生产点能够获得病毒抗原和所需数量的兔子，疫苗确实可以在现场生产。基层兽医干部将经典猪瘟病毒注射到健康的兔子体内。在正常情况下，病毒不会杀死兔子，只会引起几种热反应。“典型热反应”是指在发烧前有24-48小时的潜伏期。兔子体温会比正常体温高1-2摄氏度，持续18小时以上。“轻度热反应”表现为24-72小时的潜伏期和0.5-1摄氏度体温升高超过12小时。“可疑热反应”的潜伏期和高温期不规则地短或长，这表明兔子的免疫系统可能对其他

⁸⁴ 中华人民共和国农业部, “关于消灭猪瘟歼灭战的通知” (1964年4月28日), 载陕西省畜牧兽医总站编:《兽医工作文件选编》(出版地不详, 1975年), 第7-9页。

pathogens rather than the injected swine fever virus. Only the rabbits that show the first two reactions can be used for the production of vaccine.⁸⁵

The appropriately infected rabbits can provide the swine fever vaccine in two ways. The easiest way was to extract blood from the heart. It was called “blood vaccine” (血苗) and could be directly injected into pigs for vaccination to induce immunity. Another method that was trickier yet more reliable was the “spleen-lymph vaccine” (脾淋苗, Figure 5). This type of vaccine was taken from the dissected rabbits’ spleen or mesenteric lymph nodes. The spleen-lymph vaccine had to be diluted 100 times with saline solution or distilled water before use. Initially, the liver and kidney were used as well. But as these two organs turned out to be instable in virus content, producing the vaccine from these organs was banned after 1963. In principle, these “on-site produced vaccines” (就地制苗) needed to be administered immediately. However, when necessary, they could be stored at 0–4 degrees Celsius for 72 hours; at 15–20 degrees for 48 hours; and at 20–25 degrees for 12 hours. Regardless of breed, size, age, sex, or pregnancy, the vaccine was administrable to all pigs. In terms of the injection site, the root of the ear or buttocks were ideal for fully grown pigs and the inner thigh of the hind legs for piglets. Pigs would be immunized four days after inoculation, and the immunity would last for a year and a half.⁸⁶

⁸⁵ “Zhuwen tuhua ruodu yimiao zhizao ji jianyan guicheng” 猪瘟疫化弱毒疫苗制造及检验规程 (August 1963), in *Xumu shouyi gongzuo cankao ziliao* 畜牧兽医工作参考资料, ed. Gansu sheng xumuting 甘肃省畜牧厅 (Lanzhou: Gansu sheng xumuting, 1964), 55–57.

⁸⁶ “Zhuwen tuhua ruodu yimiao zhizao ji jianyan guicheng,” 60–62.

是病原体而不是注射的猪瘟病毒。只有表现出前两种反应的兔子才能用于疫苗生产。⁸⁵

适当感染的兔子可以通过两种方式提供猪瘟疫苗。最简单的方法是从心脏中提取血液。它被称为“血苗”（血苗），可以直接注射到猪体内进行疫苗接种以诱导免疫。另一种更棘手但更可靠的方法是“脾淋苗”（脾淋苗，图5）。这种疫苗取自解剖兔子的脾脏或肠系膜淋巴结。脾淋苗在使用前必须用生理盐水或蒸馏水稀释100倍。最初，肝脏和肾脏也用于制苗。但由于这两种器官的病毒含量不稳定，1963年后禁止从这些器官生产疫苗。原则上，这些“就地制苗”（就地制苗）需要立即使用。但是，必要时，它们可以在0-4摄氏度下储存72小时；在15-20摄氏度下储存48小时；在20-25摄氏度下储存12小时。无论品种、大小、年龄、性别或怀孕与否，疫苗都适用于所有猪。就注射部位而言，成年猪的耳根或臀部是理想选择，仔猪的后腿内侧是理想选择。猪在接种后四天即可免疫，免疫力可持续一年半。⁸⁶

⁸⁵ “Zhuwen tuhua ruodu yimiao zhizao ji jianyan guicheng” 猪瘟疫化弱毒疫苗制造及检验规程 (August 1963), in *Xumu shouyi gongzuo cankao ziliao* 畜牧兽医工作参考资料, ed. Gansu sheng xumuting 甘肃省畜牧厅 (Lanzhou: Gansu sheng xumuting, 1964), 55–57.

⁸⁶ “朱文图画若杜疫苗制造及检验规程，” 60–62。



这是采取家兔脾脏和肠间淋巴结的情形。
农业电影社稿

Figure 5. A vaccine producer collecting samples of spleen and lymph nodes from a rabbit. Source: *Renmin Ribao* 人民日报 (December 18, 1957).

Among many other tasks of the CPT program, the central government stressed the local production and use of the rabbit-attenuated swine fever vaccine, as eliminating the disease became one of the top priorities in the veterinary realm during the Third Five-Year Plan period (1963–1968).⁸⁷ Accordingly, local veterinary cadres and prevention workers participated in extra training sessions to learn knowledge about the particular vaccine. For example, in 1964 prevention workers from all the communes and brigades in Gansu Province were required to attend workshops on the making of the rabbit-attenuated swine fever vaccine organized by the prefecture- and county-level veterinary authorities. They were seven to ten-day workshops and focused on the techniques of

⁸⁷ Zhonghua renmin gongheguo nongyebu, “Guanyu zai disange wunian jihua qinei jibenshang xiaomie zhuwen de zhishi,” 50 and 53.



这是采取家兔脾脏和肠间淋巴结的情形。
农业电影社稿

图 5。疫苗生产者正在从兔子身上采集脾脏和淋巴结样本。来源：《人民日报》（1957年12月 e: 18日）。

在CPT项目的诸多任务中，中央政府强调了兔源弱毒猪瘟疫苗的本地生产和使用，因为在第三个五年计划期间（1963-1968），消除猪瘟成为兽医领域的首要任务之一。⁸⁷ 因此，地方兽医干部和防疫人员参加了额外的培训课程，学习有关该疫苗的知识。例如，1964年，甘肃省所有公社和生产队的防疫人员都被要求参加由地市级和县级兽医部门组织的兔源弱毒猪瘟疫苗制作讲习班。这些讲习班为期七到十天，重点是

⁸⁷ 中华人民共和国农业部，《关于在第三个五年计划期内基本上消灭猪瘟的指示》，第50和53页。

making and injecting the vaccine. The workshop attendees were further asked to go back to their collectives and find “the educated youths whose family background, thoughts, and labor attitude are good and who love the veterinary work and have a certain level of cultural refinement,” thereby training more grassroots prevention workers and “making the technology take root in every production team.” Gansu provincial leaders could remain optimistic in fighting swine fever since approximately 20,000 prevention workers and more than 5,000 veterinarians across the province were active in the “relatively sound grassroots veterinary organizations.”⁸⁸

Nevertheless, swine fever never completely disappeared and was still rampant in the 1970s. In neighboring Shaanxi Province, for instance, collective-based mass vaccination continued to be emphasized in 1972 in tandem with the slogans of “self-breeding and self-raising” and “keeping sheds clean and disinfected.” The province proudly reported that 6,920,000 doses of the rabbit-attenuated swine fever vaccine were produced on site at the sub-county level and 12,000,000 pigs were vaccinated by grassroots veterinary workers within the year. Shaanxi recorded a 83% vaccination rate for the spring regular inoculation and 76.8% for the autumn. By the end of 1972, swine fever basically disappeared in 1,328 communes, which represented more than half of all the communes in the province. Despite these impressive numbers, which were perceivable only to the provincial or upper bureaucrats, swine fever reemerged in May 1973. The disease was never eradicated both during and after the Maoist era. Veterinary workers in communes and lower collectives kept doing their job.⁸⁹

⁸⁸ “Guanyu jiben xiaomie zhuwen de yijian.”

⁸⁹ Shaanxi sheng geming weiyuanhui nonglinju 陕西省革命委员会农林局, “Zhuanfa sheng xumu shouyi zongzhan guanyu jinkuai shixian quansheng jiben xiaomie zhuwen de yijian de tongzhi” 转发省畜牧兽医总站关于尽快实现全省基本消灭猪瘟的意见的通知 (May 4, 1973), in *Shouyi gongzuo wenjian xuanbian* 兽医工作文件选编, ed. Shaanxi sheng xumu shouyi zongzhan 陕西省畜牧兽医总站 (N.p., 1975), 14–18.

制造和注射疫苗。研讨会与会者还被要求回到他们的集体中，寻找“家庭背景好、思想好、劳动态度好、热爱兽医工作、有一定文化修养的知识青年”，从而培养更多的基层防疫人员，“让技术在每个生产队扎根”。甘肃省领导对防治猪瘟保持乐观，因为全省约有2万名防疫人员和5000多名兽医活跃在“比较健全的基层兽医组织”中。⁸⁸

然而，猪瘟从未完全消失，在20世纪70年代仍然猖獗。例如，在邻近的陕西省，1972年继续强调以集体为基础的大规模疫苗接种，同时提出“自繁自养”和“保持猪舍清洁消毒”的口号。该省自豪地报告说，县级以下生产了692万剂兔化猪瘟疫苗，基层兽医人员在一年内为1200万头猪接种了疫苗。陕西省春季常规接种疫苗率为83%，秋季为76.8%。到1972年底，猪瘟在1328个公社基本消失，这占全省公社总数的一半以上。尽管这些令人印象深刻的数字只有省级或更高级别的官员才能感知到，但猪瘟在1973年5月再次出现。在毛泽东时代及其之后，这种疾病从未被根除。公社和下级集体的兽医人员继续他们的工作。⁸⁹

⁸⁸ “关于基本消灭猪瘟的意见。”

⁸⁹ 陕西省革命委员会农林局，“转发省畜牧兽医总站关于尽快实现全省基本消灭猪瘟的意见的通知”（1973年5月4日），载陕西省畜牧兽医总站编《兽医工作文件选编》（无出版地，1975年），第14-18页。

Although this story of the on-site production of swine fever vaccine foregrounds the proactive roles of local collectives in fighting animal infectious diseases, it is hard to tell the effectiveness of such vaccines. Above all, it is doubtful that the quality control was rigid. Maintaining and specifying groups of rabbits healthy enough for vaccine production must not have been easy or self-evident. So did rightly going through every stage of the procedure of making vaccines out of rabbits’ bodies. Also, given that the mass vaccination campaigns in the Maoist period for both humans and nonhumans tended to prioritize the sheer number of the vaccinated at any cost,⁹⁰ and that local veterinary cadres often complained about the improper distribution of vaccines running out of shelf life,⁹¹ it was highly likely that countless invalid vaccines were injected into communal livestock.

Moreover, this model of on-site produced swine fever vaccine seems not to have been commonly replicated in the making of vaccines against other major animal diseases. Certainly, grassroots vaccine manufacturing posts could not deal with more intricate technologies. Therefore, lower communities repeatedly asked upper-level authorities to supply, for example, vaccines against swine erysipelas and swine pneumonia far ahead of their expiration date.⁹²

In general, the presence of the central or provincial governments in the process of mass animal vaccination was rather unremarked. What was problematic was not the state’s excessive intervention into the life of nonhumans and the subsistence of local people that was inextricable

⁹⁰ Brazelton, *Mass Vaccination*, especially 123–143.

⁹¹ Pingle xian nonglin jishu zhidaozhan, “Guanyu 1953 nian xumu shouyi gongzuo zongjie”; “Yulin zhuanqu 1970 nian jihua xuyao gezhong chuqin yijunmiao de baogao” 玉林专区 1970 年计划需要各种畜禽疫苗的报告 (August 16, 1969), GYMA, 030-010-038-03.

⁹² “Yulin zhuanqu 1970 nian jihua xuyao gezhong chuqin yijunmiao de baogao.”

尽管这个关于现场生产猪瘟疫苗的故事突出了地方集体在对抗动物传染病方面的积极作用，但很难判断这些疫苗的有效性。最重要的是，其质量控制是否严格令人怀疑。维持和指定足够健康的兔群用于疫苗生产一定不容易或不言而喻。同样，正确地完成从兔子体内制造疫苗的每个程序阶段也并非易事。此外，考虑到毛泽东时代针对人类和非人类的大规模疫苗接种运动往往不惜一切代价优先考虑接种数量之多⁹⁰，以及地方兽医干部经常抱怨疫苗分发不当导致过期⁹¹，因此很可能无数无效疫苗被注射到集体牲畜体内。

此外，这种现场生产猪瘟疫苗的模式似乎并未在生产其他主要动物疾病疫苗时普遍复制。当然，基层疫苗生产点无法处理更复杂的技术。因此，下级社区反复要求上级部门提前供应例如猪丹毒和猪肺炎疫苗。⁹²

总的来说，中央或省级政府在大规模动物疫苗接种过程中的存在相当不引人注目。问题不在于国家对非人类生命和当地人民生计的过度干预，这种干预是密不可分的

⁹⁰ Brazelton, *Mass Vaccination*, 特别是 123–143。

⁹¹ 平乐县农林技术指导站，“关于 1953 年畜牧兽医工作总结”；“玉林专区 1970 年计划需要各种畜禽疫苗苗的报告” 玉林专区 1970 年计划需要各种畜禽疫苗苗的报告 (1969 年 8 月 16 日), GYMA, 030-010-038-03。

⁹² “玉林专区 1970 年计划需要各种畜禽疫苗苗的报告。”

from those animals. It was the state’s indifference if not incompetence and its constant shifting of its responsibility to communes and lower collectives.

The Choice between Treatment and Culling

In Maoist China, both upper-level governments and local collectives had no doubt about protecting collective livestock as shared property. It is not incorrect to say that the Communist Party’s agrarian policies, especially land reform and collectivization, harmed animals in rural villages.⁹³ But I stress that it is equally worth paying attention to how rural residents who had to adapt to the ways of living in collectivized communities strove to fix the problems and take care of farm animals for the sake of their communal survival.⁹⁴ The central and provincial governments had no reason to oppose such local efforts. The State Council therefore emphasized in 1955 that under the new conditions of agricultural collectivization, breeding and protecting young livestock in collectives were of paramount importance for the general development of agriculture. It was not difficult to see that this directive was in line with some key aspects of the CPT such as “self-breeding and self-raising” and self-sufficiency in mass animal vaccination.⁹⁵

Going one step further, local collectives were very conservative in slaughtering animals. On the ground, only veterinary cadres and veterinarians who worked at local veterinary stations could decide which animals were to be killed. It was their duty to “correctly handle old, feeble, or

⁹³ Braden, “Serve the People.”

⁹⁴ See Chapter Four of this dissertation.

⁹⁵ Zhonghua renmin gongheguo guowuyuan 中华人民共和国国务院, “Guanyu baohu youchu de zhishi” 关于保护幼畜的指示 (December 17, 1955), in *Xumu gongzuo wenjian xuanbian* 畜牧工作文件选编, ed. Shaanxi sheng xumu shouyi zongzhan 陕西省畜牧兽医总站 (N.p., 1974), 83–85.

来自这些动物。这是国家漠不关心，甚至可以说是无能，并且不断将其责任推卸给公社和下级集体。

治疗与扑杀的选择

在毛泽东时代的中国，上级政府和地方集体都毫无疑问地将保护集体牲畜视为共同财产。说中国共产党的农业政策，特别是土地改革和集体化，损害了农村地区的动物，这并非不正确。⁹³ 但我强调，同样值得关注的是，那些不得不适应集体化社区生活方式的农村居民，为了他们的共同生存，如何努力解决问题和照顾农场动物。⁹⁴ 中央和省级政府没有理由反对这种地方努力。因此，国务院在1955年强调，在农业集体化的新条件下，集体中幼畜的繁殖和保护对于农业的总体发展至关重要。不难看出，这项指令与CPT的一些关键方面相符，例如“自繁自养”和大规模动物疫苗接种的自给自足。⁹⁵

更进一步，地方集体在宰杀动物方面非常保守。在当地，只有兽医干部和在当地兽医站工作的兽医才能决定哪些动物应该被宰杀。他们的职责是“正确处理年老、体弱或

⁹³ Braden, “Serve the People.”

⁹⁴ 参见本论文第四章。

⁹⁵ 中华人民共和国国务院, 《关于保护幼畜的指示》（1955年12月17日），载陕西省畜牧兽医总站编：《畜牧工作文件选编》（无出版地，1974年），第83-85页。

disabled” animals to “maintain the peoples’ interests.” Without a slaughter permit issued by these local experts, in principle no one in the collectives could kill animals. And killable livestock were strictly limited to the old, feeble, and disabled.⁹⁶ This tendency hardly changed, even in the midst of the infamous Great Leap Famine in 1959–1961. The Gansu Provincial Committee of the Communist Party in 1960 reaffirmed its commitment to “protecting people as well as livestock for the sake of production” (保人保畜, 以利生产) and required communes and lower collectives to “slaughter, systematically under the guidance of leaders, a certain number of sheep that need to be culled and give the mutton and lamb to the masses in famine areas, thereby increasing the meat supply and protecting the peoples’ physical strength.” Out of the 8,020,000 sheep and goats in the entire province, the committee only allowed the cull of 500,000 to 600,000, no more than 10% of the ovine population in pastoral districts and 7% in agricultural districts.⁹⁷

The hesitancy about culling collective animals was found in the context of animal disease control. While promoting vaccination against swine fever, veterinary cadres ran into local peasants who regarded inoculated pigs’ loss of appetite as a sign of serious vaccine side-effects. The villagers therefore asked them to issue a slaughter permit or even culled a few pigs arbitrarily. The cadres had to “patiently educate them repetitively” that the vaccination should not be the excuse for the reckless slaughter of livestock.⁹⁸ Of course, there were moments when culling appeared inevitable.

However, for impoverished collectives, animals, if not animal bodies, were too precious to simply be

⁹⁶ Guangxi sheng nonglinting 广西省农林厅, “Guanyu shouyi renyuan ying jin keneng zhixing zaisha niuzhi de jianyan qianzheng de hanfu” 关于兽医人员应尽可能执行宰杀牛只得检验签证的函复 (September 22, 1953), GARA, X163-001-0241-0015.

⁹⁷ Zhonggong Gansu sheng weiyuanhui 中共甘肃省委员会, “Guanyu zaisha yipi taotai yangzhi de tongzhi” 关于宰杀一批淘汰羊只的通知 (December 13, 1960), GPA, 192-005-1033-0016.

⁹⁸ Pingle xian nonglin jishu zhidaozhan, “Guanyu 1953 nian xumu shouyi gongzuo zongjie.”

残疾” 动物，以“维护人民的利益”。原则上，未经这些地方专家颁发的屠宰许可证，集体中任何人都不得宰杀动物。可宰杀的牲畜严格限于年老、体弱和残疾的动物。⁹⁶ 即使在1959年至1961年臭名昭著的大跃进饥荒期间，这种趋势也几乎没有改变。1960年，中共甘肃省委重申了“保人保畜，以利生产”的承诺（保人保畜，以利生产），并要求公社和下级集体“在领导的指导下，有计划地宰杀一定数量需要淘汰的羊，将羊肉分发给饥荒地区群众，从而增加肉类供应，保护人民的体力”。在全省802万只绵羊和山羊中，省委只允许宰杀50万至60万只，牧区不超过绵羊总数的10%，农区不超过7%。⁹⁷

在动物疫病控制中，人们对扑杀集体饲养动物的犹豫不决显而易见。在推广猪瘟疫苗接种时，兽医干部遇到了一些当地农民，他们认为接种疫苗的猪食欲不振是疫苗严重副作用的迹象。因此，村民们要求他们发放屠宰许可证，甚至随意扑杀了一些猪。干部们不得不“耐心反复地教育他们”，接种疫苗不应成为随意屠宰牲畜的借口。⁹⁸ 当然，有时扑杀似乎不可避免。然而，对于贫困的集体来说，动物，即使不是动物尸体，也太珍贵了，不能简单地被

⁹⁶ 广西省农林厅, 《关于兽医人员应尽可能执行宰杀牛只得检验签证的函复》（1953年9月22日）, GARA, X163-001-0241-0015。

⁹⁷ 中共甘肃省委员会, “关于宰杀一批淘汰养殖的通知” 一批淘汰羊只的通知 (1960年12月13日), GPA, 192-005-1033-0016。

⁹⁸ 平乐县农林技术指导站, “关于1953年畜牧兽医工作总结。”

discarded. Even though animal parts and tissue were compromised by epizootic diseases, they were still useful for the communal metabolic circle. Local veterinary workers and peasants cautiously sorted out the edible parts from culled or dead pigs which suffered from relatively mild epizootics such as swine pneumonia or erysipelas. The meat blocks, liver, heart, lung, and kidney were deemed good to eat if properly cooked. The stomach, intestine, bone, and blood could be selectively used as fertilizer. Only other useless parts should be appropriately incinerated and buried underground.⁹⁹

Moreover, there is evidence that the state policy toward mass culling of infected animals was twofold. On the one hand, central, provincial, and municipal governments were less reluctant to kill animals grown in urban distribution lines of animal protein, which were typically comprised of state-owned farms, temporary sheds run by municipal food companies, and supply and marketing cooperatives (供销社) in major cities. But on the other hand, they were more prudential in culling pigs raised by and for local collectives. In September 1954, swine fever broke out first in a farm owned by the Beijing municipal food company and soon spread out to other state-owned farms. By November, 4,200 pigs had died of the epidemic and 3,600 pigs were urgently culled as the city could not secure vaccines in time. In December, the situation had not yet turned around although more than 10,000 pigs had already been slaughtered. The Beijing Municipal Government eventually decided to cull the remaining 23,900 in the farms it directly managed.¹⁰⁰ In the same year, swine fever also prevailed in Guangxi as well and the Guangxi Provincial Government similarly ordered

⁹⁹ If pigs contracted anthrax, a more dangerous disease that can infect humans, the entire body must be thoroughly burned out. “Dali fangzhi zhu chuanranbing de chubu yijian.”

¹⁰⁰ “Zhuwen yanzhong” 猪瘟严重 (November 16, 1954), BMA, 001-006-01027-00028; “Zhuwen jixu manyan” 猪瘟继续蔓延 (December 14, 1954), BMA, 001-006-01027-00029; “Fengtai qu fenfang siyang maozhu dabufen chuanran zhuyi siwang” 丰台区粉房饲养毛猪大部分传染猪疫死亡 (November 20, 1954), BMA, 046-002-00442-00238.

被丢弃。即使动物的身体部位和组织受到动物流行病的损害，它们仍然对公共代谢循环有用。当地兽医和农民小心翼翼地因猪肺炎或丹毒等相对轻微的动物流行病而被扑杀或死亡的猪中分拣出可食用的部分。如果烹饪得当，肉块、肝脏、心脏、肺和肾脏都被认为是可食用的。胃、肠、骨骼和血液可以有选择地用作肥料。只有其他无用的部分才应适当焚烧并埋在地下。⁹⁹

此外，有证据表明，国家对大规模扑杀受感染动物的政策是双重的。一方面，中央、省和市政府不太情愿扑杀在城市动物蛋白分销线中饲养的动物，这些分销线通常由国有农场、市食品公司经营的临时棚屋以及主要城市的供销社组成。但另一方面，他们在扑杀由当地集体饲养并为其饲养的猪时更为谨慎。1954年9月，猪瘟首先在北京一家市食品公司拥有的农场爆发，并很快蔓延到其他国有农场。到11月，已有4200头猪死于疫情，3600头猪被紧急扑杀，因为该市未能及时获得疫苗。到12月，尽管已有1万多头猪被宰杀，但情况尚未好转。北京市政府最终决定扑杀其直接管理的农场中剩余的23900头猪。¹⁰⁰同年，猪瘟也在广西流行，广西省政府也同样下令

⁹⁹ 如果猪感染了炭疽这种可能感染人类的更危险的疾病，则必须彻底焚烧整个猪体。“大力防治猪传染病的初步意见。”

¹⁰⁰ “Zhuwen yanzhong” 猪瘟严重 (November 16, 1954), BMA, 001-006-01027-00028; “Zhuwen jixu manyan” 猪瘟继续蔓延 (December 14, 1954), BMA, 001-006-01027-00029; “Fengtai qu fenfang siyang maozhu dabufen chuanran zhuyi siwang” 丰台区粉房饲养毛猪大部分传染猪疫死亡 (November 20, 1954), BMA, 046-002-00442-00238.

the mass culling of pigs grown on state-owned farms for the purpose of providing cities in the province with meat and sending animal products to the nearby large outport, Guangzhou.¹⁰¹

However, the same document by the Guangxi Government communicated different instructions to the local collectives. It required each collective to remain vigilant against the swine fever. Veterinary workers and the masses in the areas where the infection had not yet been detected needed to reinforce the general management of animal caretaking, especially keeping individual animals and their sheds clean and disinfected. In the areas already hit by the epizootic, in addition to promptly deploying quarantine and disinfection, the provincial government explicitly “prohibited [collectives] from indiscreetly killing infected pigs and circulating the untreated meat made of the infected.” These cases imply that collective-owned pigs were less killable than their state-owned counterparts. It was perhaps because the former were integral to the self-sufficiency and survival of grassroots collectives, whereas the latter were regarded as more easily disposable as the upper-level state apparatus was capable of replacing infected animals with new healthy ones to maintain its stock for urban meat distribution.¹⁰²

If it was less likely that local veterinary workers culled pigs in their collectives, the same was the case for plowing oxen, which were usually more expensive and seen as indispensable for agricultural production. A good example can be found in Gansu Province where FMD rampaged in 1964–65. As soon as they heard that the disease was spreading in neighboring Xinjiang, all people’s communes in the province embarked on an effort to vaccinate a total of 352,151 cattle. However, FMD eventually penetrated 91 communes and infected 17,737 oxen. Among them, 14,220 were

¹⁰¹ Guangxi sheng nongyeting 广西省农业厅, “Guanyu jiaqiang zhuwen fangzhi de tongzhi” 关于加强猪瘟防治的通知 (January 4, 1954), GARA, X163-001-0390-0077.

¹⁰² Guangxi sheng nongyeting, “Guanyu jiaqiang zhuwen fangzhi de tongzhi.”

大规模扑杀国有农场饲养的生猪，目的是为省内城市提供肉类，并向附近的广州大港口输送畜产品。¹⁰¹

然而，广西省政府的同一份文件向地方集体传达了不同的指示。它要求每个集体对猪瘟保持警惕。在尚未发现感染的地区，兽医工作者和群众需要加强动物饲养的总体管理，特别是保持个体动物及其棚舍的清洁和消毒。在已经爆发疫情的地区，除了及时部署隔离和消毒外，省政府明确“禁止[集体] 随意宰杀病猪并流通未经处理的病猪肉”。这些案例表明，集体所有的猪比国有猪更不容易被宰杀。这可能是因为前者对于基层集体的自给自足和生存至关重要，而后者则被认为更容易被处理掉，因为上层国家机构能够用新的健康动物替换受感染的动物，以维持其用于城市肉类分配的库存。¹⁰²

如果说当地兽医人员不太可能在其集体中扑杀生猪，那么对于耕牛来说也是如此，耕牛通常更昂贵，被认为是农业生产不可或缺的。一个很好的例子可以在甘肃省找到，该省在1964-65年间口蹄疫肆虐。当他们听说疾病在邻近的新疆蔓延时，该省所有人民公社都开始努力为总计352,151头牛接种疫苗。然而，口蹄疫最终渗透到91个公社，感染了17,737头牛。其中，14,220头是

¹⁰¹ 广西省农业厅，《关于加强猪瘟防治的通知》（1954年1月4日），GARA, X163-001-0390-0077。

¹⁰² 广西省农业厅，《关于加强猪瘟防治的通知》。

bovine workers in collectives under Maqu County. Local cadres and veterinary workers in these collectives decided to isolate the infected from the healthy oxen and intensively heal and take care of the former rather than culling them out. That year, only 84 cattle died or were unavoidably slaughtered due to the epizootic, which accounted for 0.5% of the number of the infected.¹⁰³ Given that the Maoist period is infamous in terms of the unreliability of its statistical data, this particular information might not be fully trustworthy.¹⁰⁴ Still, it at least shows that grassroots collectives wanted to convince their superiors that they were successful in responding to FMD without losing too many of their cattle. I have not found any evidence that local veterinary workers were in fact enthusiastic for culling and covered it up in their reports to upper-level governments.

As we have seen, mass culling was not typically a part of the collective-based CPT program during the Maoist period. Treatment, not slaughtering, as its name signaled, was, with prevention, at the core of the scheme. How then did local veterinary workers treat and cure livestock that had contracted epizootic and zoonotic diseases? The two main remedies were serums and traditional medicines. The Ministry of Agriculture compiled local communes’ therapeutic practices and standardized the treatments for major animal infectious diseases in 1959. According to the handbook edited by the Ministry and circulated to every commune, the FMD serum extracted from cattle recovered from the disease was a must-use. When it came to anthrax, two subcutaneous or intravenous injections of 150 to 300 milliliters of the serum with an interval of 10 hours was the most common method to heal herds. It was complemented by 1,000–2,000 units of penicillin per kilogram when available. Likewise, swine fever serum was widely used for treating the disease.¹⁰⁰

¹⁰³ Gansu sheng renmin weiyuanhui, “Guanyu xiaomie shengchu koutiyi gongzuo zongjie baogao.”

¹⁰⁴ On the development of statistics in the 1950s, see Arunabh Ghosh, *Making It Count: Statistics and Statecraft in the Early People’s Republic of China* (Princeton: Princeton University Press, 2020).

玛曲县集体中的牛群。这些集体中的地方干部和兽医决定将受感染的牛与健康的牛隔离，并对前者进行集中治疗和护理，而不是扑杀它们。那一年，只有84头牛因疫病死亡或被迫宰杀，占受感染牛只总数的0.5%。¹⁰³考虑到毛泽东时代统计数据不可靠的恶名，这一具体信息可能不完全可信。¹⁰⁴尽管如此，它至少表明基层集体希望说服上级，他们在应对口蹄疫方面取得了成功，而没有损失太多牛只。我没有发现任何证据表明当地兽医实际上热衷于扑杀，并在向上级政府的报告中隐瞒了这一点。

正如我们所看到的，大规模扑杀通常不是毛泽东时代集体化CPT项目的一部分。治疗，而不是宰杀，正如其名称所示，与预防一起，是该计划的核心。那么，当地兽医是如何治疗和治愈感染了流行病和人畜共患病的牲畜的呢？两种主要的治疗方法是血清和传统药物。农业部于1959年汇编了地方公社的治疗实践，并对主要动物传染病的治疗进行了标准化。根据农业部编辑并分发给每个公社的手册，从康复牛只中提取的口蹄疫血清是必须使用的。对于炭疽病，最常见的治疗方法是皮下或静脉注射150至300毫升血清，间隔10小时，注射两次。如果条件允许，辅以每公斤1000-2000单位的青霉素。同样，猪瘟血清也广泛用于治疗该病。¹⁰⁰

¹⁰³ 甘肃省人民委员会，“关于消灭牲畜口蹄疫工作总结报告。”

¹⁰⁴ 关于20世纪50年代统计学的发展，参见Arunabh Ghosh, *Making It Count: Statistics and Statecraft in the Early People’s Republic of China*（普林斯顿：普林斯顿大学出版社，2020）。

milliliters of the serum per 50 kilograms of pigs was the recommended dose, and the second shot should follow the next day. Other than penicillin and sulfanilamide, all other drugs had proved ineffective.¹⁰⁵

Serums, however, were effective yet expensive. As of 1950, 1 milliliter of the swine fever serum was equivalent to 500 grams of millet.¹⁰⁶ Therefore, central and provincial governments asked local communities to find a way to take advantage of what was available on the ground which, in most cases, meant “using traditional medicines and trying experimental treatments.”¹⁰⁷ In 1959, for example, Tantang Commune of Guixian County, Guangxi, just like other grassroots communities, was launching the mass scientific experiment campaign.¹⁰⁸ As a part of the campaign, thousands of youths, peasants, and veterinary workers tried to familiarize themselves with livestock artificial insemination technologies and experimented with traditional Chinese medicines that were expected to be effective against swine fever.¹⁰⁹

The case of the commune in Guangxi was hardly unique. With Guangxi, more provinces and municipalities including Sichuan, Shanghai, Hebei, Shanxi, Shandong, Zhejiang, Fujian, and Yunnan pushed their communes and lower collectives to test traditional remedies and collect local medicinal

¹⁰⁵ Nongyebu xumu shouyiju, *Renmin gongshe shouyi gongzuo shouce*, 48, 52, and 69.

¹⁰⁶ Zhou Fengming 周凤鸣, “Beijing shi jiaoqu gongzuowei bagao jiaoqu zhuwen fasheng qingkuang” 北京市郊区工作委报告郊区猪瘟发生情况 (March 24, 1950), BMA, 009-001-00125-00007.

¹⁰⁷ Xibei junzheng weiyuanhui, “Jieshao 51 nian koutiyi fangzhi jingyan.”

¹⁰⁸ On the mass scientific experiment movement, see Schmalzer, *Red Revolution, Green Revolution*, 3–6.

¹⁰⁹ Chong Guo 崇国, “Tansuo zengchan de tujing: Tantang gongshe kexue yanjiu xingcheng qunzhong yundong” 探索增产的途径: 覃塘公社科学研究形成群众运动, *Renmin Ribao* 人民日报 (December 4, 1959).

每50公斤猪推荐剂量为{v1}毫升血清，第二天应注射第二针。除青霉素和磺胺外，所有其他药物均已证明无效。¹⁰⁵

然而，血清有效但昂贵。截至1950年，1毫升猪瘟血清相当于500克小米。¹⁰⁶ 因此，中央和省级政府要求地方社区利用当地现有资源，这在大多数情况下意味着“使用传统药物和尝试实验性治疗”。¹⁰⁷ 例如，1959年，广西贵县坦塘公社与其他基层社区一样，正在开展群众性科学实验运动。¹⁰⁸ 作为运动的一部分，成千上万的青年、农民和兽医工作者试图熟悉牲畜人工授精技术，并试验了预计对猪瘟有效的传统中药。¹⁰⁹

广西公社的案例绝非个例。除了广西，包括四川、上海、河北、山西、山东、浙江、福建和云南在内的更多省市也推动其公社和下级集体测试传统疗法并收集当地药材

¹⁰⁵ 农业部畜牧兽医局，《人民公社兽医工作手册》，第48、52、69页。

¹⁰⁶ 周凤鸣，“北京市郊区工作委报告郊区猪瘟发生情况”（1950年3月24日），BMA, 009-001-00125-00007。

¹⁰⁷ 西北军政委员会，“介绍51年口蹄疫防治经验。”

¹⁰⁸ On the mass scientific experiment movement, see Schmalzer, *Red Revolution, Green Revolution*, 3–6.

¹⁰⁹ 崇国，“探索增产的途径：覃塘公社科学研究形成群众运动”，《人民日报》（1959年12月4日）。

herbs to treat pigs.¹¹⁰ As was the case with the on-site rabbit-attenuated swine fever vaccine, the demand to secure effective traditional veterinary medicines locally was justified in the name of communal self-sufficiency. Some additional slogans such as “self-collecting, self-manufacturing” (自采自制) or “self-producing, self-using” (自产自用) appeared in official documents.¹¹¹ Sichuan’s Wusheng County was particularly touted by China’s biggest state media, *People’s Daily*, as exemplary in this regard. In 1959 folk veterinarians, veterinary cadres, prevention workers, and the masses from its 35 different communes eventually “discovered” about 90 “wild medicinal materials” that proved effective for treating pig diseases. Thanks to such communal bottom-up efforts, the newspaper article noted that not a single pig in the county died from swine fever that year.¹¹²

Conclusion

Local veterinary workers—namely lower-ranking veterinary cadres, epidemic prevention workers, folk veterinarians, and caretakers in communes, brigades, and teams—shaped the Comprehensive Prevention and Treatment against contagious animal diseases. While carrying out lockdown, quarantine, and disinfection, vaccinating countless animals, and healing rather than culling infected animals, they resiliently grappled with the upper-level governments’ demands to make biopower work on their behalf in addition to being self-sufficient in maintaining communal animal power. On

¹¹⁰ Jiangxi sheng nongyeting zhongshouyi shiyansuo 江西省农业厅中兽医实验所, “1957 nian shiyan zhongyao peifang jianjie” 1957 年实验中药配方简介, *Minjian shouyi tongxun* 民间兽医通讯 4 (March 1958): 20–21; “Baozheng yangzhu shengchan gaosudu fazhan, gedi jiti yangchang dazhua fangyi zhushe.”

¹¹¹ “Wusheng xian jiben xiaomie zhude zhuyao yibing” 武胜县基本消灭猪的主要疫病, *Renmin Ribao* 人民日报 (February 2, 1960); “Baozheng yangzhu shengchan gaosudu fazhan, gedi jiti yangchang dazhua fangyi zhushe.”

¹¹² “Wusheng xian jiben xiaomie zhude zhuyao yibing.”

草药治疗猪瘟。¹¹⁰ 与现场兔化猪瘟疫苗的情况一样，以社区自给自足的名义，确保当地获得有效的传统兽药的需求是合理的。官方文件中出现了一些额外的口号，如“自采自制”（自采自制）或“自产自用”（自产自用）。¹¹¹ 四川武胜县尤其被中国最大的官方媒体《人民日报》誉为这方面的典范。1959年，来自该县35个不同公社的民间兽医、兽医干部、防疫人员和群众最终“发现”了大约90种“野生药材”，这些药材被证明对治疗猪病有效。报纸文章指出，由于这种社区自下而上的努力，当年该县没有一头猪死于猪瘟。¹¹²

结论

地方兽医工作者——即基层兽医干部、防疫人员、民间兽医以及公社、大队和生产队的饲养员——共同塑造了动物传染病的综合防治体系。他们在实施封锁、隔离和消毒，为无数动物接种疫苗，以及治疗而非扑杀受感染动物的同时，顽强地应对着上级政府的要求，即让生物权力为他们服务，并实现社区动物力量的自给自足。在

¹¹⁰ Jiangxi sheng nongyeting zhongshouyi shiyansuo 江西省农业厅中兽医实验所, “1957 nian shiyan zhongyao peifang jianjie” 1957 年实验中药配方简介, *Minjian shouyi tongxun* 民间兽医通讯 4 (March 1958): 20–21; “Baozheng yangzhu shengchan gaosudu fazhan, gedi jiti yangchang dazhua fangyi zhushe.”

¹¹¹ “武胜县基本消灭猪的主要疫病”，《人民日报》（1960年2月2日）；“保证养猪生产高速发展，各地集体养殖场大抓防疫注射。”

¹¹² “武胜县基本消灭猪的主要疫病。”

the one hand, this chapter urges that we fully recognize such proactive roles of local communes and lower collectives. On the other, it argues that the so-called state, or central and provincial governments, did not properly intervene into the life of nonhumans in collectives and the subsistence of rural residents closely connected to those animals.

After Mao’s death in 1976, the three-decade social experiment of collective economy ended, and people’s communes and lower collectives across the nation were gradually dismantled between 1978 and 1985. The CPT, too, came to an end. Some former barefoot veterinarians and prevention workers, the fruits of mass science, took advantage of their expertise and became entrepreneurial animal doctors in their villages. There were no longer collective animal farms and communally owned livestock. Individual animal owners paid for veterinary service. The sense of communal bonds and common interests, accordingly, was slipping away. If an epizootic disease spread in a region, individuals would take responsibility only for what they possessed. If one failed to save their animal assets and their livelihood got worse, they might leave their home for cities in the hopes of getting a job at a factory. Reflecting such postsocialist changes, the Guangxi Zhuang Autonomous Region Government promulgated new legislation on the health of livestock and poultry in 1980. It specified the transfer of responsibility for epizootic and zoonotic disease control from collectives to individual livestock owners. For the first time, as far as I know, culling and slaughtering came to be overtly included in juxtaposition with “treating infected animals in quarantine” in this kind of official document.¹¹³

¹¹³ “Guangxi Zhuangzu zizhiqu jiachu jiaqin jianyi zanxing guize” 广西壮族自治区家畜家禽检疫暂行规则 (March 29, 1980), in *Guangxi shouyi fangjian tongxun* 广西兽医防检通讯 1, ed. Guangxi zhuangzu zizhiqu shouyi fangyi jianyi zhan 广西壮族自治区兽医防疫检疫站 (Nanning: Guangxi qiao yinshuachang, 1981), 1–3.

一方面，本章强调我们应充分认识地方公社和下级集体所发挥的这种积极作用。另一方面，本章认为所谓的国家，即中央和省级政府，并未适当干预集体中非人类的生存以及与这些动物密切相关的农村居民的生计。

1976年毛泽东逝世后，长达三十年的集体经济社会实验宣告结束，全国各地的人民公社和下级集体在1978年至1985年间逐步解体。CPT也随之终结。一些前赤脚兽医和防疫工作者，作为群众科学的成果，利用他们的专业知识，成为村里的创业型动物医生。不再有集体动物农场和公有牲畜。个体动物主人支付兽医服务费用。因此，集体纽带和共同利益的意识逐渐消逝。如果一个地区爆发流行病，个人只对自己拥有的东西负责。如果一个人未能保住他们的动物资产，生计恶化，他们可能会离开家乡前往城市，希望在工厂找到一份工作。为反映这种后社会主义的变化，广西壮族自治区政府于1980年颁布了新的畜禽健康法规。该法规明确规定，流行病和人畜共患病控制的责任从集体转移到个体畜主。据我所知，这是首次在此类官方文件中，扑杀和屠宰被明确地与“隔离治疗受感染动物”并列提及。¹¹³

¹¹³ “广西壮族自治区家畜家禽检疫暂行规则” (1980年3月29日)，载《广西兽医防检通讯》1，广西壮族自治区兽医防疫检疫站编（南宁：广西区贸桥印刷厂，1981），1-3页。

This story of the CPT in Maoist China could be seen as another tragicomedy of the dogmatic state's pursuit of multispecies biopower in the contexts of a collective economy which denied private ownership and of populist amateurism that can easily lead to anti-intellectualism. I hope to revisit, however, the grassroots community-based nature of the CPT from a present-day perspective that increasingly values the idea that humans and nonhumans co-inhabit a shared ecology, if not a commune, of the planet. Local veterinary workers, as we have seen, did not necessarily protect their animal patients sheerly to serve the arguably formidable Communist Party and the state. They did know that their subsistence was inevitably connected to the health of animals. Under given conditions, they therefore tried their best to keep their invaluable animal companions alive with their knowledge and labor so that they could preserve their own multispecies communities, or “more-than-people's communes.” In so doing, the communities could carry on. If so, this communal initiative in the prevention of and response to animal diseases may deserve more than oblivion.

Local communities acted as more-than-people's communes over the course of dealing with infectious animal diseases. The multispecies aspects of the communes could also be foregrounded in more everyday contexts. Local veterinarians as grassroots experts were at the forefront of not only the CPT, but also mundane practices. In the next chapter, I zoom in on the world of folk animal healers and delve into how the veterinary expertise necessary for more-than-people's communes was constructed and codified roughly in the first decade of the PRC.

毛泽东时代中国CPT的故事可以被视为教条主义国家在集体经济背景下追求多物种生物权力的又一出悲喜剧，这种集体经济否认私有制，而民粹主义的业余精神很容易导致反智主义。然而，我希望从当今的视角重新审视CPT的基层社区性质，因为当今社会越来越重视人类和非人类共同居住在一个共享生态系统（如果不是一个公社）中的理念。正如我们所见，当地兽医工作者保护他们的动物病人不一定仅仅是为了服务于可以说强大的共产党和国家。他们确实知道自己的生存与动物的健康密不可分。在既定条件下，他们因此尽力用自己的知识和劳动来维持他们宝贵的动物伙伴的生命，以便他们能够保护自己的多物种社区，或者说“超越人类的公社”。通过这样做，这些社区得以延续。如果是这样，这种在动物疾病预防和应对方面的社区倡议可能值得我们铭记，而不仅仅是遗忘。

在应对动物传染病的过程中，地方社区扮演了“超人类”公社的角色。公社的多物种特性也可以在日常的语境中得到强调。作为基层专家的当地兽医不仅站在CPT（合作社生产队）的最前沿，也站在日常实践的最前沿。在下一章中，我将深入探讨民间动物治疗师的世界，并研究在中华人民共和国成立的第一个十年左右，构建和编纂“超人类”公社所需的兽医专业知识的过程。

Chapter Two:
Science for the Commune

Introduction

When the Chinese Communist Party (CCP) came to power in 1949, it presented itself as a group of revolutionaries creating a new modern nation and bidding farewell to China’s “feudal” or “reactionary” past. However, the Communists soon realized that the mission was much more difficult than they thought it would be. The CCP lacked human and material resources in many realms, including veterinary medicine. While Western-trained veterinary scientists (西兽医) were rare, folk animal healers (民间兽医) who used acupuncture and herbal medicines to treat farm animals accounted for about 95% of China’s veterinary workforce until the late 1950s. It seemed clear to many CCP leaders that without these healers, the state could not build a nationwide modern veterinary system and contain the spread of infectious diseases among animals. The revolutionaries had to turn to what they saw as the remnant of the “old society.” They soon embarked on organizing and reforming the folk veterinarians in tandem with the better-known project of agricultural collectivization.¹¹⁴

This chapter examines how China’s folk animal healers, formerly private practitioners, were transformed into collective veterinarians (社内兽医 or 集体兽医) in the 1950s. This process of collectivizing local veterinary personnel and their knowledge was a prerequisite for the codification

¹¹⁴ Zhou Enlai 周恩来, “Guowuyuan guanyu jiaqiang minjian shouyi gongzuo de zhishi” 国务院关于加强民间兽医工作的指示 (January 5, 1956), Gansu Provincial Archives (hereafter GPA), 222-001-0108-0001; Zhonggong zhongyang zhengzhiju 中共中央政治局, *1956 nian dao 1967 nian quanguo nongye fazhan gangyao cao'an* 1956 年到 1967 年全国农业发展纲要草案 (Beijing: Renmin chubanshe, 1956).

第二章：
为公社服务的科学

引言

1949年中国共产党（CCP）执政时，它将自己描绘成一群正在创建新现代国家并告别中国“封建”或“反动”过去的革命者。然而，共产党人很快意识到这项任务比他们想象的要困难得多。中国共产党在许多领域都缺乏人力和物力资源，包括兽医学。虽然受西方培训的兽医科学家（西兽医）很少，但直到20世纪50年代末，使用针灸和草药治疗农场动物的民间动物治疗师（民间兽医）仍占中国兽医劳动力的约95%。许多中共领导人清楚地认识到，如果没有这些治疗师，国家就无法建立全国性的现代兽医体系并控制动物传染病的传播。革命者不得不求助于他们眼中的“旧社会”残余。他们很快着手组织和改革民间兽医，同时开展了更广为人知的农业集体化项目。¹¹⁴

本章探讨了中国的民间动物治疗师（以前是私人执业者）在20世纪50年代如何转变为集体兽医（社内兽医或 集体兽医）。这种将地方兽医人员及其知识集体化的过程是编纂的先决条件

¹¹⁴ 周恩来, “国务院关于加强民间兽医工作的指示”（1956年1月5日），甘肃省档案馆（以下简称GPA），222-001-0108-0001；中共中央政治局，《1956年到1967年全国农业发展纲要草案》（北京：人民出版社，1956年）。

of what came to be called Traditional Chinese Veterinary Medicine (TCVM), a specific form of modernized Chinese veterinary medicine rooted in Maoism. In so doing, I show that collectivism, often regarded by China historians as the fundamental cause of the infamous tragedies such as the Great Leap Famine (1959–1961) and the Cultural Revolution (1966–1976), could facilitate knowledge production. With this case study on folk veterinary medicine, this chapter complicates the ideas that social, political factors in socialist countries, unlike their liberal democratic counterparts, had a negative impact on the construction of science¹¹⁵ and that Maoism was inherently hostile to science.¹¹⁶

The Earlier Encounters, 1949–1956

The CCP’s attitude to folk veterinarians had been ambivalent since the Yan’an period (1935–1948).¹¹⁷ On the one hand, the Party was suspicious of indigenous animal healers, doubting both their expertise and their political loyalty. On the other, the Party saw the inevitability of mobilizing them to control animal diseases in the absence of Western veterinary practitioners in rural

¹¹⁵ Nikolai Kremontsov, *Stalinist Science* (Princeton: Princeton University Press, 1997); Loren R. Graham, *What Have We Learned About Science and Technology from the Russian Experience?* (Stanford: Stanford University Press, 1998); Laurence Schneider, *Biology and Revolution in Twentieth-century China* (Lanham: Rowman and Littlefield, 2003); Danian Hu, *China and Albert Einstein: The Reception of the Physicist and His Theory in China, 1917–1979* (Cambridge, Mass.: Harvard University Press, 2005); Chunjuan Nancy Wei and Darryl E. Brock eds., *Mr. Science and Chairman Mao’s Cultural Revolution: Science and Technology in Modern China* (Lanham: Lexington Books, 2013).

¹¹⁶ On the relations between Maoism and science, see Wei and Brock eds., *Mr. Science and Chairman Mao’s Cultural Revolution*; Sigrid Schmalzer, *Red Revolution, Green Revolution: Scientific Farming in Socialist China* (Chicago: University of Chicago Press, 2016); Rui Kunze and Marc Andre Matten, *Knowledge Production in Mao-era China: Learning from the Masses* (Lanham: Lexington Books, 2021); Danian Hu, “Despite or Due to the Cultural Revolution: The Development of Chinese Science, Technology, and Medicine in the 1960s and 1970s,” *Endeavor* 43.3 (September 2017): 78–84.

¹¹⁷ On how Yan’an and the Shaan-Gan-Ning Border Region “accidently” became the cradle of the Chinese revolution, see Joseph Esherick, *Accidental Holy Land: The Communist Revolution in Northwest China* (Oakland: University of California Press, 2022).

的产物，这种兽医形式被称为传统中兽医药（TCVM），是根植于毛泽东思想的现代化中兽医药的一种具体形式。通过这样做，我表明集体主义（中国历史学家常将其视为大跃进饥荒（1959-1961）和文化大革命（1966-1976）等臭名昭著的悲剧的根本原因）可以促进知识生产。通过对民间兽医学的案例研究，本章使以下观点复杂化：社会主义国家的社会政治因素，与自由民主国家不同，对科学的构建产生了负面影响¹¹⁵，以及毛泽东思想本质上是敌视科学的。¹¹⁶

早期接触，1949-1956

自延安时期（1935-1948）以来，中共对民间兽医的态度一直模棱两可。¹¹⁷一方面，党对本土动物治疗师持怀疑态度，怀疑他们的专业知识和政治忠诚度。另一方面，在农村地区缺乏西方兽医的情况下，党认为动员他们控制动物疾病是不可避免的。

¹¹⁵ Nikolai Kremontsov, *Stalinist Science* (Princeton: Princeton University Press, 1997); Loren R. Graham, *What Have We Learned About Science and Technology from the Russian Experience?* (Stanford: Stanford University Press, 1998); Laurence Schneider, *Biology and Revolution in Twentieth-century China* (Lanham: Rowman and Littlefield, 2003); Danian Hu, *China and Albert Einstein: The Reception of the Physicist and His Theory in China, 1917–1979* (Cambridge, Mass.: Harvard University Press, 2005); Chunjuan Nancy Wei and Darryl E. Brock eds., *Mr. Science and Chairman Mao’s Cultural Revolution: Science and Technology in Modern China* (Lanham: Lexington Books, 2013).

¹¹⁶ 关于毛泽东思想与科学的关系，参见魏和布罗克主编的《科学先生与毛主席的文化大革命》；西格丽德·施马尔泽的《红色革命，绿色革命：社会主义中国的科学农业》（芝加哥：芝加哥大学出版社，2016）；昆泽和马滕的《毛泽东时代中国的知识生产：向群众学习》（兰哈姆：莱克星顿图书，2021）；胡大年，“尽管或由于文化大革命：20世纪60年代和70年代中国科学、技术和医学的发展”，《奋进》43.3（2017年9月）：78-84页。

¹¹⁷ 关于延安和陕甘宁边区如何“偶然”成为中国革命的摇篮，参见约瑟夫·埃舍里克，《偶然的圣地：中国西北的共产主义革命》（奥克兰：加州大学出版社，2022）。

revolutionary bases. In 1944, Chairman Mao Zedong expressed astonishment that “both human and livestock mortalities in Shaan-Gan-Ning Border Region are extremely high.” He acknowledged that the problem might not be solved by relying only on “new doctors” (新医), although they “were more marvelous than ‘old doctors’ (旧医) for sure.” Chairman Mao, therefore, urged his comrades to “unite” with the old doctors and “old-style veterinarians” (旧式兽医) and “help them be advanced” to save people and animals in the cradle of the Chinese Revolution.¹¹⁸

In August 1949, on the eve of the founding of the New People’s Republic, Le Tianyu,¹¹⁹ Dean of the College of Agricultural Science at North China University, echoed Mao in emphasizing the importance of “the reform of old-style veterinarians” and the gradual “scientization” (科学化) of their knowledge. By doing so, Le argued, the Communists could take advantage of the expertise of folk veterinarians at a time when “training a large number of [modern] veterinarians is not achievable in a short period of time” and when “western drugs and sera are very much lacking.”¹²⁰ That is to say, in early 1950, the state still saw local veterinarians as the object of “scientific” reform, if not a secondary expedient. Before and after the birth of the new People’s Republic of China, the Communists reluctantly approached folk healers in the hope of mobilizing them and taking advantage of their skills.

¹¹⁸ Mao Zedong 毛泽东, “Wenhua gongzuo zhong de tongyi zhanxian” 文化工作中的统一战线 (October 30, 1944), *Mao Zedong Xuanji* 毛泽东选集 3 (Beijing: Renmin Chubanshe, 1991), 1011–1013.

¹¹⁹ On Le Tianyu’s active role in supporting Lysenkoism in China, see Schneider, *Biology and Revolution in Twentieth-century China*, 104–108, 117–163.

¹²⁰ Le Tianyu 乐天宇, “Jiefangqu de nongye kexue gongzuo” 解放区的农业科学工作, *Renmin Ribao* 人民日报 (August 27, 1949).

革命根据地。1944年，毛泽东主席惊讶地表示，“无论是人还是

陕甘宁边区的牲畜死亡率极高。”他承认，仅仅依靠“新医”可能无法解决这个问题，新医，尽管他们“肯定比‘旧医’更了不起”。旧医因此，毛主席敦促他的同志们与老医生和“旧式兽医”团结起来，旧式兽医并“帮助他们进步”，以拯救中国革命摇篮中的人民和动物。¹¹⁸

1949年8月，在新中国成立前夕，华北大学农学院院长乐天宇¹¹⁹响应毛泽东的号召，强调“旧式兽医的改造”和他们知识的逐步“科学化”的重要性。乐天宇认为，通过这样做，共产党人可以利用民间兽医的专业知识，因为当时“在短时间内培养大量[现代]兽医是不可能实现的”，而且“西药和血清非常缺乏”。¹²⁰也就是说，在1950年初，国家仍然将地方兽医视为“科学”改革的对象，即使不是次要的权宜之计。在新中国成立前后，共产党人勉强接触民间医生，希望动员他们并利用他们的技能。

¹¹⁸ 毛泽东，“文化工作中的统一战线”（1944年10月30日），《毛泽东选集》第3卷（北京：人民出版社，1991年），第1011–1013页。

¹¹⁹ 关于乐天宇在中国支持李森科主义的积极作用，参见施耐德（Schneider）的《20世纪中国的生物学与革命》，第104–108、117–163页。

¹²⁰ 乐天宇，“解放区的农业科学工作”，《人民日报》（1949年8月27日）。

The leaders in Guangxi, China’s southwestern most province, adopted Beijing’s view of local animal healers. In 1951, the Guangxi Provincial Government designated Lingui and Lingchuan as testing counties for organizing folk veterinarians. The cadres first counted the number of animal healers in the two counties. They found 65, many of whom were not full-time veterinarians, but “half-veterinarian-half-peasants” who performed both agricultural labor and veterinary work. The cadres admitted these 65 folk veterinarians would have to provide their services to all 354 villages. They hence assigned each veterinarian four to six villages, not allowing them to freely work in a few preferred marketplaces.¹²¹

The cadres soon expressed their contempt for folk veterinarians, complaining that they “were not refined politically and culturally” and that “they were not enthusiastic enough about [the ideal of] ‘Serving the People.’” Wu Zongliao from Lingui was one of these veterinarians. When a woman visited Wu’s place to ask him to treat her ill cow, he yelled at her: “Am I not at home? Find me when I am at work!” The Communists cadres considered Wu “arrogant.” Another folk veterinarian from Lingui, Pan Zhubao, was steeped in elitist “privilege,” according to the local cadres. Concerned with his “individualistic self-interest,” Pan dared to expect the local government to buy him a new horse so that he could ride it when making his veterinary rounds.¹²²

Some of the local Communist cadres who were dissatisfied with folk veterinarians, nonetheless, focused more on how to reform their thoughts and knowledge than on blaming them.

¹²¹ Guangxi sheng nonglinting 广西省农林厅, “Guangxi sheng xunlian he zuzhi zhongshouyi de jingyan” 广西省训练和组织中兽医的经验 (1952), Guangxi Zhuang Autonomous Region Archives (hereafter GARA), X163-001-0119-0009.

¹²² Guangxi sheng nonglinting, “Guangxi sheng xunlian he zuzhi zhongshouyi de jingyan”; Guilin zhuanqu nonglin gongzuozhan jiachu baoyusuo 桂林专区农林工作站家畜保育所, “Guilin zhuanqu zuzhi zhongshouyi qingkuang baogao” 桂林专区组织中兽医情况报告 (1952), GARA, X163-001-0117-0008.

中国西南省份广西的领导人采纳了北京对当地动物治疗师的看法。1951年，广西省政府指定临桂和灵川为组织民间兽医的试点县。干部们首先统计了这两个县的动物治疗师数量。他们发现有65人，其中许多人并非全职兽医，而是“半兽医半农民”，他们既从事农业劳动，也从事兽医工作。干部们承认，这65名民间兽医必须为所有354个村庄提供服务。因此，他们给每位兽医分配了四到六个村庄，不允许他们在少数几个偏爱的市场自由工作。¹²¹

干部们很快就表达了对民间兽医的蔑视，抱怨他们“政治上和文化上都不够成熟”，并且“对[‘为人民服务’的]理想不够热情”。临桂的吴宗辽就是其中一位兽医。当一位妇女到吴家请他治疗生病的牛时，他冲她喊道：“我不在家吗？等我上班的时候再来找我！” 共产党干部认为吴“傲慢”。据当地干部称，另一位来自临桂的民间兽医潘竹宝则沉浸在精英主义的“特权”中。潘竹宝“个人主义的私利”让当地干部担忧，他竟然期望当地政府给他买一匹新马，以便他在出诊时可以骑乘。¹²²

一些对民间兽医不满的当地共产党员干部，更关注如何改造他们的思想和知识，而不是指责他们。

¹²¹ 广西省农林厅, 《广西省训练和组织中兽医的经验》（1952）, 广西壮族自治区档案馆（以下简称GARA）, X163-001-0119-0009。

¹²² 广西省农林厅, 《广西省训练和组织中兽医的经验》; 桂林专区农林工作站家畜保育所, 《桂林专区组织中兽医情况报告》（1952）, GARA, X163-001-0117-0008。

The leaders of Ulanqab Banner, Inner Mongolia, for example, accused folk veterinarians in Ulanqab of being politically “conservative” and only interested in enriching themselves. The leaders attributed this “thought problem” of animal healers to the previous “feudal” political systems in which veterinarians were of low social status and treated with disrespect by the “counterrevolutionary” rulers. In other words, folk veterinarians’ ideological backwardness was not their fault, but a result of their socioeconomic circumstances. As such, since the new Communist government would strengthen the solidarity among the state, the veterinarians, and the villagers, the cadres were optimistic about the prospects for reform.¹²³

In this vein, the Ulanqab leaders organized learning sessions to educate folk veterinarians in Marxist political economy and improve their veterinary expertise. The cadres invited both seasoned veterinarians and healthy young men with little veterinary experience to the sessions. The experienced animal healers from time to time exchanged their secret remedies with each other, but they assumed a wait-and-see attitude and were not willing to share their expertise with young students for free. On top of that, as the veterinary cadres who were supposed to be instructors “were short of capabilities and experiences,” the sessions never went smooth. Nevertheless, the state cadres tried their best to “increase the students’ passion for learning by less discussing the theories and more carrying out practices and by avoiding jargons and using easy colloquial words.” These encounters at the grassroots gradually closed the distance between the state and folk veterinarians.¹²⁴

¹²³ Wulanhua qu zhengfu 乌兰花区政府, “Wumeng wulanhua qu siziwang qi lianhe juban zhongshouyi jiangxihui zongjie baogao” 乌盟乌兰花区四子王旗联合举办中兽医讲习会总结报告 (April 15, 1952), Inner Mongolia Bayannuur Municipal Archives (hereafter IMBMA), 026-02-0017-001.

¹²⁴ Wulanhua qu zhengfu, “Wumeng wulanhua qu siziwang qi lianhe juban zhongshouyi jiangxihui zongjie baogao.”

例如，内蒙古乌兰察布旗的领导人指责乌兰察布的民间兽医在政治上“保守”，只顾自己发财。领导人将动物治疗师的这种“思想问题”归因于以前的“封建”政治制度，在这种制度下，兽医社会地位低下，受到“反革命”统治者的不尊重。换句话说，民间兽医的思想落后不是他们的错，而是他们的社会经济环境造成的。因此，由于新的共产党政府将加强国家、兽医和村民之间的团结，干部们对改革前景持乐观态度。¹²³

本着这种精神，乌兰察布的领导人组织了学习班，对民间兽医进行马克思主义政治经济学教育，并提高他们的兽医专业知识。干部们邀请了经验丰富的兽医和缺乏兽医经验的健康青年参加学习班。经验丰富的动物治疗师不时地互相交流他们的秘方，但他们采取观望态度，不愿意免费与年轻学生分享他们的专业知识。最重要的是，由于本应担任指导员的兽医干部“能力和经验不足”，学习班从未顺利进行。尽管如此，国家干部们还是尽力“通过少讲理论多实践、避免使用行话和使用通俗易懂的口语来增加学生的学习热情”。这些基层接触逐渐拉近了国家与民间兽医之间的距离。¹²⁴

¹²³ 乌兰花区政府, “乌盟乌兰花区四子王旗联合举办中兽医讲习会总结报告” (1952年4月15日), 内蒙古巴彦淖尔市档案馆 (以下简称IMBMA), 026-02-0017-001。

¹²⁴ 乌兰花区政府, “乌盟乌兰花区四子王旗联合举办中兽医讲习会总结报告”。

Some folk veterinarians, too, took the initiative in forging connections with the state. In the northwest where the nomadic way of living was prevalent and therefore folk veterinarians were especially influential,¹²⁵ the animal doctors established the Association for Chinese Veterinarians in 1952. Through this organization, veterinarians spearheaded the “patriotic livestock protection movement” in response to the state’s call.¹²⁶ In the meantime, Yu Xiushui, one of the best known folk veterinarians in Huxian County, Shaanxi Province, believed that animal doctors needed not only their own professional associations, but also better clinical organizations. Hence, Yu and five of his colleagues in the county invested 30 *yuan* each to establish a “veterinary union clinic” (兽医联合诊所) in 1952. They used their union clinic to provide neighboring villages with affordable veterinary checkups. The county government later granted a loan of 250 *yuan* to Yu’s clinic to encourage this business model.¹²⁷

Summarizing these interactions between the state and folk veterinarians, in 1953, the Northwest Great Administrative Area Government which had jurisdiction over the provinces of Shaanxi, Gansu, Ningxia, Qinghai, and Xinjiang reported to Beijing on the general situation of its animal healers. By October 1953, there were about 5,000 Han Chinese folk veterinarians—2,276 in Shaanxi; 2,000 in Gansu; approximately 700 in Ningxia, Qinghai, and Xinjiang—while the exact number of ethnic minority healers remained unknown. With the exception of a few doctors who ran

¹²⁵ On the longue durée history of the region, see James A. Millward, *Eurasian Crossroads: A History of Xinjiang* (New York: Columbia University Press, 2007); Peter C. Perdue, *China Marches West: The Qing Conquest of Central Eurasia* (Cambridge, Mass., Harvard University Press, 2005).

¹²⁶ “Xibei zuzhi zhongshouyi canjia aiguo baochu yundong” 西北组织中兽医参加爱国保畜运动, *Renmin Ribao* 人民日报 (March 15, 1953).

¹²⁷ Yu Xiushui 余修水, “Huxian diyi shouyi lianhe zhenliao suo suozhang Yu Xiushui de fayan” 鄂县第一兽医联合诊所所长余修水的发言, in *Shaanxi sheng zhongshouyi zuotanhui huikan* 陕西省中兽医座谈会汇刊, ed. Shaanxi sheng nongyeting xumuju 陕西省农业厅畜牧局 (N.p., 1956), 27–28.

一些民间兽医也主动与国家建立联系。在游牧生活方式盛行、民间兽医影响力尤其大的西北地区，¹²⁵动物医生于1952年成立了中华兽医协会。通过该组织，兽医们响应国家的号召，率先开展了“爱国牲畜保护运动”。¹²⁶与此同时，陕西省户县最著名的民间兽医之一余秀水认为，动物医生不仅需要自己的专业协会，还需要更好的临床组织。因此，余秀水和他的五位同事于1952年每人投资30元，成立了“兽医联合诊疗所”兽医联合诊疗所。他们利用联合诊所为邻近村庄提供负担得起的兽医检查。县政府后来向余秀水的诊所提供了250元贷款，以鼓励这种经营模式。¹²⁷

总结国家与民间兽医之间的这些互动，1953年，管辖陕西、甘肃、宁夏、青海和新疆等省份的西北大行政区政府向北京汇报了其动物治疗师的总体情况。截至1953年10月，约有5000名汉族民间兽医——其中陕西2276名；甘肃2000名；宁夏、青海和新疆约700名——而少数民族治疗师的具体人数不详。除少数医生经营的诊所外，

¹²⁵ On the longue durée history of the region, see James A. Millward, *Eurasian Crossroads: A History of Xinjiang* (New York: Columbia University Press, 2007); Peter C. Perdue, *China Marches West: The Qing Conquest of Central Eurasia* (Cambridge, Mass., Harvard University Press, 2005).

¹²⁶ “西北组织中兽医参加爱国保畜运动”，《人民日报》（1953年3月15日）。

¹²⁷ 余修水，“鄂县第一兽医联合诊疗所所长余修水的发言”，载陕西省农业厅畜牧局编《陕西省中兽医座谈会汇刊》（出版地不详，1956），第27-28页。

private animal clinics in cities, most of them were scattered across rural and pastoral expanses.

“More than half of all the veterinarians were illiterate. They learned veterinary medicine from local masters or have treated animal diseases by uncritically following healing customs or relying on simple remedies passed down in their families. Yet they do have certain experiences and skills.”¹²⁸ In the pre-revolutionary period, almost all of the veterinarians “had no social foothold and were destitute.” They therefore became “conservative and superstitious,” suffering from self-depreciation or pure self-interest. To guide these folk veterinarians, the Northwest government argued that the Communist state should “stand in solidarity with (团结), educate (教育), and utilize (使用)” them.¹²⁹

To advance this policy, the Northwest government first conferred honors upon a few veterinarians who had distinguished themselves by their contributions to the anti-animal-diseases campaigns. Such “active and excellent folk veterinarians” now became state veterinary cadres or even Communist Party members. At the same time, learning from local precedents, the Northwest government recognized the models of the veterinary association and union clinic as the official format for organizing even more rank-and-file animal healers. By December 1952, 3,400 of the 5,000 folk veterinarians in the northwest were affiliated either with a local veterinary association or one of 107 union clinics. To consolidate the relationship between the state and folk animal healers, the Northwest government insisted that first and foremost, the economic conditions of the latter should be improved. If county governments mobilized the veterinarians in public work such as

¹²⁸ Xibei xingzheng weiyuanhui xumuju 西北行政委员会畜牧局, “Xibei qu sannianlai zuzhi zhongshouyi gongzuo gaikuang he jinhou de gongzuo yijian” 西北区三年来组织中兽医工作概况和今后的工作意见 (October 30, 1953), GPA, 222-001-0033-0005.

¹²⁹ Xibei xingzheng weiyuanhui xumuju, “Xibei qu sannianlai zuzhi zhongshouyi gongzuo gaikuang he jinhou de gongzuo yijian.”

城市里的私人动物诊所，其中大部分散布在农村和牧区。“超过一半的兽医是文盲。他们从当地师傅那里学习兽医，或者不加批判地遵循治疗习俗，或者依靠家族传承的简单疗法来治疗动物疾病。然而，他们确实拥有一定的经验和技能。”¹²⁸ 在革命前时期，几乎所有的兽医“都没有社会立足之地，一贫如洗。”因此，他们变得“保守和迷信”，遭受着自我贬低或纯粹的私利。为了引导这些民间兽医，西北政府认为共产党国家应该“团结（团结）、教育（教育）和利用（使用）”他们。¹²⁹

为了推行这项政策，西北政府首先对少数在抗动物疾病运动中表现突出的兽医授予荣誉。这些“积极优秀的民间兽医”现在成为国家兽医干部，甚至成为共产党员。同时，借鉴当地先例，西北政府将兽医协会和联合诊所模式确认为组织更多普通动物治疗师的官方形式。截至1952年12月，西北地区5000名民间兽医中有3400名隶属于当地兽医协会或107家联合诊所之一。为了巩固国家与民间动物治疗师之间的关系，西北政府坚持认为，首先必须改善后者的经济条件。如果县政府动员兽医参与公共工作，例如

¹²⁸ 西北行政委员会畜牧局, “西北区三年来组织中兽医工作概况和今后的工作意见” (1953年10月30日), GPA, 222-001-0033-0005.

¹²⁹ 西北行政委员会畜牧局, “西北区三年来组织中兽医工作概况和今后的工作意见”。

regular disinfection, they had to be properly paid. Local cadres also should prevent villagers from obtaining no-cost services from union clinics because the veterinarians who worked for the clinics were civil servants.¹³⁰

Years later, these interactions between the new Communist state and folk veterinarians culminated in the central government’s first formal recognition of the veterinarians’ social role and status. On January 5, 1956, Premier Zhou Enlai promulgated “The State Council’s Directive on Reinforcing the Task on Folk Veterinarians,” which would have a huge impact on the fate of China’s traditional animal healers. Unlike the fewer than ten thousand modern veterinarians in major cities, most of the hundred and eighty thousand folk veterinarians were “working people” who had been living as “half peasants–half [animal] doctors” in the countryside. Although they had not been lucky enough to receive formal education before the rise of the Communists, the folk veterinarians were able to heal animals with acupuncture and herbal therapies. Premier Zhou noted that “they are experienced in treating animal diseases, trusted by the peasant masses, and have made significant contributions to protecting and developing the livestock [industry].”¹³¹ Folk veterinarians, therefore, deserved the attention of the CCP and the People’s Republic. In this vein, the State Council suggested the slogan of “solidarity, utilization, education, and improvement”; the party and state would stand in solidarity with (团结), utilize (使用), educate (教育), and improve (提高) the animal healers.¹³²

¹³⁰ Xibei xingzheng weiyuanhui xumuju, “Xibei qu sannianlai zuzhi zhongshouyi gongzuo gaikuang he jinhou de gongzuo yijian.”

¹³¹ Zhou Enlai, “Guowuyuan guanyu jiaqiang minjian shouyi gongzuo de zhishi.”

¹³² Zhou Enlai, “Guowuyuan guanyu jiaqiang minjian shouyi gongzuo de zhishi.”

定期消毒，他们必须得到适当的报酬。地方干部还应防止村民从工会诊所获得免费服务，因为在诊所工作的兽医是公务员。¹³⁰

多年后，新生的共产主义国家与民间兽医之间的这些互动最终促成了中央政府首次正式承认兽医的社会作用和地位。1956年1月5日，周恩来总理颁布了《国务院关于加强民间兽医工作的指示》，该指示将对中国传统动物治疗师的命运产生巨大影响。与大城市中不到一万名现代兽医不同，十八万民间兽医中的大多数是“劳动人民”，他们一直以“半农半[兽]医”的身份生活在农村。尽管他们在共产党崛起之前没有机会接受正规教育，但民间兽医能够用针灸和草药疗法治疗动物。周总理指出，“他们对治疗动物疾病经验丰富，深受农民群众信任，为保护和发展畜牧[业]做出了重大贡献。”¹³¹因此，民间兽医值得中共及中华人民共和国的关注。为此，国务院提出了“团结、利用、教育、提高”的口号；党和国家将与动物治疗师团结、使用、教育和提高。¹³²

¹³⁰ 西北行政委员会畜牧局，“西北区三年来组织兽医工作概况和今后的工作意见。”¹³¹ 周恩来，“国务院关于加强民间兽医工作的指示。”¹³² 周恩来，“国务院关于加强民间兽医工作的指示。”

From Folk to Collective Veterinarians, 1956–1962

Soon after the cordial relationship between the state and folk veterinarians was set by the State Council in 1956, the state came to regard the animal doctors as useful resources for fulfilling the prime task in the second half of the decade: agricultural collectivization. Now the local governments asked folk veterinarians in union clinics to contract with nearby agricultural cooperatives for regular livestock checkups and veterinary services. Veterinary union clinics in Langshan County, Inner Mongolia, for example, signed such contracts with 70% of the county’s cooperatives by June 1956.¹³³ The coverage increased to 78% by the end of 1957.¹³⁴ By September 1958, in the midst of the Great Leap Forward, almost every local cooperative in Inner Mongolia had access to veterinary services. By combining the tasks of organizing folk veterinarians and accelerating agricultural collectivization, governments in the region were able to report that they had achieved the transition “from private to collective veterinary practices; from narrowly focusing on treatment to emphasizing both treatment and prevention; and from having patients visit a few clinics in village centers to having veterinarians go down to cooperatives.”¹³⁵

Although the veterinary union clinics and veterinary associations continued to function until the late 1950s, provincial governments saw these pre-collectivist ways of organizing folk

¹³³ Langshan xian zhengfu 狼山县政府, “Wei baosong woxian minjian shouyi zuotan huiyi de zongjie” 为报送我县民间兽医座谈会议的总结 (June 26, 1956), IMBMA, 026-01-0068-033.

¹³⁴ Langshan xian xumu shouyi gongzuozhan 狼山县畜牧兽医工作站, “Guanyu woxian 57 nian minjian shouyi gongzuo zongjie baogao” 关于我县 57 年民间兽医工作总结报告 (January 8, 1958), IMBMA, 026-02-0334-002.

¹³⁵ Neimenggu zizhiqu xumuting 内蒙古自治区畜牧厅, “Guanyu cujin minjian shouyi gongzuo dayuejin de han 关于促进民间兽医工作大跃进的函 (May 7, 1958), IMBMA, 122-03-0058-009.

From Folk to Collective Veterinarians, 1956–1962

1956年国务院确立国家与民间兽医之间的友好关系后不久，国家开始将兽医视为完成该十年后半段首要任务——农业集体化——的有用资源。现在，地方政府要求联合诊所的民间兽医与附近的农业合作社签订合同，提供定期牲畜检查和兽医服务。例如，内蒙古杭锦后旗的兽医联合诊所到1956年6月已与该县70%的合作社签订了此类合同。¹³³ 到1957年底，覆盖率增至78%。¹³⁴ 到1958年9月，在大跃进期间，内蒙古几乎每个地方合作社都能获得兽医服务。通过将组织民间兽医和加速农业集体化两项任务结合起来，该地区政府得以报告他们已实现了“从私人兽医诊所到集体兽医诊所；从狭隘地侧重治疗到治疗和预防并重；以及从患者到乡村中心少数诊所就诊到兽医下乡到合作社”的转变。¹³⁵

尽管兽医联合诊所和兽医协会一直运作到 20 世纪 50 年代末，但省级政府认为这些前集体主义的民间组织方式

¹³³ 狼山县政府, 《为报送我县民间兽医座谈会议的总结》（1956年6月26日），IMBMA, 026-01-0068-033。

¹³⁴ 狼山县畜牧兽医工作站, 《关于我县57年民间兽医工作总结报告》（1958年1月8日），IMBMA, 026-02-0334-002。

¹³⁵ 内蒙古自治区畜牧厅, 《关于促进民间兽医工作大跃进的函》（1958年5月7日），IMBMA, 122-03-0058-009。

veterinarians as “transitional.” They sought to transform individual animal healers into collective veterinarians (社内兽医 or 入社兽医). Chen Yushan, a high-ranking provincial cadre of Gansu who supervised agriculture and animal husbandry, acknowledged in August 1956 that the union clinics in the province had acquired certain “collectivist characteristics.” However, it was only “a format of organizing [local veterinarians] in the transitional period.” As Gansu Province would enter the full-swing socialist, collectivist stage with the emergence of advanced cooperatives in the next few years, Chen argued that Communist cadres at the grassroots should encourage folk veterinarians to “voluntarily” join the cooperatives. Then the veterinarians would be exclusively responsible for all veterinary services in a collective, partnering with caretakers and animal diseases prevention workers.¹³⁶

Similarly, the provincial government of Inner Mongolia reaffirmed in September 1957 that following the State Council’s 1956 directive, it would continue to consolidate local veterinary associations and union clinics so that they would remain core venues for the “strengthening of solidarity [between the state and veterinarians], the exchange of veterinary experiences, and the enhancement of political theories and technical expertise.” However, the Inner Mongolian government ordered local cadres to “mobilize some portion of folk veterinarians and have them join agricultural or pastoral production cooperatives, thereby letting them take charge of the veterinary work in cooperatives.” The provincial leaders also warned local cadres to be considerate of these folk-turned-collective veterinarians. They admitted that in the past, some collective leaders had wrongly assigned veterinarians to agricultural field labor or did not set a reasonable veterinary wage,

¹³⁶ Chen Yushan 陈玉山, “Chen Yushan futingzhang zai Gansu sheng diyici minjian shouyi daibiao huiyi shangde zongjie fayan” 陈玉山副厅长在甘肃省第一次民间兽医代表会议上的总结发言 (August 11, 1956), GPA, 222-001-0120-0007.

兽医是“过渡性的”。他们试图将个体动物治疗师转变为集体兽医（社内兽医或入社兽医）。甘肃省负责农业和畜牧业的高级干部陈玉山在1956年8月承认，该省的联合诊所已经具备了某些“集体主义特征”。然而，这只是“在过渡时期组织[地方兽医]的一种形式”。陈玉山认为，随着甘肃省在未来几年内出现高级合作社，将进入全面社会主义集体主义阶段，因此基层共产党员应鼓励民间兽医“自愿”加入合作社。届时，兽医将专门负责集体中的所有兽医服务，与饲养员和动物疾病预防工作者合作。¹³⁶

同样，内蒙古自治区政府在1957年9月重申，根据国务院1956年的指示，将继续巩固地方兽医协会和联合诊所，使其仍然是“加强国家与兽医之间团结、交流兽医经验、提高政治理论和技术专长”的核心场所。然而，内蒙古政府命令地方干部“动员一部分民间兽医，让他们加入农业或畜牧业生产合作社，从而让他们负责合作社的兽医工作”。省级领导还警告地方干部要体谅这些由民间转为集体的兽医。他们承认，过去一些集体领导错误地将兽医分配到农业田间劳动，或者没有设定合理的兽医工资，

¹³⁶ 陈玉山，“陈玉山副厅长在甘肃省第一次民间兽医代表会议上的总结发言”（1956年8月11日），甘肃省档案馆，222-001-0120-0007。

which “discouraged the [former] folk veterinarians’ work proactivity.” Hence, the provincial government argued that once folk veterinarians joined collectives, they should be first and foremost in charge of the veterinary work. In this manner, private folk veterinary practitioners gradually became full-time collective animal doctors in the late 1950s and early 1960s.¹³⁷

As the collectivization of folk veterinary experts intensified, local access to veterinary services expanded. An unexpected effect of this trend was that more local peasants reported veterinary malpractice to cadres. In 1957, rural residents at the outskirts of the city of Tianjin complained that when administering acupuncture and intravenous injections, some folk animal doctors failed to disinfect their needles, causing pustules to form on their animal patients. In worse cases, livestock contracted tetanus. After one folk doctor accidentally killed a pig by bloodletting, the owner of the pig filed an official grievance against the collective cadres and demanded compensation. When faced with these cases, local cadres reported to their superiors that some animal healers “were not humble; did not trust science; and pretended to know diseases that they actually did not know and recklessly treated ill animals.” The Communist cadres’ underlying doubt about folk veterinary knowledge beneath the slogan of “solidarity” resurfaced.¹³⁸

Folk veterinary knowledge needed to be, in the words of the State Council, “improved.” To do so, the Tianjin leaders insisted on making union clinics and new collective veterinary stations a forum for the exchange of experiences and skills. Specifically, the cadres asked folk veterinarians to abandon the “traditional conservatism” that treated veterinary knowledge as something “privately

¹³⁷ Neimenggu renmin weiyuanhui 内蒙古人民委员会, “Guanyu jiaqiang minjian shouyi gongzuo de zhishi” 关于加强民间兽医工作的指示 (September 10, 1957), IMBMA, 026-02-0334-026.

¹³⁸ “Tianjin shi minjian shouyi gongzuo huiyi baogao” 天津市民间兽医工作会议报告 (1957), Tianjin Municipal Archives (hereafter TMA), 401206800-X0165-Y-000348-009.

这“阻碍了[前]民间兽医的工作积极性”。因此，省政府认为，一旦民间兽医加入集体，他们就应该首先负责兽医工作。通过这种方式，私人民间兽医从业者在1950年代末和1960年代初逐渐成为全职集体动物医生。¹³⁷

随着民间兽医专家集体化程度的加深，当地获得兽医服务的机会也随之增加。这种趋势的一个意想不到的结果是，更多的当地农民向干部举报兽医渎职行为。1957年，天津市郊区的农村居民抱怨说，一些民间兽医在进行针灸和静脉注射时，未能对针头进行消毒，导致他们的动物患者身上形成脓疱。更糟糕的是，牲畜感染了破伤风。一名民间医生在放血时意外地杀死了一头猪后，猪的主人向集体干部提出了正式申诉，并要求赔偿。面对这些案例，当地干部向上级汇报说，一些动物治疗师“不谦虚；不相信科学；假装知道他们实际上不知道的疾病，并鲁莽地治疗生病的动物”。共产党干部在“团结”口号下对民间兽医知识的潜在怀疑再次浮现。¹³⁸

用国务院的话来说，民间兽医知识需要“改进”。为此，天津领导人坚持将联合诊所和新的集体兽医站作为交流经验和技能的场所。具体来说，干部们要求民间兽医放弃将兽医知识视为“私人”的“传统保守主义”

¹³⁷ 内蒙古人民委员会, 《关于加强民间兽医工作的指示》(1957年9月10日), 内蒙古自治区档案馆, 026-02-0334-026。

¹³⁸ 《天津市民间兽医工作会议报告》(1957), 天津市档案馆(以下简称TMA), 401206800-X0165-Y-000348-009。

owned” and “exclusively passed down from grandfathers to fathers to sons.” It should not be “family medical knowledge that cannot be handed down to outsiders.” Instead, folk veterinarians would have to share their knowledge with other veterinary workers and peasants in collectives.¹³⁹

Other provinces similarly emphasized the disclosure and sharing of folk remedies that were claimed to be effective. In Inner Mongolia, those who “are not willing to teach others one’s veterinary medical experiences” were criticized as lacking the spirit of solidarity. The cadres also argued that such “conservative” folk veterinarians were more corruptible.¹⁴⁰ Also, the Inner Mongolian leaders prescribed that the cadres and veterinarians who worked in collective veterinary stations must collect any classical veterinary texts and any other written sources about secret, experience-based therapies. Now local collective veterinary stations were required to function as a repository of scattered folk veterinary knowledge that would be communally owned and shared.¹⁴¹

The expected roles of veterinary institutions which worked with or within local collectives became clearer by 1958. According to a report written by the Jiangxi Provincial Department of Agriculture, veterinary cadres and former folk animal healers in people’s communes had already organized numerous intensive veterinary training sessions to produce a new generation of collective veterinarians. We will discuss these young animal doctors in depth in the next chapter. Collective veterinary stations also should spearhead the comprehensive collecting and summarizing of folk veterinary knowledge and expertise to lay the foundation for the wholesale imminent scientization of

¹³⁹ “Tianjin shi minjian shouyi gongzuo huiyi baogao.”

¹⁴⁰ Langshan xian zhengfu, “Wei baosong woxian minjian shouyi zuotan huiyi de zongjie.”

¹⁴¹ Neimenggu renmin weiyuanhui, “Guanyu jiaqiang minjian shouyi gongzuo de zhishi.”

拥有”和“专属地从祖父传给父亲再传给儿子”。它不应该是“不能传给外人的家族医学知识”。相反，民间兽医必须与集体中的其他兽医工作者和农民分享他们的知识。

¹³⁹

其他省份也同样强调公开和分享那些据称有效的民间疗法。在内蒙古，那些“不愿意将自己的兽医经验传授给他人”的人被批评为缺乏团结精神。干部们还认为，这种“保守”的民间兽医更容易被腐蚀。¹⁴⁰ 此外，内蒙古领导人规定，在集体兽医站工作的干部和兽医必须收集所有古典兽医文献以及其他关于秘密的、基于经验的疗法的书面资料。现在，地方集体兽医站被要求作为分散的民间兽医知识的储存库，这些知识将由集体拥有和共享。¹⁴¹

到1958年，与地方集体合作或在地方集体内部运作的兽医机构的预期作用变得更加清晰。根据江西省农业厅的一份报告，人民公社的兽医干部和前民间动物治疗师已经组织了多次密集的兽医培训，以培养新一代的集体兽医。我们将在下一章深入讨论这些年轻的动物医生。集体兽医站还应带头全面收集和总结民间兽医知识和专业技能，为即将到来的全面科学化奠定基础

¹³⁹ “天津市民间手工艺工作会议报告。”

¹⁴⁰ 郎山县政府，“为报送我县民间手工艺座谈会议的总结。”¹⁴¹ 内蒙古人民委员会，“关于加强民间手工艺工作的指示。”

“fatherland’s veterinary scholarship.”¹⁴² Responding to this provincial policy, counties such as Nanchang and Anyuan closed union clinics and transferred all of the county’s former folk veterinarians into commune veterinary stations in December 1958. The counties also discovered 576 medicinal herbs that were known to be effective treatments for certain animal diseases, 46 “secret remedies,” and 244 “commonly used local therapies.”¹⁴³

The collectivization of folk veterinary experts and expertise was completed in the second half of the 1950s. What ensued was the massive extraction and “improvement” of unsystematized local remedies and healing experiences. But how exactly could the state and collective veterinarians scientize such knowledge? Here the Communist Party’s precedential engagement with traditional “human medicine” (人医) functioned as a reference. In the following sections, I examine how Traditional Chinese Medicine (TCM) influenced what came to be known as Traditional Chinese Veterinary Medicine (TCVM) and how local animal healers participated in the state project of modernizing folk veterinary medicine.

The Scientization of Folk Veterinary Knowledge

According to Kim Taylor, TCM was institutionalized between 1953 and 1956.¹⁴⁴ At that time, Mao ratified the CCP’s Chinese medicine (中医) policy with the catchphrase, “doctors of Western

¹⁴² Jiangxi sheng nongyeting 江西省农业厅, “Jiangxi sheng minjian shouyi gaikuang” 江西省民间兽医概况, *Minjian shouyi tongxun* 民间兽医通讯 7 (September 1958), 9–13.

¹⁴³ “Minjian shouyi gongzuo dongtai” 民间兽医工作动态, *Minjian shouyi tongxun* 民间兽医通讯 9 (December 1958), 78–79.

¹⁴⁴ Kim Taylor, *Chinese Medicine in Early Communist China, 1945–1963: A Medicine of Revolution* (London: RoutledgeCurzon, 2005), 63.

“祖国兽医奖学金。”¹⁴² 为响应这项省级政策，南昌、安远等县于1958年12月关闭了联合诊所，并将全县原有的民间兽医全部调入公社兽医站。这些县还发现了576种已知对某些动物疾病有效的药草、46种“秘方”和244种“常用地方疗法”。¹⁴³

民间兽医专家和专业知识的集体化在20世纪50年代下半叶完成。随之而来的是对非系统化的地方疗法和治疗经验的大规模提取和“改进”。但国家和集体兽医究竟如何才能使这些知识科学化呢？在这里，中国共产党此前对传统“人医”的介入起到了参考作用。在接下来的章节中，我将探讨中医（TCM）如何影响了后来被称为中兽医（TCVM）的形成，以及地方动物治疗师如何参与了国家现代化民间兽医的项目。

民间兽医知识的科学化

根据金·泰勒的说法，中医在1953年至1956年间被制度化。¹⁴⁴当时，毛泽东批准了中共的中医政策，并提出了“西医”的口号，

¹⁴² 江西省农业厅，“江西省民间兽医概况”，民间兽医通讯 7 (1958年9月)，9–13页。

¹⁴³ “民间兽医工作动态”，民间兽医通讯 9 (1958年12月)，78–79页。

¹⁴⁴ 金·泰勒，《早期共产主义中国的中国医学，1945-1963：一场革命的医学》（伦敦：RoutledgeCurzon, 2005），63页。

medicine study Chinese medicine” (西医学习中医). With it, Mao stressed that Western medicine practitioners should learn traditional healing methods from some “learned Chinese medicine practitioners” (有学问的中医) and that both ought to participate in the modernization of Chinese medicine.¹⁴⁵ In other words, high-ranking folk doctors and Western-trained doctors jointly made Chinese medicine credible, if not scientific.

With the recognition of practical roles of folk veterinarians and the precedent of constructing TCM, the CCP attempted to transform traditional veterinary knowledge into a legitimate field of science serving the proletariat. Following the precedent that “doctors of Western medicine study Chinese medicine” within human medicine, this task was assigned to renowned folk veterinary masters “who have relatively higher-level techniques and some extent of culture;” veterinarians of Western veterinary medicine; and veterinary cadres.¹⁴⁶ These different veterinary experts shall “research and summate the expertise and experience of folk veterinarians in a planned and systematic way on the basis of new scientific theories and methods.” They were also expected to “proactively introduce and disseminate therapeutic examples that actually work” and, if possible, someday, “create a new veterinary school tradition of China” on their own.¹⁴⁷

Against this backdrop, four individuals appeared on the stage of history. Gao Guojing (高国景, 1893–1964) from Shanxi Province and Cui Diseng (崔涤僧, 1885–1966) from Shaanxi Province

¹⁴⁵ Mao Zedong 毛泽东, “1954 nian dui zhongyi gongzuo de zhishi” 1954 年对中医工作的指示 (July 30, 1954), *Mao Zedong Sixiang Wansui* 毛泽东思想万岁 (Wuhan: N.p., 1968). Accessed April 3, 2022, <https://www.marxists.org/chinese/maozedong/1968/3-055.htm>.

¹⁴⁶ Zhou Enlai, “Guowuyuan guanyu jiaqiang minjian shouyi gongzuo de zhishi.”

¹⁴⁷ Jiang Cisheng 蒋次升, “Jicheng he fayang zuguo shouyixue yichan” 继承和发扬祖国兽医学遗产, *Renmin Ribao* 人民日报 (May 8, 1959).

西医学习中医”。毛泽东强调，西医应该向“有学问的中医”学习传统治疗方法，两者都应该参与中医药的现代化。¹⁴⁵ 换句话说，高级民间医生和受过西方培训的医生共同使中医药变得可信，即使不科学。

随着民间兽医实际作用的认可和中医建设的先例，中国共产党试图将传统兽医知识转化为服务于无产阶级的合法科学领域。遵循人类医学领域“西医学习中医”的先例，这项任务被分配给“技术水平较高、有一定文化程度”的著名民间兽医大师；西兽医；以及兽医干部。¹⁴⁶ 这些不同的兽医专家应“在新的科学理论和方法的基础上，有计划、有系统地研究和总结民间兽医的专长和经验”。他们还应“积极介绍和传播实际有效的治疗案例”，如果可能的话，有朝一日“自行开创中国兽医新学派传统”。¹⁴⁷

在此背景下，四个人物登上了历史舞台。他们是来自山西省的高国景（1893-1964）和来自陕西省的崔涤僧（1885-1966）

¹⁴⁵ Mao Zedong 毛泽东, “1954 nian dui zhongyi gongzuo de zhishi” 1954 年对中医工作的指示 (1954年7月30日), *Mao Zedong Sixiang Wansui* 毛泽东思想万岁 (武汉: N.p., 1968). 2022年4月3日访问, <https://www.marxists.org/chinese/maozedong/1968/3-055.htm>.

¹⁴⁶ Zhou Enlai, “Guowuyuan guanyu jiaqiang minjian shouyi gongzuo de zhishi.”

¹⁴⁷ 蒋次升, “继承和发扬祖国兽医学遗产”, 《人民日报》(1959年5月8日)。

were folk veterinary masters whose knowledge and skills were revered by villagers. Gao and Cui were living sourcebooks whose vernacular knowledge would be selectively translated into the language of modern science. As we will see in this and following sections, Gao and Cui were so intellectually and politically resilient that they were willing to participate in the process of “improving” what they knew.¹⁴⁸

Jiang Cisheng (蒋次升, 1914–2004) was a “Western” veterinarian who was sympathetic to socialist ideals, although he never joined the CCP.¹⁴⁹ He earned his Doctor of Veterinary Medicine degree from Iowa State University in 1948 and dedicated himself to working with folk veterinary masters like Gao and Cui for the making of TCVM. Yu Chuan (于船, 1924–2005) was the leading Communist cadre specializing in veterinary medicine. Before the revolution, he had studied modern veterinary medicine at the Army Academy of Veterinary Medicine (formerly Beiyang Equine Medicine School).¹⁵⁰ He started to work for the CCP as a veterinary expert in 1947, and his loyalty to the Party and Maoism compensated for folk veterinarians’ possible lack of political commitment.¹⁵¹

¹⁴⁸ On the life of Gao Guojing and Cui Diseng, see Yu Chuan 于船, Wu Xuecong 吴学聪, and Jin Zhongye 金重冶, “Zongjie minjian shouyi jingyan wei fayang zuguo shouyixue yichan de zhongyao huanjie” 总结民间兽医经验为发扬祖国兽医学遗产的重要环节 (1956), in Yu Chuan 于船, *Yu Chuan Wenji* 于船文集 (Beijing: Zhongguo nongye daxue chubanshe, 2003), 169–172; “Quanguo minjian shouyi zuotanhui jieshu” 全国民间兽医座谈会结束, *Renmin Ribao* 人民日报 (September 14, 1956).

¹⁴⁹ Jiang was a member of the Jiusan Society or Nine-Three Academic Society (九三学社), one of the eight legally recognized non-Communist political parties in the PRC, until his death. Jiusan xueshe Zhejiang sheng weiyuanhui 九三学社浙江省委员会, “Jiang Cisheng xiansheng shishi” 蒋次升先生逝世, accessed April 3, 2022, http://www.zjjs.org/news_show.php?ShowId=13671.

¹⁵⁰ On the history of Army Academy of Veterinary Medicine, see Li Qingshan 李青山, “Zhongguo jindai (1840–1949 nian) shouyi gaodeng jiaoyu suyuan ji fazhan” 中国近代(1840–1949年)兽医高等教育溯源及发展, Ph.D. diss., China Agricultural University, 2015.

¹⁵¹ On Yu Chuan’s life and career, see Yu Chuan, *Yu Chuan Wenji*, 563–576.

是民间兽医大师，他们的知识和技能受到村民的尊敬。高和崔是活生生的知识宝库，他们的乡土知识将被选择性地转化为现代科学的语言。正如我们将在本节和以下章节中看到的那样，高和崔在智力和政治上都具有很强的韧性，他们愿意参与“改进”他们所知知识的过程。¹⁴⁸

蒋次升（蒋次升，1914-2004）是一位同情社会主义理想的“西方”兽医，尽管他从未加入中国共产党。¹⁴⁹他于1948年获得爱荷华州立大学兽医学博士学位，并致力于与高和崔等民间兽医大师合作，共同创建中兽医。于船（于船，1924-2005）是兽医学领域的共产主义领导干部。在革命之前，他曾在陆军兽医学院（前身为北洋马医学校）学习现代兽医学。¹⁵⁰他于1947年开始为中国共产党担任兽医专家，他对党和毛泽东思想的忠诚弥补了民间兽医可能缺乏的政治承诺。¹⁵¹

¹⁴⁸ On the life of Gao Guojing and Cui Diseng, see Yu Chuan 于船, Wu Xuecong 吴学聪, and Jin Zhongye 金重冶, “Zongjie minjian shouyi jingyan wei fayang zuguo shouyixue yichan de zhongyao huanjie” 总结民间兽医经验为发扬祖国兽医学遗产的重要环节 (1956), in Yu Chuan 于船, *Yu Chuan Wenji* 于船文集 (Beijing: Zhongguo nongye daxue chubanshe, 2003), 169–172; “Quanguo minjian shouyi zuotanhui jieshu” 全国民间兽医座谈会结束, *Renmin Ribao* 人民日报 (September 14, 1956).

¹⁴⁹ 姜直到去世前都是九三学社（九三学社）的成员，九三学社是中华人民共和国八个合法非共产党政党之一。九三学社浙江省委员会九三学社浙江省委员会，“蒋次升先生逝世” 蒋次升先生逝世，2022年4月3日访问，http://www.zjjs.org/news_show.php?ShowId=13671。

¹⁵⁰ 关于军事兽医学院的历史，参见李青山，“中国近代（1840-1949年）兽医高等教育溯源及发展”，博士论文，中国农业大学，2015年。

¹⁵¹ 关于喻川的生平与事业，参见喻川，《喻川文集》，563-576页。

The state expected the collaboration among Gao, Cui, Jiang, and Yu to lead to the scientization of folk veterinary medicine.

Gao and Cui knew that if their expertise were to be fully recognized, they had to adopt the language of modern science. It seemed clear that the state and veterinary scientists did not equate vernacular veterinary with modern veterinary science. Nevertheless, they understood that their new counterparts did not want to disparage the folk veterinarians’ profound experiences and techniques. Thus, Gao, Cui, Jiang, and Yu agreed that their common mission would be to discard the arguably unscientific parts of folk veterinary medicine and improve the rest by using the scientific method. This left one question unanswered: what was scientific in this process? The state could not answer this question. The folk veterinarians and their new colleagues—Western-trained veterinarians like Jiang and veterinary cadres like Yu—would figure it out together.

It is worth noting that all the four figures of interest, Gao Guojing, Cui Diseng, Jiang Cisheng, and Yu Chuan, considered veterinary medicine not as a field of basic science, but as an empirical, applied medical science.¹⁵² The *raison d'être* of TCVM was above all to protect farm animals and to increase agricultural production, not to defend and scientifically validate the traditional Chinese medical epistemology per se. As such, the veterinarians were not expected to ascertain, for example, the reality of meridian channels and the physiological or neurological mechanism of

¹⁵² Jiang Cisheng 蒋次升, “Xuexi zuguo shouyi xueshu de chubu tihui he xinde” 学习祖国兽医学术的初步体会和心得, *Zhongshouyixue zhuanli ziliao ji yanjiu baogao huikan* 中兽医学专题资料及研究报告汇刊 1, ed. Zhongguo shouyi xuehui zhongshouyi xiaozu 中国兽医学会中兽医小组 (Beijing: Kexue chubanshe, 1957), 11–12.

国家期望高、崔、蒋、余之间的合作能够促成民间兽医学的科学化。

高和崔深知，如果他们的专业知识要得到充分认可，就必须采用现代科学的语言。国家和兽医科学家显然没有将乡土兽医与现代兽医科学等同起来。然而，他们明白，他们的新同行并不想贬低民间兽医的深刻经验和技能。因此，高、崔、蒋、余一致认为，他们的共同使命是剔除民间兽医学中那些可以说是不科学的部分，并利用科学方法改进其余部分。这留下了一个悬而未决的问题：在这个过程中，什么是科学的？国家无法回答这个问题。民间兽医和他们的新同事——像蒋这样的受过西方培训的兽医以及像余这样的兽医干部——将共同解决这个问题。

值得注意的是，高国景、崔涤僧、蒋次生和余传这四位主要人物都认为兽医学不是一门基础科学，而是一门经验性的应用医学科学。¹⁵² 中兽医的存在理由首先是为了保护农场动物和提高农业生产，而不是为了捍卫和科学验证中医认识论本身。因此，兽医们不被期望确定，例如，经络的真实性以及经络的生理或神经机制。

¹⁵² 蒋次升, “学习祖国兽医学术的初步体会和心得”, 中兽医学专题资料及研究报告汇刊 1, 中国兽医学会中兽医小组编 (北京: 科学出版社, 1957), 11-12页。

acupuncture. The purpose of the scientization for which these veterinary elites were responsible was to prove the clinical efficacy of Chinese veterinary medicine.¹⁵³

The first step of verifying the traditional therapeutics was to bring concepts from premodern veterinary classics into line with modern physiology and pathology.¹⁵⁴ For example, Cui Diseng argued that some types of *ding* from one of the early modern veterinary classics, *Collection for Treating the Horse* (*Yuanheng liaomaji*, c. 1608),¹⁵⁵ were identical to the inflammation caused by drilling for setting up harness and bridle.¹⁵⁶ Similarly, when he explained his knowledge of horse constipation, Gao Guojing first consented to adopt the Western veterinary paradigm and then unraveled his knowledge about causes, symptoms, treatments, and post-treatment nursing care accordingly. Then

¹⁵³ Jiang Cisheng, “Xuexi zuguo shouyi xueshu de chubu tihui he xinde,” 26;

Jiang Cisheng 蒋次升 and Yang Ruo 杨若, “Zuzhi bianxie siben zhongshouyi shuji de shouhuo he tihui” 组织编写四本中兽医书籍的收获和体会, in *Yanjiu ziliao huibian* 研究资料汇编 2, ed. Zhongguo nongye kexueyuan zhongshouyi yanjiusuo 中国农业科学院中兽医研究所 (Lanzhou: Zhongguo nongye kexueyuan zhongshouyi yanjiusuo, 1963), 191.

¹⁵⁴ I recognize that the approach of TCVM elites like Jiang, Gao, Cui, and Yu to epistemological differences between Chinese medicine and modern biomedicine is only one of many ways to grapple with the very conundrum. Therefore, my focus on the pioneers’ efforts to translate Chinese medical categories into biomedical ones does not mean to deny the idea that some parts of the former cannot be translated to the latter. Marta Hanson documents the alterity and non-static nature of Chinese medical nosology which defies the simplistic categorization based on modern medicine. Comparing more broadly the ways in which Chinese and Greek medicine construe the “expressiveness of the body,” Shigehisa Kuriyama explores how the two medical systems historically diverged from each other. Marta Hanson, *Speaking of Epidemics in Chinese Medicine: Disease and the Geographic Imagination in Late Imperial China* (New York: Routledge, 2012); Shigehisa Kuriyama, *The Expressiveness of the Body and the Divergence of Greek and Chinese Medicine* (New York: Zone Books, 1999).

¹⁵⁵ Yu Benyuan 喻本元 and Yu Bengheng 喻本亨, *Yuan heng liao ma ji* 元亨疗马集 (Beijing: Nongye chubanshe, 1957). Also see David W. Ramey and Bernard E. Rollin, *Complementary and Alternative Veterinary Medicine Considered* (Ames: Iowa State Press, 2008), 29–30.

¹⁵⁶ Cui Diseng 崔涤僧, Li Wenhua 李文化, et al., “Shiyong zhongyao bianzheng zhiliao maluo dingzheng de yanjiu baogao” 试用中药辩证治疗马骡疗症的研究报告, in *Zhongshouyi kexue jishu ziliao xuanji* 中兽医科学技术资料选集, ed. Zhongguo nongye kexueyuan zhongshouyi yanjiusuo 中国农业科学院中兽医研究所 (Lanzhou: Zhongguo nongye kexueyuan zhongshouyi yanjiusuo, 1964), 96–100.

针灸。这些兽医精英负责的科学化旨在证明中国兽医学的临床疗效。¹⁵³

验证传统疗法的第一步是将前现代兽医经典的观念与现代生理学和病理学相结合。¹⁵⁴ 例如，崔涤僧认为，早期现代兽医经典之一《元亨疗马集》（约1608年）¹⁵⁵中的某些类型的疔与为安装挽具和缰绳而钻孔引起的炎症相同。¹⁵⁶ 同样，当他解释他对马匹便秘的认识时，高国经首先同意采用西方兽医范式，然后相应地阐述了他关于病因、症状、治疗和治疗后护理的知识。然后

¹⁵³ 蒋次生， “学习祖国兽医学术的初步体会和心得” ， 26；

Jiang Cisheng 蒋次升 and Yang Ruo 杨若, “Zuzhi bianxie siben zhongshouyi shuji de shouhuo he tihui” 组织编写四本中兽医书籍的收获和体会, in *Yanjiu ziliao huibian* 研究资料汇编 2, ed. Zhongguo nongye kexueyuan zhongshouyi yanjiusuo 中国农业科学院中兽医研究所 (Lanzhou: Zhongguo nongye kexueyuan zhongshouyi yanjiusuo, 1963), 191.

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¹⁵⁵ 喻本元和喻本亨， 《元亨疗马集》（北京： 农业出版社， 1957）。另见 David W. Ramey 和 Bernard E. Rollin， 《*Complementary and Alternative Veterinary Medicine Considered*》 （Ames： Iowa State Press， 2008） ， 29-30。

¹⁵⁶ Cui Diseng 崔涤僧, Li Wenhua 李文化, et al., “Shiyong zhongyao bianzheng zhiliao maluo dingzheng de yanjiu baogao” 试用中药辩证治疗马骡疗症的研究报告, in *Zhongshouyi kexue jishu ziliao xuanji* 中兽医科学技术资料选集, ed. Zhongguo nongye kexueyuan zhongshouyi yanjiusuo 中国农业科学院中兽医研究所 (Lanzhou: Zhongguo nongye kexueyuan zhongshouyi yanjiusuo, 1964), 96–100.

he presented statistical evidence that his therapeutic methods had an 86 percent cure rate.¹⁵⁷ With an emphasis on clinical implication and utility, the Mater veterinarians aimed to reinterpret the foundational contents of premodern veterinary classics in a way that contemporary readers could understand and apply.

The next step was to bring the “pattern differentiation and treatment determination” (辨证论治), the hallmark of TCM, to the veterinary realm.¹⁵⁸ Chinese medicine practitioners and intellectuals had developed the unique notion of “pattern of disorder” (证) which does not neatly correspond to the biomedical concept of disease.¹⁵⁹ While categories of disease reflect a set of specific causes and cures based on germ theories,¹⁶⁰ “patterns” or “syndromes” are patternized constellations of manifested individual symptoms. In principle, if one can discern a pattern in the bodies of sick animals, one can determine effective treatments to ease off the pattern of symptoms without knowing the name and cause of the disease in question.¹⁶¹

¹⁵⁷ Gao Guojing 高国景, “Zhiliao qiwozheng de jingyan zhengli” 治疗起卧症的经验整理, in *Zhongsouyixue zhuanti zhiliao ji yanjiu baogao huikan* 1, ed. Zhongguo shouyi xuehui zhongshouyi xiaozu, 54–59.

¹⁵⁸ Volker Scheid, *Chinese Medicine in Contemporary China: Plurality and Synthesis* (Durham: Duke University Press, 2002), 200–237; Lei, *Neither Donkey Nor Horse*, 181–190.

¹⁵⁹ Zhen Cheng 甄橙, *Bing yu zheng de duiqi: fansi 18 shiji de yixue* 病与证的对峙: 反思 18 世纪的医学 (Beijing: Beijing daxue chubanshe, 2007).

¹⁶⁰ Charles E. Rosenberg, “The Tyranny of Diagnosis: Specific Entities and Individual Experience,” in *Our Present Complaint: American Medicine, Then and Now* (Baltimore: Johns Hopkins University Press, 2007).

¹⁶¹ In this sense, according to Sean Lei, the pattern may not serve “as truthful representations of disease entities” yet could be “a necessary tool for utilizing supposedly valuable therapeutics” and “circumvent[ing] the threat of the germ theory of diseases.” He also argues that “‘pattern differentiation and treatment determination’ was not merely the expression of an alternative theory of disease; more importantly, it represented the rise of a philosophical alternative to the representationist conception of realism” and that “the point of realism is not to represent the reality out there but to intervene in the world right here.” Lei, *Neither Donkey Nor Horse*, 183, 188–189.

他提供了统计证据，证明他的治疗方法治愈率达 86%。¹⁵⁷ Mater 兽医强调临床意义和实用性，旨在以当代读者能够理解和应用的方式重新诠释前现代兽医经典的奠基内容。

下一步是将中医的标志性特征“辨证论治”引入兽医领域。¹⁵⁸ 中医从业者和知识分子发展出了独特的“证”概念，它与生物医学的疾病概念并不完全对应。¹⁵⁹ 疾病类别反映了一系列基于细菌理论的特定病因和治疗方法，¹⁶⁰ 而“证”或“症候群”是表现出的个体症状的模式化组合。原则上，如果能在病畜体内辨别出一种模式，就可以确定有效的治疗方法来缓解症状模式，而无需知道相关疾病的名称和病因。¹⁶¹

¹⁵⁷ Gao Guojing 高国景, “Zhiliao qiwozheng de jingyan zhengli” 治疗起卧症的经验整理, in *Zhongsouyixue zhuanti zhiliao ji yanjiu baogao huikan* 1, ed. Zhongguo shouyi xuehui zhongshouyi xiaozu, 54–59.

¹⁵⁸ Volker Scheid, 《当代中国的中国医学：多元与综合》（杜克大学出版社，2002），200-237页；雷，《非驴非马》，181-190页。

¹⁵⁹ 甄橙，病与证的对峙：反思 18 世纪的医学（北京：北京大学出版社，2007）。

¹⁶⁰ Charles E. Rosenberg, “The Tyranny of Diagnosis: Specific Entities and Individual Experience”，载于*Our Present Complaint: American Medicine, Then and Now*（巴尔的摩：约翰霍普金斯大学出版社，2007）。

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As Chinese veterinary medicine shares basic theories on organisms’ physiology and pathology with Chinese medicine, Gao Guojing and Cui Diseng were familiar with “pattern differentiation and treatment determination.” It was a reasonable framework to explain the ways in which folk veterinarians identified therapeutic methods based on their diagnosis of the “patterns” of disorder. Cui helped Jiang Cisheng understand this point.¹⁶² Before long, Jiang argued that Chinese veterinary therapeutics was different from Western veterinary treatments that are reductively aimed towards specific causes. Jiang then acknowledged that TCVM could provide valid treatments without the strong basis of bacteriological etiology.¹⁶³ Now the folk master and the veterinary scientist came to agree that scientifically proving traditional veterinary knowledge meant verifying the efficacy of acupunctural and herbal medical treatments, among others, derived from “pattern differentiation.”

The notion of “pattern,” however, did not replace all disease categories.¹⁶⁴ Patterns and diseases were intertwined on the clinical ground. The conceptional pluralism lasted in clinical practice. Rather than separating the three categorical systems, Gao, Cui, Jiang, and Yu bypassed this arguably manageable uncertainty about the definition of disease, and left it to their colleagues in the field of human medicine. Instead, they focused on empirical experimentation with folk veterinary therapeutics in the hope of making TCVM a legitimate field of applied medical science.

¹⁶² Jiang Cisheng 蒋次升, “Zhongxi shouyi jiehe de huigu yu duice” 中西兽医结合的回顾与对策, *Zhongshouyixue 中兽医学杂志* 1992.4 (December 1992): 1.

¹⁶³ Jiang, “Xuexi zuguo shouyi xueshu de chubu tihui he xinde,” 24–26.

¹⁶⁴ Also see Lei, *Neither Donkey Nor Horse*, 184.

由于中国兽医学与中医学在生物体生理病理学方面共享基本理论，高国景和崔涤生熟悉“辨证论治”。这是一个合理的框架，可以解释民间兽医根据对疾病“证型”的诊断来确定治疗方法的方式。崔帮助蒋次生理解了这一点。¹⁶²不久，蒋次生提出，中国兽医治疗学不同于还原性地针对特定病因的西方兽医治疗。蒋次生随后承认，中兽医可以在没有细菌病因学强大基础的情况下提供有效的治疗。¹⁶³现在，民间大师和兽医科学家都同意，科学地证明传统兽医知识意味着验证针灸和草药治疗等源自“辨证论治”的疗效。

然而，“证型”的概念并没有取代所有的疾病类别。¹⁶⁴在临床实践中，证型和疾病是相互交织的。这种概念上的多元性在临床实践中持续存在。高、崔、蒋、俞没有将这三个分类系统分开，而是绕过了这种关于疾病定义的可控不确定性，将其留给了他们在人类医学领域的同事。相反，他们专注于对民间兽医疗法进行经验性实验，希望使中兽医成为应用医学科学的合法领域。

¹⁶² 蒋次升，“中西兽医结合的回顾与对策”，中兽医学杂志1992.4（1992年12月）：1。¹⁶³ 蒋，“学习祖国兽医学术的初步体会和心得”，24-26。¹⁶⁴ 另见雷，《Neither Donkey Nor Horse》，184。

Making TCVM a Socialist Science

The folk veterinarians and Western-trained scientists now had a vision of how to place traditional veterinary knowledge into the framework of modern science. It was a similar process to the modernization of Chinese medicine in the first half of the twentieth century, which historian of medicine Sean Lei explained as a process of making “mongrel medicine” that was “neither donkey nor horse.”¹⁶⁵ However, the revolutionary politics of the time pushed the veterinary elites to pay attention another layer of translation: reframing modernized folk veterinary medicine in socialist terms. Facing this particular challenge, folk masters learned from their collaboration with veterinary cadres like Yu Chuan.

Gao Guojing, Cui Diseng, Jiang Cisheng, and Yu Chuan first tried to associate their new intellectual project of making TCVM with Chairman Mao’s famous theory of practice. According to Mao, “correct ideas” and true knowledge always come from “social practice, the struggle for production, the class struggle, and scientific experiment,” all of which must be initiated by working people in rice paddy fields or on shop floors.¹⁶⁶ Hence, when Jiang Cisheng defined Chinese veterinary medicine as the “summation of our nation’s working people’s experiences in the fight against livestock diseases,”¹⁶⁷ he claimed that it might be able to carve out its niche in socialist China because it was based on what Mao termed “correct ideas.”¹⁶⁸

¹⁶⁵ Lei, *Neither Donkey Nor Horse*.

¹⁶⁶ Mao Zedong 毛泽东, “Where Do Correct Ideas Come From?” (May 1963), in *Selected Readings from the Works of Mao Tsetung* (Beijing: Waiwen chubanshe, 1971), 502–504.

¹⁶⁷ Jiang Cisheng 蒋次升, “Dang de zhongshouyi zhengce de shengli” 党的中兽医政策的胜利, *Renmin Ribao* 人民日报 (April 15, 1960).

¹⁶⁸ On the importance of “experience” (经验) as embodied knowledge in China, see Farquhar, *Knowing Practice*, 1–7. Sigrid Schmalzer points out that the “experience paradigm” in Maoist China, however, tended to preclude “recognition

将中兽医打造成社会主义科学

民间兽医和受过西方训练的科学家现在有了一个愿景，那就是如何将传统兽医知识纳入现代科学的框架。这与20世纪上半叶中医药的现代化过程类似，医学史学家雷祥麟将其解释为一种“不驴不马”的“杂种医学”的形成过程。¹⁶⁵然而，当时的革命政治促使兽医精英们关注另一个层面的转化：用社会主义术语重新构建现代化的民间兽医。面对这一特殊挑战，民间大师们从与俞川等兽医干部的合作中学习。

高国京、崔涤僧、姜次生和俞川首先尝试将他们将中兽医打造成新知识体系的计划与毛主席著名的实践论联系起来。毛泽东认为，“正确的思想”和真知总是来源于“社会实践、生产斗争、阶级斗争和科学实验”，所有这些都必须由稻田或车间的劳动人民发起。¹⁶⁶因此，当姜次生将中兽医定义为“我国劳动人民在与牲畜疾病作斗争中的经验总结”时，¹⁶⁷他声称它可能能够在社会主义中国开辟自己的利基市场，因为它基于毛泽东所说的“正确思想”。¹⁶⁸

¹⁶⁵ 雷，《非驴非马》。

¹⁶⁶ 毛泽东，“人的正确思想是从哪里来的？”（1963年5月），载《毛泽东著作选读》（北京：外文出版社，1971），502-504页。

¹⁶⁷ 蒋次升，“党的中兽医政策的胜利”，《人民日报》（1960年4月15日）。

¹⁶⁸ 关于“经验”作为中国具身知识的重要性，参见Farquhar，《*Knowing Practice*》，1-7页。Sigrid Schmalzer指出，然而，毛泽东时代的“经验范式”往往排除了“认知”

However, being “correct” was not enough. Chinese veterinary medicine now needed to be reformed to a “dialectically materialist” science. Dialectical materialism is a form of materialism which posits that “all nature can be explained in terms of matter and energy.”¹⁶⁹ As such, it refutes idealism that postulates metaphysical principles to explain the natural world. Nevertheless, it is also differentiated from “mechanistic materialism,” “a belief that all phenomena in nature, including human behavior, can ultimately be explained in terms of the simplest interactions of matter.”¹⁷⁰ Instead, dialectical materialists suggest that there are “‘levels of being’ in nature, such as the physical, the biological, and the social” from below and that a series of quantitative changes in the lower levels can dialectically cause qualitative changes in the upper levels.¹⁷¹ In this sense, matter and energy in the most fundamental level of the physical are both components of the driving force of the natural world, including its biological and social levels. At the same time, due to qualitative leaps between levels, “social laws cannot be reduced to biological laws, and biological laws cannot be reduced to physiochemical laws.”¹⁷² In sum, dialectical materialism is a philosophy of science veering between the extremes of mechanistic materialism and metaphysical idealism. In the Soviet Union, it was sanctioned by Lenin, Trotsky, Bukharin, and Stalin as the correct, Bolshevik way of the natural and social sciences.¹⁷³

of the ways in which technologies are embedded in culture.” Sigrid Schmalzer, “Layer upon Layer: Mao-Era History and the Construction of China’s Agricultural Heritage,” *East Asian Science, Technology and Society* 13.3 (September 2019): 420.

¹⁶⁹ Loren R. Graham, *Science in Russia and the Soviet Union: A Short History* (Cambridge: Cambridge University Press, 1990), 100.

¹⁷⁰ Graham, *Science in Russia and the Soviet Union*, 100–101.

¹⁷¹ Graham, *Science in Russia and the Soviet Union*, 101.

¹⁷² Graham, *Science in Russia and the Soviet Union*, 101.

¹⁷³ For more information about dialectical materialism, see Graham, *Science in Russia and the Soviet Union*, 99–120; Helena Sheehan, *Marxism and the Philosophy of Science: A Critical History* (London: Verso, 2017 [1985]), 21–66. Also see Jongsik

然而，“正确”还不够。中兽医现在需要改革为一门“辩证唯物主义”科学。辩证唯物主义是一种唯物主义形式，它认为“所有自然现象都可以用物质和能量来解释”。¹⁶⁹ 因此，它驳斥了假设形而上学原理来解释自然世界的唯心主义。然而，它也不同于“机械唯物主义”，“一种认为自然界中所有现象，包括人类行为，最终都可以用最简单的物质相互作用来解释的信念”。¹⁷⁰ 相反，辩证唯物主义者认为自然界中存在“‘存在层次’，例如物理、生物和社会”，并且下层的一系列量变可以辩证地导致上层的质变。¹⁷¹ 从这个意义上说，物理最基本层面的物质和能量都是自然世界（包括其生物和社会层面）驱动力的组成部分。同时，由于层次之间的质的飞跃，“社会规律不能简化为生物规律，生物规律不能简化为物理化学规律”。¹⁷² 总之，辩证唯物主义是一种介于机械唯物主义和形而上学唯心主义极端之间的科学哲学。在苏联，它被列宁、托洛茨基、布哈林和斯大林批准为自然科学和社会科学的正确布尔什维克方式。¹⁷³

of the ways in which technologies are embedded in culture.” Sigrid Schmalzer, “Layer upon Layer: Mao-Era History and the Construction of China’s Agricultural Heritage,” *East Asian Science, Technology and Society* 13.3 (September 2019): 420.

¹⁶⁹ Loren R. Graham, *Science in Russia and the Soviet Union: A Short History* (Cambridge: Cambridge University Press, 1990), 100.

¹⁷⁰ Graham, *Science in Russia and the Soviet Union*, 100–101.

¹⁷¹ 格雷厄姆，《俄罗斯和苏联的科学》，101页。

¹⁷² 格雷厄姆，《俄罗斯和苏联的科学》，101。

¹⁷³ 有关辩证唯物主义的更多信息，请参阅 Graham, *Science in Russia and the Soviet Union*, 99–120; Helena Sheehan, *Marxism and the Philosophy of Science: A Critical History* (London: Verso, 2017 [1985]), 21–66。另请参阅 Jongsik

Any scientific endeavors in Maoist China, too, needed to be labeled as dialectical material.

As one of leading veterinary cadres in the CCP, Yu Chuan fluently explained the history of

biological and medical sciences with a focus on the binary between materialism and idealism. Yu

insisted that Charles Darwin and Robert Virchow laid the foundation for scientific materialism in

Western science, refuting the hitherto prevalent “metaphysical” views such as vitalism and

humoralism. Yet, it was not until Friedrich Engels’ formulation of dialectical materialism that

Western science matured into scientific materialism.¹⁷⁴

Yu Chuan saw the same tension between materialism and idealism in the history of

biological and medical sciences in East Asia. The Yin-Yang theory and the Five Phases theory, he

asserted, were “primitive materialist,” whereas “shamanistic medical concepts” were idealist. Yu

continued that since diverse forms of folk veterinary medicine in China had developed under the

influences of both primitive materialism and idealism for thousands of years, they were not yet

thoroughly materialist. Therefore, Yu promoted the idea that folk veterinarians as well as veterinary

scientists and cadres should reform the traditional veterinary medicine by eliminating idealist

elements and enhancing primitive materialist components by using modern scientific methods. This

was how TCVM could become a full-fledged materialist science.¹⁷⁵

Christian Yi, “Dialectical Materialism Serves Voluntarist Productivism: The Epistemic Foundation of Lysenkoism in Socialist China and North Vietnam,” *Journal of the History of Biology* 54 (2021): 517–519.

¹⁷⁴ Yu Chuan 于船, “binglixue zhongde xibao bingli xueshuo” 病理学中的细胞病理学论 (1963), in *Yu Chuan wenji*, 141–144. Also, On the influence of Engels on the scientific community in China, see Sigrid Schmalzer, *The People’s Peking Man: Popular Science and Human Identity in Twentieth-century China* (Chicago: University of Chicago Press, 2008), 55–85.

¹⁷⁵ Yu Chuan, Wu Xuecong, and Jin Zhongye, “Zongjie minjian shouyi jingyan wei fayang zuguo shouyixue yichan de zhongyao huanjie”; Yu Chuan 于船, “Zhongxi shouyi jibing guandian de fazhan gaikuang” 中西兽医疾病观点的发展概况 (1961), in *Yu Chuan wenji*, 145; Yu Chuan 于船, “Zhongshouyixue de xueshu sixiang” 中兽医学的学术思想 (1959), in *Yu Chuan wenji*, 149–163.

毛泽东时代的中国，任何科学探索也都需要被贴上辩证唯物主义的标签。作为中共兽医战线的领导干部之一，喻川流利地阐述了生物医学史，重点关注唯物主义和唯心主义之间的二元对立。喻川坚持认为，查尔斯·达尔文和罗伯特·魏尔啸为西方科学中的科学唯物主义奠定了基础，驳斥了迄今为止流行的“形而上学”观点，如生机论和体液论。然而，直到弗里德里希·恩格斯提出辩证唯物主义，西方科学才成熟为科学唯物主义。¹⁷⁴

喻川认为东亚生物医学史中也存在唯物主义和唯心主义之间的相同张力。他断言，阴阳学说和五行学说是“原始唯物主义”，而“萨满教医学概念”则是唯心主义。喻川接着说，由于中国各种形式的民间兽医学在原始唯物主义和唯心主义的影响下发展了数千年，它们尚未完全唯物主义化。因此，喻川提倡民间兽医以及兽医科学家和干部应通过消除唯心主义元素，并利用现代科学方法增强原始唯物主义成分来改革传统兽医学。这就是中兽医如何才能成为一门成熟的唯物主义科学。¹⁷⁵

Christian Yi, “Dialectical Materialism Serves Voluntarist Productivism: The Epistemic Foundation of Lysenkoism in Socialist China and North Vietnam” , *Journal of the History of Biology* 54 (2021): 517–519。

¹⁷⁴ 于船, “病理学中的细胞病理学论” (1963), 载于于船文集, 第141-144页。另见, 关于恩格斯对中国科学界的影响, Sigrid Schmalzer, 《人民的北京人: 20世纪中国的科普与人类认同》(芝加哥: 芝加哥大学出版社, 2008), 第55-85页。

¹⁷⁵ 于船、吴学聪、金中业, “总结民间兽医经验为发扬祖国兽医学遗产的重要环节”; 于船, “中西兽医疾病观点的发展概况” (1961), 载于于船文集, 第145页; 于船, “中兽医学的学术思想” (1959), 载于于船文集, 第149-163页。

In accordance with such directionality, master veterinarians, veterinary scientists, and cadres should take the materialist “essence” and expel the idealist “chaff and dust” from the rich pool of traditional knowledge and practices.¹⁷⁶ Then, they instilled “useful parts of Western veterinary medicine” into Chinese veterinary medicine, thereby lifting the latter “up to the level of modern science.”¹⁷⁷ New scientific discoveries inspired by folk veterinary medicine would then “enrich Western veterinary medicine.” Ultimately, the two branches of veterinary medicine should coalesce into a single integrated “new veterinary school” of China to “better serve the people.”¹⁷⁸ The CCP fully supported this project, calling it the “integration of Chinese and Western veterinary medicines.”

Taking all these threads into consideration, the final phase of the authorization of folk veterinary healing was to set up and carry out experiments to verify the clinical efficacy of treatments in question. In fact, a similar approach was taken for scientizing Chinese medicine in the Republican period. However, Nationalist scientists and Western medicine practitioners called for repealing Chinese medical theories and only preserving herbal therapies. Without preserving traditional pharmacological theories, they conducted experiments with medicinal plants or traditional drugs on ill animals. If a treatment worked in clinical trials, the experimenters identified the active component

¹⁷⁶ Jin Zhongye 金重冶 and Yu Chuan 于船, “Zhide zhongshi de liangben zhongshouyi zhuzuo” 值得重视的两本中兽医著作 (1959), in *Yu Chuan wenji*, 514–515.

¹⁷⁷ Jin Zhongye and Yu Chuan, “Zhide zhongshi de liangben zhongshouyi zhuzuo”; Jiang Cisheng, “Xuexi zuguo shouyi xueshu de chubu tihui he xinde”; Jiang Ciseng 蒋次升, “Zhongshouyi tuan jie hezuo zhengli fayang zuguo shouyixue yichan wei xumu shengchan fuwu” 中兽医团结合作整理发扬祖国兽医学遗产为畜牧生产服务 (1959), GPA, 222-001-0223-0005.

¹⁷⁸ Jiang Cisheng, “Xuexi zuguo shouyi xueshu de chubu tihui he xinde”; Jiang Cisheng, “Zhongshouyi tuan jie hezuo zhengli fayang zuguo shouyixue yichan wei xumu shengchan fuwu.”

根据这一方向性，兽医大师、兽医学家和干部应从丰富的传统知识和实践中汲取唯物主义的“精髓”，剔除唯心主义的“糟粕和尘埃”。¹⁷⁶ 然后，他们将“西方兽医学的有用部分”注入中国兽医学，从而将后者“提升到现代科学的水平”。¹⁷⁷ 受民间兽医学启发的新科学发现将“丰富西方兽医学”。最终，兽医学的两个分支应合并为中国单一的综合“新兽医学院”，以“更好地为人民服务”。¹⁷⁸ 中共全力支持这一项目，称之为“中西兽医结合”。

综合考虑所有这些线索，民间兽医治疗授权的最后阶段是建立并进行实验，以验证相关治疗的临床疗效。事实上，民国时期中医药的科学化也采取了类似的方法。然而，国民党科学家和西医从业者呼吁废除中医理论，只保留草药疗法。在不保留传统药理学理论的情况下，他们对患病动物进行了药用植物或传统药物的实验。如果某种治疗在临床试验中有效，实验者就会确定其活性成分

¹⁷⁶ 金重冶、于船：《值得重视的两本中兽医著作》（1959），载《于船文集》，第514—515页。

¹⁷⁷ Jin Zhongye and Yu Chuan, “Zhide zhongshi de liangben zhongshouyi zhuzuo”; Jiang Cisheng, “Xuexi zuguo shouyi xueshu de chubu tihui he xinde”; Jiang Ciseng 蒋次升, “Zhongshouyi tuan jie hezuo zhengli fayang zuguo shouyixue yichan wei xumu shengchan fuwu” 中兽医团结合作整理发扬祖国兽医学遗产为畜牧生产服务 (1959), GPA, 222-001-0223-0005.

¹⁷⁸ 蒋次生：《学习祖国兽医学学术的初步体会和心得》；蒋次生：《中兽医团结合作整理发扬祖国兽医学遗产为畜牧生产服务》。

and capitalize upon it to make a mass-producible pharmaceutical commodity.¹⁷⁹ Gao, Cui, Jiang, and Yu, under Mao’s flag, distinguished themselves from their Nationalist predecessors by recognizing the TCVM’s defining principle, the “pattern differentiation and treatment determination,” in the name of materialist science and incorporating it into their experimental research design.

In their joint experiment on the treatment of porcine asthma, Gao Guojing and Jiang Cisheng collected pigs with asthma and classified them by “differentiated pattern.” Each group was subdivided again into a control subgroup and three treatment subgroups. Respectively for these three groups, Gao and Jiang prescribed “determined treatments” of indigenous drug and acupuncture; a Western antibiotic (oxytetracycline); and a combination of Chinese and Western therapeutic methods. To judge the curative efficacy, Gao and Jiang used an X-ray machine to visualize the change in the lung lesions.¹⁸⁰

This research design became a model for other folk veterinarians and veterinary scientists. Folk veterinarians from the northeast, for example, conducted an experiment on the remedial effect of acupuncture for equine indigestion, following Gao and Jiang. They treated with acupuncture 94 horses which exhibited symptoms that included loss of appetite, the decrease of intestinal peristalsis, and diarrhea. Within two days of the treatment, 91 horses stopped showing symptoms.¹⁸¹ In another

¹⁷⁹ Lei, *Neither Donkey Nor Horse*, 193–221; Zhu Fei 朱绯 and Zhu Guannan 朱冠楠, “Feizhi zhongyi an dui zhongshouyi fazhan de yingxiang yu shouyi guoyao zhiliao yanjiusuo de chuangan” 废止中医案对中兽医发展的影响与兽医国药治疗研究所的创办, *Zhongguo nongshi* 2017.2 (March 2017): 43–50.

¹⁸⁰ Gao Guojing 高国景, Jiang Cisheng 蒋次升, et al., “Guanyu zhu qichuanbing zhenliao he fangzhi fangfa de shiyan baogao” 关于猪气喘病诊疗和防治的实验报告, in *Yanjiu zhiliao huibian* 2, ed. Zhongguo nongye kexueyuan zhongshouyi yanjiusuo, 109–113.

¹⁸¹ Yang Zhenzhong 杨进中 and Zhang Qi 张棋, “Zhenjiu zhiliao maluo qianpang shanshang, chanwantong, pixu weiruo, shangshui qiwo, niu duzang wu zhong jibing chubu guancha baogao” 针灸治疗马骡前膀闪伤, 缠腕痛, 脾虚胃弱, 伤水起卧, 牛肚张五种疾病初步观察报告, in *Yanjiu zhiliao huibian* 2, ed. Zhongguo nongye kexueyuan zhongshouyi yanjiusuo, 71.

并利用它来生产可大规模生产的医药商品。¹⁷⁹ 高、崔、蒋、余在毛泽东的旗帜下，通过以唯物主义科学的名义，认识到中医兽医的定义原则——“辨证论治”，并将其纳入他们的实验研究设计中，从而与他们的国民党前辈区分开来。

在高国景和蒋次生的猪哮喘治疗联合实验中，他们收集了患有哮喘的猪，并根据“辨证分型”进行分类。每个组又细分为一个对照亚组和三个治疗亚组。高和蒋分别给这三个组开出了“确定治疗”的本土药物和针灸；一种西药抗生素（土霉素）；以及中西医结合的治疗方法。为了判断疗效，高和蒋使用X光机观察肺部病变的变化。¹⁸⁰

这种研究设计成为其他民间兽医和兽医学家的典范。例如，东北的民间兽医效仿高和蒋，进行了一项针灸治疗马匹消化不良的补救效果实验。他们用针灸治疗了94匹出现食欲不振、肠蠕动减少和腹泻等症状的马。治疗两天内，91匹马停止出现症状。¹⁸¹ 在另一项研究中，

¹⁷⁹ Lei, *Neither Donkey Nor Horse*, 193–221; 朱绯和朱冠楠, “废止中医案对中兽医发展的影响与兽医国药治疗研究所的创办”, 中国农史 2017.2 (2017年3月): 43–50.

¹⁸⁰ 高国景、蒋次升等, “关于猪气喘病诊疗和防治方法的实验报告”, 载《研究资料汇编》2, 中国农业科学院中兽医研究所编, 109–113页。

¹⁸¹ 杨进中和张棋, “针灸治疗马骡前膀闪伤, 缠腕痛, 脾虚胃弱, 伤水起卧, 牛肚张五种疾病初步观察报告”, 载《研究资料汇编》2, 中国农业科学院中兽医研究所编, 第71页。

experiment to verify the efficacy of a traditional decoction for sheep constipation, experiments measured the sheep’s pattern of rumen motility or the regular movement of the ruminants’ first compartment of the stomach before and after taking the decoction. By counting the numbers of rumen contraction, they concluded that the decoction indeed facilitated rumen motility and therefore stimulated digestion.¹⁸²

Like any scientific experiment, these experiments on folk therapeutics sometimes failed to prove given hypotheses. But as meaningful data gradually accumulated, veterinary elites like Gao, Cui, and Jiang could sort out the healing methods that had proven effective. It was particular experimental research design featuring the methodology of “pattern differentiation and treatment determination,” herbal therapies and acupuncture, Western drugs, and high technologies such as X-ray that led the veterinary experimenters to believe that Chinese veterinary medicine could be not only scientific, but also indeed effective. This project of transforming folk veterinary medicine into TCVM was a patchwork of translating, bridging, rearranging, and streamlining. The process inevitably retained some uncertainty and arbitrary selection.¹⁸³ Nevertheless, as results of the well-designed experiments accumulated, the more scientifically authorized the TCVM therapeutics became.

¹⁸² Liang Zhaonian 梁兆年, Yu Shunxiang 余顺祥, et al., “Zhongshouyi zhiliao luoma jiezheng youxiao fangji: Jiuqu chengqi tang, jiuqu qijieyin de yaoli zuoyong chubu yanjiu” 中兽医治疗骡马结症有效方剂: 酒曲承气汤, 酒曲七结饮的药理作用初步研究, in *Zhongshouyi kexue jishu ziliao xuanji*, ed. Zhongguo nongye kexueyuan zhongshouyi yanjiusuo, 125.

¹⁸³ Linda Barnes similarly admits that pre-1850 Westerners who tried to tease out Chinese medicine “introduced distortions and false equations” and “selectively appropriated pieces of Chinese practice, and then rewrote them.” To her, then, the useful question is: “Why these particular misunderstandings?” Linda L. Barnes, *Needles, Herbs, Gods, and Ghosts: China, Healing, and the West to 1848* (Cambridge, Mass.: Harvard University Press, 2007), 5.

为了验证一种治疗绵羊便秘的传统汤剂的疗效，实验测量了绵羊在服用汤剂前后瘤胃蠕动模式，即反刍动物胃的第一部分有规律的运动。通过计算瘤胃收缩的次数，他们得出结论，该汤剂确实促进了瘤胃蠕动，从而刺激了消化。¹⁸²

与任何科学实验一样，这些关于民间疗法的实验有时未能证明给定的假设。但随着有意义的数据逐渐积累，高、崔、蒋等兽医精英得以筛选出已被证明有效的治疗方法。正是这种以“辨证论治”方法论、草药疗法和针灸、西药以及X射线等高科技为特色的特殊实验研究设计，使兽医实验者相信中国兽医学不仅是科学的，而且确实是有效的。将民间兽医学转化为中兽医学的过程是一个翻译、桥接、重组和精简的拼凑过程。这个过程不可避免地保留了一些不确定性和任意选择。¹⁸³然而，随着精心设计的实验结果的积累，中兽医治疗方法变得更具科学权威性。

¹⁸² 梁兆年、余顺祥等，“中兽医治疗骡马结症有效方剂：酒曲承气汤、酒曲七结饮的药理作用初步研究”，载中国农业科学院中兽医研究所编：《中兽医科学技术资料选集》，第125页。

¹⁸³ 琳达·巴恩斯（Linda Barnes）也承认，1850年前试图梳理中医的西方人“引入了扭曲和错误的等式”，并“有选择地挪用了中医的某些部分，然后对其进行了改写”。对她来说，有用的问题是：“为什么会有这些特殊的误解？” Linda L. Barnes, *Needles, Herbs, Gods, and Ghosts: China, Healing, and the West to 1848* (Cambridge, Mass.: Harvard University Press, 2007), 5.

The Codification of TCVM, 1956–1962

While the construction of TCVM was in progress, folk masters, Western scientists, and veterinary cadres were keen on building a prestigious research institute dedicated exclusively to TCVM. Their efforts bore fruit in June 1958 when the first national conference for the research on TCVM was held in Lanzhou, Gansu Province. During the conference, the veterinary authorities decided that the Chinese Academy of Agricultural Sciences (CAAS) would expand and reorganize its northwestern branch in Lanzhou into the Institute of Traditional Chinese Veterinary Medicine (ICTVM) by July 1958 (Figure 6). While Chen Yushan, a provincial leader who oversaw agriculture, took the highest position, Gao Guojing, Cui Diseng, and Jiang Cisheng were nominated as co-vice directors of the ITCVM.¹⁸⁴ Jiang praised the ITCVM as “the first ever national-level research institute for Chinese veterinary medicine in Chinese history” in which “Chinese veterinary medicine practitioners participate” directly.¹⁸⁵

¹⁸⁴ Zhongguo nongye kexueyuan Lanzhou xumu yu shouyao yanjiusuo suozhi bianzuan weiyuanhui 中国农业科学院兰州畜牧与兽药研究所所志编纂委员会, ed., *Zhongguo nongye kexueyuan Lanzhou xumu yu shouyao yanjiusuo suozhi 1958–2008* 中国农业科学院兰州畜牧与兽药研究所所志 1958–2008 (Beijing: Zhongguo nongye kexue jishu chubanshe, 2008), 4.

¹⁸⁵ Jiang Cisheng, “Zhongshouyi tuanjie hezuo zhengli fayang zuguo shouyixue yichan wei xumu shengchan fuwu.”

TCVM 的法典化, 1956–1962 年

在TCVM的建设过程中，民间大师、西方科学家和兽医干部都热衷于建立一个专门从事TCVM的著名研究所。1958年6月，在甘肃兰州召开了第一次全国TCVM研究会议，他们的努力取得了成果。会议期间，兽医主管部门决定，中国农业科学院（CAAS）将于1958年7月前将其在兰州的西北分院扩建并改组为中国传统兽医学研究所（ICTVM）（图6）。虽然负责农业的省级领导陈玉山担任最高职位，但高国经、崔迪生和蒋次生被提名为ICTVM的共同副所长。¹⁸⁴蒋赞扬ICTVM是“中国历史上第一个国家级中兽医研究所”，其中“中兽医从业者”直接参与。¹⁸⁵

¹⁸⁴ Zhongguo nongye kexueyuan Lanzhou xumu yu shouyao yanjiusuo suozhi bianzuan weiyuanhui 中国农业科学院兰州畜牧与兽药研究所所志编纂委员会, ed., *Zhongguo nongye kexueyuan Lanzhou xumu yu shouyao yanjiusuo suozhi 1958–2008* 中国农业科学院兰州畜牧与兽药研究所所志 1958–2008 (Beijing: Zhongguo nongye kexue jishu chubanshe, 2008), 4.

¹⁸⁵ Jiang Cisheng, “Zhongshouyi tuanjie hezuo zhengli fayang zuguo shouyixue yichan wei xumu shengchan fuwu.”



Figure 6. The Inaugural General Meeting of the ITCVM of CAAS on July 1, 1958 featuring Jiang Cisheng (the sixth from the left in the first row), Cui Diseng (the eighth), Yu Chuan (the eleventh), and Gao Guojing (the thirteenth). Source: Zhongguo nongye kexueyuan Lanzhou xumu yu shouyao yanjiusuo suozhi bianzuan weiyuanhui 中国农业科学院兰州畜牧与兽药研究所志编纂委员会, ed., *Zhongguo nongye kexueyuan Lanzhou xumu yu shouyao yanjiusuo suozhi*, 1958–2008 中国农业科学院兰州畜牧与兽药研究所志, 1958–2008 (Beijing: Zhongguo nongye kexue jishu chubanshe, 2008), iii.

As soon as it came into being, the ITCVM, as the institutional foothold for the veterinary elites, spearheaded a series of publication projects in an effort to codify TCVM. Another national research conference on TCVM took place at the ITCVM in January 1959. The participants reached an agreement that the ITCVM should compile and publish a set of orthodox TCVM monographs.¹⁸⁶ Among them were *Traditional Chinese Veterinary Acupuncture and Moxibustion* (1959),¹⁸⁷ *Traditional*

¹⁸⁶ Jiang Cisheng, “Zhongshouyi tuanjie hezuo zhengli fayang zuguo shouyixue yichan wei xumu shengchan fuwu.”

¹⁸⁷ Zhongguo nongye kexueyuan zhongshouyi yanjiusuo 中国农业科学院中兽医研究所, *Zhongshouyi Zhenjiuxue* 中兽医针灸学 (Beijing: Nongye chubanshe, 1959).



图 6. 中国农业科学院中兽医研究所成立大会，1958 年 7 月 1 日，出席者包括蒋次生（前排左六）、崔涤生（左八）、余传（左十一）和高国景（左十三）。资料来源：中国农业科学院兰州畜牧与兽药研究所志编纂委员会编，《中国农业科学院兰州畜牧与兽药研究所志，1958–2008》（北京：中国农业科学技术出版社，2008），iii。

中兽医研究所一经成立，便作为兽医精英的机构立足点，率先开展了一系列出版项目，以期将中兽医规范化。1959年1月，中兽医研究所召开了另一次全国中兽医研究会议。与会者达成一致，认为中兽医研究所应编写并出版一套正统的中兽医专著。¹⁸⁶其中包括《中兽医针灸学》(1959)、¹⁸⁷《中兽医》

¹⁸⁶ 蒋次生，“中兽医团结合作整理发扬祖国兽医学遗产为畜牧生产服务。”

¹⁸⁷ 中国农业科学院中兽医研究所，中兽医针灸学 (北京：农业出版社，1959)。

Chinese Veterinary Pharmacy (1959),¹⁸⁸ *Traditional Chinese Veterinary Diagnostics* (1962),¹⁸⁹ and *Traditional Chinese Veterinary Therapeutics* (1962).¹⁹⁰ In June 1959, fourteen of twenty co-authors and reviewers, including Jiang Cisheng, Cui Diseng, and Gao Guojing, gathered in Lanzhou, Gansu province to compile *TCV Acupuncture and Moxibustion*. Like Jiang, Cui, and Gao, the other co-authors were Western-trained veterinary scientists, veterinary cadres, or seasoned folk veterinarians who worked at veterinary institutes in Jiangxi, Jilin, Shanxi, and Ningxia.

The fourteen veterinary elites spent a month “collectively reviewing” the preliminary materials which were collected and compiled by local veterinarians and “collectively writing” the manuscript of *TCV Acupuncture and Moxibustion*. The monograph covers the characteristics and brief history of acupuncture and moxibustion; basic meridian theories of TCM; styles, techniques, and precautions of using acupuncture and moxibustion for animals; animal-specific acupunctural devices (Figure 7); acupunctural points of horses, cattle, pigs, mules, and donkeys; and detailed clinical records of acupunctural therapeutics. The co-authors hoped that this book could contribute to the scientific recognition of the “TCVM theoretical system as basis and folk veterinarians’ experiences as backbone.”¹⁹¹

¹⁸⁸ Zhongguo nongye kexueyuan zhongshouyi yanjiusuo 中国农业科学院中兽医研究所, *Shouyi Zhongyaoxue* 兽医中药学 (Beijing: Nongye chubanse, 1959).

¹⁸⁹ Zhongguo nongye kexueyuan zhongshouyi yanjiusuo 中国农业科学院中兽医研究所, *Zhongshouyi Zbenduanxue* 中兽医诊断学 (Beijing: Nongye chubanse, 1962).

¹⁹⁰ Zhongguo nongye kexueyuan zhongshouyi yanjiusuo 中国农业科学院中兽医研究所, *Zhongshouyi zhibiliaoxue* 中兽医治疗学 (Beijing: Nongye chubanse, 1962).

¹⁹¹ The co-authors candidly admitted that they could not yet fully include the “modern medicine-based theoretical research on acupuncture and moxibustion and the most recent achievements about acupuncture and moxibustion facilitated by scientific instruments.” Zhongguo nongye kexueyuan zhongshouyi yanjiusuo, *Zhongshouyi Zhenjiuxue*, 4.

中兽医药（1959），¹⁸⁸ 中兽医诊断学（1962），¹⁸⁹ 和 中兽医治疗学（1962）。¹⁹⁰ 1959年6月，包括蒋次生、崔涤生和高国经在内的20位合著者和审稿人中的14位齐聚甘肃兰州，共同编纂《中兽医针灸学》。与蒋、崔、高一样，其他合著者都是在江西、吉林、山西和宁夏的兽医研究所工作的西医兽医科学家、兽医干部或经验丰富的民间兽医。

这十四位兽医精英花了一个月的时间“集体审阅”当地兽医收集和整理的初步材料，并“集体撰写”了《中兽医针灸学》的手稿。该专著涵盖了针灸的特点和简史；中医基本经络理论；动物针灸的风格、技术和注意事项；动物专用针灸器械（图7）；马、牛、猪、骡、驴的针灸穴位；以及针灸治疗的详细临床记录。合著者希望这本书能有助于科学界认可“以中兽医理论体系为基础，以民间兽医经验为骨干”的观点。

¹⁸⁸ 中国农业科学院中兽医研究所，《兽医中药学》（北京：农业出版社，1959）。¹⁸⁹ 中国农业科学院中兽医研究所，《中兽医诊断学》（北京：农业出版社，1962）。¹⁹⁰ 中国农业科学院中兽医研究所，《中兽医治疗学》（北京：农业出版社，1962）。

¹⁹¹ 合著者坦率地承认，他们尚不能完全纳入“基于现代医学的针灸理论研究以及科学仪器促进的最新针灸成果”。中国农业科学院中兽医研究所，《中兽医针灸学》，第4页。

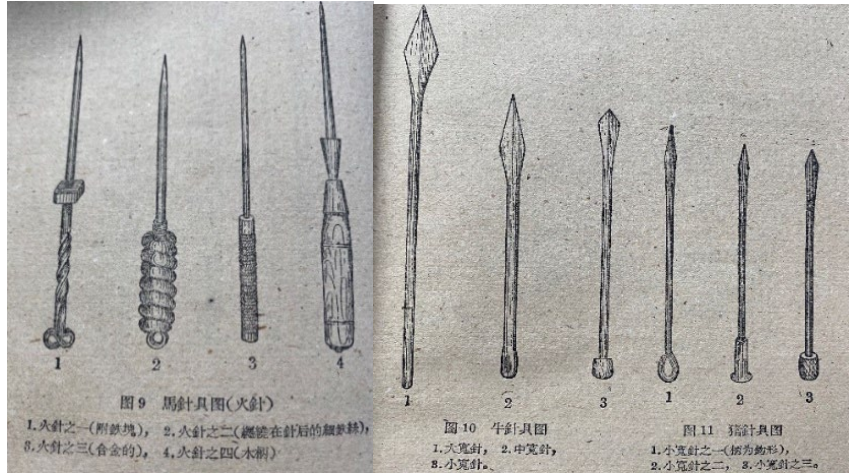


Figure 7. Acupunctural needles for horses, cattle, and pigs. Source: Zhongguo nongye kexueyuan zhongshouyi yanjiusuo 中国农业科学院中兽医研究所, *Zhongsouyi Zhenjiuxue* 中兽医针灸学 (Beijing: Nongye chubanshe, 1959), 44 and 46.

Jiang Cisheng, Cui Diseng, and eighteen other veterinary specialists from Jilin, Jiangxi, Gansu, Guangxi, and Sichuan completed the second monograph, *Traditional Chinese Veterinary Pharmacy*. In July 1959, they reviewed early drafts compiled by local veterinarians in Jilin and Jiangxi.¹⁹² The monograph includes the names, places of origin, and medicinal parts of 346 basic medicinal herbs; how to decoct; doses and possible combinations. *TCV Pharmacy* became the definitive pharmacopeia of the Maoist era.¹⁹³

The third monograph, *Traditional Chinese Veterinary Diagnostics*, was completed in June 1960 and published in 1962. Cui, Gao, and Jiang participated in the compilation. *TCV Diagnostics* reaffirmed that TCVM would serve the CCP’s policy of the “full-scale development of livestock industry with a focus on pig raising.” It reinforced the theoretical foundation of TCVM such as the

¹⁹² Zhongguo nongye kexueyuan zhongshouyi yanjiusuo, *Shouyi Zhongyaoxue*, 5–6.

¹⁹³ Jiang Cisheng and Yang Ruo, “Zuzhi bianxie siben zhongshouyi shuji de shouhuo he tihui,” 189.

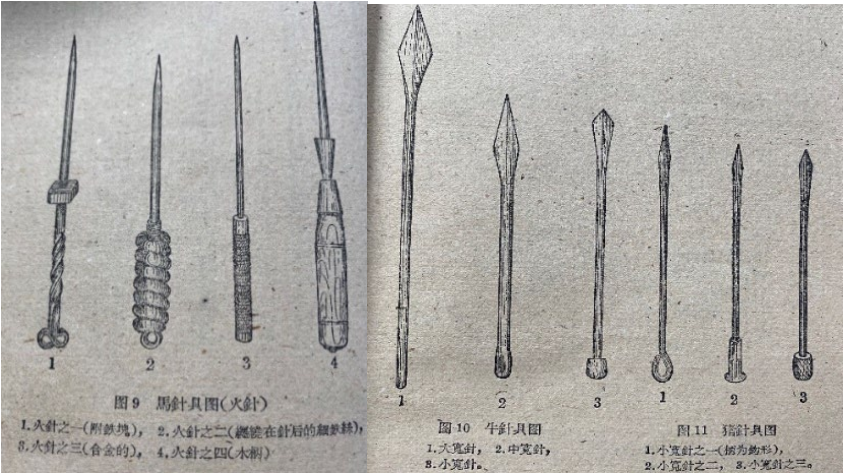


图 7。马、牛、猪针灸针。来源：中国农业科学院中兽医研究所，《中兽医针灸学》（北京：农业出版社，1959），第44页和第46页。

吉林、江西、甘肃、广西和四川的江次生、崔迪生及其他十八位兽医专家完成了第二部专著，即《中兽医方剂学》。1959年7月，他们审阅了吉林和江西当地兽医编写的早期草稿。¹⁹² 该专著包括346种基本药材的名称、产地和药用部位；如何煎煮；剂量和可能的组合。《中兽医方剂学》成为毛泽东时代权威的药典。¹⁹³

第三部专著《中兽医诊断学》于1960年6月完成，并于1962年出版。崔、高、江参与了编纂。《中兽医诊断学》重申中兽医将服务于中国共产党的“以养猪为重点，全面发展畜牧业”的政策。它强化了中兽医的理论基础，例如

¹⁹² 中国农业科学院中兽医研究所，《兽医中药学》，5–6页。

¹⁹³ 蒋次生、杨若，“组织编写四本中兽医书籍的收获和体会”，189页。

Yin and Yang; the Five Phases; the physiology based on *Qi*, Blood (血), Essence (精), Mind (神), the *Zang-fu* organs (脏腑), and the meridian channels;¹⁹⁴ the pathology of Eight Patterns (八证); and the diagnostics of Four Diagnoses (四诊). Although these theories are mostly derived from TCM, TCVM theories also have unique elements especially as to diagnostics that are differentiated from human medicine. Since animal patients cannot express their symptoms and pain in words, animal doctors should be sophisticated in how to observe, listen to, and touch animal bodies. *TCV Diagnostics*, for example, systematizes theories on how to read animals’ shapes and postures (形); the color of tongue and teeth rather than skin, which is covered by fur; and sounds of whining, growling, and breathing.¹⁹⁵

The last monograph of the set, *Traditional Chinese Veterinary Therapeutics*, was completed by twenty-nine co-authors including Cui, Gao, and Jiang in December 1961 and published in 1962. *TCV Therapeutics* summarizes the clinical implications of TCVM theories presented in the previous three monographs. Centering on the “Pattern Differentiation and Treatment Determination,” it formulized the sets of etiology, symptoms, diagnosis, treatments, annotations of each treatment, nursing, and prevention measures for the three hundred most common animal diseases.¹⁹⁶

These four books, according to Jiang Cisheng, were the results of the “first-ever attempt in Chinese history that all the [national] capacity is mobilized under the auspices of the central

¹⁹⁴ On the physiology of Chinese medicine, also see Ted J. Kaptchuk, *The Web That Has No Weaver: Understanding Chinese Medicine* (London: Rider Books, 1983).

¹⁹⁵ Zhongguo nongye kexueyuan zhongshouyi yanjiusuo, *Zhongshouyi Zhenduanxue*, especially Part 1 (上篇).

¹⁹⁶ Zhongguo nongye kexueyuan zhongshouyi yanjiusuo, *Zhongshouyi zhihiao xue*.

阴阳；五行；基于气、血（血）、精（精）、神（神）、脏腑器官（脏腑）和经络的生理学；¹⁹⁴八证（八证）的病理学；以及四诊（四诊）的诊断学。尽管这些理论大多源自中医，但中兽医理论也有其独特的元素，尤其是在诊断方面，这与人类医学有所不同。由于动物患者无法用语言表达其症状和疼痛，因此兽医应精通如何观察、倾听和触摸动物身体。中兽医诊断学，例如，系统化了关于如何解读动物的形状和姿势（形）；舌头和牙齿的颜色而非被毛覆盖的皮肤；以及呜咽、咆哮和呼吸的声音的理论。¹⁹⁵

该系列丛书的最后一本专著，即《中兽医治疗学》，由包括崔、高、蒋在内的29位合著者于1961年12月完成，并于1962年出版。《中兽医治疗学》总结了前三本专著中提出的中兽医理论的临床意义。该书以“辨证论治”为中心，针对三百种最常见的动物疾病，制定了病因、症状、诊断、治疗、每种治疗的注释、护理和预防措施。¹⁹⁶

据蒋次生所说，这四本书是“中国历史上首次在中央主持下调动全国力量”的成果

¹⁹⁴ 关于中医生理学，另见 Ted J. Kaptchuk, *The Web That Has No Weaver: Understanding Chinese Medicine* (伦敦: Rider Books, 1983)。

¹⁹⁵ 中国农业科学院中兽医研究所，《中兽医诊断学》，尤其是第一部分（上篇）。¹⁹⁶ 中国农业科学院中兽医研究所，《中兽医治疗学》。

government to write a set of monographs on Chinese veterinary medicine.”¹⁹⁷ They laid a strong foundation for codifying and delineating the orthodox understandings of TCVM. While some parts of these publications were simply for archiving fragmentary and unwritten veterinary knowledge that would have otherwise been lost or died out, other parts explicitly reflected the co-authors’ desire to construct a coherent, sophisticated set of new knowledge that was meant to be both scientific and socialist. With the completion of this publication project in 1962, TCVM as state-sanctioned knowledge became established.

Conclusion

The interactions between folk veterinarians and the Communist Party were ambivalent before and after the founding of the new People’s Republic in 1949. On the one hand, the state cadres appreciated the contributions of local animal healers to the early anti-animal diseases campaigns and their efforts to protect livestock. On the other, the modernizing revolutionaries had reservations about the credibility of folk veterinarians’ expertise. As such, over the course of agricultural collectivization in the 1950s, the state organized the animal doctors who had formerly been private practitioners into state-sanctioned veterinary associations and union clinics and asked them to serve the people with a selfless, collectivist mindset. These folk veterinarians were eventually transformed into collective veterinarians who would be in charge of each commune’s or brigade’s veterinary station. This process of collectivizing veterinary personnel was also a process of disclosing countless “secret” remedies and healing experiences that had been exclusively transmitted within family

¹⁹⁷ Jiang Cisheng and Yang Ruo, “Zuzhi bianxie siben zhongshouyi shuji de shouhuo he tihui,” 189.

政府编写一套关于中国兽医的专著。”¹⁹⁷ 他们为编纂和描绘中医兽医的正统理解奠定了坚实的基础。这些出版物中的一些部分仅仅是为了存档那些原本会丢失或消亡的零散的、不成文的兽医知识，而另一些部分则明确反映了合著者构建一套连贯、精妙的新知识的愿望，这些知识旨在既科学又社会主义。随着1962年该出版项目的完成，中医兽医作为国家认可的知识得以确立。

结论

在1949年新中国成立前后，民间兽医与中国共产党之间的互动是矛盾的。一方面，国家干部赞赏地方动物治疗师对早期动物疾病防治运动的贡献以及他们保护牲畜的努力。另一方面，现代化革命者对民间兽医专业知识的可信度持保留态度。因此，在20世纪50年代农业集体化过程中，国家将以前的私人执业动物医生组织成国家认可的兽医协会和联合诊所，并要求他们以无私的集体主义心态为人民服务。这些民间兽医最终转变为集体兽医，负责每个公社或大队的兽医站。这种兽医人员集体化的过程，也是一个揭示无数“秘方”和治疗经验的过程，这些秘方和经验以前只在家族内部传承。

¹⁹⁷ 蒋次生、杨若：《组织编写四本兽医书籍的收获和体会》，第189页。

networks. This communal circulation of knowledge laid the foundation for the project of scientizing folk veterinary medicine.

It was renowned folk veterinary masters like Gao Guojing and Cui Diseng who translated folk veterinary knowledge into the language of modern science with the help from and collaboration with Western-trained veterinary scientists such as Jiang Cisheng and Communist cadres in the veterinary realm like Yu Chuan. These veterinary experts together constructed an orthodox set of new hybrid veterinary science which came to be called TCVM. It was framed in accordance with the Maoist and Marxist philosophy of science, namely theory of practice and dialectical materialism. And at a more technical level, the most salient strategy for constructing TCVM was to come up with a particular experimental research design that incorporated various elements, putting together “pattern differentiation and treatment determination,” acupuncture, herbal medicine, medical chemicals, X-ray devices, serum test, biopsy, autopsy and the like. Despite some uncertainty and arbitrary selection embedded in the process, as experimental data was accumulated, folk veterinarians and other veterinary experts could confirm a series of formulaic healing protocols. The state had only sketchy guidelines regarding this intellectual project. Veterinary elites took the initiative.

The codification of TCVM was complete with the establishment of the Institute of TCVM under the CAAS in 1958 and the publication of four orthodox monographs: *Traditional Chinese Veterinary Acupuncture and Moxibustion* (1959), *Traditional Chinese Veterinary Pharmacy* (1959), *Traditional Chinese Veterinary Diagnostics* (1962), and *Traditional Chinese Veterinary Therapeutics* (1962). By August 1963, the top leaders in Beijing were confident enough in the scientific validity of TCVM and practical values of local veterinarians. As such, they reconfirmed the 1956 policy of “solidarity,

network. This knowledge's public circulation laid the foundation for the project of scientizing folk veterinary medicine.

正是像高国经和崔迪僧这样著名的民间兽医大师，在姜次生等受西方训练的兽医科学家以及余传等兽医领域的共产党员的帮助和合作下，将民间兽医知识转化成了现代科学的语言。这些兽医专家共同构建了一套正统的新型混合兽医科学，即后来的中兽医。它是在毛泽东思想和马克思主义科学哲学，即实践论和辩证唯物主义的指导下形成的。在技术层面，构建中兽医最显著的策略是提出一种特殊的实验研究设计，该设计融合了各种元素，将“辨证论治”、针灸、草药、医用化学品、X射线设备、血清检测、活检、尸检等结合在一起。尽管在此过程中存在一些不确定性和随意选择，但随着实验数据的积累，民间兽医和其他兽医专家能够确认一系列程式化的治疗方案。国家对这项知识项目只有粗略的指导方针。兽医精英们主动承担了这项任务。

1958年，中国农业科学院（CAAS）成立了中兽医研究所，并出版了四部正统专著：《中国传统兽医针灸》（1959年）、《中国传统兽医中药》（1959年）、《中国传统兽医诊断学》（1962年）和《中国传统兽医治疗学》（1962年），中兽医的编纂工作至此完成。到1963年8月，北京的最高领导人对中兽医的科学有效性和地方兽医的实用价值充满信心。因此，他们再次确认了1956年的“团结”政策，

utilization, education, and improvement.”¹⁹⁸ The staunch support of the central government allowed a few veterinary elites to enjoy the high socio-intellectual status in the mid-1960s.

While the new veterinary hybrid science emerged at the initiative of Gao Guojing, Cui Diseng, Jiang Cisheng, Yu Chuan, and their elite colleagues between 1956 and 1963, a much larger number of rank-and-file folk-turned-collective veterinarians were at the forefront of the everyday clinical work. Many of them trained the next generations of local animal doctors by taking their own apprentices or teaching at different forms of governmental short-term veterinary sessions. TCVM as a proletarian science, after all, could not be isolated from veterinarians, peasants, and animals on the ground; it needed to serve local communes, brigades, and teams. The next chapter delves into communal veterinary stations where these ordinary veterinarians served ill animal patients and fellow peasants with TCVM.

¹⁹⁸ Zhou Enlai 周恩来, “Guowuyuan guanyu minjian shouyi gongzuo de jue ding” 国务院关于民间兽医工作的决定 (August 10, 1963), BMA, 145-001-00308-001.

利用、教育和改进。”¹⁹⁸ 中央政府的坚定支持使少数兽医精英在20世纪60年代中期享有较高的社会知识地位。

虽然新的兽医混合科学在1956年至1963年间在高国景、崔迪生、蒋次生、余川及其精英同事的倡议下出现，但更多的普通民间兽医和集体兽医处于日常临床工作的前沿。他们中的许多人通过收徒或在不同形式的政府短期兽医培训班授课，培养了下一代当地动物医生。中医兽医作为一门无产阶级科学，毕竟不能脱离兽医、农民和动物；它需要为当地公社、大队和生产队服务。下一章将深入探讨这些普通兽医在中医兽医站为患病动物和农民提供服务的情况。

¹⁹⁸ 周恩来, “国务院关于民间兽医工作的决定” (1963年8月10日), BMA, 145-001-00308-001.

Chapter Three:
Grassroots Experts

Introduction

Concepts of “meritocracy” and “anti-intellectualism” which emerged in the Anglo-American public sphere in the 1950s and 60s captured one of the central questions of the preceding and following periods: How should the state and society address the socio-intellectual gaps among citizens? The participation of diverse individuals in public affairs had expanded throughout the twentieth century, especially during and after the wars. A thorough democratization of society, accordingly, seemed to be required to counter the allegedly old-fashioned principles of organizing society such as aristocratism and nepotism. But at the same time, facing more complex and professional issues of society, the public often came to feel pressured to obey the advice of experts without understanding why. How and to what degree public decision-making was to be democratized hence remained controversial.¹⁹⁹

On one side stood a strong belief in elite experts in diverse fields. On the other, a growing number of people started to accept the idea that there is no difference in the intellectual potential and capacity between experts and non-experts. The former found its ways in meritocracy, roughly defined as the idea that social disparities derived from differences in merit are natural and even fair. Some of extreme advocates of the latter led to anti-intellectualism, denying every sense of difference between experts and non-experts and believing that so-called specialists can be entirely replaced by

¹⁹⁹ Michael Young, *The Rise of the Meritocracy, 1870–2033: An Essay on Education and Equality* (London: Thames and Hudson, 1958); Richard Hofstadter, *Anti-Intellectualism in American Life* (New York: Vintage Books, 1963).

第三章：
草根专家

导言

20世纪50年代和60年代在英美公共领域出现的“精英管理”和“反智主义”概念，抓住了前一时期和后一时期的一个核心问题：国家和社会应如何解决公民之间的社会智力差距？在整个20世纪，特别是在战争期间和战后，不同个体参与公共事务的范围不断扩大。因此，社会彻底民主化似乎是必要的，以对抗据称过时的社会组织原则，如贵族主义和裙带关系。但与此同时，面对更复杂和专业的社会问题，公众常常感到有压力，不得不听从专家的建议，却不明白原因。因此，公共决策如何以及在多大程度上实现民主化仍然存在争议。¹⁹⁹

一方面，人们坚信各个领域的精英专家。另一方面，越来越多的人开始接受这样一种观点，即专家和非专家在智力潜能和能力上没有区别。前者体现在精英管理中，大致定义为社会差异源于功绩差异是自然甚至公平的观念。后者的一些极端倡导者导致了反智主义，否认专家和非专家之间存在任何意义上的差异，并认为所谓的专家可以完全被取代

¹⁹⁹ 迈克尔·扬，《精英的崛起，1870-2033：论教育与平等》（伦敦：泰晤士与赫德逊出版社，1958）；理查德·霍夫施塔特，《美国生活中的反智主义》（纽约：复古图书出版社，1963）。

learned amateurs. Since then, there have been multifarious debates on the spectrum between elitism and egalitarianism in Western societies.²⁰⁰

The Western terms “meritocracy” and “anti-intellectualism” did not pass through the Bamboo Curtain in the mid-twentieth century. Contemporaries in Mao-era China, however, grappled with the same question that those concepts raised. Perhaps people in China did so in a more radical way than the Western counterparts as the Chinese Communist Party (CCP) officially opposed any kinds of inequality and inequity and represented the peasantry and workers. As a result, the CCP supported the “mass line” in almost every aspect of social life including natural sciences. Pursuing this ideal of “mass science,” Mao Zedong tended to regard intellectuals, experts, and technocrats as prone to capitalism and elitist arrogance, whereas the masses were praised as the Alpha and the Omega that “true” knowledge originates in and eventually serves. In other words, the capacity of revolutionary laboring people (“red”) was apparently prioritized over the professional role of elites (“expert”) in Maoist China.²⁰¹

Paying attention to this notion of mass science, we may expect that China can be characterized as an anti-scientific, anti-intellectual, and thoroughly egalitarian society. Historians of science in China, however, have argued that Mao and the CCP, like its Nationalist predecessors, accepted science and technology as fundamental means to improve the Chinese nation. If so, Maoist

²⁰⁰ Jacques Rancière, *The Ignorant Schoolmaster: Five Lessons in Intellectual Emancipation*, trans. Kristin Ross (Stanford: Stanford University Press, 1991); Jan-Werner Müller, *What Is Populism?* (Philadelphia: University of Pennsylvania Press, 2016); Susan Jacoby, *The Age of American Unreason in a Culture of Lies* (New York: Vintage Books, 2018 [2008]); Naomi Oreskes, *Why Trust Science?* (Princeton: Princeton University Press, 2019); Michael Sandel, *The Tyranny of Merit: What’s Become of the Common Good?* (New York: Farrar, Straus and Giroux, 2020).

²⁰¹ On Mao’s description of “red” and “expert” and his call for “being both red and expert” (又红又专), see Mao Zedong 毛泽东, “Guanyu nongye wenti” 关于农业问题 (October 9, 1957), *Mao Zedong Wenji* 毛泽东文集 vol. 7 (Beijing: Renmin chubanshe, 1999), 306–311. Also see Sigrid Schmalzer, *Red Revolution, Green Revolution: Scientific Farming in Socialist China* (Chicago: University of Chicago Press, 2016), 34–38.

有学识的业余爱好者。自那时起，西方社会就精英主义和平均主义之间的光谱展开了各种辩论。²⁰⁰

“精英管理”和“反智主义”这两个西方术语在20世纪中期并未传入中国。然而，毛泽东时代的中国人却在努力解决这些概念所提出的相同问题。也许中国人比西方同行做得更彻底，因为中国共产党（CCP）官方反对任何形式的不平等和不公平，并代表农民和工人。因此，中国共产党在社会生活的几乎所有方面，包括自然科学，都支持“群众路线”。为了追求“大众科学”的理想，毛泽东倾向于将知识分子、专家和技术官僚视为容易滋生资本主义和精英傲慢的人，而群众则被赞扬为“真正”知识的起源和最终服务对象。换句话说，在毛泽东时代的中国，革命劳动人民（“红”）的能力显然优先于精英（“专家”）的专业作用。²⁰¹

关注“大众科学”这一概念，我们可能会认为中国是一个反科学、反智、彻底的平等主义社会。然而，中国的科学史学家认为，毛泽东和中国共产党，就像其国民党前身一样，接受科学技术作为改善中华民族的基本手段。如果是这样，毛泽东思想

²⁰⁰ 雅克·朗西埃，《无知的教师：智力解放的五个教训》，克里斯汀·罗斯译（斯坦福：斯坦福大学出版社，1991）；扬·维尔纳·米勒，《什么是民粹主义？》（费城：宾夕法尼亚大学出版社，2016）；苏珊·雅各比，《谎言文化中美国非理性时代》（纽约：Vintage Books，2018 [2008]）；内奥米·奥雷斯克斯，《为何信任科学？》（普林斯顿：普林斯顿大学出版社，2019）；迈克尔·桑德尔，《精英的暴政：共同利益何去何从？》（纽约：Farrar, Straus and Giroux，2020）。

²⁰¹ 关于毛泽东对“又红又专”的描述和号召，参见毛泽东，《关于农业问题》（1957年10月9日），毛泽东文集第7卷（北京：人民出版社，1999），第306-311页。另见Sigrid Schmalzer，《红色革命，绿色革命：社会主义中国的科学农业》（芝加哥：芝加哥大学出版社，2016），第34-38页。

China was not anti-scientific; it just sought a different science.²⁰² Then were the Chinese people in the Maoist era anti-elite and anti-intellectual in the name of revolutionary egalitarianism? This is the central question that this chapter examines by studying local veterinarians in the high Maoist period (1958–1976). The sections to come will show that there is no simple answer to the question. In Maoist China, elitism and meritocracy coexisted with egalitarian fervor and anti-intellectualism.

Stratified Veterinary Education, 1958–1966

As the State Council emphasized the mobilization of folk veterinarians in the grassroots veterinary work in 1956 and the project of constructing Traditional Chinese Veterinary Medicine (TCVM) went smoothly, the central and provincial governments started to systematize diverse veterinary educational programs in the late 1950s. Because there were too few Western veterinary practitioners to cover the vast rural regions, the state paid particular attention to training new generations of veterinarians, on the one hand, and “improving the technical level” of existing animal healers, on the other. Roughly between 1958 and 1966, veterinary education was divided into introductory programs for novices and professional reeducation sessions for those who had already practiced. These educational opportunities were either apprenticeships led by “Chinese veterinary medicine practitioners who has advanced technical expertise” within communes or lower collectives or governmental programs organized by provincial, district-, or county-level authorities.²⁰³

²⁰² Schmalzer, *Red Revolution, Green Revolution*, 22–26; Rui Kunze and Marc Andre Matten, *Knowledge Production in Mao-era China: Learning from the Masses* (Lanham: Lexington Books, 2021), the Introduction and Chapter One.

²⁰³ Guo Haien 国海恩, “Tigao zhongshouyi zhiliang bing peiyang xinsheng liliang” 提高中兽医质量并培养新生力量 (January 1960), Gansu Provincial Archives (hereafter GPA), 116-001-0198-0011.

中国并非反科学；它只是寻求一种不同的科学。²⁰²那么，毛泽东时代的中国人是否以革命平等主义之名反精英、反知识分子？这是本章通过研究高毛泽东时期（1958-1976）的当地兽医来探讨的核心问题。接下来的章节将表明，这个问题没有简单的答案。在毛泽东时代的中国，精英主义和英才教育与平等主义热情和反智主义并存。

分层兽医教育，1958-1966

1956年，国务院强调动员民间兽医参与基层兽医工作，中兽医（TCVM）建设项目进展顺利，中央和省级政府于1950年代后期开始系统化各种兽医教育项目。由于西兽医从业者太少，无法覆盖广阔的农村地区，国家一方面特别重视培养新一代兽医，另一方面则“提高”现有动物治疗师的“技术水平”。大约在1958年至1966年间，兽医教育分为针对新手的入门课程和针对已有实践经验者的专业再教育课程。这些教育机会要么是由公社或下级集体内“具有先进技术专长的中兽医从业者”领导的学徒制，要么是由省、地区或县级政府机构组织的政府项目。²⁰³

²⁰² Schmalzer, *Red Revolution, Green Revolution*, 22–26; Rui Kunze and Marc Andre Matten, *Knowledge Production in Mao-eraChina: Learning from the Masses* (Lanham: Lexington Books, 2021), 导言和第一章。

²⁰³ 郭海恩, “提高中兽医质量并培养新生力量”（1960年1月），甘肃省档案馆（以下简称GPA），116-001-0198-0011。

The early veterinary apprenticeship at the grassroots was modeled after human medical training. Rural youths aged 16 to 35 with “pure political thoughts” and an upper elementary or middle school education were eligible to become apprentices. After three to five years of apprenticeship, apprentices were believed to have enough expertise to be able to practice on their own. Mentors were supposed to teach apprentices clinical knowledge such as how to diagnose and prescribe, the characteristics of local medicinal herbs, basic scientific knowledge, and political ideology. Apprentices could not withdraw from the program they were required to “absolutely follow” their mentors’ tutelage. Nor could they arbitrarily leave the training workplace or engage in “indiscreet behavior.” The mentor-apprentice relationship was strictly hierarchical.²⁰⁴

Jiangxi Province spearheaded more advanced apprenticeship programs in June 1958. Instead of being taught by animal doctors in one’s hometown, some qualified young villagers were invited to county-level veterinary clinics to study veterinary theory and practice. While reading foundational texts such as *Yuanheng Brothers’ Collection for Treating the Horse* (元亨疗马集, c. 1608), *Practical Veterinary Acupuncture*, and *Traditional Chinese Veterinary Pharmacy*, they helped their mentors in the clinical settings and learned diagnostics, animal acupuncture, castration, and herbology. Once they finished the apprenticeship, they returned to their own collectives and worked there as a professional veterinarian. These former apprentices and their home collectives paid an honorarium to mentors and county clinics. These types of training gave grassroots youths a better opportunity than at-home-collectives apprenticeships to learn higher-level veterinary skills and theories.²⁰⁵

²⁰⁴ “Xingan xian zhongshouyi dai tudi zanxing banfa” 新干县中兽医带徒弟暂行办法, *Minjian shouyi tongxun* 民间兽医通讯 3 (January 1958): 5–7.

²⁰⁵ Jiangxi sheng nongyeting 江西省农业厅, “Guanyu duo dai tudi peiyang zhongshouyi jishu rencai de tongzhi” 关于多带徒弟培养中兽医技术人才的通知 (June 16, 1958), *Minjian shouyi tongxun* 民间兽医通讯 6 (July 1958): 7.

基层兽医的早期学徒制是仿照人类医学培训模式。16至35岁、具有“纯洁政治思想”和高小或初中文化程度的农村青年有资格成为学徒。经过三到五年的学徒期，学徒被认为拥有足够的专业知识，可以独立执业。导师应该教授学徒临床知识，如如何诊断和开药、当地草药的特性、基本科学知识和政治思想。学徒不能退出该项目，他们被要求“绝对服从”导师的教导。他们也不能随意离开培训场所或从事“轻率行为”。师徒关系是严格的等级制度。²⁰⁴

1958年6月，江西省率先推行了更先进的学徒计划。一些合格的年轻村民被邀请到县级兽医诊所学习兽医理论和实践，而不是由家乡的兽医教授。他们阅读了诸如《元亨疗马集》(元亨疗马集，约1608年)、《实用兽医针灸学》和《中兽医方剂学》等基础文本，同时在临床环境中协助导师，学习诊断、动物针灸、阉割和草药学。学徒期满后，他们回到自己的集体，在那里担任专业兽医。这些前学徒和他们的家乡集体向导师和县诊所支付了酬金。这些类型的培训为基层青年提供了比在家乡集体学徒制更好的机会，以学习更高水平的兽医技能和理论。²⁰⁵

²⁰⁴ “新干县中兽医带徒弟暂行办法”，民间兽医通讯 3（1958年1月）：5-7。

²⁰⁵ 江西省农业厅，“关于多带徒弟培养中兽医技术人才的通知”（1958年6月16日），民间兽医通讯 6（1958年7月）：7。

The early 1960s witnessed many governmental short-term veterinary trainings for peasants, even though different forms of apprentice programs remained the major gateway to becoming veterinarians. Prefecture- and county-level governments even in remote Guangxi were able to host introductory sessions, rather than leaving primary veterinary education in the hands of local senior veterinarians. By October 1963, several counties under Yulin District of Guangxi had 122 veterinary stations staffed with 356 veterinarians. Ninety percent of these local veterinarians were trained through three- to six-month sessions run by Yulin Prefecture Government. And such local education was hardly a one-time event. Some of them had completed in extra rounds of similar training sessions through which 108 of the 356 veterinarians could be considered “technically advanced.” Although the prefecture cadres complained that the local veterinarians were too reliant upon “convenient” western drugs and therefore fell short of mastering TCVM,²⁰⁶ they seemed satisfied to have locally trained a sufficient number of veterinarians.²⁰⁷

The government of Bobai County, a county in Yulin Prefecture, launched its own veterinary classes specializing in diagnosing diseases of livestock and poultry and administering vaccines. Each commune selected one or two peasants from a “good class background” (e.g., poor and lower-middle peasants, People’s Liberation Army veterans), a strong work ethic, and passion for veterinary medicine and sent them to the county seat where they received seven days of training. The material support was meager. The trainees needed to bring their own food stamps, clothes, bedding, and a mosquito net; the county government provided meals and reimbursed the trainees’ transportation

²⁰⁶ For example, see Xiaoping Fang, *Barefoot Doctors and Western Medicine in China* (Rochester: University of Rochester Press, 2012).

²⁰⁷ “Yulin diqu 1963 nian qu minjian shouyizhan gongzuo zongjie” 玉林地区 1963 年区民间兽医站工作总结 (1964), Guangxi Yulin Municipal Archives (hereafter GYMA), 064-001-009-02.

20世纪60年代初，政府为农民举办了许多短期兽医培训，尽管不同形式的学徒项目仍然是成为兽医的主要途径。即使是偏远的广西，地市级和县级政府也能够举办入门课程，而不是将初级兽医教育交给当地资深兽医。到1963年10月，广西玉林地区下辖的几个县共有122个兽医站，配备了356名兽医。其中90%的当地兽医是通过玉林地区政府举办的为期三到六个月的培训班培养出来的。这种地方教育绝非一次性事件。其中一些人已经完成了额外的类似培训课程，通过这些课程，356名兽医中有108人可以被认为是“技术先进的”。尽管地区干部抱怨当地兽医过于依赖“方便”的西药，因此未能掌握中兽医，²⁰⁶但他们似乎对当地培养了足够数量的兽医感到满意。²⁰⁷

玉林地区博白县政府开办了自己的兽医班，专门诊断牲畜和家禽疾病并接种疫苗。每个公社从“出身好”（如贫下中农、解放军退伍军人）、工作努力、热爱兽医的农民中挑选一到两名，送到县城接受为期七天的培训。物质支持微薄。学员需要自带粮票、衣物、被褥和蚊帐；县政府提供伙食并报销学员的交通费。

²⁰⁶ 例如，参见方小平，《中国的赤脚医生与西方医学》（罗切斯特：罗切斯特大学出版社，2012年）。

²⁰⁷ “玉林地区1963年区民间兽医站工作总结”（1964），广西玉林市档案馆（以下简称GYMA），064-001-009-02。

and miscellaneous fees.²⁰⁸ Depending on what kinds of introductory programs the training enrolled in, ranging from at-home-collective apprenticeships to governmental training sessions, these budding animal doctors at the grassroots acquired a different degree of expertise and professional confidence.

In the late 1950s and early 1960s, local governments were concerned more with the professionalization of their local veterinarians than with training new ones, as the former were considered more readily available. In June 1958, the Jiangxi Provincial Department of Agriculture organized a one-year-and-half reading and research session for full-time, advanced veterinarians who were active in provincial, district-, county-, and commune-level veterinary institutions. Among them were Western-trained veterinarians who were required by the state to study TCVM under to the slogan of “Western veterinarians learn Chinese veterinary medicine (西兽医学习中兽医).” The participants were expected to master three classical texts, *The Yellow Emperor’s Inner Classic* (黄帝内经), *Yuanheng Brothers’ Collection for Treating the Horse*, and *The Compendium of Materia Medica* (本草纲目) and more latest TCVM monographs such as *Traditional Chinese Veterinary Pharmacy* and *Practical Veterinary Acupuncture* by the end of 1959. As these participants were themselves experienced veterinarians, this session had no designated instructors. Instead, they read and discussed the texts together.²⁰⁹

²⁰⁸ Bobai xian renmin weiyuanhui 博白县人民委员会, “Guanyu juban minjian xumu shouyi ren yuan xunlianban de tongzhi” 关于举办民间畜牧兽医人员训练班的通知 (May 12, 1964), Guangxi Bobai County Archives (hereafter GBCA), 9-2-010-15.

²⁰⁹ Jiangxi sheng xumu shouyi gongzuozhe xiehui 江西省畜牧兽医工作者协会, “Guanyu zuzhi zhongshouyi yewu xuexi de tongzhi” 关于组织中兽医业务学习的通知 (June 18, 1958), *Minjian shouyi tongxun* 民间兽医通讯 6 (July 1958): 8–10.

以及杂费。²⁰⁸ 根据所参加的入门培训项目类型，从家庭集体学徒制到政府培训班，这些基层初出茅庐的动物医生获得了不同程度的专业知识和职业信心。

在20世纪50年代末和60年代初，地方政府更关心的是其地方兽医的专业化，而不是培训新的兽医，因为前者被认为更容易获得。1958年6月，江西省农业厅为活跃在省、区、县、公社各级兽医机构的全职高级兽医组织了一期为期一年半的阅读和研究班。其中包括国家要求学习中兽医的西兽医，口号是“西兽医学习中兽医（西兽医学习中兽医）”。学员们被要求掌握三部经典著作：《黄帝内经》（黄帝内经）、《元亨疗马集》和《本草纲目》（本草纲目），以及截至1959年底的最新中兽医专著，如《中兽医方剂学》和《兽医针灸学》。由于这些学员本身就是经验丰富的兽医，因此本次培训没有指定讲师。相反，他们一起阅读和讨论文本。²⁰⁹

²⁰⁸ 博白县人民委员会, “关于举办民间畜牧兽医人员训练班的通知”（1964年5月12日），广西博白县档案馆（以下简称GBCA），9-2-010-15。

²⁰⁹ 江西省畜牧兽医工作者协会, “关于组织中兽医业务学习的通知”（1958年6月18日），民间兽医通讯 6（1958年7月）：8-10。

Around at the same time, Jiangxi Provincial Government offered a three-month program to rank-and-file folk veterinarians and part-time animal doctors who mostly practiced traditional therapeutics. It attracted 40 participants.²¹⁰ In the spirit of the “integration of Chinese and Western veterinary medicine” (中西兽医结合), the participants took courses on not only epidemiology, physiology, animal anatomy, veterinary internal medicine and surgery, but also on TCVM diagnostics, TCVM pharmacy, and animal acupuncture. The director of the program, Liu Ming’an, noted that although the trainees varied in the depth and breadth of their prior knowledge, the sessions worked well because “the culturally refined helped the culturally backward and the experienced shared tips with the less-experienced.” Through this training, local veterinarians, Liu concluded, were able to acquire “new knowledge and new skills.”²¹¹

In addition to the technical training, this professionalization program was especially attentive to the social stature and self-esteem of grassroots animal doctors. In Liu Ming’an’s words, animal healers indeed had been regarded in the “counterrevolutionary era” of the first half of the twentieth century as “lowlives” (下流人物) whose vocation was “menial work (下贱工作).” But Liu emphasized that under Communist rule, they came to enjoy the “glorious title of the trainees of the Jiangxi’s first veterinary professionalization program” and become the “vanguard of veterinary work in our province.”²¹² Similarly, Jin Zhongye, a high-ranking veterinary cadre from the central Ministry of Agriculture who visited Jiangxi to celebrate the completion of the program, reassured the trainees

²¹⁰ Jiangxi sheng nongyeting 江西省农业厅, “Dingqi juban diyiqi zhongshouyi jinxiuban” 定期举办第一期中兽医进修班, *Minjian shouyi tongxun* 民间兽医通讯 4 (March 1958): 56.

²¹¹ Liu Ming’an 刘明安, “Jiangxi sheng diyijie zhongshouyi jinxiuban jieye dianli shangde zhongjie baogao” 江西省第一届中兽医进修班结业典礼上的总结报告, *Minjian shouyi tongxun* 民间兽医通讯 7 (September 1958): 50–53.

²¹² Liu Ming’an, “Jiangxi sheng diyijie zhongshouyi jinxiuban jieye dianli shangde zhongjie baogao,” 51–52.

大约在同一时间, 江西省政府为普通民间兽医和兼职动物医生（他们大多从事传统治疗）提供了一个为期三个月的项目。该项目吸引了40名参与者。²¹⁰本着“中西兽医结合”的精神, 中西兽医结合, 学员们不仅学习了流行病学、生理学、动物解剖学、兽医内科和外科等课程, 还学习了中兽医诊断学、中兽医药物学和动物针灸。项目负责人刘明安指出, 尽管学员们之前的知识深度和广度各不相同, 但培训班效果很好, 因为“文化程度高的帮助文化程度低的, 经验丰富的与经验不足的分享经验”。刘总结道, 通过这次培训, 当地兽医能够掌握“新知识和新技能”。²¹¹

除了技术培训, 这个专业化项目还特别关注基层动物医生的社会地位和自尊。用刘明安的话来说, 在二十世纪上半叶的“反革命时代”, 动物治疗师确实被视为“下流人物”, 他们的职业是“下贱工作”。但刘强调, 在共产党的领导下, 他们获得了“江西省首届兽医专业化项目学员的光荣称号”, 并成为“我省兽医工作的先锋”。²¹²同样, 中央农业部高级兽医干部金中业在访问江西庆祝项目完成时, 向学员们保证

²¹⁰ 江西省农业厅, “定期举办第一期中兽医进修班”, 民间兽医通讯 4 (1958年3月): 56。

²¹¹ Liu Ming’an 刘明安, “Jiangxi sheng diyijie zhongshouyi jinxiuban jieye dianli shangde zhongjie baogao” 江西省第一届中兽医进修班结业典礼上的总结报告, *Minjian shouyi tongxun* 民间兽医通讯 7 (September 1958): 50–53.

²¹² Liu Ming’an, “Jiangxi sheng diyijie zhongshouyi jinxiuban jieye dianli shangde zhongjie baogao,” 51–52.

that they no longer needed to suffer from an “inferiority complex (自卑感).” Jin proclaimed that “knowing how many characters [i.e. bookish learning] cannot be the only benchmark to measure how culturally sophisticated one is.” Hence, China’s folk veterinarians who already have rich clinical experiences and effective therapeutic tools such as acupuncture, Jin argued, “could well surpass the level of international veterinary experts.”²¹³

Nonetheless, the gap between local folk doctors and elite veterinarians did exist in Maoist China. So did educational elitism. Gansu Provincial Government ran a more prestigious veterinary program for talented practicing veterinarians mainly from high-level institutions in agriculture and the military. The Provincial Department of Animal Husbandry designed a six-month professionalization program. The program started to train the first cohort of 20 veterinarians in the spring of 1958 and planned to educate 200 specialists over the next five years.²¹⁴ But before long in 1959, the department became more ambitious and expanded the program into a separate veterinary educational institution, Gansu TCVM School, offering a full-year curriculum.²¹⁵ The provincial government was dissatisfied that the School which did not have its own campus had to borrow classrooms from the Institute of TCVM (ITCVM). Expecting the School to accommodate 1,000

²¹³ Jin Zhongye 金重冶, “Zai yidijie zhongshouyi jinxiuban jieye dianli shangde jianghua” 在第一届中兽医进修班结业典礼上的讲话, *Minjian shouyi tongxun* 民间兽医通讯 7 (September 1958): 54–55.

²¹⁴ Gansu sheng bianzhi weiyuanhui 甘肃省编制委员会, “Guanyu xumuting juban minjian shouyi jinxiuban de baogao” 关于畜牧厅举办民间兽医进修班的报告 (March 27, 1958), GPA, 145-008-0030-0001; Gansu sheng renmin weiyuanhui 甘肃省人民委员会, “Guanyu tongyi juban minjian shouyi jinxiuban de pifu” 关于同意举办民间兽医进修班的批复 (March 26, 1958), GPA, 145-009-0031-0012.

²¹⁵ Gansu sheng renmin weiyuanhui 甘肃省人民委员会, “Guanyu zhongshouyi jinxiuban gaishe wei zhongshouyi xuexiao de pifu” 关于中兽医进修班改设为中兽医学校的批复 (November 26, 1958), GPA, 177-005-0226-0007.

他们不再需要遭受“自卑感（自卑感）”的困扰。金宣称，“知道多少字[即书本知识]不能是衡量一个人文化修养的唯一标准。”因此，金认为，中国民间兽医已经拥有丰富的临床经验和针灸等有效的治疗工具，“完全可以超越国际兽医专家的水平。”²¹³

尽管如此，在毛泽东时代的中国，地方民间医生和精英兽医之间确实存在差距。教育精英主义也同样存在。甘肃省政府为主要来自农业和军事领域高级机构的优秀执业兽医开设了一个更负盛名的兽医项目。省畜牧厅设计了一个为期六个月的专业化项目。该项目于1958年春开始培训第一批20名兽医，并计划在未来五年内培养200名专家。²¹⁴ 但不久之后，在1959年，该厅变得更有雄心，将该项目扩展为一个独立的兽医教育机构——甘肃中医兽医学校，提供为期一年的课程。²¹⁵ 省政府对该学校没有自己的校区，不得不向中医兽医研究所（ITCVM）借用教室感到不满。预计该学校将容纳1,000名学生

²¹³ 金重冶，“在第一届中兽医进修班结业典礼上的讲话”，民间兽医通讯 7（1958年9月）：54-55。

²¹⁴ 甘肃省编制委员会，“关于畜牧厅举办民间兽医进修班的报告”（1958年3月27日），甘肃省档案馆，145-008-0030-0001；甘肃省人民委员会，“关于同意举办民间兽医进修班的批复”（1958年3月26日），甘肃省档案馆，145-009-0031-0012。

²¹⁵ 甘肃省人民委员会，“关于中兽医进修班改设为中兽医学校的批复”（1958年11月26日），甘肃省档案馆，177-005-0226-0007。

students and 100 staff in two years, the government created a new campus for to the School in 1961.²¹⁶

Along with the institutional support from the provincial government, students of Gansu TCVM School received a more systematized veterinary education. The No. 614 Cohort of the School who registered on April 2, 1960 and completed ahead of schedule on February 9, 1961, for example, comprised 51 veterinarians. Only two were recruited from local collectives; the remaining 49 were military veterinary officials who had some knowledge and clinical experience of Western veterinary science. The eight-month “theory” curriculum was well structured: 162 hours for political classes, 216 hours for premodern veterinary classics, 287 hours for the basic principles of TCVM and diagnostics, 217 hours for TCVM pharmacy, 100 hours for veterinary acupuncture. Once the students had completed their study of theory, they were sent to major veterinary institutions in Gansu, Beijing, Hebei, and Jiangxi for a two-month “practice.”²¹⁷ It seems obvious that veterinary students who enrolled in this School had an advantage over, for instance, those who were apprentices in a commune and the participants in a county-level seven-day training session.

The elitism in the veterinary education was prevalent until the eve of the Cultural Revolution. Advanced veterinarians who had taught at veterinary educational programs all over the country were invited to a four-month pedagogical workshop in Hebei TCVM School in 1961.²¹⁸ A few selected local veterinarians across Hebei Province attended a three-month intensive practical

²¹⁶ Gansu sheng xumuting 甘肃省畜牧厅, “Guanyu zhongshouyi xuexiao jiben jianshe gongzuo de anpai yijian” 关于中兽医学校基本建设工作的安排意见 (April 27, 1960), GPA, 144-002-0281-0007.

²¹⁷ Gansu sheng xumuting 甘肃省畜牧厅, “Zhongshouyi xuexiao 614 ban jiaoxue gongzuo zongjie” 中兽医学校 614 班教学工作总结 (February 7, 1961), GPA, 216-002-1918-0031.

²¹⁸ Gansu sheng xumuting 甘肃省畜牧厅, “Xuanpai jiaoshi canjia zhongshouyi xunlianban de tongzhi” 选派教师参加中兽医训练班的通知 (April 24, 1961), GPA, 216-002-1912-0007.

两年内，学校的学生和教职工达到100人，政府于1961年为学校创建了新校区。²¹⁶

在甘肃省政府的制度支持下，甘肃兽医学校的学生接受了更加系统化的兽医教育。例如，该校于1960年4月2日注册并于1961年2月9日提前完成学业的614期学员，共有51名兽医。其中只有两人是从当地集体招募的；其余49人是具有一定西方兽医知识和临床经验的军事兽医官员。为期八个月的“理论”课程结构完善：政治课162小时，前现代兽医经典216小时，中兽医基本原理和诊断287小时，中兽医药物217小时，兽医针灸100小时。学生完成理论学习后，被派往甘肃、北京、河北和江西的主要兽医机构进行为期两个月的“实践”。²¹⁷ 显然，在这所学校就读的兽医学生比那些在公社当学徒和参加县级七天培训班的人更有优势。

兽医教育中的精英主义盛行，直到文化大革命前夕。1961年，全国各地兽医教育项目的资深兽医教师应邀到河北中医兽医学校参加为期四个月的教学研讨会。²¹⁸ 河北省内一些选定的地方兽医参加了为期三个月的强化实践

²¹⁶ 甘肃省畜牧厅, 《关于中兽医学校基本建设工作的安排意见》（1960年4月27日），甘肃省档案馆, 144-002-0281-0007。

²¹⁷ 甘肃省畜牧厅, 《中兽医学校614班教学工作总结》（1961年2月7日），甘肃省档案馆, 216-002-1918-0031。

²¹⁸ Gansu sheng xumuting 甘肃省畜牧厅, “Xuanpai jiaoshi canjia zhongshouyi xunlianban de tongzhi” 选派教师参加中兽医训练班的通知 (April 24, 1961), GPA, 216-002-1912-0007.

training from September to November 1965 and the results of the clinical experiments that they carried out during the session were published as a book.²¹⁹ Even in March 1966, right before the outbreak of the Cultural Revolution in May, the newly established TCVM major at Gansu Agricultural University was recruiting students in the hope of laying the foundation of the elitist veterinary higher education.²²⁰

In the pre-Cultural Revolution period, veterinary education was accessible to grassroots actors. Yet it was hierarchical. There were several ways to become a professional veterinarian. Some of those opportunities extended to the masses in communes or lower collectives, in accordance with the state’s official support of the ideal of “mass science.” The state was keen on raising the social status of veterinarians. Nonetheless, as the next section will show, many non-elite animal healers saw a discrepancy between the promise of socialist egalitarianism and the stratified reality. In the process of becoming and working as an animal doctor, benefits, grievances, hubris and diffidence were unequally distributed.

Enduring Disparities, 1960–1966

In addition to the educational disparity, the instability of local veterinary institutions in the first half of the 1960s threatened some animal doctors’ social position. The stark shortage of food during the Great Leap Famine (1959–1961) forced grassroots collectives to prioritize staple farming over

²¹⁹ Hebei sheng Ding xian zhongshouyi xuexiao 河北省定县中医学校, ed., *Zhong zhuan ban 1965 niandu shengchan shixi lincuang ziliao buji* 中专班 1965 年度生产实习临床资料汇集 (Ding xian: Hebei sheng Ding xian zhongshouyi xuexiao, 1966).

²²⁰ Gansu nongye daxue 甘肃农业大学, “Chengbao guanyu zhongshouyi zhuan ye shizi peibei de yijian” 呈报关于中医专业师资配备的意见 (March 23, 1966), GPA, 216-001-1279-0015.

从1965年9月到11月进行的培训，以及他们在培训期间进行的临床实验结果，都以书籍的形式出版。²¹⁹ 即使在1966年3月，也就是5月文化大革命爆发前夕，甘肃农业大学新成立的兽医专业仍在招收学生，希望能为精英兽医高等教育奠定基础。²²⁰

在文化大革命前夕，兽医教育对基层人员开放。然而，它却是分层的。成为一名专业兽医有多种途径。其中一些机会延伸到公社或较低集体中的大众，这与国家官方支持“大众科学”的理想相符。国家热衷于提高兽医的社会地位。然而，正如下一节将展示的，许多非精英的动物治疗师发现社会主义平等主义的承诺与分层的现实之间存在差异。在成为和从事动物医生工作的过程中，利益、不满、傲慢和自卑的分布是不平等的。

持续存在的差距，1960–1966

除了教育方面的差异，20 世纪 60 年代上半叶地方兽医机构的不稳定性也威胁到了一些动物医生的社会地位。大跃进饥荒（1959-1961 年）期间，粮食严重短缺，迫使基层集体优先发展主食种植，而非

²¹⁹ 河北省定县中医学校编：《中专班 1965 年度生产实习临床资料汇集》（定县：河北省定县中医学校，1966 年）。

²²⁰ 甘肃农业大学，“呈报关于中医专业师资配备的意见”（1966 年 3 月 23 日），GPA，216-001-1279-0015。

animal husbandry and derange veterinary stations. Before the famine, 3,097 full-time veterinarians worked at 1,093 veterinary institutions across Gansu Province in 1959. By 1961, the number had dropped to 1,700. The knowledge and roles of veterinarians were, at least temporarily, regarded as secondary, and many veterinarians were mobilized to agricultural field labor. Accordingly, about half of the local veterinary stations were closed; only 533 out of 1,093 remained open. The famine made the veterinary medicine less pressing.²²¹

To resolve the problem of minimizing the importance of veterinary work, the Gansu Provincial Department of Agriculture and Animal Husbandry ordered the restoration the local veterinary institutions in 1962.²²² Such attempt was in line with the change in central politics. Mao took responsibility for the failure of the Great Leap Forward and retreated from the front lines. Reform-minded Liu Shaoqi and Deng Xiaoping instead held power and enforced the “readjustment” program, partially decollectivizing the rural economy, allowing local peasants to privately own a small plot of field and some livestock, and reintroducing markets and material incentives. Local animal healers, too, had to readjust to the new political environment.²²³

Let us now examine the situation of veterinarians in Gaotai County, Gansu Province in November 1962. Reflecting the readjustment policy, the county decollectivized the county-owned farm animals and sent them down to communes and brigades. The remaining 13 veterinarians, five apprentices, and other material resources of its five county-level veterinary hospitals were

²²¹ Gansu sheng nongmuting 甘肃省农牧厅, “Guanyu minjian shouyi wenti de yijian” 关于民间兽医问题的意见 (December 24, 1962), GPA, 216-001-0978-0021.

²²² Gansu sheng nongmuting, “Guanyu minjian shouyi wenti de yijian.”

²²³ On the changes in central policies in the first half of the 1960s, see Roderick MacFarquhar, *The Origins of the Cultural Revolution Vol. 3: The Coming of the Cataclysm, 1961-1966* (Oxford: Oxford University Press, 1997); Roderick MacFarquhar and Michael Schoenhals, *Mao’s Last Revolution* (Cambridge, Mass.: Belknap Press of Harvard University Press, 2006).

畜牧业和兽医站陷入混乱。饥荒发生前，1959年甘肃省有1093个兽医机构，共有3097名全职兽医。到1961年，这一数字已降至1700人。兽医的知识和作用，至少是暂时性的，被认为是次要的，许多兽医被动员到农业田间劳动。因此，大约一半的当地兽医站被关闭；1093个兽医站中只有533个仍然开放。饥荒使得兽医工作变得不那么紧迫。²²¹

为了解决兽医工作重要性被最小化的问题，甘肃省农牧厅于1962年下令恢复地方兽医机构。²²² 这种尝试与中央政治的变化是一致的。毛泽东为大跃进的失败承担了责任，并从前线撤退。改革派刘少奇和邓小平掌握了权力，并实施了“调整”计划，部分地解除了农村经济的集体化，允许当地农民私有小块土地和一些牲畜，并重新引入市场和物质激励。当地的动物治疗师也必须适应新的政治环境。²²³

现在来看看1962年11月甘肃省高台县兽医的状况。为响应调整政策，该县将县属农场牲畜下放到公社和生产队。县属五所兽医院的13名兽医、5名学徒以及其他物资

²²¹ 甘肃省农牧厅，《关于民间兽医问题的意见》（1962年12月24日），GPA，216-001-0978-0021。

²²² 甘肃省农牧厅，《关于民间兽医问题的意见》。

²²³ 关于20世纪60年代上半叶中央政策的变化，参见Roderick MacFarquhar, *The Origins of the Cultural Revolution Vol. 3: The Coming of the Cataclysm, 1961-1966* (Oxford: Oxford University Press, 1997); Roderick MacFarquhar and Michael Schoenhals, *Mao’s Last Revolution* (Cambridge, Mass.: Belknap Press of Harvard University Press, 2006)。

redistributed to nine communal stations. Each communal veterinary station was now required to be self-sufficient in accounting. The payment of salaries and the rationing of food and necessities were to be done within the budget of the station with limited support from the state. Veterinarians’ salaries depended on their expertise and work performance. As such, far from the ideal of socialist egalitarianism, members of communal and brigade-level veterinary stations now had to grapple with frequent clashes of interests among members of their staff.²²⁴

The following year in 1963, the central Ministry of Agriculture reaffirmed Gansu Province’s approach to readjusting local veterinary stations. The Ministry then asked each veterinary station to reorganize itself with a focus on the technical quality of veterinary cadres and veterinarians. Although the stations had been understaffed after the Great Famine, the Ministry postulated that thanks to rigorous veterinary educational programs, the stations could “gradually replenish” veterinary workers who had “a certain level of expertise.” As a last resort, they might reserve some less qualified workers.²²⁵

Against this backdrop, the socioeconomic conditions of veterinarians in Gansu were much more diversified by January 1964. As Table 1 shows, there were three groups of local veterinarians based on the employment condition: (1) full-time veterinarians who worked at collective veterinary stations; (2) professional veterinarians who privately practiced and were autonomous from the collective economy; and (3) part-time veterinarians in collectives, or “half-peasant-half-veterinarians”

²²⁴ Gansu sheng nongmuting 甘肃省农牧厅, “Guanyu zhuanfa Gaotai xian jiceng zhongshouyi zuzhi qingkuang baogao” 关于转发高台县基层中兽医组织情况报告 (November 13, 1962), GPA, 124-001-0024-0026.

²²⁵ Zhonghua renmin gongheguo nongyebu 中华人民共和国农业部, “Guanyu zhengdun he chongshi xumu shouyizhan de zhishi” 关于整顿和充实畜牧兽医站的指示 (January 3, 1963), in *Xumu shouyi gongzuo cankao ziliao* 畜牧兽医工作参考资料, ed. Gansu sheng xumuting 甘肃省畜牧厅 (Lanzhou: Gansu sheng xumuting, 1964), 44.

重新分配到九个公社兽医站。每个公社兽医站现在都必须在核算方面自给自足。工资支付、食物和必需品的配给都必须在兽医站的预算内进行，国家只提供有限的支持。兽医的工资取决于他们的专业知识和工作表现。因此，公社和生产大队兽医站的成员现在不得不对员工之间频繁的利益冲突，这与社会主义平均主义的理想相去甚远。²²⁴

次年，即1963年，中央农业部重申了甘肃省调整地方兽医站的做法。农业部随后要求每个兽医站进行重组，重点关注兽医干部和兽医的技术素质。尽管大饥荒后兽医站人手不足，但农业部推断，由于严格的兽医教育计划，兽医站可以“逐步补充”具有“一定专业水平”的兽医工作者。作为最后手段，他们可能会保留一些资历较浅的工作人员。

²²⁵

在此背景下，截至1964年1月，甘肃省兽医的社会经济状况更加多样化。如表1所示，根据就业条件，当地兽医分为三类：（1）在集体兽医站工作的全职兽医；（2）私人执业且独立于集体经济的专业兽医；（3）集体中的兼职兽医，或称“半农半兽医”。

²²⁴ 甘肃省农牧厅, “关于转发高台县基层中兽医组织情况报告” (1962年11月13日), 甘肃省档案馆, 124-001-0024-0026.

²²⁵ Zhonghua renmin gongheguo nongyebu 中华人民共和国农业部, “Guanyu zhengdun he chongshi xumu shouyizhan de zhishi” 关于整顿和充实畜牧兽医站的指示 (January 3, 1963), in *Xumu shouyi gongzuo cankao ziliao* 畜牧兽医工作参考资料, ed. Gansu sheng xumuting 甘肃省畜牧厅 (Lanzhou: Gansu sheng xumuting, 1964), 44.

(半农半医). The proportion of each group in the total veterinary population varied by sub-provincial prefecture or municipality. The income disparities were remarkable, ranging from double (Jinquan Prefecture) to 40 times (Wudu Prefecture).

Prefecture/ Municipality	Total no. of vets	No. of vets in District or communal institutions	No. of vets in Brigade- and team- level institutions	No. of decollectivized, private practitioners	No. of part- time veterinarians ("half- peasant- half-vet")	Lowest income (yuan)	Highest income (yuan)
Wudu	297	115	3	60	119	15	628
Wuwei	220	142	14	5	59	21	126
Pingliang	622	333	23	16	250	10	60
Tianshui	923	570	38	84	231	7	120
Lanzhou	90	60	1	29	0	30	180
Jiuquan	137	92	5	12	28	28	56
Gannan	108	10	0	16	82	N/A	N/A
Linxia	198	36	4	59	99	15	66
Zhangye	190	105	38	15	32	20	104
Qingyang	452	174	109	33	136	18	60
Dingxi	665	311	130	107	117	8	75
Total	3902	1948	365	436	1153	N/A	N/A

Table 1. Numbers and Incomes of Local Veterinarians in Gansu Province, 1964. Source: Gansu sheng xumuting 甘肃省畜牧厅, “Quansheng minjian shouyi qingkuang tongji huizongbiao” 全省民间兽医情况统计汇总表 (January 5, 1964), GPA, 222-001-0290-0001.

There is no specific information in this archival document about the interrelationships between where one worked, how much one could earn, what kind of the education one received, and how advanced one’s veterinary skills were. But it can be assumed that while veterinarians in collective veterinary stations might receive institutional benefits such as rationing and social welfare, private practitioners might be self-confident animal healers whose educational background was deemed better or whose expertise was widely recognized within communities, thereby being able to earn a better income. Half-peasant-half-veterinarians, about one-fourth of all veterinarians in the

(半农半医)。每个群体在兽医总人口中的比例因省辖地级市或县级市而异。收入差距显著，从两倍（金泉县）到40倍（武都县）不等。

县/市 市	总计 兽医人数 no.	兽医人数 区 or 公共 机构	兽医数量 在旅- and 团队- 级别 机构	No. of decollectivized, 私人 从业者	兼职人数 time 兽医 (“半农 半兽” 半兽医”)	最低 收入 (元)	最高 收入 (元)
Wudu	297	115	3	60	119	15	628
武威	220	142	14	5	59	21	126
平凉	622	333	23	16	250	10	60
天水	923	570	38	84	231	7	120
兰州	90	60	1	29	0	30	180
酒泉	137	92	5	12	28	28	56
甘南	108	10	0	16	82	N/A	N/A
临夏	198	36	4	59	99	15	66
张掖 e	190	105	38	15	32	20	104
庆阳	452	174	109	33	136	18	60
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这份档案文件没有具体说明工作地点、收入、所受教育以及兽医技能水平之间的相互关系。但可以推断，集体兽医站的兽医可能获得配给和社会福利等制度性福利，而私人执业者可能是自信的动物治疗师，他们的教育背景被认为更好，或者他们的专业知识在社区中得到广泛认可，从而能够获得更好的收入。半农半兽医，约占所有兽医的四分之一

province, were likely to enjoy little prestige and social authority, although they were indispensable in maintaining the grassroots veterinary services. What seems clear is that throughout the first half of the 1960s, a form of meritocracy emerged in the local veterinary realm and disproportionately produced winners and losers.²²⁶

Autobiographic records of low-profile veterinarians support that they had difficulty establishing their authority among villagers. These veterinarians made certain achievements in or exemplary contributions to their collectives and became entitled to share their experiences with the rest of the population. Although the narrative arc of these autobiographies is extremely propagandistic and triumphalist, the authors tended to highlight the hardship they went through in their earlier career so that their eventual success would seem more dramatic. Reading between the lines, we can get a sense of how local veterinarians with limited social capital struggled to gain their footing in the community.²²⁷

Xu Yougan was a poor-peasant-turned-veterinarian practicing in Yangkeng Commune, Zhejiang Province. He participated in the animal disease control activities at home in the early 1950s, which led him to enroll in a six-month veterinary training offered by the prefecture government in 1956. Xu completed the program and returned to his home collective to work as a part-time veterinarian. He might have expected some compliments from his relatives and neighbors. But they showed no respect for the local veterinary education that Xu had received. Xu was asked, “why are you working in this business? Cooping yourself up in a pigpen every day, what kind of

²²⁶ Gansu sheng xumuting 甘肃省畜牧厅, “Quansheng minjian shouyi qingkuang tongji huizongbiao” 全省民间兽医情况统计汇总表 (January 5, 1964), GPA, 222-001-0290-0001.

²²⁷ On critical reading of propagandistic materials in the Mao-era, see Sigrid Schmalzer, “Beyond Bias: Critical Analysis and Layered Reading of Mao-Era Sources,” *Positions: Asia Cultures Critique* 29.4 (November 2021): 759–782; Aminda Smith, “Foreword: The Maoism of PRC History,” *Positions: Asia Cultures Critique* 29.4 (November 2021): 659–674.

的兽医，可能几乎没有什么声望和社会权威，尽管他们在维持基层兽医服务方面不可或缺。显而易见的是，在整个20世纪60年代上半叶，地方兽医领域出现了一种精英管理形式，并不成比例地产生了赢家和输家。²²⁶

低调兽医的自传记录表明，他们很难在村民中树立权威。这些兽医在他们的集体中取得了一定的成就或做出了堪称典范的贡献，并有权与其他人分享他们的经验。尽管这些自传的叙事弧线极具宣传性和胜利主义色彩，但作者倾向于强调他们早期职业生涯中所经历的艰辛，以便他们的最终成功显得更具戏剧性。字里行间，我们可以感受到社会资本有限的当地兽医如何在社区中努力站稳脚跟。²²⁷

许友干是浙江省洋坑公社的一名贫农出身的兽医。他于20世纪50年代初在家乡参与了动物疫病防治活动，这促使他于1956年参加了由地方政府提供的为期六个月的兽医培训。许友干完成了培训并回到家乡集体担任兼职兽医。他可能期待着亲戚和邻居的赞扬。但他们对许友干所接受的当地兽医教育不屑一顾。许友干被问到：“你为什么要做这行？每天把自己关在猪圈里，这算什么

²²⁶ 甘肃省畜牧厅，“全省民间兽医情况统计汇总表”（1964年1月5日），甘肃省档案馆，222-001-0290-0001。

²²⁷ 关于毛泽东时代宣传材料的批判性阅读，参见 Sigrid Schmalzer，“超越偏见：毛泽东时代资料的批判性分析和分层阅读”，*Positions: Asia Cultures Critique* 29.4（2021年11月）：759–782；Aminda Smith，“前言：中华人民共和国历史中的毛泽东主义”，*Positions: Asia Cultures Critique* 29.4（2021年11月）：659–674。

achievement could you ever make?” After five years of hard work, Xu was promoted to a full-time veterinarian in a prefecture-level veterinary station in 1961. Even so, he still faced disdain from peasants. Some of them waved off the veterinarian’s expertise, saying “pigs are dying before long anyway. What on earth is the use of all this?” If Xu thought that he deserved better, he never said anything about it. He might have rejected such thinking as “capitalist.”²²⁸

Qiao Ruilan was not stellar in her career too. She was one of few female graduates of Baoding Agricultural University. After finishing the degree in August 1964, Qiao started her career as a veterinary cadre at the Bureau of Agriculture and Forestry of Xushui County Government in Hebei Province. In March 1965, she figured out that others from her cohort received 36 *yuan* for a month, while she got only 29.5 *yuan*. It was not clear if she filed an official grievance about this pay differential. In any case, she was soon sent to the veterinary station of Suicheng Commune a month later. Qiao did not mention being demoted. She was immediately called to the clinical scene and endured the scrutiny of sharp-eyed villagers who were accustomed to distinguishing a competent animal doctor from an incompetent one.²²⁹

Her animal patients were also difficult. At first, Qiao was ashamed that she had trouble administering medications. She worried that she would not be able to “live up to the hope of the Party and the people.” Then she remembered the famous tale of a foolish but determined old man who was moving a mountain (愚公移山). Following his example, while learning from books how to

²²⁸ Xu You’gan 徐佑淦, “Quanxin quanyi wei shengchu baojian shiye fuwu” 全心全意为牲畜保健事业服务 (1962), Zhejiang sheng 1962 niandu nongye xianjin danwei he laodong mofan daibiao huiyi wenjian 浙江省 1962 年度农业先进单位 and 劳动模范代表会议文件, Unpublished document.

²²⁹ Qiao Ruilan 乔瑞兰, “Yi zhuxi sixiang guashuai, zuo hongse de shouyi jishuyuan” 以主席思想挂帅, 做红色的兽医技术员 (circa 1967), Baoding diqu xumu xianjin danwei xianjin gongzuozhe daibiao huiyi fayan cailiao 保定地区先进单位先进工作者代表会议发言材料, Unpublished document.

你还能做出什么成就？” 经过五年的努力，徐于1961年被提升为地级兽医站的专职兽医。即便如此，他仍然面临农民的鄙视。他们中的一些人对兽医的专业知识不屑一顾，说：“猪反正迟早要死。这一切到底有什么用？” 如果徐认为他应该得到更好的待遇，他从未提及。他可能将这种想法斥为“资本主义”。²²⁸

乔瑞兰的职业生涯也并非一帆风顺。她是保定农业大学为数不多的女毕业生之一。1964年8月毕业后，乔开始在河北省徐水县政府农林局担任兽医干部。1965年3月，她发现同批同学每月工资36元，而她只有29.5元。她是否就这种工资差异提出正式申诉尚不清楚。无论如何，她很快于一个月后被派往遂城公社兽医站。乔没有提及被降职。她立即被召到临床现场，忍受着眼光敏锐的村民的审视，这些村民习惯于区分称职和不称职的兽医。²²⁹

她的动物病人也很难搞。起初，乔为自己难以

用药而感到羞愧。她担心自己无法“不辜负党和人民的希望”。后来她想起了愚公移山的著名故事。她效仿愚公，一边从书本中学习如何

²²⁸ 徐佑淦, “全心全意为牲畜保健事业服务” (1962), 浙江省1962年度农业先进单位和劳动模范代表会议文件, 未出版文件。

²²⁹ Qiao Ruilan 乔瑞兰, “Yi zhuxi sixiang guashuai, zuo hongse de shouyi jishuyuan” 以主席思想挂帅, 做红色的兽医技术员 (circa 1967), Baoding diqu xumu xianjin danwei xianjin gongzuozhe daibiao huiyi fayan cailiao 保定地区先进单位先进工作者代表会议发言材料, Unpublished document.

handle livestock, Qiao did not mind apprenticing herself to the other veterinarians in the commune.

One day, some villagers rushed to the veterinary station because a pregnant donkey was having a difficult delivery. Qiao was the only veterinarian available, as her senior colleagues were traveling.

She had never treated donkeys and “worried that if she fail[ed], the authority of the entire veterinary station would be discredited.” The villagers, however, insisted that she take care of the donkey.

When she arrived, many other villagers surrounded the patient, impatiently waiting. Qiao diagnosed that the unborn foal was already dead. She was at a loss, and in desperation contacted a physician in the communal hospital. Qiao knew that she had assisted in the delivery of human infants. The human doctor was able to do what the animal doctor needed, and the mother donkey was saved.²³⁰

Both Xu Yougan and Qiao Ruilan were less successful products of mass science. The ideal itself did not automatically earn the peasants’ trust. Nor could it conceal the social cleavages among veterinarians. As such, the social stratification and conflicts of interest remained a constant through the Great Leap Famine and the ensuing readjustment. What was unique in Maoist China, however, was a violent upheaval which aimed to nullify the unequal distribution of social capital in the first 17 years of the People’s Republic of China (PRC): the Cultural Revolution.

Upending Hierarchies, 1966–1976

In May 1966, the Great Proletarian Cultural Revolution began. It lauded the “poor and lower-middle peasants” (贫下中农) as the true driver of the revolution which would eliminate all the evils of the past. The post-GLF readjustment in the first half of the 1960s was redefined as the

²³⁰ Qiao Ruilan, “Yi zhuxi sixiang guashuai, zuo hongse de shouyi jishuyuan.”

处理牲畜，乔不介意拜公社里其他兽医为师。一天，一些村民冲到兽医站，因为一头怀孕的驴难产。乔是唯一在场的兽医，因为她的资深同事都出差了。她从未治疗过驴，并“担心如果她失败[了]，整个兽医站的权威都会受到损害。”然而，村民们坚持让她照顾这头驴。当她到达时，许多其他村民围着这头驴，焦急地等待着。乔诊断出未出生的驴驹已经死了。她束手无策，绝望中联系了公社医院的一位医生。乔知道她曾协助接生过人类婴儿。这位人类医生能够做到动物医生需要做的事情，母驴得救了。²³⁰

许有干和乔瑞兰都是群众科学不太成功的产物。理想本身并不能自动赢得农民的信任。它也无法掩盖兽医之间的社会分歧。因此，社会分层和利益冲突在大跃进饥荒和随后的调整中始终存在。然而，毛泽东时代的中国独特之处在于一场旨在消除中华人民共和国成立前17年社会资本不平等分配的剧烈动荡：文化大革命。

Upending Hierarchies, 1966–1976

1966年5月，无产阶级文化大革命开始。它赞扬“贫下中农”是革命的真正推动者，将消除过去的一切弊端。20世纪60年代上半叶大跃进后的调整被重新定义为

²³⁰ 乔瑞兰，“以主席思想挂帅，做红色的兽医技术员。”

counterrevolutionary years under Liu Shaoqi, the “capitalist roader.” As in every other realm, the Cultural Revolution attempted to upend the value system which had dominated the veterinary profession. Now poor and lower-middle peasants would reject the allegedly “capitalistic” meritocracy and take back the “great power in veterinary medicine (兽医大权)” in their hands. In this section, I examine how the Cultural Revolution reshuffled the local veterinary spheres.

The first three years of the Cultural Revolution (1966–1969) were extremely chaotic. In the midst of the uncontrolled uprisings of Red Guards and rebellious workers, the local governmental apparatus, including provincial governments, were subverted in many places.²³¹ In 1969, military force brought the widespread disorder to an end, and local governments were gradually restored under the new name of “revolutionary committee.”²³²

Zhejiang Province was one of the provinces that recovered from this turmoil and tried to quickly resume its veterinary affairs. On August 15, 1969, Zhejiang Provincial Revolutionary Committee explained how the Cultural Revolution’s reform in the veterinary realm would look. It proclaimed that “prioritizing only work competence” (i.e., work in command, 业务挂帅) and following the “expert line” (专家路线) were “counterrevolutionary” and “revisionist.” All veterinary workers should prioritize political preparedness over professional competence and expertise (i.e., politics in command, 政治挂帅) and strictly follow the “mass line.” To do so, poor and lower-middle peasants themselves needed to become “barefoot animal doctors” (赤脚兽医) who would

²³¹ Andrew G. Walder, *China Under Mao: A Revolution Derailed* (Cambridge, Mass.: Harvard University Press, 2015); Andrew G. Walder, *Agents of Disorder: Inside China’s Cultural Revolution* (Cambridge, Mass.: The Belknap Press of Harvard University Press, 2019).

²³² Walder and Dong recently complicates this conventional periodization of the Cultural Revolution and argues that the military intervention exacerbated and prolonged local factional conflicts. Andrew G. Walder and Dong Guoqiang, *A Decade of Upheaval: The Cultural Revolution in Rural China* (Princeton: Princeton University Press, 2021).

刘少奇这个“走资派”领导下的反革命时期。正如在其他所有领域一样，文化大革命试图颠覆主导兽医行业的价值体系。现在，贫下中农将拒绝所谓的“资本主义”精英管理，将“兽医大权”重新掌握在自己手中。在本节中，我将探讨文化大革命如何重组地方兽医领域。

文化大革命的头三年（1966-1969）极其混乱。在红卫兵和造反工人失控的起义中，包括省级政府在内的许多地方政府机构被颠覆。²³¹1969年，军事力量结束了普遍的混乱，地方政府在“革命委员会”的新名称下逐渐恢复。²³²

浙江省是摆脱这场动乱并试图迅速恢复兽医事务的省份之一。1969年8月15日，浙江省革命委员会解释了文化大革命在兽医领域的改革将如何进行。它宣称“只重视工作能力”（即业务挂帅）和遵循“专家路线”是“反革命”和“修正主义”。所有兽医工作者都应将政治准备置于专业能力和专业知识之上（即政治挂帅），并严格遵循“群众路线”。为此，贫下中农自己需要成为“赤脚兽医”，他们将

²³¹ Andrew G. Walder, *China Under Mao: A Revolution Derailed* (马萨诸塞州剑桥：哈佛大学出版社，2015 年)；Andrew G. Walder, *Agents of Disorder: Inside China’s Cultural Revolution* (马萨诸塞州剑桥：哈佛大学贝尔纳普出版社，2019 年)。

²³² Walder 和 Dong 最近使文化大革命的这种传统分期复杂化，并认为军事干预加剧并延长了地方派系冲突。Andrew G. Walder 和 Dong Guoqiang, *A Decade of Upheaval: The Cultural Revolution in Rural China* (普林斯顿：普林斯顿大学出版社，2021 年)。

continue the agricultural labor and practice as part-time veterinarians. This new type of veterinarian would hold the “great power in veterinary medicine” and “help, enlighten, and reform” the “old-fashioned veterinarians.”²³³

Once the provincial directive was issued in 1969, counties and communes followed suit. But some grassroots communities were exceptional. Zhanqi Commune in Yinxian County seemed to be one of the collectives where the Cultural Revolution arrived at veterinary stations as early as 1966. “Poor and lower-middle peasants” overthrew the commune veterinary station and built a new one. The advocates of the Cultural Revolution complained that veterinarians in the former veterinary station “monopolized the expertise” (i.e., technical blockade, 技术封锁) and “suppressed” and “excluded” low-profile veterinary workers in the commune. They insisted that those “old” veterinarians had become depraved because they did not perform agricultural labor alongside poor and lower-middle peasants. They therefore decided to abolish the position of full-time veterinarian. The commune’s new veterinary station was staffed exclusively by with part-time peasants-turned-veterinarians.²³⁴

An even more brutal way to humiliate established local veterinarians was to stigmatize them as “capitalist roaders.” Barefoot veterinarians in Zhanqi Commune condemned private practitioners renowned for the excellent services, accusing them of benefitting from the “free market” and

²³³ Zhejiang sheng geming weiyuanhui shengchan zhihuizu 浙江省革命委员会生产指挥组, “Quansheng xumu shengchan gongzuo huiyi jiyao” 全省畜牧生产工作会议纪要 (August 15, 1969) in *Xumu shouyi gongzuo ziliao* 畜牧兽医工作资料 1, ed. Zhejiang sheng geming weiyuanhui shengchan zhihuizu nonglinju 浙江省革命委员会生产指挥组农林局 (N.p.: Zhejiang sheng geming weiyuanhui shengchan zhihuizu nonglinju, 1971), 14–22.

²³⁴ Zhanqi gongshe geming yinong yiyi shouyizhan 占岐公社革命亦农亦医兽医站, “Women pinxia zhongnong yiding yao peiyang ziji de shouyi jishu duiwu” 我们贫下中农一定要培养自己的兽医技术队伍 (October 1969), Yin xian nongye xue dazhai jingyan jiaoliu dahui ziliao 鄞县农业学大寨经验交流大会资料, Unpublished document.

继续从事农业劳动，并兼职兽医。这种新型兽医将拥有“兽医界的巨大权力”，并“帮助、启发和改造”那些“老式兽医”。²³³

Once the provincial directive was issued in 1969, counties and communes followed suit. But some grassroots communities were exceptional. Zhanqi Commune in Yinxian County seemed to be one of the collectives where the Cultural Revolution arrived at veterinary stations as early as 1966. “Poor and lower-middle peasants” overthrew the commune veterinary station and built a new one. The advocates of the Cultural Revolution complained that veterinarians in the former veterinary station “monopolized the expertise” (i.e., technical blockade, 技术封锁) and “suppressed” and “excluded” low-profile veterinary workers in the commune. They insisted that those “old” veterinarians had become depraved because they did not perform agricultural labor alongside poor and lower-middle peasants. They therefore decided to abolish the position of full-time veterinarian. The commune’s new veterinary station was staffed exclusively by with part-time peasants-turned-veterinarians.²³⁴

一种更残酷的羞辱当地资深兽医的方式是给他们贴上“走资派”的标签。战旗公社的赤脚兽医谴责那些以优质服务闻名的私人执业兽医，指责他们从“自由市场”中获利，并且

²³³ Zhejiang sheng geming weiyuanhui shengchan zhihuizu 浙江省革命委员会生产指挥组, “Quansheng xumu shengchan gongzuo huiyi jiyao” 全省畜牧生产工作会议纪要 (August 15, 1969) in *Xumu shouyi gongzuo ziliao* 畜牧兽医工作资料 1, ed. Zhejiang sheng geming weiyuanhui shengchan zhihuizu nonglinju 浙江省革命委员会生产指挥组农林局 (N.p.: Zhejiang sheng geming weiyuanhui shengchan zhihuizu nonglinju, 1971), 14–22.

²³⁴ Zhanqi gongshe geming yinong yiyi shouyizhan 占岐公社革命亦农亦医兽医站, “Women pinxia zhongnong yiding yao peiyang ziji de shouyi jishu duiwu” 我们贫下中农一定要培养自己的兽医技术队伍 (October 1969), Yin xian nongye xue dazhai jingyan jiaoliu dahui ziliao 鄞县农业学大寨经验交流大会资料, Unpublished document.

enriching themselves.²³⁵ About a decade later, in October 1975, commune veterinary stations established during the Cultural Revolution held a nationwide conference to discuss their local experiences. Most stations reported having spurned “malefactors who sneaked into the animal husbandry and veterinary stations” during the Cultural Revolution. Veterinarians who had been denounced placed “money in command” (金钱挂帅) and defrauded the poor and lower-middle peasants of their pennies.²³⁶ We cannot tell how valid these denunciations were. But we do know that the Cultural Revolution did provide hitherto underprivileged veterinary workers with an opportunity and a set of languages to attack their more privileged colleagues.

Although the Cultural Revolutionists’ frustration at the socio-intellectual disparity in the local veterinary world was understandable, abolishing full-time veterinary jobs, suppressing private practitioners, and denying the values of professional competence and expertise were certainly anti-intellectual actions. If they were to proceed in this direction, they needed to convince their fellow commune members—both sympathetic and hostile to the Cultural Revolution’s overturning of the preexisting hierarchies—that their communities would be fine without capable veterinary experts who were deemed “old-fashioned” or “revisionist.” Here Maoist voluntarism came to the fore.

The Cultural Revolutionists in Zhanqi Commune declared that new barefoot half-peasant-half-veterinarians might be “technically untenable temporarily.” As such, it would be likely that “the masses distrust” barefoot animal doctors. Nonetheless, the lack of veterinary expertise could be overcome as long as these determined new veterinarians did their best. According to Mao, “although

²³⁵ Zhanqi gongshe geming yinong yiye shouyizhan, “Women pinxia zhongnong yiding yao peiyang ziji de shouyi jishu duiwu.”

²³⁶ Zhonghua renmin gongheguo nongyebu 中华人民共和国农业部, “Quanguo gongshe xumu shouyizhan jingyan jiaoliu zuotanhui jianbao” 全国公社畜牧兽医站经验交流座谈会简报 (October 18, 1975), in *Shouyi gongzuo wenjian xuanbian* 兽医工作文件选编, ed. Shaanxi sheng xumu shouyi zongzhan 陕西省畜牧兽医总站 (N.p., 1975), 108–113.

中饱私囊。²³⁵ 大约十年后, 即1975年10月, 文革期间成立的公社兽医站举行了全国会议, 讨论各地经验。大多数兽医站报告称, 在文革期间, 他们唾弃了“混入畜牧兽医站的坏分子”。被谴责的兽医“金钱挂帅”, 骗取贫下中农的血汗钱。²³⁶ 我们无法判断这些谴责的真实性。但我们确实知道, 文化大革命为以前处于劣势的兽医工作者提供了一个机会和一套语言, 来攻击他们那些享有更多特权的同事。

尽管文化大革命者对地方兽医界社会智力差距的沮丧是可以理解的, 但废除全职兽医工作、压制私人执业者以及否定专业能力和专业知识价值, 无疑是反智行为。如果他们要朝这个方向发展, 他们需要说服他们的公社成员——无论是同情还是敌视文化大革命推翻现有等级制度的人——他们的社区在没有那些被认为是“老式”或“修正主义者”的有能力的兽医专家的情况下也能安然无恙。毛泽东的唯意志论在此凸显出来。

战旗公社的文化大革命者宣称, 新的赤脚半农半兽医可能“暂时技术上站不住脚”。因此, “群众不信任”赤脚兽医的可能性很大。然而, 只要这些坚定的新兽医尽力而为, 兽医专业知识的缺乏是可以克服的。毛泽东认为, “尽管

²³⁵ 战旗公社革命亦农亦医兽医站, “我们贫下中农一定要培养自己的兽医技术队伍。”

²³⁶ 中华人民共和国农业部, 《全国公社畜牧兽医站经验交流座谈会简报》(1975年10月18日), 载陕西省畜牧兽医总站编《兽医工作文件选编》(无出版地, 1975), 第108-113页。

the weapon is an important factor in winning a war, [...] the determinant is not the material but the human.” There was no better way to brush up the “human factor,” of great courage and determination, than to read Mao’s texts. The political learning of Maoism and the application of its lessons to the everyday life, therefore, would be the key to success as a veterinarian. This was why barefoot veterinarians had to place “politics in command,” not “work competence in command” or “economy in command.”²³⁷

At the same time, the advocates of the Cultural Revolution appropriated the famous slogan of “prevention rather than treatment” (防重于治). As Chapter I shows, the slogan has been popular before the Cultural Revolution. Some barefoot animal doctors, however, concluded that whereas counterrevolutionaries in the Liu Shaoqi’s faction put treatment first in the readjustment period, loyal Maoists prioritized prevention over treatment only after the Cultural Revolution, thereby saving hundreds of thousands of livestock. One implication of drawing a false dichotomy between the pre-1966 period that sought “treatment rather than prevention” and the post-1966 era of “prevention rather than treatment” was the obscuring of the barefoot veterinarians’ limited healing skills. Preventative measures such as administering regular vaccinations and quarantining sick animals were easier for amateur veterinary workers to do than mastering diagnostics and therapeutics. As treatment was overshadowed by prevention, the difference between those who could and could not heal animals became less salient.²³⁸

²³⁷ Zhanqi gongshe geming yinong yiyi shouyizhan, “Women pinxia zhongnong yiding yao peiyang ziji de shouyi jishu duiwu.” As for Mao’s original text that may undergird this point, see Mao Zedong 毛泽东, “Where Do Correct Ideas Come From?” (May 1963) in *Selected Readings from the Works of Mao Tsetung* (Peking: Foreign Languages Press, 1971), 502–504.

²³⁸ Zu Xingguo 祖兴国, “Re’ai xumu shiye, dang pinxia zhongnong fangxin de chijiao shouyi: Jilin sheng Panshi xian Mingcheng gongshe Xialu dadui xumu shouyisuo chijiao shouyi Zu Xingguo zishu” 热爱畜牧事业, 当贫下中农放心

武器是赢得战争的重要因素, [……] 决定因素不是物质, 而是人。” 没有比阅读毛泽东的著作更好的方式来重塑 “人” 这一重要因素, 即巨大的勇气和决心。因此, 毛泽东思想的政治学习及其在日常生活中的应用, 将是兽医取得成功的关键。这就是为什么赤脚兽医必须把 “政治放在首位”, 而不是 “工作能力放在首位” 或 “经济放在首位”。²³⁷

At the same time, the advocates of the Cultural Revolution appropriated the famous slogan of “prevention rather than treatment” (防重于治). As Chapter I shows, the slogan has been popular before the Cultural Revolution. Some barefoot animal doctors, however, concluded that whereas counterrevolutionaries in the Liu Shaoqi’s faction put treatment first in the readjustment period, loyal Maoists prioritized prevention over treatment only after the Cultural Revolution, thereby saving hundreds of thousands of livestock. One implication of drawing a false dichotomy between the pre-1966 period that sought “treatment rather than prevention” and the post-1966 era of “prevention rather than treatment” was the obscuring of the barefoot veterinarians’ limited healing skills. Preventative measures such as administering regular vaccinations and quarantining sick animals were easier for amateur veterinary workers to do than mastering diagnostics and therapeutics. As treatment was overshadowed by prevention, the difference between those who could and could not heal animals became less salient.²³⁸

²³⁷ 战旗公社革命亦农亦医兽医站, “我们贫下中农一定要培养自己的兽医技术队伍。” 至于毛泽东可能支持这一观点的原文, 请参见毛泽东, 《人的正确思想是从哪里来的? 》(1963年5月), 载于《毛泽东著作选读》(北京: 外文出版社, 1971年), 第502-504页。

²³⁸ 祖兴国, “热爱畜牧事业, 当贫下中农放心的赤脚兽医: 吉林省磐石县明城公社下路大队畜牧兽医所赤脚兽医祖兴国自述”

Barefoot animal doctors embodied and benefited from the changes in the veterinary profession during the Cultural Revolution. The term “barefoot animal doctors” was derived from its counterpart in human medicine: barefoot doctors.²³⁹ Most barefoot veterinarians were former prevention workers in production brigades or teams; they usually continued to work as a half-peasant-half-veterinarian in their sub-communal collectives.²⁴⁰ Compared to the diverse veterinary training programs in the pre-Cultural Revolution era, the grassroots training sessions for barefoot veterinarians trained a larger number of animal healers in a shorter period of time. For example, 588 new barefoot animal doctors completed the five-day barefoot veterinary trainings in Lanxi County, Zhejiang Province alone in June 1970.²⁴¹ Similarly, Quxian County, Zhejiang also saw 1,183 veterinarians in the next year.²⁴² Given that a prefecture which covered about a dozen counties only had 90 to 650 veterinarians in 1964 (Table I), the numbers of new barefoot veterinarians are impressive. The barefoot veterinary education also tended to focus on swine disorders, reflecting the

的赤脚兽医: 吉林省磐石县明城公社下鹿大队畜牧兽医所赤脚兽医祖兴国自述 (March 26, 1973), Jilin diqu xumu fangyi jingyan jiaoliuhui cailiao 吉林地区 畜牧防疫经验交流会材料, Unpublished document.

²³⁹ Fang, *Barefoot Doctors and Western Medicine in China*; Miriam Gross, “Between Party, People, and Profession: The Many Faces of the ‘Doctor’ during the Cultural Revolution,” *Medical History* 62.3 (July 2018): 333–359.

²⁴⁰ Zhanqi gongshe geming yinong yiyi shouyizhan, “Women pinxia zhongnong yiding yao peiyang ziji de shouyi jishu duiwu”; Zhejiang sheng geming weiyuanhui shengchan zhihuizu, “Quansheng xumu shengchan gongzuo huiyi jiyao,” 21.

²⁴¹ Duan Lunjing 段伦敬, “Zhonggong Lanxi xianwei shuji Duan Lunjing tongzhi zai sheng yangzhu gongzuo zuotanhui shang de huibao” 中共兰溪县委书记段伦敬同志在省养猪工作座谈会上的汇报 (June 30, 1970), in *Xumu shouyi gongzuo ziliao* 1, ed. Zhejiang sheng geming weiyuanhui shengchan zhihuizu nonglinju, 57.

²⁴² Qu xian geming weiyuanhui 衢县革命委员会, “Jianjue luoshi dang de fangzhen zhengce, jixu dali fazhan yangzhu shiye” 坚决落实党的方针政策, 继续大力发展养猪事业 (circa June 1971), in *Xumu shouyi gongzuo ziliao* 1, ed. Zhejiang sheng geming weiyuanhui shengchan zhihuizu nonglinju, 62.

赤脚兽医体现并受益于文化大革命期间兽医专业的变化。“赤脚兽医”一词来源于人类医学中的对应词: 赤脚医生。²³⁹大多数赤脚兽医是生产大队或生产队的前防疫员; 他们通常继续在半公共集体中从事半农半兽医的工作。²⁴⁰与文化大革命前兽医培训项目的多样性相比, 赤脚兽医的基层培训班在更短的时间内培训了更多的动物治疗师。例如, 仅1970年6月, 浙江省兰溪县就有588名新的赤脚兽医完成了为期五天的赤脚兽医培训。²⁴¹同样, 次年浙江省衢县也有1183名兽医。²⁴²考虑到1964年一个涵盖约十几个县的地区只有90到650名兽医(表一), 新赤脚兽医的数量令人印象深刻。赤脚兽医教育也倾向于关注猪病, 这反映了

的赤脚兽医: 吉林省磐石县明城公社下鹿大队畜牧兽医所赤脚兽医祖兴国自述 (March 26, 1973), Jilin diqu xumu fangyi jingyan jiaoliuhui cailiao 吉林地区 畜牧防疫经验交流会材料, Unpublished document.

²³⁹ Fang, *Barefoot Doctors and Western Medicine in China*; Miriam Gross, “Between Party, People, and Profession: The Many Faces of the ‘Doctor’ during the Cultural Revolution” , *Medical History* 62.3 (2018年7月) : 333–359.

²⁴⁰ Zhanqi gongshe geming yinong yiyi shouyizhan, “Women pinxia zhongnong yiding yao peiyang ziji de shouyi jishu duiwu”; Zhejiang sheng geming weiyuanhui shengchan zhihuizu, “Quansheng xumu shengchan gongzuo huiyi jiyao,” 21.

²⁴¹ Duan Lunjing 段伦敬, “Zhonggong Lanxi xianwei shuji Duan Lunjing tongzhi zai sheng yangzhu gongzuo zuotanhui shang de huibao” 中共兰溪县委书记段伦敬同志在省养猪工作座谈会上的汇报 (June 30, 1970), in *Xumu shouyi gongzuo ziliao* 1, ed. Zhejiang sheng geming weiyuanhui shengchan zhihuizu nonglinju, 57.

²⁴² Qu xian geming weiyuanhui 衢县革命委员会, “Jianjue luoshi dang de fangzhen zhengce, jixu dali fazhan yangzhu shiye” 坚决落实党的方针政策, 继续大力发展养猪事业 (circa June 1971), in *Xumu shouyi gongzuo ziliao* 1, ed. Zhejiang sheng geming weiyuanhui shengchan zhihuizu nonglinju, 62.

government’s prioritization of pigs over other animals,²⁴³ and further stressed the use of the resources that were available on the ground.²⁴⁴

Most veterinarians who started their career before the Cultural Revolution were called “old-fashioned veterinarians” (旧式兽医) or “old veterinarians” (老兽医), regardless of their background and skills. From the official perspective that barefoot veterinarians were expected to have, these senior veterinarians were the object of political reform and even class struggle. It can be imagined from what we know about the massive violence during the Cultural Revolution that numerous veterinarian who had practiced before the Cultural Revolution were indeed persecuted and many were dead by the late 1960s.²⁴⁵ Unfortunately, it is extremely difficult to obtain the documentary evidence of this devastation, as the Chinese government restricts access to archives on these campaigns. Instead, I turn to what we can learn about how “old veterinarians” survived the Cultural Revolution and what the relationship between “old” and barefoot veterinarians looked like.

Kou Tiande was a seasoned veterinarian who worked in the veterinary station in Donglou Commune of Xinxian County, Shanxi Province. He conducted a self-criticism on December 27, 1966, recorded as an unpublished document. According to his self-defensive note, Kou was born to a poor peasant family. He did not mention what kind of professional training he received but

²⁴³ See Chapter Four of this dissertation.

²⁴⁴ The slogans of “combining the native and the foreign, regarding the native as primary” (土洋结合, 以土为主) and “using the crude and simple methods, doing things with locally available materials” (因陋就简, 就地取材) represented this directionality. Zhejiang sheng geming weiyuanhui xuefang lingdao xiaozu 浙江省革命委员会血防领导小组, “Guanyu qianduan gengniu xuefang gongzuo qingkuang he jinhou yijian de baogao” 关于前段耕牛血防工作情况和今后意见的报告 (February 23, 1971), in *Xumu shouyi gongzuo ziliao* 1, ed. Zhejiang sheng geming weiyuanhui shengchan zhihuizu nonglinju, 42; Shanxi sheng Fanshi xian shouyiyuan 山西省繁峙县兽医院, “Keyan huodong 1975 nian de wancheng qingkuang he 1976 nian de gongzuo jihua” 科研活动 1975 年的完成情况和 1976 年的工作计划 (circa 1975), Unpublished document.

²⁴⁵ Walder, *Agents of Disorder*; Walder and Dong, *A Decade of Upheaval*.

政府将猪置于其他动物之上的优先地位，²⁴³并进一步强调了利用现有资源的重要性。²⁴⁴

大多数在文化大革命前开始职业生涯的兽医都被称为“旧式兽医”（旧式兽医）或“老兽医”（老兽医），无论其背景和技能如何。从赤脚兽医应有的官方视角来看，这些资深兽医是政治改革乃至阶级斗争的对象。从我们对文化大革命期间大规模暴力的了解可以想象，许多在文化大革命前执业的兽医确实受到了迫害，许多人在1960年代末期已经死亡。²⁴⁵不幸的是，由于中国政府限制查阅这些运动的档案，因此很难获得这场灾难的文献证据。相反，我将探讨我们能了解到“老兽医”如何在文化大革命中幸存下来，以及“老兽医”与赤脚兽医之间的关系是怎样的。

寇天德是一位经验丰富的兽医，在山西省忻县东楼公社兽医站工作。他于1966年12月27日进行了自我批评，记录为一份未发表的文件。根据他的自辩记录，寇出生于一个贫农家庭。他没有提及自己接受过何种专业培训，但

²⁴³ 参见本论文第四章。

²⁴⁴ “土洋结合，以土为主”和“因陋就简，就地取材”的口号代表了这一方向性。浙江省革命委员会血防领导小组，“关于前段耕牛血防工作情况和今后意见的报告”（1971年2月23日），载于浙江省革命委员会生产指挥组农林局编，《畜牧兽医工作资料》1，第42页；山西省繁峙县兽医院，“科研活动1975年的完成情况和1976年的工作计划”（约1975年），未发表文件。

²⁴⁵ Walder, *Agents of Disorder*; Walder 和 Dong, *A Decade of Upheaval*.

seemed to have become a local veterinarian in the late 1950s or early 1960s. He worked in the county’s veterinary hospital until 1962 when he was transferred to Donglou Commune’s station. Kou confessed that at that time, he “could not understand what ‘Serving the People’ meant.” He just followed the value of *si* (私)—the private or the self—and took care of his parents, his wife, and a small plot of field that his family owned (自留地). Kou was not willing to leave the station and tour around brigades and teams to check up on the livestock. Even though villagers called on him, he did not respond “when he was unhappy, it was too far away, the weather was bad, the illness sounded too serious, and what was expected to do sounded unfamiliar.”²⁴⁶

Kou Tiande admitted that in the first half of the 1960s, he had been cynical about studying Maoism. The leaders of the commune bought Mao’s writings and organized study groups in 1963. Kou reluctantly attended a group yet did not read Mao. In August 1965 when the Socialist Education Movement started in Donglou, a group of cadres from above visited Kou’s veterinary station and encouraged him to study “politics.” He “felt very disgusted.” As a veterinarian, “as long as one can treat livestock well, everything should be fine. Why should one study Mao or something?” But, according to the autography, Kou was soon inspired by the devotional cadres who patiently tolerated him. Kou must have known that the readers of his self-criticism wanted him to depart from the attitude represented by “work competence in command” and “expert line.”²⁴⁷

It was hard to determine whether it was fair to characterize Kou in his earlier career as the so-called Liu-Shaoqi-style revisionist. But it might not be wise to try to exculpate him. Instead, Kou

²⁴⁶ Kou Tiande 寇天德, “Mao Zedong sixiang zhaoliang le wo de xin” 毛泽东思想照亮了我的心 (December 27, 1966), Shanxi sheng Xin xian Donglou gongshe shouyi Kou Tiande zishu 山西省忻县东楼公社兽医寇天德自述, Unpublished document, 1.

²⁴⁷ Kou Tiande, “Mao Zedong sixiang zhaoliang le wo de xin,” 2.

在20世纪50年代末或60年代初，他似乎已成为当地的一名兽医。他在县兽医院工作，直到1962年被调到东楼公社的兽医站。寇坦言，当时他“不明白‘为人民服务’是什么意思”。他只是遵循私的价值观，照顾他的父母、妻子和家里拥有的一小块自留地。寇不愿意离开兽医站，也不愿意到大队和生产队巡视检查牲畜。即使村民叫他，他也不回应，“当他不高兴时，路太远，天气不好，病情听起来太严重，以及预期要做的事情听起来很陌生。”²⁴⁶

寇天德承认，在20世纪60年代上半叶，他对学习毛泽东思想持怀疑态度。公社领导在1963年购买了毛泽东著作并组织了学习小组。寇不情愿地参加了一个小组，但没有阅读毛泽东著作。1965年8月，当社会主义教育运动在东楼开始时，一群上级干部来到寇的兽医站，鼓励他学习“政治”。他“感到非常厌恶”。作为一名兽医，“只要能把牲畜治好，一切都应该没问题。为什么要学习毛泽东思想或其他什么？”但是，根据自传，寇很快就被那些耐心容忍他的虔诚干部所鼓舞。寇一定知道，他的自我批评的读者希望他摆脱“业务挂帅”和“专家路线”所代表的态度。²⁴⁷

很难确定将寇在早期职业生涯中定性为所谓的刘少奇式修正主义者是否公平。但试图为他开脱可能不明智。相反，寇

²⁴⁶ 寇天德，“毛泽东思想照亮了我的心”（1966年12月27日），山西省忻县东楼公社兽医寇天德自述，未出版文献，第1页。

²⁴⁷ 寇天德，“毛泽东思想照亮了我的心”，第2页。

tried to convince those who had forced him into self-criticism that he had converted to Maoism, valuing the idea of “breaking the private and establishing the public” (破私立公). It was his long-held dream, Kou said, to work as a veterinarian in his own hometown so that he could support his family while remaining close to them. But Donglou Commune fortunately asked him to stay despite his “political” problems. Remembering what Mao had said about the Canadian martyr Dr. Norman Bethune who “never benefits himself and professionally benefits others (毫不利己, 专门利人),” Kou humbly decided to prioritize the public (staying in Donglou) over the private (returning to the home collective). Although more diverse personal experiences need to be examined, Kou Tiande’s autobiographic narrative seems to show how “old veterinarians” came to terms with the reforms of the Cultural Revolution in the grassroots veterinary spheres.²⁴⁸

The career of Zu Xingguo, in contrast, may epitomize the rise of a barefoot veterinarian. Zu was working for Xialu Brigade in Mingcheng Commune of Panshi County, Jilin Province. On March 26, 1973, he explained how he could become a model veterinary worker. According to his short autobiography, he was a returned youth who graduated from a middle school in a nearby city and came back to his rural hometown in 1963. Between 1963 and 1965, he taught reading to poor and lower-middle peasants by using Mao’s writings. He then worked as a part-time animal diseases prevention worker, while doing everyday farming work. This earlier career made Zu a promising candidate for training as a barefoot veterinarian. In 1970, brigade leaders recommended him for a two-month barefoot veterinary session hosted by the commune. After completing the program, Zu returned to serve the brigade veterinary station as a barefoot veterinarian.²⁴⁹

²⁴⁸ Kou Tiande, “Mao Zedong sixiang zhaoliang le wo de xin,” 3–4.

²⁴⁹ Zu Xingguo, “Re’ai xumu shiye, dang pinxia zhongnong fangxin de chijiao shouyi,” 1–3.

试图说服那些强迫他进行自我批评的人，他已经皈依了毛泽东思想，珍视“破私立公”的理念（破私立公）。寇说，他长久以来的梦想是在家乡当一名兽医，这样他就可以在养家糊口的同时，也能和家人保持亲近。但东楼公社幸运地要求他留下，尽管他有“政治”问题。想起毛泽东关于加拿大烈士诺尔曼·白求恩大夫“毫不利己，专门利人”的说法（毫不利己，专门利人），寇谦逊地决定将公共利益（留在东楼）置于个人利益（回到家乡集体）之上。尽管需要审视更多不同的个人经历，但寇天德的自传叙述似乎展示了“老兽医”如何在基层兽医领域接受文化大革命的改革。²⁴⁸

相比之下，祖兴国的职业生涯可能象征着赤脚兽医的崛起。祖兴国当时在吉林省磐石县明城公社下路大队工作。1973年3月26日，他解释了自己如何成为一名模范兽医工作者。根据他的简短自传，他是一名返乡青年，毕业于附近城市的一所中学，并于1963年回到家乡农村。1963年至1965年间，他通过毛泽东著作教贫下中农识字。之后，他一边从事日常农业劳动，一边担任兼职动物疾病防治员。早期的职业生涯使祖兴国成为一名有前途的赤脚兽医培训候选人。1970年，大队领导推荐他参加公社举办的为期两个月的赤脚兽医培训班。完成培训后，祖兴国回到大队兽医站担任赤脚兽医。²⁴⁹

²⁴⁸ 寇天德，《毛泽东思想照亮了我的心》，第3-4页。

²⁴⁹ 祖兴国，《热爱畜牧事业，当贫下中农放心的赤脚兽医》，第1-3页。

Even at the height of the Cultural Revolution in the 1970s, barefoot animal doctors lacked confidence. Administrative cadres and peasants in Zu’s brigade “did not trust” them and “superstitiously believed in old veterinarians.” Some people more aggressively opined that “Zu Xingguo is our collective’s tenderfoot. Only after studying at the commune veterinary station for two months, he came back and is now about to work as a vet in our new [brigade] station. Does this make any sense?” Disheartened, Zu conceded. “It is inevitable even for old veterinarians who graduated agricultural university to make mistakes. I am just a middle school graduate and received a two-month veterinary training. I bear no comparison with them. [...] With heavy worries, my heart sinks to the bottom.”

But the model barefoot veterinarian’s autobiography must not end here. Zu was nevertheless touched by the encouragement of passionate local cadres, and Mao’s texts on Bethune and the story of the old man moving a mountain. This kind of support was deeply voluntarist and gave him “unlimited power.” Feeling more confident, Zu resolved not to be a “shoe-wearing veterinarian” (穿鞋兽医) who would care only about self-interest and fame, but instead a barefoot veterinarian who would sacrifice himself for the poor and lower-middle peasants.²⁵⁰

Zu Xingguo’s disappointment notwithstanding, there is no evidence that he participated in persecuting “old veterinarians” like Kou Tiande in his collective. Rather, like Miriam Gross’s argument that elite doctors from cities and barefoot doctors developed a symbiotic relationship in the countryside,²⁵¹ Zu and “old veterinarians” seemed to benefit from each other. Zu occasionally visited the commune veterinary station where he learned new therapeutics from the cautious “old

²⁵⁰ Zu Xingguo, “Re’ai xumu shiye, dang pinxia zhongnong fangxin de chijiao shouyi,” 3–4, 8.

²⁵¹ Gross, “Between Party, People, and Profession.”

即使在20世纪70年代文化大革命的高潮时期，赤脚兽医也缺乏信心。祖的生产队的行政干部和农民“不信任”他们，并“迷信地相信老兽医”。一些人更激烈地表示，“祖兴国是我们集体的嫩脚。只在公社兽医站学习了两个月，他就回来了，现在就要在我们新的[生产队]站当兽医了。这有道理吗？”祖沮丧地承认。“即使是农业大学毕业的老兽医也难免犯错。我只是个中学生，接受了两个月的兽医培训。我无法与他们相比。[……]忧心忡忡，我的心沉到了谷底。”

但模范赤脚兽医的自传绝不能到此为止。祖仍然被热情的当地干部的鼓励，以及毛泽东关于白求恩的著作和愚公移山的故事所感动。这种支持是深刻的志愿主义，给了他“无限的力量”。祖感到更有信心，决心不做只关心自身利益和名声的“穿鞋兽医”（穿鞋兽医），而是做一名为贫下中农牺牲自己的赤脚兽医。²⁵⁰

尽管祖兴国感到失望，但没有证据表明他参与迫害集体中的“老兽医”，如寇天德。相反，正如米里亚姆·格罗斯（Miriam Gross）所说，城市里的精英医生和赤脚医生在农村发展出共生关系一样，²⁵¹ 祖和“老兽医”似乎也互惠互利。祖偶尔会去公社兽医站，在那里他从谨慎的“老”兽医那里学到新的治疗方法。

²⁵⁰ 祖兴国，“热爱畜牧事业，当贫下中农放心的赤脚兽医”，第3-4、8页。

²⁵¹ 格罗斯，“在党、人民和专业之间。”

veterinarians” there who had hunkered down. Whenever Zu encountered difficult cases, he asked the “old veterinarians” for assistance.²⁵² From the end of “old veterinarians,” sharing their knowledge with barefoot animal doctors would have been a way to show the people in their collectives that they opposed the “technical blockade.” Kou, too, criticized his fellow “old veterinarian” who did not want to teach people how to castrate swine. He said that such a “conservative” attitude was no longer tolerable.²⁵³ Barefoot and “old” veterinarians found a way to coexist.

The Veterinarian’s Way under High Maoism

As we have seen, although the ideal of mass science was the norm in Maoist China, there was a considerable gap in the socio-intellectual hierarchy among local veterinarians. The reforms of the Cultural Revolution in the veterinary realm aimed to end this disparity by putting first the political resolution and willpower over what arguably was a capitalistic meritocracy. It seems that there was no systematic brake which could prevent the Cultural Revolution from degenerating into anti-intellectualism and vandalism. It also seems that it failed in its aspiration to eliminate inequality. Despite the state’s extensive support, barefoot veterinarians who at best were semi-professionals were not able to entirely replace the veterinarians of the pre-Cultural Revolution era. Many local people and even cadres still had much more faith in the “old veterinarians” than in the half-peasant-half-veterinarians who were to replace them. To prove themselves, barefoot veterinarians first and

²⁵² Zu Xingguo, “Re’ ai xumu shiye, dang pinxia zhongnong fangxin de chijiao shouyi,” 2–3, 7.

²⁵³ Kou Tiande, “Mao Zedong sixiang zhaoliang le wo de xin,” 5–7.

那里的“老兽医”们都蛰伏起来。每当祖遇到疑难病例时，他都会向这些“老兽医”寻求帮助。²⁵² 对“老兽医”们来说，与赤脚兽医分享知识是向集体中的人们表明他们反对“技术封锁”的一种方式。寇也批评了他的“老兽医”同事，因为他不愿意教人们如何给猪阉割。他说，这种“保守”的态度是不能再容忍的。²⁵³ 赤脚兽医和“老”兽医找到了一种共存的方式。

高毛泽东时代下的兽医之道

正如我们所见，尽管群众科学的理想在毛泽东时代的中国是常态，但在地方兽医的社会智力等级中存在相当大的差距。文化大革命在兽医领域的改革旨在通过将政治决心和意志置于可以说是资本主义的精英管理之上，来结束这种差异。似乎没有系统的制动可以阻止文化大革命退化为反智主义和破坏行为。它似乎也未能实现其消除不平等的愿望。尽管国家提供了广泛支持，但充其量只是半专业人士的赤脚兽医未能完全取代文革前的兽医。许多当地人甚至干部仍然对“老兽医”比对那些将要取代他们的半农半兽医更有信心。为了证明自己，赤脚兽医首先要

²⁵² Zu Xingguo, “Re’ ai xumu shiye, dang pinxia zhongnong fangxin de chijiao shouyi,” 2–3, 7.

²⁵³ 寇天德，“毛泽东思想照亮了我的心”，5-7页。

foremost needed to earn the trust of the other members of the collective. There was no other way to be accepted as a genuine veterinarian but to learn to heal animal patients.

Perhaps the barefoot veterinarian Zu Xingguo knew this. During his regular check-up tour in 1970, Zu stopped by the Third Production Team of Xialu Brigade where there was an ailing horse, which walked with a slight limp and had no fever. Zu diagnosed systemic rheumatism and treated accordingly for the next two days. Instead of improving, the horse developed a fever. At that point, Zu “became anxious that the sick horse may die and that he would inflict damage on the production team.” Lacking confidence, Zu spoke with the horse’s caretaker and offered to take the animal to the commune veterinary station where there were seasoned veterinarians with better resources. The caretaker was afraid that the commune station was too far for the sick horse to travel. “We want you to keep treating this horse. You should do it more boldly! Even if the animal dies, we won’t resent you for sure.”²⁵⁴

Zu felt encouraged and took responsibility for his patient’s health. He consulted other caretakers and draymen about the horse’s diet, recent work schedule, and the like. One of them informed Zu that the horse had tussled with another one in the stable a few days earlier. Zu, once again, closely examined the horse and spotted an inflamed bite wound on its right rear leg. After six days of dedicated treating and nursing for six days, the patient made a complete recovery. Although Zu Xingguo was not the best veterinarian, continuing to encounter and treat the animal patients in his brigade enabled him to develop and grow as an animal doctor. His neighbors noticed.²⁵⁵

²⁵⁴ Zu Xingguo, “Re’ai xumu shiye, dang pinxia zhongnong fangxin de chijiao shouyi,” 7–8.

²⁵⁵ Zu Xingguo, “Re’ai xumu shiye, dang pinxia zhongnong fangxin de chijiao shouyi,” 7–8.

首先需要赢得集体其他成员的信任。只有学会治疗动物病人，才能被接纳为真正的兽医。

也许赤脚兽医祖兴国深谙此道。1970年，祖兴国在例行巡诊时，来到下路大队第三生产队，那里有一匹生病的马，走路有点跛，但没有发烧。祖兴国诊断为全身性风湿病，并据此治疗了两天。然而，马的病情非但没有好转，反而发烧了。那时，祖兴国“开始担心这匹病马可能会死，那样他就会给生产队造成损失”。由于缺乏信心，祖兴国与马的饲养员交谈，提出将这匹动物带到公社兽医站，那里有经验丰富的兽医和更好的资源。饲养员担心公社兽医站太远，病马无法承受旅途。“我们希望你继续治疗这匹马。你应该更大胆地去做！即使动物死了，我们也绝不会怨恨你。”²⁵⁴

祖感到很受鼓舞，并对病马的健康负责。他向其他饲养员和马车夫咨询了马的饮食、最近的工作安排等。其中一人告诉祖，几天前这匹马在马厩里和另一匹马打过架。祖再次仔细检查了这匹马，发现它的右后腿上有一个发炎的咬伤。经过六天的精心治疗和护理，这匹病马完全康复了。尽管祖兴国不是最好的兽医，但不断接触和治疗他所在大队的动物病人，使他得以成长为一名动物医生。他的邻居们都注意到了这一点。²⁵⁵

²⁵⁴ Zu Xingguo, “Re’ai xumu shiye, dang pinxia zhongnong fangxin de chijiao shouyi,” 7–8.

²⁵⁵ 祖兴国，“热爱畜牧事业，当贫下中农放心的赤脚兽医”，7-8。

Similarly, an unnamed barefoot veterinarian in Chengdong Brigade, Chengguan Commune in Gaoping County, Shanxi Province was working near pigpens. One day, the veterinarian noticed that a sow that recently given birth to seven piglets had stopped eating and become emaciated. What was more serious, according to the doctor, was that she had stopped lactating, which would endanger the lives of her litter and the brigade’s hog production. The veterinarian had seen this symptom before. In the past, the veterinarian attempted to promote digestion with herbal remedies, but to no avail. The doctor also tried out a TCVM decoction which was known as an appetite stimulant and a multivitamin shot. “Both Chinese and Western means did not work,” and pigs were still not eating enough. After some time had passed without any treatment being successful, the animals went into estrus. As soon as they became pregnant, their appetites returned.²⁵⁶

Encountering another case of this mysterious “postnatal inappetence,” the barefoot veterinarian double-checked whether the sow had caught a cold, but her temperature was normal. Expecting that the next estrus cycle might solve the problem, the desperate doctor offered a bowl of seaweed soup to the sow, which she sniffed and then quickly consumed. After eating seaweed soup for the next few days, the lactation resumed. The barefoot animal doctor reported this to Gaoping County Veterinary Hospital. The director of the county hospital then conducted a clinical survey, which led veterinarians in communes and brigades in Gaoping to examine 40 similar cases. The local veterinarians concluded that an iodine deficiency caused the loss of appetite. In the final report by the county veterinary hospital, Chengdong Brigade’s barefoot veterinarian, although anonymized under the Cultural Revolution’s preference for socialist collectivism and avoidance of the glorification of anyone except Mao, was honored as a provider of the first case of the swine

²⁵⁶ Gaoping xian shouyiyuan 高平县兽医院, ed., *Zhongshouyi linchuang keji ziliao huibian* 中兽医临床科技资料汇编 (N.p., 1975), 3–5.

同样，山西省高平县城关公社城东大队一位不具名的赤脚兽医在猪圈附近工作。一天，这位兽医发现一头刚生了七只小猪的母猪停止进食，变得消瘦。更严重的是，据医生说，她停止了泌乳，这将危及她产下的小猪的生命和大队的生猪生产。这位兽医以前见过这种症状。过去，这位兽医曾尝试用草药促进消化，但无济于事。医生还尝试了一种中兽医汤剂，这种汤剂被认为是开胃药和多种维生素注射剂。“中西医结合都没有奏效”，猪仍然吃得不够。经过一段时间的治疗都没有成功后，这些动物进入了发情期。一旦它们怀孕，食欲就恢复了。²⁵⁶

遇到另一起这种神秘的“产后厌食症”病例，赤脚兽医再次检查了母猪是否感冒，但她的体温正常。绝望的医生预计下一个发情周期可能会解决问题，于是给母猪提供了一碗海带汤，母猪闻了闻，然后迅速喝光了。在接下来的几天里，母猪吃了海带汤后，泌乳恢复了。赤脚兽医将此事报告给了高平县兽医院。县医院院长随后进行了一项临床调查，这使得高平县各公社和大队的兽医检查了40个类似病例。当地兽医得出结论，碘缺乏导致了食欲不振。在高平县兽医院的最终报告中，城东大队的赤脚兽医，尽管在文化大革命时期，为了社会主义集体主义和避免除了毛泽东之外的任何人的荣耀而被匿名化，但她作为首例猪病病例的提供者而受到表彰

²⁵⁶ 高平县兽医院编，《中兽医临床科技资料汇编》（无出版地，1975），第3-5页。

postnatal inappetence and praised as a model worker who exemplified the strength of the barefoot animal doctor program.²⁵⁷

Some barefoot veterinarians gained recognition from even more advanced colleagues. Meng Shiqing was a barefoot sheep doctor in Shurigentai Brigade, Herisitai Commune, Plain Blue Banner of Yang League, Inner Mongolia. Meng was originally a youth who had been sent-down from the city and knew nothing about ovine acupuncture. While handling large numbers of sheep as part of his everyday labor in pastoral Inner Mongolia, he saw many sick animals. He was willing to help them, but there were few veterinarians in his small brigade. Therefore, Meng resolved to teach himself to treat sheep with acupuncture since he did not have enough Western medicine. He consulted textbooks on human and bovine acupuncture and got a sense of where the acupuncture points were and the remedial effects of each. Based on this knowledge, Meng tried to stick a needle into the most likely acupuncture points on a sheep’s body.²⁵⁸

An acupunctural point on nose of human and cattle, *renzhong* (人中), for example, is for healing a stomachache or cold. Meng repeatedly experimented with sheep who had caught cold. He eventually figured out that a sheep’s *renzhong* is at the upper center of the nose where wooly and bald parts meet (Figure 8). Within a couple of years, he had completed his own therapeutics and became famous in his collective. Meng’s reputation spread to veterinary scientists in the Inner Mongolia Academy of Agriculture in 1973. They invited him to come to the academy to demonstrate his skills. Once Meng’s therapeutics seemed to work, the scientists rigorously verified their efficacy through autopsy. At that point, the scientists decided to codify Meng’s acupunctural knowledge and circulate

²⁵⁷ Gaoping xian shouyiyuan, ed., *Zhongshouyi linchuang keji ziliao huibian*, 3–5.

²⁵⁸ Meng Shiqing 孟世清 and Neimenggu nongmu xueyuan shouyixi 内蒙古农牧学院兽医系, *Zhenjin zhiliao yang bing de jingyan zhengli* 针灸治疗羊病的经验整理 (N.p.: Neimenggu nongmu xueyuan shouyixi, 1979), 1–2.

²⁵⁷ Gao平县兽医院, 编, 中兽医临床科技资料汇编, 3–5.

²⁵⁸ 孟世清和内蒙古农牧学院兽医系, 针灸治疗羊病的经验整理 (无出版地: 内蒙古农牧学院兽医系, 1979), 1-2页。e

产后食欲不振，并被誉为赤脚兽医计划的模范工作者。²⁵⁷

一些赤脚兽医甚至得到了更高级同事的认可。孟世清是内蒙古镶黄旗赫里西台公社舒日根台大队的一名赤脚羊医。孟原本是一名从城市下放的青年，对羊针灸一无所知。在内蒙古牧区日常劳动中，他接触了大量的羊，看到了许多生病的动物。他愿意帮助它们，但他的小队里兽医很少。因此，孟决心自学用针灸治疗羊，因为他没有足够的西药。他查阅了人体和牛针灸的教科书，了解了穴位的位置和每个穴位的治疗效果。根据这些知识，孟尝试将针扎入羊身上最可能的穴位。²⁵⁸

例如，人或牛鼻子上 的一个穴位，人中，可以治疗胃痛或感冒。孟反复用感冒的羊做实验。他最终发现羊的人中位于鼻子上方中心，即有毛和无毛部分的交界处（图8）。几年之内，他完成了自己的疗法，并在他的集体中声名鹊起。1973年，孟的名声传到了内蒙古农业科学院的兽医科学家那里。他们邀请他到科学院展示他的技能。孟的疗法似乎奏效后，科学家们通过尸检严格验证了其疗效。那时，科学家们决定将孟的针灸知识编纂成典并推广。

it to veterinarians in entire Inner Mongolia. Although Meng was found to have made a few errors which were corrected based on the materials published by the Institute of TCVM, he was hailed as an exemplary barefoot veterinarian and as a symbol of mass science.²⁵⁹

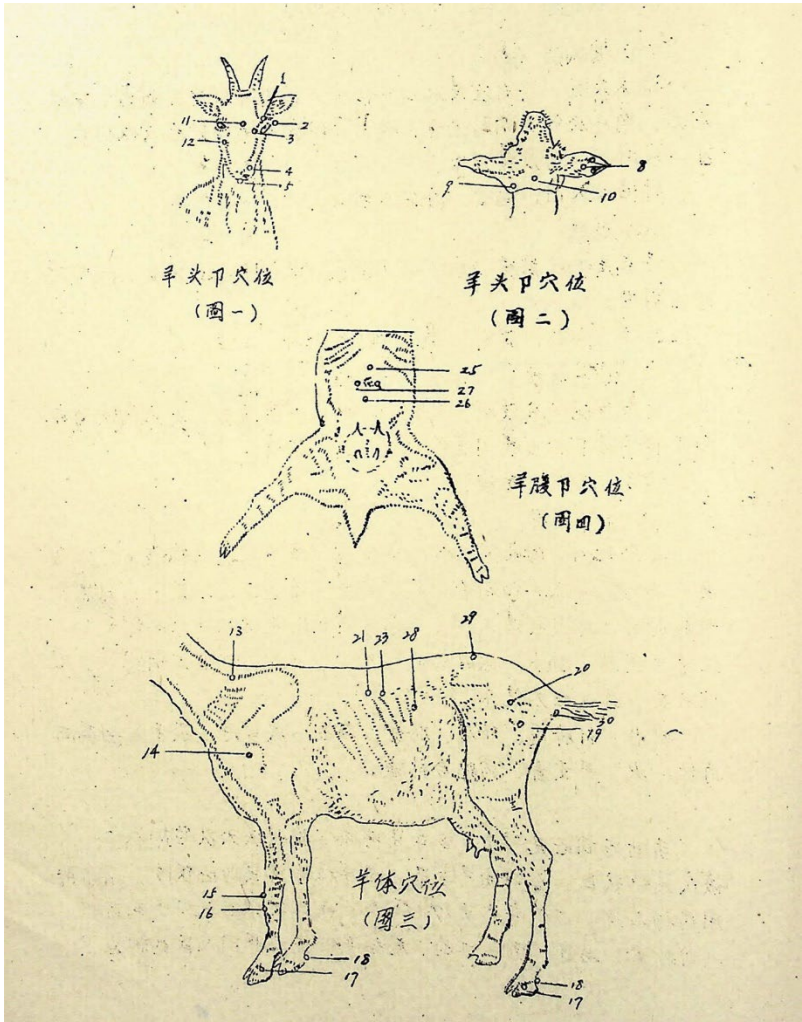


Figure 8. Meng Shiqing’s map of ovine acupunctural points. Source: Meng Shiqing 孟世清 and Neimenggu nongmu xueyuan shouyixi 内蒙古农牧学院兽医系, *Zhenjiu zhiliao yang bing de jingyan zhengli* 针灸治疗羊病的经验整理 (N.p.: Neimenggu nongmu xueyuan shouyixi, 1979), 8.

²⁵⁹ Meng Shiqing and Neimenggu nongmu xueyuan shouyixi, *Zhenjiu zhiliao yang bing de jingyan zhengli*, 1–2 and 9–10.

它被推广给内蒙古全体兽医。尽管孟被发现犯了一些错误，这些错误根据中医兽医研究所发布的材料得到了纠正，但他仍被誉为模范赤脚兽医和群众科学的象征。²⁵⁹

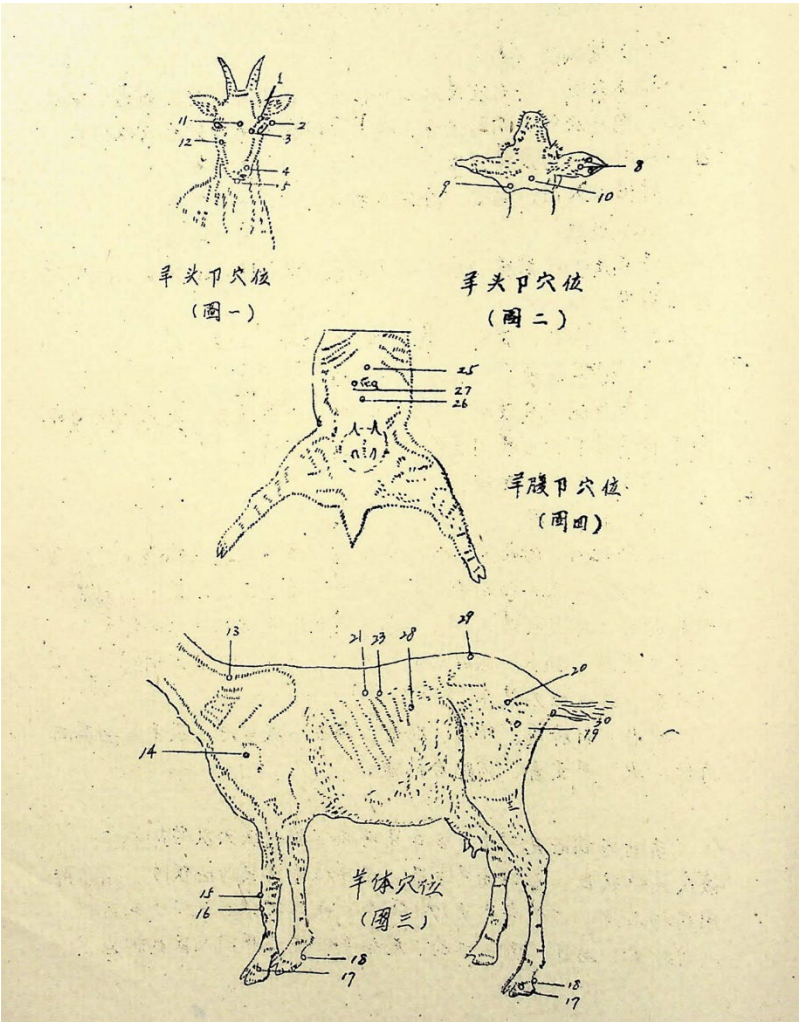


图8。孟世清的绵羊针灸穴位图。来源：孟世清和内蒙古农牧学院兽医系，《针灸治疗羊病的经验整理》（无出版地：内蒙古农牧学院兽医系，1979），第8页。

²⁵⁹ 孟世清和内蒙古农牧学院兽医系，《针灸治疗羊病的经验整理》，第1-2页和第9-10页。

Both barefoot veterinarians and many other types of animal healers were willing to adhere to the rule of protecting and treating animals. Among them were grassroots veterinarians from the before the era of the Cultural Revolution, higher-level veterinarians in suburban areas, and ethnic minority veterinary experts. These animal doctors assumed a range of political burdens in the anti-intellectual climate of the Cultural Revolution. For the sake of their survival, veterinarians who were regarded as the politically suspicious by the state needed to show the ethos of the Cultural Revolution. While not challenging the cause of subverting the hierarchy, they appropriated the voluntarist and practice-centric approach to immerse themselves in the clinical work. They demonstrated a strong commitment to trial and error on filthy farms, just like the old man who moved a mountain. As long as these veterinarians held their ground, local peasants would consult them as experts who could restore their animals to health, regardless of how rigorously “politics in command” silenced “work in command” and the “expert line” was rejected by the “mass line” in official discourse. In other words, after the radical, anti-intellectual first years of the Cultural Revolution, now the social meaning of Maoist voluntarism changed from the idea that would enable the replacement of “old veterinarians” by barefoot veterinarians to a politically safe excuse to focus on the clinical work without having to be seen as a supporter of either meritocracy or anti-intellectualism.

Kou Tiande, the “old” veterinarian, proved his worth by throwing himself into clinical practice. In December 1966, in addition to participating in field labor, he saw more than 40 livestock per day. Peasants and caretakers were satisfied with him because he had no problem curing common swine and bovine disorders. The one veterinary service that Kou could not provide was castrating boars. To “serve the people,” he decided to learn how. He contacted and shadowed anyone who could teach him. He observed 70 to 80 times and then practiced it himself a few times. When Kou

赤脚兽医和许多其他类型的动物治疗师都愿意遵守保护和治疗动物的原则。其中包括文化大革命前的基层兽医、郊区的高级兽医以及少数民族兽医专家。这些动物医生在文化大革命的反智氛围中承担了一系列政治负担。为了生存，被国家视为政治上可疑的兽医需要表现出文化大革命的精神。他们没有挑战颠覆等级制度的事业，而是采用了一种志愿主义和以实践为中心的方法，沉浸在临床工作中。他们表现出对肮脏农场上反复试验的强烈承诺，就像那个移山的老人一样。只要这些兽医坚守阵地，当地农民就会向他们咨询，将他们视为能够恢复动物健康的专家，无论“政治挂帅”在官方话语中如何严格地压制“业务挂帅”，以及“专家路线”如何被“群众路线”所否定。换句话说，在文化大革命激进的反智主义初期之后，毛泽东思想中的志愿主义的社会意义从“赤脚兽医取代‘老兽医’”的理念，转变为一个政治上安全的借口，即专注于临床工作，而无需被视为精英主义或反智主义的支持者。

兽医寇天德通过投身临床实践证明了自己的价值。1966年12月，他除了参加田间劳动外，每天还诊治40多头牲畜。农民和饲养员对他很满意，因为他能治愈常见的猪牛疾病。寇天德唯一不能提供的兽医服务是阉割公猪。为了“为人民服务”，他决定学习如何阉割。他联系并跟随任何能教他的人。他观察了七八十次，然后自己练习了几次。当寇天德

came across a pig which needed to be desexed, he volunteered to do it. The pig’s caretaker turned him down, saying that “you don’t know how to do it, right? [...] If you haphazardly castrate the pig, you only torment it.” Yet Kou believed that he could do it. “I have learned it. Let me give it a shot today.” The procedure went well. Before long, the news that Kou was able to neuter boars was in the air, and people from the commune and affiliated brigades called on him to perform the service. Over the next year and a half, he castrated more than 1,000 pigs in his commune.²⁶⁰

In 1971, even highly advanced veterinary scientists in Beijing Municipal Veterinary Hospital joined countless barefoot animal doctors, avoiding the “expert line” and championing indefatigable practice in a voluntarist fashion. What the 16 veterinarians in the hospital did was to treat critical animal patients transferred from communes and brigades on the outskirts of the capital. They must have been aware that if they identified themselves as high-profile specialists, they would face severe criticism. Therefore, when they were required to summarize their work experiences in front of the audience of poor and lower-middle peasants, all that they talked about was how Maoism helped them master the surgery for equine hernia.²⁶¹

Equine hernia was hard to treat. Before the Cultural Revolution, the Beijing veterinarians used to transfer all animals diagnosed with this refractory disease to the China Agricultural University Veterinary Hospital. As the university hospital had been relocated in the midst of the political turmoil, they had to take on the cases. At first, some of the 16 veterinarians “did not feel up to treating it well” and suggested inviting an experienced veterinarian from outside. But others,

²⁶⁰ Kou Tiande, “Mao Zedong sixiang zhaoliang le wo de xin,” 4–7.

²⁶¹ Beijing shi shouyiyuan 北京市兽医院, “Xuexi bianzheng weiwufa, Gan chuang jinqu po nanguan” 学习辩证唯物法, 敢闯禁区破难关 (May 1971), Beijing shi nonglin xitong huo xue huo yong Mao Zedong sixiang jiji fenzi daibiao dahui dianxing cailiao 北京市农林系统活学活用毛泽东思想积极分子代表大会典型材料, Unpublished document, 1–2.

遇到一头需要阉割的猪，他自告奋勇地去做了。猪的饲养员拒绝了他，说：“你不会做，对吧？[……] 如果你胡乱阉割猪，只会折磨它。”但寇相信他能做到。“我学过。今天让我试一试。”手术很顺利。不久，寇能阉割公猪的消息传开了，公社和附属大队的人都请他提供服务。在接下来的一年半里，他在公社里阉割了1000多头猪。²⁶⁰

1971年，即使是北京市兽医院的高级兽医科学家也加入了无数赤脚兽医的行列，避开“专家路线”，以志愿者的精神倡导不懈实践。这家医院的16位兽医所做的是治疗从首都郊区公社和大队转来的危重动物病人。他们一定意识到，如果他们自称是高级专家，就会面临严厉的批评。因此，当他们被要求在贫下中农听众面前总结工作经验时，他们所谈论的都是毛泽东思想如何帮助他们掌握了马疝气手术。²⁶¹

马疝气很难治疗。文化大革命前，北京的兽医们常常将所有被诊断出患有这种难治之症的动物转送至中国农业大学兽医院。由于该大学医院在政治动荡中已迁址，他们不得不接手这些病例。起初，16位兽医中的一些人“觉得自己无法很好地治疗”，并建议邀请一位经验丰富的外部兽医。但其他人，

²⁶⁰ 寇天德，《毛泽东思想照亮了我的心》，第4-7页。

²⁶¹ 北京市兽医院，《学习辩证唯物法，敢闯禁区破难关》（1971年5月），北京市农林系统活学活用毛泽东思想积极分子代表大会典型材料，未出版文献，第1-2页。

recalling Mao’s messages that “the question of the human [will or determination] is the most fundamental issue,” insisted that they solve the problem by themselves. “Although we lack experience, we can increase it only through practice. Although our technical level is not enough, it can be enhanced through working.” The surgery was not easy, though. The Beijing veterinarians repeatedly practiced it for quite a long time. It still “sometimes worked yet the number of deaths were not small.” At this moment, according to the veterinarians, they realized that they had only valued the “external factors” at the expense of the “internal factors.” The former meant the veterinarians’ surgical mastery, while the latter referred to their faith in animals’ own will and ability to recover from the disease. Once they firmly grasped both factors and continued practicing, the veterinarians could report that they had made a breakthrough and save hundreds of horses. Now voluntarist practice, along with a sufficient number of sick animals, became magic words which made it possible for veterinarians to continue to do their job.²⁶²

The expertise and historical experiences of ethnic minority veterinarians in broader border frontier areas are largely unknown. Some minorities such as Mongolians and Tibetans have been preserving distinct traditional medical theories and practices, known respectively as Traditional Mongolian Medicine (蒙医) and Traditional Tibetan Medicine (藏医). Both are different from what is known as Traditional Chinese Medicine (中医).²⁶³ Given that horses, sheep, goats, and other animals, are indispensable for the nomadic minorities, it is not surprising that they have developed

²⁶² Beijing shi shouyiyuan, “Xuexi bianzheng weiwufa, Gan chuang jinqu po nanguan,” 2–5.

²⁶³ On traditional Tibetan medicine, see Geoffrey Samuel, “Introduction: Medicine and Healing in Tibetan Societies,” *East Asian Science, Technology, and Society* 7.3 (September 2013): 335–351.

回顾毛泽东 “人的[意志或决心] 问题是最根本的问题” 的指示，坚持他们自己解决问题。 “虽然我们缺乏经验，但只有通过实践才能增加经验。虽然我们的技术水平不够，但可以通过工作来提高。” 然而，手术并不容易。北京的兽医们反复练习了很长时间。它仍然 “有时有效，但死亡人数并不少”。此时，据兽医们说，他们意识到他们只重视 “外部因素”，而牺牲了 “内部因素”。前者指兽医的手术掌握能力，后者指他们对动物自身意志和从疾病中恢复能力的信心。一旦他们牢牢掌握了这两个因素并继续实践，兽医们就可以报告他们取得了突破，并挽救了数百匹马。现在，志愿主义实践，以及足够数量的病畜，成了让兽医们能够继续工作的魔咒。²⁶²

少数民族兽医在边境地区所具备的专业知识和历史经验鲜为人知。一些少数民族，如蒙古族和藏族，一直保留着独特的传统医学理论和实践，分别被称为蒙医和藏医。这两种医学都与我们所熟知的中医不同。²⁶³ 鉴于马、绵羊、山羊和其他动物对游牧民族来说不可或缺，他们发展出

²⁶² 北京市兽医员，《学习辩证唯物法，敢闯禁区破难关》，2-5。

²⁶³ 关于藏族传统医学，参见 Geoffrey Samuel, “Introduction: Medicine and Healing in Tibetan Societies,” *East Asian Science, Technology, and Society* 7.3 (2013 年 9 月): 335–351。

their own traditional systems of veterinary knowledge.²⁶⁴ However, it could be dangerous to claim such a uniqueness in the Cultural Revolution era. Ethnocentrism was easily seen as antithetical to the universality of class struggle and the ideal of socialist internationalism. Chinese Muslims from Linxia Hui Autonomous Prefecture, Gansu Province that I introduce here were among the minority folk veterinarians whose status was precarious. And they, like Kou Tiande and other high-profile veterinarians in Beijing, wanted to convince outsiders that they were working hard in accordance with practice-based voluntarism.²⁶⁵

Ma Haifeng was a Hui Muslim veterinarian working in the veterinary station of Sanshilipu Commune, Hezheng County. On his clinical tour to affiliated brigades and teams, Ma encountered a seven-year-old cow. Villagers informed him that the cow had given birth to a calf before becoming infertile four years previously. About that time, the cow started to have xeroderma, losing her coat. Ma was confused. Can xeroderma cause sterility? He doubted it. Therefore, he administered several local medicinal herbs and TCM decoctions that were expected to be effective in treating infertility. After a certain amount of time, the mating season passed, and it became obvious that all therapies failed. He had no idea what to do next. But livestock sterility was fatal to communal self-sufficiency and could not be ignored. As a commune veterinarian, Ma had to emulate Chairman Mao and the old man. He became attentive to the cow’s skin disease, although he had no assurance that curing xeroderma would return her fecundity. According to Ma’s veterinary expertise and clinical experience, dryness on the surface can be caused by excessive moistness inside the body. Hence, to

²⁶⁴ For example, see Yu Chuan 于船, “Xizang shouyixue fazhan shilüe” 西藏兽医学发展史略, in *Yu Chuan Wenji* 于船文集 (Beijing: Zhongguo nongye daxue chubanshe, 2003), 32–34.

²⁶⁵ Linxia Huizu zizhizhou xumu shouyi gongzuozhan 临夏回族自治州畜牧兽医工作站, “Gansu sheng Linxia Huizu zizhizhou zhongshouyi jingyan jieshao” 甘肃省临夏回族自治州中兽医经验介绍 (April 1976), Unpublished document.

他们自己的传统兽医知识体系。²⁶⁴ 然而，在文化大革命时期，声称这种独特性可能是危险的。民族中心主义很容易被视为与阶级斗争的普遍性和社会主义国际主义理想相悖。我在这里介绍的来自甘肃省临夏回族自治州的中国穆斯林，是少数地位不稳定的民间兽医之一。他们和寇天德以及北京其他知名兽医一样，希望让外界相信他们正在努力工作，遵循基于实践的自愿主义。²⁶⁵

马海峰是河政县三十里铺公社兽医站的一名回族兽医。在对附属大队和生产队进行临床巡诊时，马海峰遇到了一头七岁大的母牛。村民告诉他，这头母牛在四年前生下一头小牛后就变得不育了。大约从那时起，这头母牛开始出现皮肤干燥，毛发脱落。马海峰感到困惑。皮肤干燥会导致不育吗？他对此表示怀疑。因此，他给母牛施用了几种当地草药和中药汤剂，希望能有效治疗不育症。一段时间后，发情期过去了，很明显所有治疗都失败了。他不知道接下来该怎么办。但是牲畜不育对公社的自给自足是致命的，不能忽视。作为一名公社兽医，马海峰必须效仿毛主席和那位老人。他开始关注母牛的皮肤病，尽管他不能保证治愈皮肤干燥就能恢复其生育能力。根据马海峰的兽医专业知识和临床经验，体表干燥可能是由体内过度潮湿引起的。因此，为了

²⁶⁴ 例如，参见于船，“西藏兽医学发展史略”，载于船文集（北京：中国农业大学出版社，2003），第32-34页。

²⁶⁵ Linxia Huizu zizhizhou xumu shouyi gongzuozhan 临夏回族自治州畜牧兽医工作站, “Gansu sheng Linxia Huizu zizhizhou zhongshouyi jingyan jieshao” 甘肃省临夏回族自治州中兽医经验介绍 (April 1976), Unpublished document.

treat xeroderma, he prescribed traditional warming therapies. In a few weeks, the areas affected by xeroderma started to grow fur. Ma was excited but careful enough to observe whether this bodily reaction of the cow would lead to another pregnancy. It eventually did. Villagers and caretakers of the cow must have been delighted to have such a skilled animal doctor like Ma in their collective.²⁶⁶

It might not be a stretch to posit that establishing and practicing the clinical competence is a universal veterinary work ethic, even in Mao-era China. But in the place where meritocratic inequality should not be tolerated and professional competence faced ambivalent reactions, this must have been framed in terms of Maoist voluntarism. This was where the Cultural Revolution in the local veterinary spheres eventually arrived. It is also worth emphasizing that animals were present in this process of addressing the question of veterinary authority. Given that the bodies and behaviors of ailing animals determined the success or failure of treatments, the consolidation of the local veterinarians’ knowledge and social status depended on animals—such as the sow who sniffed at seaweed soup and the sheep with a cold that accepted acupunctural treatment.

Conclusion

This chapter traces how local veterinarians struggled to stabilize their social status and establish their professional authority in their collectives in the 1960s and 70s. They had to face up the complicated realities in which conflicting norms and ideals such elitism, anti-intellectualism, meritocracy, and egalitarianism were intertwined. Looking at how the veterinarians came to terms with such

²⁶⁶ Linxia Huizu zizhizhou xumu shouyi gongzuozhan, “Gansu sheng Linxia Huizu zizhizhou zhongshouyi jingyan jieshao,” 4–5.

治疗皮肤干燥症，他开了传统的温热疗法。几周后，受皮肤干燥症影响的区域开始长出毛发。马很高兴，但又足够小心，观察奶牛的这种身体反应是否会导致再次怀孕。最终确实如此。村民和奶牛的饲养员一定很高兴他们的集体里有马这样一位技艺高超的兽医。

266

可以这样说，即使在毛泽东时代的中国，建立和实践临床能力也是一种普遍的兽医职业道德。但在一个不应容忍精英不平等、专业能力面临矛盾反应的地方，这一定是以毛泽东主义的唯意志论来构建的。这就是地方兽医领域文化大革命最终的走向。同样值得强调的是，在解决兽医权威问题的过程中，动物也参与其中。鉴于患病动物的身体和行为决定了治疗的成功或失败，地方兽医知识和社會地位的巩固依赖于动物——例如嗅海带汤的母猪和接受针灸治疗的感冒绵羊。

结论

本章追溯了当地兽医在 20 世纪 60 年代和 70 年代如何努力稳定其社会地位并在其集体中建立专业权威。他们不得不面对复杂的现实，其中精英主义、反智主义、英才教育和平等主义等相互冲突的规范和理想交织在一起。考察兽医如何应对这些

²⁶⁶ 临夏回族自治州畜牧兽医工作站， 《甘肃省临夏回族自治州中兽医经验介绍》， 4-5。

complexity can open a window onto a larger historical trajectory of Maoist China’s approach to the question of socio-intellectual disparities.

The Great Famine was not the only legacy of agricultural collectivization and the Great Leap Forward. They also resulted in mass science, turning thousands of ordinary peasants into full-time animal doctors who were supposed to work for communes and brigades in the late 1950s and early 1960s. Although clearly inspired by Maoist egalitarianism, the local veterinary education, either apprenticeships or governmental short trainings, was meritocratic if not elitist. This was not particularly a product of the 1960s readjustment policy, notwithstanding the fact that it indeed became more outspoken in the period, but an inherent characteristic of the Maoist mass scientific veterinary education. The socio-intellectual capital was disproportionately distributed among veterinarians in the first 17 years of the PRC.

The Cultural Revolution in the local veterinary realm can be construed as an anti-intellectual attempt to overturn the socio-intellectual hierarchy and eliminate the disparities in expertise, social fame, and economic status. The advocates of the Cultural Revolution argued that the ordinary masses of poor and lower-middle peasants should take back the “great power in veterinary medicine” from a few “capitalistic,” “revisionist,” and “counterrevolutionary” animal doctors who followed the “expert line” instead of the “mass line” and placed “work competence in command” and “economics in command” above “politics in command.” Full-time veterinary positions in communal and brigade-level veterinary stations were abolished in the belief that they could be replaced by part-time semi-amateurs, or barefoot veterinarians. Even if such replacement could not take place right away, it would happen in the end, as long as barefoot veterinarians adhered to the Maoist voluntarist idea that constant practice on the clinical ground with an iron will must pay off.

复杂性可以为我们了解毛泽东时代中国处理社会智力差异问题的历史轨迹打开一扇窗户。

大饥荒并非农业集体化和大跃进的唯一遗产。它们还导致了大众科学的兴起，将成千上万的普通农民转变为全职兽医，这些兽医本应在1950年代末和1960年代初为公社和生产队工作。尽管明显受到毛泽东平均主义的启发，但地方兽医教育，无论是学徒制还是政府短期培训，都是精英主义的，即使不是精英主义的。这并非1960年代调整政策的产物，尽管在此期间它确实变得更加突出，但它是毛泽东时代大众科学兽医教育的内在特征。在中华人民共和国成立的头17年里，社会智力资本在兽医中分配不均。

地方兽医领域的文化大革命可以被解读为一种反智主义的尝试，旨在推翻社会知识等级制度，消除专业知识、社会声望和经济地位方面的差异。文化大革命的倡导者认为，贫下中农的普通群众应该从少数“资本主义”、“修正主义”和“反革命”的兽医手中夺回“兽医大权”，这些兽医奉行“专家路线”而非“群众路线”，将“业务挂帅”和“经济挂帅”置于“政治挂帅”之上。公社和生产队级别的兽医站的全职兽医职位被废除，因为人们认为他们可以被兼职的半专业人员或赤脚兽医取代。即使这种取代不能立即发生，只要赤脚兽医坚持毛泽东的唯意志论思想，即在临床实践中以钢铁般的意志不断实践必将取得成功，那么最终也会实现。

It turned out that the political expediency and violence of the Cultural Revolution was by no means a cure-all for socio-intellectual inequalities. Even under high Maoism, to be recognized as a decent veterinarian in one's community where many livestock suffered from diverse illnesses, the healer must have shown clinical competence and gained the trust of other peasants. This hard reality spared the lives of many talented veterinary practitioners trained in the pre-Cultural Revolution era and enabled their coexistence with barefoot animal doctors. In the latter years of the Cultural Revolution when the resentment of the underprivileged lost its momentum and massive destruction had brought about disillusionment, Maoist voluntarism unexpectedly provided both “old” and barefoot veterinarians with a politically safe excuse. With it, all types of veterinarians could immerse themselves in their clinical work.

Animals were integral in this eventual compromise made in the middle ground between meritocracy and anti-intellectualism. Animal patients showed their agency when they chose treatments and therapies. If living animals determined the success or failure of every veterinary treatment, veterinarians' socio-intellectual status and authority, based on the accumulation of such clinical interactions, depended on them, too. Animal patients themselves might not be able to heal the frustration of doctors in Maoist China. However, the animal healers, despite all their dissatisfaction, would have found a way to affirm their existence in more-than-people's commune when they tried their best to save the lives of animals who were in pain right in front of them.

事实证明，文化大革命的政治权宜之计和暴力绝非解决社会智力不平等的万灵药。即使在毛泽东思想盛行时期，要想在一个牲畜饱受各种疾病困扰的社区中被公认为一名合格的兽医，治疗者也必须表现出临床能力并赢得其他农民的信任。这一严峻的现实挽救了许多在文革前受训的有才华的兽医的生命，并使他们能够与赤脚兽医共存。在文革后期，当弱势群体的怨恨失去动力，大规模破坏带来幻灭时，毛泽东的自愿主义出人意料地为“老”兽医和赤脚兽医提供了一个政治上安全的借口。有了它，所有类型的兽医都可以沉浸在他们的临床工作中。

动物是这种最终妥协中不可或缺的一部分，这种妥协是在精英管理和反智主义之间的中间地带达成的。动物病患在选择治疗方案时表现出了它们的主体性。如果活着的动物决定了每一次兽医治疗的成功或失败，那么兽医的社会智力地位和权威（基于此类临床互动的积累）也取决于它们。动物病患本身可能无法治愈毛泽东时代中国医生的挫败感。然而，这些动物治疗者，尽管有种种不满，当他们尽力挽救眼前痛苦的动物的生命时，他们会在超越人民公社的范围内找到一种方式来肯定自己的存在。

Chapter Four:
The Ideology of Care

Introduction

In 1965, Zhu Yulan found herself standing in bovine droppings. As an animal caretaker, she was responsible for the collectively owned livestock in her production team in Yucheng County, Henan Province. The cattle shed was literally a slurry of muck. As the “stench stung [her] nose,” she could not easily wade through the excrement and secretly thought that it was definitely “not a lifetime job.” But she was surrounded by other caretakers and poor peasants whose “whole bodies were covered in cow dung.” Reading these colleagues’ countenance and recalling Chairman Mao’s teaching of “having a firm will and being fearless in self-sacrifice,” Zhu needed to be willing to “roll in the waste and thereby cultivate the red mind.”²⁶⁷

Zhu Yulan’s story appeared on the *People’s Daily*, China’s largest state-controlled newspaper, on March 7, 1969. Although this story contained unrealistic advice—the cowshed must be clean, and excrement removed at once—and perhaps is not even compelling as propaganda, her resolution to cultivate the “red mind” (红心) was hardly unique. Thousands of similar stories on Maoist animal caretakers were circulated in the preceding and following periods. Most of them are unmistakably propagandistic and didactic. But in light of what we know about rural China under Mao, this ideological work on animal care seems counterintuitive and even unnecessary. Were animals even able to survive agricultural collectivization in the mid-1950s? Or was not it the case that almost all livestock perished in the Great Leap Famine and the Cultural Revolution? What was the purpose of

²⁶⁷ “Zai Mao Zedong sixiang buyu xia zhuzhuang chengzhang: Ji huixiang zhishi qingnian nü siyangyuan Zhu Yulan” 在毛泽东思想哺育下茁壮成长: 记回乡知识青年女饲养员朱玉兰, *Renmin Ribao* 人民日报 (March 7, 1969).

第四章：
The Ideology of Care

导言

1965年，朱玉兰发现自己站在牛粪中。作为一名动物饲养员，她负责河南省虞城县生产队集体所有的牲畜。牛棚里简直是一片泥泞。当“臭味刺痛[她]的鼻子”时，她无法轻易地趟过排泄物，心里暗想这绝对“不是一份终身职业”。但她周围都是其他饲养员和贫农，他们“全身都沾满了牛粪”。朱玉兰看着这些同事的神情，回想起毛主席“下定决心，不怕牺牲”的教导，她必须愿意“在粪便中打滚，从而培养红心”。²⁶⁷

朱玉兰的故事刊登在1969年3月7日中国最大的官方报纸《人民日报》上。尽管这个故事包含不切实际的建议——牛棚必须干净，粪便必须立即清除——而且作为宣传可能也缺乏说服力，但她培养“红心”的决心绝非独一无二。在此之前和之后，成千上万个关于毛泽东时代动物饲养员的类似故事广为流传。其中大多数都带有明显的宣传和说教性质。但鉴于我们对毛泽东时代中国农村的了解，这种关于动物饲养的思想工作似乎是反直觉的，甚至是多余的。动物在20世纪50年代中期的农业集体化中还能存活下来吗？或者说，是不是几乎所有牲畜都在大跃进饥荒和文化大革命中灭绝了？这样做的目的是什么？

²⁶⁷ “在毛泽东思想哺育下茁壮成长：记回乡知识青年女饲养员朱玉兰”，《人民日报》（1969年3月7日）。

all this propaganda? At least, they may remind scholars of modern China that caretakers and nonhuman animals in Maoist China have received little attention, despite the possibility that the human-animal relationship was an indelible component of rural communities.



Figure 9. Adolescent caretakers and animals in Changning Commune, Yuci County, Shanxi Province, 1960. Source: Author’s Collection.

This chapter examines the workings of Maoist ideology in animal care by analyzing propagandistic sources on “model caretakers”—especially, unpublished memoirs and self-criticism documents written by individual caretakers, articles in the *People’s Daily* (人民日报), and archival materials from Beijing and Tianjin—as well as oral historical data. I argue that animals were integral to the state’s imperative to build a socialist collective economy and to impose moral transformations on rural residents. In the Chinese Communist Party’s (CCP) quest to form communities based on

所有这些宣传？至少，它们可以提醒现代中国的学者，毛泽东时代的中国，饲养员和非人类动物很少受到关注，尽管人与动物的关系可能是农村社区不可磨灭的组成部分。



图 9。 1960 年山西省榆次县长宁公社的青少年饲养员和动物。来源：作者收藏。

本章通过分析关于“模范饲养员”的宣传资料——特别是饲养员个人撰写的未发表的回忆录和自我批评文件、《人民日报》上的文章以及来自北京和天津的档案材料——以及口述历史资料，考察了毛泽东思想在动物护理中的运作方式。我认为，动物对于国家建立社会主义集体经济和对农村居民进行道德改造的必要性至关重要。在中国共产党（CCP）建立基于

shared ownership and public responsibility, animals were framed as the common property of collectives. And by taking good care of farm animals, caretakers (and individual peasants) were expected to overcome individual selfishness and serve the greater common good.

Collectivization, Animals, and Caretakers in the Mid-1950s

The socialist revolution in rural China manifested itself as a series of radical socioeconomic policies, mainly land reforms in 1949–1953 followed by agricultural collectivization between 1953 and 1985. The CCP destroyed the landlord class and boldly redistributed land ownership. The majority of poor Chinese peasants came to have a sense of possessing their own lands, houses, and animals for the first time.²⁶⁸ But not long after, the Communists deprived the peasants of their private property and collectivized the country’s rural economy. The party-state attempted to maximize its capacity to control and extract an agrarian surplus for the sake of primitive accumulation to support rapid industrialization.²⁶⁹ The revolutionaries launched an ambitious social experiment to build a collective economy based on shared ownership and collective responsibility. Gradually incorporated into “production teams (elementary cooperatives),” “production brigades (advanced cooperatives),” or “people’s communes,” peasants were now required to synchronize individual, cooperative, and national interests.

²⁶⁸ Yi Wu, *Negotiating Rural Land Ownership in Southwest China: State, Village, Family* (Honolulu: University of Hawaii Press, 2016); Matthew Noellert, *Power Over Property: The Political Economy of Communist Land Reform in China* (Ann Arbor: University of Michigan Press, 2020).

²⁶⁹ Schurmann Franz, *Ideology and Organization in Communist China* (Berkeley: University of California Press, 1966); Ho-fung Hung, *The China Boom: Why China Will Not Rule the World* (New York: Columbia University Press, 2015); Tiejun Wen 温铁军, *Ten Crises: The Political Economy of China’s Development, 1949-2020* (Singapore: Palgrave Macmillan, 2021).

在共同所有权和公共责任的框架下，动物被视为集体的共同财产。通过悉心照料农场动物，饲养员（和个体农民）被期望克服个人私利，服务于更大的公共利益。

20世纪50年代中期的集体化、动物与饲养员

中国农村的社会主义革命表现为一系列激进的社会经济政策，主要是1949年至1953年的土地改革，随后是1953年至1985年的农业集体化。中国共产党摧毁了地主阶级，大胆地重新分配了土地所有权。大多数贫苦的中国农民第一次有了拥有自己的土地、房屋和动物的感觉。²⁶⁸但不久之后，共产党剥夺了农民的私有财产，并对全国农村经济进行了集体化。党国试图最大限度地控制和榨取农业剩余，以实现原始积累，支持快速工业化。²⁶⁹革命者发起了一项雄心勃勃的社会实验，旨在建立一个基于共同所有权和集体责任的集体经济。农民们逐渐被纳入“生产队（初级社）”、“生产大队（高级社）”或“人民公社”，现在被要求将个人、合作社和国家利益同步起来。

²⁶⁸ 易武，《中国西南农村土地所有权谈判：国家、乡村、家庭》（檀香山：夏威夷大学出版社，2016）；马修·诺勒特，《财产权力：中国共产主义土地改革的政治经济学》（安娜堡：密歇根大学出版社，2020）。

²⁶⁹ 弗朗茨·舒尔曼，《共产主义中国的意识形态与组织》（伯克利：加州大学出版社，1966）；洪福丰，《中国繁荣：中国为何不会统治世界》（纽约：哥伦比亚大学出版社，2015）；温铁军，《十大危机：中国发展的政治经济学，1949-2020》（新加坡：帕尔格雷夫·麦克米伦，2021）。

Both human peasants and nonhuman animals underwent collectivization. Draft animals—oxen, donkeys, horses, and mules—and meat animals—pigs, sheep, chickens, ducks, geese, and rabbits—had always been an integral part of everyday life in rural China. In the early stage of collectivization, the CCP encouraged animal owners to “voluntarily” join and donate animals to cooperatives. In some cases, local Communist cadres implemented a more cautious and transitional means to collectivize animals that was called “privately owning and feeding, collectively borrowing and using.” Either way, the animals were to be expropriated as cooperatives’ “public property.”²⁷⁰

The collectivization of animals, however, did not go smoothly. It was not because the Communists were unable to overcome the opposition from landlords’ militant resistance and rich farmers’ wait-and-see attitude. The coercive and violent land reform and collectivization uprooted pre-revolutionary private ownership of farm animals. From the perspective of the state, the real challenge came with the “absence of experience and institution.” For instance, all 85 animals in an elementary cooperative in Boye County, Hebei were already concentrated in 1954. Yet the author of this local report argued that no one knew the proper way to run collective animal caretaking. Peasants fed and kept animals that they did not own by rule of thumb. They also “indiscreetly beat and overworked” animals.²⁷¹ In 1955, a *People’s Daily* editor noted that “everyone still lacked the practice and experience of collectively keeping and using draft animals; lacked the idea of collectivism (集体主义); and lacked the animal caretaking and managing regulations that are

²⁷⁰ Jiangxi sheng Nanchang zhuan yuan gongshu nongye ke 江西省南昌专员公署农业科, ed., *Nongyeshe jiachu siyangyuan xuexi cankao ziliao* 农业社家畜饲养员学习参考资料 (N.p., Jiangxi sheng Nanchang zhuan yuan gongshu nongye ke, 1955), 32–37.

²⁷¹ Jiangxi sheng Nanchang zhuan yuan gongshu nongye ke, *Nongyeshe jiachu siyangyuan xuexi cankao ziliao*, 37–41.

人类农民和非人类动物都经历了集体化。役畜——牛、驴、马和骡子——以及肉畜——猪、羊、鸡、鸭、鹅和兔——一直是中国农村日常生活中不可或缺的一部分。在集体化初期，中共鼓励动物主人“自愿”加入合作社并捐赠动物。在某些情况下，地方共产党员干部采取了一种更谨慎和过渡性的方式来集体化动物，这种方式被称为“私有私养，集体借用”。无论哪种方式，这些动物 都将被征用为合作社的“公共财产”。²⁷⁰

然而，动物的集体化并不顺利。这并非因为共产党人无法克服地主武装抵抗和富农观望态度的反对。强制和暴力的土地改革和集体化根除了革命前农场动物的私有制。从国家的角度来看，真正的挑战来自于“缺乏经验和制度”。例如，河北省博野县一个初级合作社的全部85头牲畜在1954年就已经集中起来。然而，这份地方报告的作者认为，没有人知道如何正确地进行集体动物饲养。农民们凭经验饲养和照料不属于他们的动物。他们还“不加区分地殴打和过度劳役”动物。²⁷¹1955年，《人民日报》的一位编辑指出，“大家对集体饲养和使用役畜仍然缺乏实践和经验；缺乏集体主义思想；缺乏动物饲养和管理规定，这些规定是

²⁷⁰ 江西省南昌专员公署农业科编，《农业社家畜饲养员学习参考资料》（无出版地：江西省南昌专员公署农业科，1955），第32-37页。

²⁷¹ 江西省南昌专员公署农业科，《农业社家畜饲养员学习参考资料》，第37-41页。

necessary.” The absence of these elements led to a situation where “not a small number of animals become emaciated and even die.”²⁷²

The suffering and death of countless livestock caused by collectivization, however, could not be simply attributed to the lack of experience. According to Peter Braden, one of the few animal historians of modern China, this resulted from the intrinsic flaw of the collectivist system. “[T]he unclear lines of responsibility in cooperatives hampered efforts to achieve accountability in the care and usage of these indispensable yet underappreciated [bovine] workers. Perverse economic incentives exposed cooperative herds to inattentive care and excessive workloads.”²⁷³ In other words, Braden argues that the introduction of the collective ownership of livestock led to the neglect and overwork of laboring animals. Moreover, as a means of implicit defiance to the reform in ownership, peasants often sold off or even killed animals, which only worsened animals’ agony in this period.²⁷⁴

Although it cannot be denied that collectivization, especially at its inception, was harmful to animal wellbeing, it is equally worth appreciating how some actors strove to ameliorate their suffering. Rather than halting the social experiment of collectivization, the revolutionaries sought

²⁷² Zhou Siping 周思平 and Jian Hong 剑虹, “Zhuyi xuanze he jiaoyu nongye hezuoshede siyangyuan” 注意选择和教育农业合作社的饲养员, *Renmin Ribao* 人民日报 (May 30, 1955). There are plenty of similar examples. For example, see Zhonggong Beijing shi weiyuanhui 中共北京市委员会, “Guanyu jiaoqu nongye hezuohua yundong xiang zhongyangde baogao” 关于郊区农业合作化运动向中央的报告 (September 30, 1955), in *Zhongguo wushi niandai chuzhongqi de zhengzhi yundong shujuku* 中国五十年代初中期的政治运动数据库. ed. Song Yongyi 宋永毅 (Hong Kong: Xianggang zhongwen daxue Zhongguo yanjiu fuwu zhongxin, 2014); Ren Yongda 任永达, “Heilongjiang sheng bufen nongcun zai jianli gaojishe zhong faxian nongmin zaisha he chumai gengchu xianxiang.” 黑龙江省部分农村在建立高级社中发现农民宰杀和出卖耕畜现象 (February 16, 1956), in *Zhongguo wushi niandai chuzhongqi de zhengzhi yundong shujuku*, ed. Song Yongyi (Hong Kong: Xianggang zhongwen daxue Zhongguo yanjiu fuwu zhongxin, 2014).

²⁷³ Peter W. Braden, “Serve the People: Bovine Experiences in China’s Civil War and Revolution, 1935–1961,” Ph.D. diss., University of California at San Diego, 2020, 129.

²⁷⁴ Braden, “Serve the People,” Chapter Three.

necessary.” The absence of these elements led to a situation where “not a small number of animalsbecome emaciated and even die.”²⁷²

然而，集体化导致无数牲畜的痛苦和死亡，不能简单地归因于经验不足。根据彼得·布雷登（Peter Braden）的说法，他是为数不多的研究中国现代动物历史的学者之一，这源于集体主义制度的内在缺陷。 “[合]作社中不明确的责任划分阻碍了在照料和使用这些不可或缺却未被充分重视的[牛] 工方面实现问责制的努力。扭曲的经济激励使合作社的牲畜面临疏忽照料和过度劳累。”²⁷³换句话说，布雷登认为，牲畜集体所有制的引入导致了役用动物的忽视和过度劳累。此外，作为对所有权改革的一种默许反抗方式，农民经常出售甚至宰杀动物，这只会加剧动物在这一时期的痛苦。²⁷⁴

尽管不可否认，集体化，尤其是在其初期，对动物的福祉造成了损害，但同样值得赞赏的是，一些参与者努力改善它们的痛苦。革命者并没有停止集体化的社会实验，而是寻求

²⁷² Zhou Siping 周思平 and Jian Hong 剑虹, “Zhuyi xuanze he jiaoyu nongye hezuoshede siyangyuan” 注意选择和教育农业合作社的饲养员, *Renmin Ribao* 人民日报 (May 30, 1955). There are plenty of similar examples. For example, see Zhonggong Beijing shi weiyuanhui 中共北京市委员会, “Guanyu jiaoqu nongye hezuohua yundong xiang zhongyangde baogao” 关于郊区农业合作化运动向中央的报告 (September 30, 1955), in *Zhongguo wushi niandai chuzhongqi de zhengzhi yundong shujuku* 中国五十年代初中期的政治运动数据库. ed. Song Yongyi 宋永毅 (Hong Kong: Xianggang zhongwen daxue Zhongguo yanjiu fuwu zhongxin, 2014); Ren Yongda 任永达, “Heilongjiang sheng bufen nongcun zai jianli gaojishe zhong faxian nongmin zaisha he chumai gengchu xianxiang.” 黑龙江省部分农村在建立高级社中发现农民宰杀和出卖耕畜现象 (February 16, 1956), in *Zhongguo wushi niandai chuzhongqi de zhengzhi yundong shujuku*, ed. Song Yongyi (Hong Kong: Xianggang zhongwen daxue Zhongguo yanjiu fuwu zhongxin, 2014).

²⁷³ Peter W. Braden, “为人民服务：1935-1961年中国内战和革命中的牛类经历”，博士论文，加州大学圣地亚哥分校，2020年，第129页。

²⁷⁴ Braden, “为人民服务”，第三章。

better ways to institutionalize it. It was at this point that “collective animal caretakers” (集体饲养员) appeared on the stage. The Boye cooperative, for example, allocated to some villagers the task of taking care of collectivized animals and animal farms in 1955.²⁷⁵ These caretakers were put in charge of animal care at the grassroots for the remaining years of the Maoist era. The propagandistic sources on “model caretakers” hint at the qualifications the state required for this new occupation.

Caretakers were first and foremost expected to be experts in livestock keeping. If prospective caretakers already had some kind of relevant knowledge and experience, they were more likely to find a job in a collective pigsty or cowshed. The work team of the Central Ministry of Agriculture’s Bureau of Animal Husbandry and Veterinary Services, after reviewing the earlier stages of collectivization in Hebei Province in 1955, recommended to recruit “the middle-aged or elderly who are experienced in raising farm animals as caretakers.”²⁷⁶ At a time when most rural residents did not have access to regular education in animal breeding or veterinary medicine, in many cases, people obtained knowledge from their family or during military service.

According to an unpublished paper produced by the Nanchang Municipal Secretariat for Swine Production Conference, Wan Fuxiu was a caretaker in Tuanjie Brigade of Nanchang County, Jiangxi. She was born to a small peasant family in a neighboring region. As a child she learned how to handle pigs by helping her mother. Soon after she married into a Tuanjie native family, the brigade leadership considered her personal background and assigned her to be a caretaker.²⁷⁷

²⁷⁵ Jiangxi sheng Nanchang zhuanyuan gongshu nongyeke, *Nongyeshe jiachu siyangyuan xuexi cankao ziliao*, 37–41.

²⁷⁶ Jiangxi sheng Nanchang zhuanyuan gongshu nongyeke, *Nongyeshe jiachu siyangyuan xuexi cankao ziliao*, 37.

²⁷⁷ Nanchang xian jiangxiang gongshe tuanjie dadui 南昌县蒋巷公社团结大队, “Yige nǚ siyangyuande chengzhang: Nanchang xian jiangxiang gongshe shacheng guanliqu tuanjie dadui Wan Fuxiu tongzhi lao siyang shengzhu shiji” 一个

更好地将其制度化。正是在此时, “集体饲养员” 登上了历史舞台。例如, 博野合作社于1955年将照料集体化动物和动物农场的任务分配给了一些村民。²⁷⁵在毛泽东时代剩余的岁月里, 这些饲养员负责基层动物的照料工作。关于“模范饲养员”的宣传资料暗示了国家对这一新职业的材质要求。

饲养员首先被期望成为畜牧业专家。如果未来的饲养员已经具备相关知识和经验, 他们更有可能在集体猪圈或牛棚找到工作。中央农业部畜牧兽医局工作组在1955年审查河北省集体化早期阶段后, 建议招募“有饲养牲畜经验的中老年人担任饲养员”。²⁷⁶在当时, 大多数农村居民无法获得正规的动物育种或兽医教育, 在许多情况下, 人们通过家庭或服兵役获得知识。

根据南昌市生猪生产会议秘书处发布的一份未出版文件, 万福秀是江西省南昌县团结大队的一名饲养员。她出生在一个邻近地区的小农家庭。小时候, 她通过帮助母亲学会了如何照料猪。她嫁到团结大队后不久, 大队领导考虑到她的个人背景, 便派她担任饲养员。²⁷⁷

²⁷⁵江西省南昌专员公署农业科, 农业社家畜饲养员学习参考资料, 37–41。

²⁷⁶ Jiangxi sheng Nanchang zhuanyuan gongshu nongyeke, *Nongyeshe jiachu siyangyuan xuexi cankao ziliao*, 37.

²⁷⁷南昌县蒋巷公社团结大队南昌县蒋巷公社团结大队, “一个女饲养员的成长: 南昌县蒋巷公社沙城管理区团结大队万福秀同志老饲养生猪事迹” 一个

Wang Bingxiang was a destitute and homeless Henan native, living off the generosity of local folk temples. He enlisted in the People’s Liberation Army (PLA) in 1940 and served as a military animal caretaker until his discharge in 1946. Wang eventually settled in a village in Yicheng County, Shanxi. When his village joined an elementary cooperative for the first time as early as 1952, he took advantage of his knowledge of managing horses and donkeys and volunteered to be a caretaker. Whenever fellow peasants praised Wang’s skills, he shook his head and humbly said: “I do not have something that can be called experience. But I do care about the collective’s animals just like the ones in my own home.”²⁷⁸

Although some prior knowledge and experience were crucial in becoming a caretaker, strong class consciousness and revolutionary vigor were also emphasized. A *People’s Daily* article discussed how to select the right persons for animal caretaking in nascent collectives. The authors prioritized the willingness to “face adversity and work hard” and a sense of responsibility over technical competence. They warned that if “protecting public interests” and “proactive working” took a backseat to “experiences and techniques of raising animals,” it might allow more former landlords and rich farmers who were politically suspicious to take charge of collectives’ animals, which might lead to sabotage. If so, more reliable caretakers must come from the ranks of the inexperienced but revolutionary poor peasantry.²⁷⁹

女饲养员的成长: 南昌县蒋巷公社沙城市管理区团结大队万福秀同志老饲养生猪事迹 (January 1960), Unpublished material.

²⁷⁸ Li Wenshan 李文珊, “Mofan siyangyuan Wang Bingxiang” 模范饲养员王秉祥, *Renmin Ribao* 人民日报 (February 12, 1955).

²⁷⁹ Zhou Siping and Jian Hong, “Zhuyi xuanze he jiaoyu nongye hezuoshede siyangyuan.”

王炳祥是河南人，家境贫寒，无家可归，靠当地民间寺庙的施舍度日。他于1940年参军，加入中国人民解放军，担任军马饲养员，直到1946年退伍。王炳祥最终在山西省翼城县的一个村庄定居下来。早在1952年，当他所在的村庄首次加入初级社时，他利用自己管理马匹和驴子的知识，自愿成为一名饲养员。每当乡亲们称赞王炳祥的技能时，他总是摇摇头，谦虚地说：“我没有什么经验可言。但我确实像对待自己家里的牲畜一样关心集体的牲畜。”²⁷⁸

虽然一些先前的知识和经验对于成为一名饲养员至关重要，但阶级意识和革命热情也同样受到强调。《人民日报》的一篇文章讨论了如何在新生集体中选择合适的动物饲养员。作者们将“面对逆境、努力工作”的意愿和责任感置于技术能力之上。他们警告说，如果“保护公共利益”和“积极工作”退居次要地位，而“饲养动物的经验和技术”占据主导，可能会让更多政治上可疑的前地主和富农负责集体的牲畜，这可能导致破坏。如果是这样，更可靠的饲养员必须来自那些没有经验但具有革命精神的贫苦农民。²⁷⁹

女饲养员的成长: 南昌县蒋巷公社沙城市管理区团结大队万福秀同志老饲养生猪事迹 (January 1960), Unpublished material.

²⁷⁸ 李文珊, “模范饲养员王秉祥”, 《人民日报》(1955年2月12日)。

²⁷⁹ 周恩平、简洪, “注意选择和教育农业合作社的饲养员”。

Hua Yinfeng was a representative poor-peasant-turned-caretaker. During the Republican era, her parents, who had fallen behind in their rent, “sold” Hua to work as a maid for a wealthy family. It was not until the Communist “liberation” and land reform that she enjoyed “human life” (人的生活) and “human treatment” (人的待遇). In 1953, she began working at a collective pigsty in her hometown, Dongyang County, Zhejiang. Some neighbors complained that keeping pigs was “too mucky, too nauseating, and not promising at all.” But she seemed determined: “There is no good or bad work. Regardless of what we do, we should do well, make a result, and contribute to our nation. This is what is the most promising.” With this story of Hua, the state might want to send the message that taking good care of the collective’s pigs was a way to support the revolution and, more importantly, to be a decent human.²⁸⁰

As early as 1951, a poor female peasant, Wang Ru started to keep calves in a prototype collective cowshed. She had never taken care of animals before, but her “political resolution” led her to the job. She worked hard and quickly became accustomed to feeding calves, drawing water, and milking cows. In the process, Wang received a great deal of attention from local cadres and was invited to attend cultural activities, thereby becoming literate. Her newly acquired ability to read newspapers, a kind of reward for her hard work with the bovines, helped her to “be more courageous and consolidate confidence.” This story may contain some propagandistic hyperbole. Though it is hard to tell how Wang “really” felt, what the state expected was obvious: Become a caretaker and realize one’s social role and responsibility.²⁸¹

²⁸⁰ Zhang Zheng 张政, “Yige nongchang nü siyangyuan de gongxian” 一个农场女饲养员的贡献, *Renmin Ribao* 人民日报 (September 13, 1955).

²⁸¹ Liu Shiyuan 刘士元, “Nü siyangyuan Wang Ru” 女饲养员王儒, *Zhongguo nongken* 1954.12 (December 1954): 29.

华银凤是贫农出身的饲养员代表。民国时期，她的父母因拖欠租金，将华“卖”给一个富裕家庭做女佣。直到共产党“解放”和土地改革后，她才过上“人的生活”（人的生活）和“人的待遇”（人的待遇）。1953年，她开始在浙江东阳县老家的集体猪圈工作。一些邻居抱怨养猪“太脏、太恶心，一点前途都没有”。但她似乎很坚定：“工作没有好坏之分。无论我们做什么，都应该做好，做出成绩，为国家做贡献。这才是最有前途的。”通过华的故事，国家可能想传达这样的信息：照顾好集体的猪是支持革命的一种方式，更重要的是，是做一个正派的人。²⁸⁰

早在1951年，贫农王茹就开始在一个集体牛棚原型中饲养牛犊。她以前从未照料过动物，但她的“政治决心”使她从事了这项工作。她努力工作，很快就习惯了喂养牛犊、打水和挤奶。在此过程中，王茹受到了当地干部的极大关注，并被邀请参加文化活动，从而学会了识字。她新获得的阅读报纸的能力（这是她辛勤照料牛群的一种回报）帮助她“更加勇敢，巩固了信心”。这个故事可能包含一些宣传性的夸张。尽管很难说王茹“真正”的感受如何，但国家期望的是显而易见的：成为一名饲养员，实现自己的社会角色和责任。²⁸¹

²⁸⁰ Zhang Zheng 张政, “Yige nongchang nü siyangyuan de gongxian” 一个农场女饲养员的贡献, *Renmin Ribao* 人民日报 (1955年9月13日)。

²⁸¹ Liu Shiyuan 刘士元, “Nü siyangyuan Wang Ru” 女饲养员王儒, 《中国农垦》1954.12（1954年12月）：29。

The state also required model caretakers to help fellow peasants accept a new collectivist animal ethic. Caretaker Wang Bingxiang often came into conflict with neighbors. He was unhappy with some overzealous peasants who overworked their draft animals. When he encountered a young man who had not watered an ox for hours, Wang forcibly took the animal back from him and scolded him. “Even though the beast did not want to drink water, you should not have skipped watering!” Wang also quarreled with someone who had loaded a donkey with bags of cotton without consulting him. He was so furious that although it was the middle of the workday, he brought the animal back to the stable to guarantee that it received food and rest. As similar conflicts happened again and again, some peasants officially filed a grievance. The model caretaker, however, convinced local leaders and fellow peasants that, as a caretaker, protecting collective property was “my duty for the entire cooperative as well as everyone and by no means something for myself.” The state sided with Wang and told his story in the *People’s Daily*.²⁸²

The Great Leap Forward and Swine Production, 1958–1962

Once the presence and labor of caretakers, at least partially, rectified the initial imperfections of collective animal keeping in the mid-1950s, the overconfident party-state launched the Great Leap Forward Movement and continued to scale up collectivization in the last years of the 1950s. The objective of the Great Leap Forward was to make a great leap in every sort of economic production, including animal husbandry. In November 1958, the Sixth Plenary Session of the Eighth Central Committee of the CCP proclaimed that the People’s Republic would focus on the simultaneous

²⁸² Li Wenshan, “Mofan siyangyuan Wang Bingxiang.”

国家还要求模范饲养员帮助其他农民接受新的集体主义动物伦理。饲养员王炳祥经常与邻居发生冲突。他对一些过度劳累役畜的农民感到不满。当他遇到一个几个小时没有给牛喂水的年轻人时，王强行把牛从他那里带回来并训斥了他。“即使这头牲畜不想喝水，你也不应该跳过喂水！”王还与一个未经他同意就给驴子装上棉花袋的人争吵。他非常愤怒，尽管当时是工作日的中间，他还是把动物带回马厩，以确保它得到食物和休息。由于类似的冲突一再发生，一些农民正式提出了申诉。然而，这位模范饲养员说服了当地领导和农民，作为一名饲养员，保护集体财产是“我为整个合作社以及每个人应尽的职责，绝不是为了我自己。”国家支持王，并在《人民日报》上讲述了他的故事。

282

大跃进与生猪生产，1958—1962

20世纪50年代中期，在饲养员的照料和劳动至少部分纠正了集体饲养牲畜的最初缺陷后，过于自信的党国发起了大跃进运动，并在20世纪50年代的最后几年继续扩大集体化规模。大跃进的目标是在包括畜牧业在内的各种经济生产中实现大跃进。1958年11月，中共八届六中全会宣布，中华人民共和国将重点关注同步发展

²⁸² 李文山，《模范饲养员王炳祥》。

development of both staple farming and livestock production.²⁸³ Mao and his comrades would have seen this move as a delayed realization of what they believed at the earlier stage of collectivization. The economy of scale that collectivization enabled was bound to provide a favorable condition for developing the livestock industry. Moreover, food for people, feed for animals, and animal manure could be managed within a cooperative in an integrated manner, which would increase the efficiency of resource allocation.²⁸⁴ People’s Communes, the product of the Great Leap Forward, were to be more-than-people’s communes from the beginning.

Among different kinds of livestock, pigs received special attention during the Great Leap Forward years, something that can be attributed to no other than Mao. On October 31, 1959, after receiving a report about the rapid increase of swine production in Wangqiansi Commune, Wuqiao County, Hebei Province, an excited Mao urged Chinese people to make a correction to the conventional order of priority in domestic animals—horses came first, followed by cattle, sheep, chickens, and dogs. Pigs were the least important. Mao “agree[d] wholeheartedly” with the idea that now “pigs must go up to the top of six categories of livestock” and that the new order should be “pigs, cattle, sheep, horses, chickens, and dogs.” Bringing up Soviet soil scientist and microbiologist Vasili Williams,²⁸⁵ Mao explained that agriculture, forestry, and animal husbandry were mutually dependent and that none of the three was dispensable for the future of the nation. He cited manure

²⁸³ Zhonggong Zhongyang 中共中央, “Guanyu 1959 nian guomin jingji jihua de jue ding” 关于 1959 年国民经济计划的决定 (December 10, 1958), in *Jianguo yilai zhongyao wenxian xuanbian* 建国以来重要文献选编 11, ed. Zhonggong zhongyang wenxian yanjiushi 中共中央文献研究室 (Beijing: Zhongyang wenxian chubanshe, 1994), 628.

²⁸⁴ Zhonghua renmin gongheguo guowuyuan 中华人民共和国国务院, “Guanyu zengchan shengzhu de zhishi” 关于增产生猪的指示 (December 7, 1955), Beijing Municipal Archives (hereafter BMA), 088-001-00323-0033.

²⁸⁵ On Williamsian soil science and microbiology, see Jongsik Christian Yi, “Dialectical Materialism Serves Voluntarist Productivism: The Epistemic Foundation of Lysenkoism in Socialist China and North Vietnam,” *Journal of the History of Biology* 54.3 (October 2021): 513–539.

粮食种植和畜牧生产都得到了发展。²⁸³毛泽东和他的同志们会认为此举是他们认为在集体化早期阶段的延迟实现。集体化带来的规模经济必然为发展畜牧业提供有利条件。此外，人的食物、动物饲料和动物粪便可以在合作社内以综合方式管理，这将提高资源配置效率。²⁸⁴人民公社是大跃进的产物，从一开始就注定是“超大型”人民公社。

在大跃进期间，在不同种类的牲畜中，猪受到了特别关注，这只能归因于毛泽东。1959年10月31日，在收到河北省吴桥县王前寺公社生猪产量快速增长的报告后，兴奋的毛泽东敦促中国人民纠正传统家畜的优先顺序——马排第一，其次是牛、羊、鸡和狗。猪是最不重要的。毛泽东“衷心同意”了“猪必须上升到六类牲畜之首”的观点，新的顺序应该是“猪、牛、羊、马、鸡和狗”。毛泽东提到了苏联土壤科学家和微生物学家瓦西里·威廉姆斯，²⁸⁵解释说农业、林业和畜牧业相互依存，三者缺一不可，关系到国家的未来。他引用了粪肥

²⁸³ 中共中央, “关于1959年国民经济计划的决定” (1958年12月10日), 载中共中央文献研究室编: 《建国以来重要文献选编》第11册 (北京: 中央文献出版社, 1994), 第628页。

²⁸⁴ 中华人民共和国国务院, “关于增产生猪的指示” (1955年12月7日), 北京档案馆 (以下简称BMA), 088-001-00323-0033。

²⁸⁵ 关于威廉姆斯土壤科学和微生物学, 参见Jongsik Christian Yi, “辩证唯物主义服务于唯意志论的生产主义: 社会主义中国和北越李森科主义的认识论基础”, 《生物学史杂志》54.3 (2021年10月): 513-539。

as an example. “The biggest sources of fertilizer in our county are pigs and other large livestock.” Mao insisted that “swine manures as organic fertilizers are ten times better than inorganic chemical fertilizers” and that, therefore, “a pig is a small-scale organic fertilizer plant.” Moreover, pigs were easy and inexpensive to raise and produced “meats, bristles, leathers, bones, and guts” which could count as raw materials for industry and export. Mao ended with expressing his hope that “every person raises a pig; every *mu* has a pig.”²⁸⁶

Once Chairman Mao championed pig raising, his other sporadic notes about pigs were rediscovered. As early as 1955, Mao stated that “pig raising is a huge issue which is related to fertilizing, meat diet, and foreign exchange earnings.” In the same document, he presented an idea that local cadres should encourage every household to raise one or more pigs in addition to developing collective hog farms.²⁸⁷ In 1956, Mao praised Shimenxiang Cooperative, Yanggu County, Shandong Province for having laid the foundation of swine production even though the villagers traditionally had not raised pigs. Since Shimenxiang set such an exemplary precedent, Mao asked, “why could not other cooperatives do the same?”²⁸⁸ These messages, as Table 1 shows, led to the nationwide growth in the swine population in the late 1950s.²⁸⁹

²⁸⁶ One *mu* is roughly equivalent to 0.165 acre. Mao Zedong 毛泽东, “Guanyu yangzhu de yi feng xin: Hebei sheng Wuqiao xian Wangqiansi renmin gongshe yangzhu jingyan yiwende pishi” 关于养猪的一封信: 河北省吴桥县王谦寺人民公社养猪经验一文的批示 (October 31, 1959), in *Xumu gongzuo wenjian xuanbian* 畜牧工作文件选编, ed. Shaanxi sheng xumu shouyi zongzhan 陕西省畜牧兽医总站 (N.p.: Shaanxi sheng xumu shouyi zongzhan, 1974), 1–2.

²⁸⁷ Mao Zedong 毛泽东, “Zheli yangle yidapi maozhu yiwen anyu” 这里养了一大批毛猪一文按语 (1955), in *Xumu gongzuo wenjian xuanbian*, ed. Shaanxi sheng xumu shouyi zongzhan, 3.

²⁸⁸ Mao Zedong 毛泽东, “Shandong Yanggu xian Shimen xiang nongye hezuoshe yangzhu baogaode pishi” 山东阳谷县石门乡农业合作社养猪报告的批示 (1956), in *Xumu gongzuo wenjian xuanbian*, ed. Shaanxi sheng xumu shouyi zongzhan, 3–4.

²⁸⁹ As Mao’s remarks suggest, pig raising was promoted for the three reasons: (1) obtaining manure; (2) supplying meat to urban areas; and (3) using animal parts as raw materials for industrial production. However, only the first reason reflected the interests of local collectives, whereas the remaining two had to do with the needs of the state. Given that

举例来说。“我们县最大的肥料来源是猪和其他大型牲畜。”毛泽东坚持认为“猪粪作为有机肥比无机化肥好十倍”，因此“一头猪就是一个小型有机肥厂”。此外，养猪既简单又便宜，还能生产“肉、鬃毛、皮革、骨头和内脏”，这些都可以作为工业和出口的原材料。毛泽东最后表达了他的希望：“每人养一头猪；每亩地有一头猪。”²⁸⁶

毛主席倡导养猪后，他关于猪的其他零星批示也被重新发现。早在1955年，毛泽东就指出“养猪是一个大问题，关系到施肥、肉食和外汇收入”。在同一份文件中，他提出了一个想法，即地方干部除了发展集体养猪场外，还应鼓励每家每户养一头或多头猪。²⁸⁷ 1956年，毛泽东赞扬了山东省阳谷县石门乡合作社，尽管村民传统上不养猪，但该社为生猪生产奠定了基础。既然石门乡树立了这样一个榜样，毛泽泽东问道：“为什么其他合作社不能这样做呢？”²⁸⁸ 如表1所示，这些信息导致了20世纪50年代末全国生猪数量的增长。²⁸⁹

²⁸⁶ One *mu* is roughly equivalent to 0.165 acre. Mao Zedong 毛泽东, “Guanyu yangzhu de yi feng xin: Hebei sheng Wuqiao xian Wangqiansi renmin gongshe yangzhu jingyan yiwende pishi” 关于养猪的一封信: 河北省吴桥县王谦寺人民公社养猪经验一文的批示 (October 31, 1959), in *Xumu gongzuo wenjian xuanbian* 畜牧工作文件选编, ed. Shaanxi sheng xumu shouyi zongzhan 陕西省畜牧兽医总站 (N.p.: Shaanxi sheng xumu shouyi zongzhan, 1974), 1–2.

²⁸⁷ 毛泽东, “这里养了一大批毛猪一文按语” (1955), 载陕西省畜牧兽医总站编：《畜牧工作文件选编》，第3页。

²⁸⁸ 毛泽东, “山东阳谷县石门乡农业合作社养猪报告的批示” (1956), 载陕西省畜牧兽医总站编：《畜牧工作文件选编》，第3-4页。

²⁸⁹ 正如毛泽东的讲话所暗示的，推广养猪有三个原因：（1）获取肥料；（2）为城市地区提供肉类；（3）将动物部分用作工业生产的原材料。然而，只有第一个原因反映了地方集体的利益，而其余两个则与国家的需求有关。鉴于此，

Year	No. of Pigs	
	Total No.	No. of Sow
1955	89,720,000	N/A
1956	84,030,000	N/A
1957	145,900,000	N/A
1958	138,290,000	N/A
1959	120,420,000	N/A
1960	82,270,000	N/A
1961	75,520,000	N/A
1962	99,970,000	12,483,000
1963	131,800,000	14,025,000
1964	152,470,000	14,115,000
1965	166,930,000	N/A
1966	193,360,000	N/A
1967	190,060,000	N/A
1968	178,630,000	N/A
1969	172,510,000	N/A
1970	206,100,000	N/A
1971	250,350,000	N/A
1972	263,680,000	N/A
1973	257,940,000	N/A
1974	260,780,000	23,122,000
1975	281,170,000	26,457,000
1976	287,250,000	28,043,000
1977	291,780,000	27,668,000

Table 2. The total number of pigs raised in China, 1955–1977. Source: Zhonghua renming gongheguo nongyebu xumu shouyisi 中华人民共和国农业部畜牧司, *Zhongguo xumuye tongji 1949–1989* 中国畜牧业统计 1949–1989 (Beijing: Zhongguo jingji chubanshe, 1990), 25.

raising pigs was extremely labor-intensive, some shrewd people at the grassroots were not always willing to keep pigs at home. Even some collectives regarded collective hog production as unhelpful for their communities. A former commune veterinarian I interviewed in Langfang, Hebei, which is very close to Beijing, told me that he and his fellow commune constituents cared more about draft animals than pigs because donkeys and mules benefited the commune, while hogs were all sent to Beijing. Author’s interview with former commune veterinarian Mr. Wen in Langfang, Hebei, May 2021. In that case, pig raising in Maoist China might not have necessarily served the interests of individual peasants as well as collectives, which is radically different from, for example, profit-driven practices of American hog industry. See Joseph Leslie Anderson, *Capitalist Pigs: Pigs, Pork, and Power in America* (Morgantown: West Virginia University Press, 2019); Alex Blanchette, *Porkopolis: American Animality, Standardized Life, and the Factory Farm* (Durham: Duke University Press, 2020).

Year	生猪数量	
	总数	母猪数量
1955	89,720,000	N/A
1956	84,030,000	N/A
1957	145,900,000	N/A
1958	138,290,000	N/A
1959	120,420,000	N/A
1960	82,270,000	N/A
1961	75,520,000	N/A
1962	99,970,000	12,483,000
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1975	281,170,000	26,457,000
1976	287,250,000	28,043,000
1977	291,780,000	27,668,000

表2。1955—1977年中国生猪饲养总量。来源：中华人民共和国农业部畜牧司，《中国畜牧业统计 *1949–1989*》（北京：中国经济出版社，1990），第25页。

养猪是劳动密集型活动，一些基层精明的人并不总是愿意在家养猪。甚至一些集体也认为集体养猪对他们的社区没有帮助。我在河北廊坊（离北京很近）采访的一位前公社兽医告诉我，他和他的公社社员更关心役畜而不是猪，因为驴和骡子对公社有利，而猪都被送往北京。作者于2021年5月在河北廊坊采访了前公社兽医温先生。在这种情况下，毛泽东时代的养猪业可能不一定符合个体农民和集体的利益，这与例如美国生猪产业的利润驱动做法截然不同。参见Joseph Leslie Anderson, 《*Capitalist Pigs: Pigs, Pork, and Power in America*》（Morgantown: West Virginia University Press, 2019）；Alex Blanchette, 《*Porkopolis: American Animality, Standardized Life, and the Factory Farm*》（Durham: Duke University Press, 2020）。

A 1960 document from the Beijing municipal government shows how local governments responded to Mao’s attention to pigs. It states that pig raising in the countryside on the outskirts of the capital should “walk on two legs,” taking “collective caretaking as primary and individual caretaking as ancillary” (集体饲养为主, 私养为辅). Each of the four administrative levels—county, commune, brigade, and team—would establish its own collective pigsties and manage them separately. At the same time, individual peasants from cooperatives were encouraged to raise their own pigs.²⁹⁰

Under this dual system of collective and individual caretaking, animal caretakers worked on collective farms and earned “work points.” But at the same time, as individual members in a cooperative, most of them started to keep at least one sow. They could sell privately grown pigs to either state-owned food companies or to their neighbors and earn money, as long as they turned in the requisition quota of manure to the state. Either way, caretakers were dependent on animals as their major or only source of income.²⁹¹ However, draft animals such as oxen, donkeys, and mules came to be categorized separately as “large livestock” (大牲畜 or 大家畜). Unlike pigs, individual households were rarely allowed to maintain draft animals.²⁹²

As collective caretaking of pigs and draft animals rose to prominence during the Great Leap Forward, so did the roles of caretakers. The ideal caretakers depicted in propagandistic materials

²⁹⁰ Zhonggong Beijing shiwei xumu lingdao xiaozu 中共北京市委畜牧领导小组. “Yangzhu qitiao jingyan, siyangyuan baxiang biao zhun” 养猪七条经验, 饲养员八项标准 (March 5, 1960), BMA, 001-014-00609-00010.

²⁹¹ Author’s interview with former caretakers Mrs. Hong and Mrs. Ling in Bobai County, Guangxi, June 2021.

²⁹² Zhonghua renmin gongheguo nongyebu dangzu 中华人民共和国农业部党组, “Guanyu dashengchu wenti de baogao” (October 21, 1961), in *Xumu shouyi gongzuo cankao ziliao* 畜牧兽医工作参考资料, ed. Gansu sheng xumuting 甘肃省畜牧厅 (Lanzhou: Gansu sheng xumuting, 1964), 5–10.

1960年北京市政府的一份文件显示了地方政府如何回应毛泽东对生猪的关注。文件中指出，首都郊区农村的养猪业应“两条腿走路”，即“集体饲养为主，个体饲养为辅”（集体饲养为主，私养为辅）。县、公社、大队、生产队这四个行政级别都将建立自己的集体猪圈并分别管理。同时，鼓励合作社的个体农民饲养自己的生猪。²⁹⁰

在这种集体和个体双重饲养制度下，饲养员在集体农场工作并获得“工分”。但与此同时，作为合作社的个体成员，他们中的大多数人开始饲养至少一头母猪。只要他们向国家上交规定的粪肥配额，就可以将私人饲养的生猪出售给国有食品公司或邻居以赚取收入。²⁹¹ 无论哪种方式，饲养员都依赖牲畜作为其主要或唯一的收入来源。然而，牛、驴、骡子等役畜被单独归类为“大牲畜”（大牲畜 或 大家畜）。与生猪不同，个体农户很少被允许饲养役畜。²⁹²

随着大跃进期间生猪和役畜的集体饲养日益突出，饲养员的角色也日益突出。宣传材料中描绘的理想饲养员

²⁹⁰ 中共北京市委畜牧领导小组。《养猪七条经验，饲养员八项标准》（1960年3月5日），BMA, 001-014-00609-00010。

²⁹¹ 作者于2021年6月在广西博白县采访了前饲养员洪女士和凌女士。

²⁹² Zhonghua renmin gongheguo nongyebu dangzu 中华人民共和国农业部党组, “Guanyu dashengchu wenti de baogao” (October 21, 1961), in *Xumu shouyi gongzuo cankao ziliao* 畜牧兽医工作参考资料, ed. Gansu sheng xumuting 甘肃省畜牧厅 (Lanzhou: Gansu sheng xumuting, 1964), 5–10.

were those who truly cherished animals. Much like their counterparts in other countries, Chinese caretakers appreciated farm animals’ instrumental value while also understanding that the animals in China were to serve the cooperatives’ interests, rather than individuals. Song Ming, a caretaker from the outskirts of Tianjin, understood that some crucial activities of cooperatives such as “plowing and sowing, transporting, milling, and so on” were “all inextricable from” taking care of animals. Hence, Song “loved animals like his own eyeballs,” and fellow cooperative constituents conferred the title of “livestock maniac” (牲口迷) on him.²⁹³ Wu Yuanhao, another Tianjin caretaker, defined livestock as the “cooperatives’ lifeblood” (社的命根子) and “cooperatives’ treasures” (社的宝贝), recognizing their importance in the collective economy. She made a firm determination in 1959 that “if the Communist Party needs, I will serve as caretaker all my life.” She tried to “ceaselessly improve the ways of caring for and managing [animals], get animals fed better, and make a bigger contribution to support the Great Leap Forward in [collective] agricultural production.”²⁹⁴ In this sense, both caretakers’ reason for being and animals’ value were situated in the context of consolidating collectivism that was at odds with the principle of private ownership.

Some caretakers seemed willing to articulate the relationship between animals and collectivism in a more grandiose fashion. Lü Pin from Liaoning clearly interpreted the meaning of the labor of collective animal caretaking in relation to bigger common good (大公) beyond individualism. After touting caretakers’ “ordinary labor” as the origin of wealth, he proclaimed that it would be “unthinkable that a person who never cares about the needs of socialist enterprise and

²⁹³ “Song Ming shi ge hao siyangyuan” 宋明是个好饲养员 (1959), Tianjin Municipal Archives (hereafter TMA), 401206800-X0283-C-000779-065.

²⁹⁴ Wu Yuanhao 吴元浩, “Woshi zenyang danghao siyangyuan de: Jiaohe xian Haocun renmin gongshe Dawupan shengchandu Wu Yuanhao” 我是怎样当好饲养员的: 交河县郝村人民公社大吴潘生产队吴元浩 (1959), TMA, 401206800-X0283-C-000779-089.

是那些真正爱护动物的人。与他们在其他国家的同行一样，中国的饲养员也重视农场动物的工具性价值，同时也明白中国的动物是为合作社的利益服务，而不是为个人服务。天津郊区的一位饲养员宋明明白，合作社的一些关键活动，如“耕种、播种、运输、磨坊等等”，“都离不开”照料动物。因此，宋“爱动物就像爱自己的眼珠子”，合作社的社员们授予他“牲口迷”的称号。²⁹³另一位天津饲养员吴元浩将牲畜定义为“社的命根子”和“社的宝贝”，认识到它们在集体经济中的重要性。她于1959年坚定决心，“如果共产党需要，我将终身担任饲养员。”她努力“不断改进饲养和管理[动物], 的方法，让动物吃得更好，为支持[集体] 农业生产中的大跃进做出更大贡献。”²⁹⁴从这个意义上说，饲养员存在的理由和动物的价值都置于巩固集体主义的背景下，这与私有制原则是相悖的。

一些饲养员似乎更愿意以一种更宏大的方式阐述动物与集体主义之间的关系。辽宁的吕品明确地将集体动物饲养劳动的意义解释为超越个人主义的大公。在宣扬饲养员的“普通劳动”是财富的源泉之后，他宣称“一个从不关心社会主义事业需求的人

²⁹³ 《宋明是个好饲养员》（1959），天津市档案馆（以下简称TMA），401206800-X0283-C-000779-065。

²⁹⁴ 吴元浩，《我是怎样当好饲养员的：交河县郝村人民公社大吴潘生产队吴元浩》（1959），TMA, 401206800-X0283-C-000779-089。

only considers individual interests can make a great result. This kind of person always wanders within the small cage of individualism (个人主义的小圈), has no lofty spirit, and only has a confined purview.” In other words, good caretakers must put the public and communal first.²⁹⁵

The demand to become a collectivist-minded caretaker, however, must have been neither easy nor natural. The somewhat counterintuitive nature of the collectivist ethics—prioritizing the common over the individual—often seemed to lead some caretakers in propaganda to express exaggerated loyalty to what elite revolutionaries designed or overreact to sarcasm. Wang Zhiwen from Yilin Commune, Funing County, Zhejiang argued that caretakers’ “revolutionary spirit of passionately loving collectives” would be essential for the “development of collective production and livestock industry.”²⁹⁶ Li Renhao from Xingshu Commune, Jin County, Liaoning was offended by neighbors who argued that “pig raising is too filthy.” Li retorted, “Filthy? Then why don’t you guys feel filthy when eating pork?! You should not see my job as filthy. My [revolutionary] mind cannot be filthy. [...] If [my body] gets filthy, it can be washed off. [In doing so, I] can bring more piglets into the world, and our cooperatives’ revenue will be increased.”²⁹⁷

Nonhuman animal agency was another source of difficulty for Maoist caretakers.²⁹⁸ Liu Qingcai from Lishu County, Jilin was a commune’s pig caretaker who was nicknamed “pig maniac”

²⁹⁵ Lǚ Pin 吕品, “Pingfan de laodong, guangrong de shiye” 平凡的劳动光荣的的事业, in *Hongse qingnian siyangyuan* 红色青年饲养员, ed. Gongqingtuan Liaoning sheng wei bangongshi 共青团辽宁省办公室 (Shenyang: Liaoning renmin chubanshe, 1960), 1–3.

²⁹⁶ “Jiangsu sheng renwei haozhao quansheng siyangyuan xuexi Wang Zhiwen re’ai jiti de geming jingshen” 江苏省人委号召全省饲养员学习王志文热爱集体的革命精神, *Renmin Ribao* 人民日报 (June 21, 1965).

²⁹⁷ Li Renhao 李仁好, “Fanzhi nengshou” 繁殖能手, in *Hongse qingnian siyangyuan*, ed. Gongqingtuan Liaoning sheng wei bangongshi, 14–18.

²⁹⁸ I broadly define the problematic concept of “animal agency” as animals’ ability to make trouble. Although animal agency is not entirely identical with human agency, such “trouble-making” would interrupt and reshape human actors’

只顾个人利益的人，是不能做出什么大成绩的。这种人永远在个人主义的小圈子里打转，没有远大的抱负，只有狭隘的眼界。” 换句话说，优秀的饲养员必须把公共和集体利益放在首位。²⁹⁵

然而，成为一个具有集体主义思想的饲养员，一定既不容易也不自然。集体主义伦理——将集体利益置于个人利益之上——多少有些反直觉，这往往导致一些宣传中的饲养员对精英革命者设计的方案表现出夸张的忠诚，或者对讽刺反应过度。浙江省阜宁县益林公社的王志文认为，饲养员“热爱集体的革命精神”对“集体生产和畜牧业的发展”至关重要。²⁹⁶ 辽宁省晋县兴墅公社的李仁浩对邻居们“养猪太脏”的说法感到不满。李反驳道：“脏？那你们吃猪肉的时候怎么不觉得脏呢？！你们不应该把我的工作看作是肮脏的。我的[革命]思想是不能肮脏的。[……]如果[我的身体]变脏了，可以洗掉。[这样做，我]可以为世界带来更多的小猪，我们合作社的收入也会增加。”²⁹⁷

非人类动物的能动性是毛主义看护者面临的另一个难题。²⁹⁸ 吉林梨树县的刘清才是一名公社的养猪员，他被戏称为“猪痴”

²⁹⁵ 吕品吕品, “平凡的劳动，光荣的事业” 平凡的劳动光荣的的事业, 载*红色青年饲养员* 红色青年饲养员, 共青团辽宁省办公室编共青团辽宁省办公室（沈阳：辽宁人民出版社，1960），1-3页。

²⁹⁶ “江苏省人委号召全省饲养员学习王志文热爱集体的革命精神” 江苏省人委号召全省饲养员学习王志文热爱集体的革命精神, 《人民日报人民日报》（1965年6月21日）。

²⁹⁷ Li Renhao 李仁好, “Fanzhi nengshou” 繁殖能手, 载于*Hongse qingnian siyangyuan*, 共青团辽宁省办公室编，14-18页。

²⁹⁸ 我将“动物能动性”这一有争议的概念广义地定义为动物制造麻烦的能力。尽管动物能动性与人类能动性不完全相同，但这种“制造麻烦”的行为会打断并重塑人类行为者的

and “regarded the collective pigsty as his own home and loved pigs like his own children.” One hog drew his attention; it had eaten nothing all day. Liu was concerned that the pig was sick. But the next day, it just foraged “normally.” This pig repeatedly showed similar behavior. Later, Liu figured out that it was a “binge-eating pig” (抢食猪) with irregular eating habits. He had to struggle with the stubborn pig to ensure that it regularly ate a fixed amount of food.²⁹⁹

Xu Xuelin was another model caretaker from a collective in the western suburb of Shanghai. One day, while Xu was cleaning the pigsty, a pig in heat suddenly escaped. Xu immediately ran after the pig. She was lucky enough to catch it but then immediately slipped and fell into a sewage gutter with the pig in her arms.³⁰⁰ Similarly, Wan Fuxiu realized that parturient sows tended to pile up hay to make nests. Because she thought that the mounds of hay could lead to piglets being suffocated or injured, she flattened them. But the pigs rebuilt them, and she had to keep flattening them. In this way, the animals were good at making their caretakers do extra work.³⁰¹

Although Maoist animal caretakers did not seem to believe in Western norms of animal rights or welfare, they tried their best, or were required, to provide pigs with outdoor activities. They believed that by doing so, “pigs would get fat more quickly and healthier.” According to an unpublished document composed perhaps for a model workers meeting in Yongfeng County,

preexisting ways of thinking, acting, and organizing the sociopolitical. For the recent discussions on animal agency, see Jason Hribal, “Animals, Agency, and Class: Writing the History of Animals from Below,” *Human Ecology Review* 14.1 (July 2007): 101–112; Susan Nance, *Entertaining Elephants: Animal Agency and the Business of the American Circus* (Baltimore: Johns Hopkins University Press, 2013); Chris Pearson, “Dogs, History, and Agency,” *History and Theory* 52.4 (December 2013): 128–145; David Gary Shaw, “The Torturer’s Horse: Agency and Animals in History,” *History and Theory* 52.4 (December 2013): 146–167.

²⁹⁹ Cai Ming 蔡明, “Yangzhu mi: ji siyangyuan Liu Qingcai” 养猪迷: 记饲养员刘清才, *Renmin Ribao* 人民日报 (December 8, 1959).

³⁰⁰ Chen Xigen 陈锡根, *Nü siyangyuan Xu Xuelin* 女饲养员徐雪林 (Shanghai: Shanghai renmin chubanshe, 1956), 7.

³⁰¹ Nanchang xian jiangxiang gongshe tuanjie dadui, “Yige nǚ siyangyuande chengzhang.”

, 并“把集体猪圈当成自己的家, 爱猪如子”。一头猪引起了他的注意; 它一整天什么也没吃。刘担心这头猪病了。但第二天, 它又“正常”地觅食了。这头猪反复表现出类似的行为。后来, 刘发现它是一头“抢食猪” (抢食猪), 饮食习惯不规律。他不得不与这头固执的猪搏斗, 以确保它定期吃到固定量的食物。²⁹⁹

徐学林是上海西郊一个集体农场的另一位模范饲养员。一天, 徐在打扫猪圈时, 一头发情的猪突然跑了出来。徐立刻追了上去。她很幸运地抓住了它, 但随即滑倒, 抱着猪掉进了污水沟里。³⁰⁰ 同样, 万福秀发现分娩的母猪喜欢堆积干草筑巢。她认为干草堆可能导致小猪窒息或受伤, 于是 she 把它们弄平了。但猪又重新筑巢, 她不得不继续把它们弄平。就这样, 这些动物很擅长让它们的饲养员做额外的工作。³⁰¹

尽管毛泽东时代的动物饲养员似乎不相信西方动物权利或福利规范, 但他们尽力或被要求为猪提供户外活动。他们认为, 这样做“猪会更快、更健康地增肥”。根据一份可能为永丰县劳模会议撰写的未发表文件,

preexisting ways of thinking, acting, and organizing the sociopolitical. For the recent discussions on animal agency, see Jason Hribal, “Animals, Agency, and Class: Writing the History of Animals from Below,” *Human Ecology Review* 14.1 (July 2007): 101–112; Susan Nance, *Entertaining Elephants: Animal Agency and the Business of the American Circus* (Baltimore: Johns Hopkins University Press, 2013); Chris Pearson, “Dogs, History, and Agency,” *History and Theory* 52.4 (December 2013): 128–145; David Gary Shaw, “The Torturer’s Horse: Agency and Animals in History,” *History and Theory* 52.4 (December 2013): 146–167.

²⁹⁹ 蔡明, 养猪迷: 记饲养员刘清才, 《人民日报》(1959年12月8日)。

³⁰⁰ 陈锡根, 《女饲养员徐雪林》(上海: 上海人民出版社, 1956), 第7页。

³⁰¹ Nanchang xian jiangxiang gongshe tuanjie dadui, “Yige nǚ siyangyuande chengzhang.”

Jiangxi, Xiong Yihe, a local caretaker, opened the pigpen at 5:30 every morning to let the pigs run. They returned for breakfast at 7:00. Then, the pigs had midday outdoor activities between 11:00 a.m. and 2:00 p.m. and dinner time at 4:30 p.m. At 7:00 p.m., the pigs played outside one more time. Xiong was, simultaneously, disciplining and being disciplined by the pigs.³⁰² Caretakers had to work with living animals for the great leap in developing communal and national livestock industry, no matter how tough their work was.

The Ideology of Animal Care in the 1960s

The intensification of collectivization culminated in the infamous Great Leap Famine in 1959–1961, which according to historians’ estimates claimed 30 million human lives nationwide.³⁰³ Although the official statistical data have been published (Table 2 and Table 3), there is no scholarly consensus on the number of animals that perished in the Famine.³⁰⁴ But this crisis did lead the state to collect the stories of local caretakers who struggled to keep their livestock fed. Shao Henglan, a 32-year-old commune caretaker in Sinan County, Guizhou, was one such caretaker. According to a *People’s Daily* article dedicated to celebrating exemplary behavior, she was prudent and devoted. Summer came without fail in the hungry year, and her commune, located in the mountains, became green. Looking at the verdant scenery and then, her limping oxen, Shao thought that winter would come, too.

³⁰² Xiong Yihe 熊义和, “Wo shi zenyang zuodao si nian bu si zhu” 我是怎样做到四年不死猪 (October 1959), Yongfeng xian tengtian guoying shangdian siyangyuan Xiong Yihe zishu 永丰县藤田国营商店饲养员熊义和自述, Unpublished material.

³⁰³ Kimberley Ens Manning and Felix Wemheuer, eds, *Eating Bitterness: New Perspectives on China’s Great Leap Forward and Famine* (Vancouver: UBC Press, 2011); Xun Zhou, ed., *The Great Famine in China, 1958-1962: A Documentary History* (New Haven: Yale University Press, 2012).

³⁰⁴ Table 2 and Table 3 suggest that the Great Leap Famine affected pigs more harshly than oxen, water buffalos, cows, horses, donkeys, mules, and camels. On the critical research on the PRC’s statistics in the 1950s, see Arunabh Ghosh, *Making It Count: Statistics and Statecraft in the Early People’s Republic of China* (Princeton: Princeton University Press, 2020).

江西当地饲养员熊贻和每天早上5:30打开猪圈，让猪跑出来。它们在7:00回来吃早餐。然后，猪在上午11:00到下午2:00之间进行中午户外活动，下午4:30吃晚餐。晚上7:00，猪又在外边玩了一次。熊贻和同时管教着猪，也被猪管教着。³⁰² 饲养员们必须与活生生的动物一起工作，以实现发展集体和国家畜牧业的巨大飞跃，无论他们的工作多么艰苦。

20世纪60年代的动物饲养理念

集体化的高潮最终导致了1959年至1961年臭名昭著的大跃进饥荒，据历史学家估计，这场饥荒在全国范围内夺走了3000万人的生命。³⁰³ 尽管官方统计数据已经公布（表2和表3），但对于在这场饥荒中死亡的动物数量，学术界尚未达成共识。³⁰⁴ 但这场危机确实促使国家收集了那些努力维持牲畜生存的当地饲养员的故事。贵州思南县32岁的公社饲养员邵恒兰就是其中一位。根据一篇旨在表彰模范行为的《人民日报》文章，她谨慎而专注。饥荒年夏天如期而至，她所在的位于山区的公社变得一片翠绿。看着翠绿的景色，再看看她跛行的牛，邵恒兰心想，冬天也快来了。

³⁰² 熊义和，“我是怎样做到四年不死猪”（1959年10月），永丰县藤田国营商店饲养员熊义和自述，未出版材料。

³⁰³ Kimberley Ens Manning and Felix Wemheuer, eds, *Eating Bitterness: New Perspectives on China’s Great Leap Forward and Famine* (Vancouver: UBC Press, 2011); Xun Zhou, ed., *The Great Famine in China, 1958-1962: A Documentary History* (New Haven: Yale University Press, 2012).

³⁰⁴ 表2和表3表明，大跃进饥荒对猪的影响比对牛、水牛、奶牛、马、驴、骡子和骆驼更严重。关于20世纪50年代中华人民共和国统计数据的批判性研究，请参见Arunabh Ghosh, *Making It Count: Statistics and Statecraft in the Early People’s Republic of China*（普林斯顿：普林斯顿大学出版社，2020）。

Whenever she was free, Shao ran into the mountain to harvest miscellaneous plants for the animals to eat. Shao dried them in the sun and kept them as silage. The pile of plants represented her hope that she and her animals would survive another harsh winter.³⁰⁵

Year	No. of Large Livestock	
	Total No.	No. of Draft Animals
1955	87,750,000	55,710,000
1956	87,730,000	54,740,000
1957	83,820,000	53,680,000
1958	77,680,000	49,920,000
1959	79,120,000	46,600,000
1960	73,360,000	41,240,000
1961	69,490,000	38,180,000
1962	70,200,000	40,180,000
1963	75,050,000	40,330,000
1964	79,430,000	41,520,000
1965	84,210,000	43,220,000
1966	87,400,000	N/A
1967	89,820,000	N/A
1968	91,790,000	N/A
1969	92,280,000	N/A
1970	94,360,000	49,350,000
1971	95,370,000	49,900,000
1972	95,760,000	51,450,000
1973	97,180,000	51,400,000
1974	97,530,000	51,460,000
1975	96,860,000	51,520,000
1976	94,980,000	50,730,000
1977	93,750,000	50,080,000

Table 3. The total number of large livestock (bovines, equines, and camels) raised in China, 1955–1977. Source: Zhonghua renming gongheguo nongyebu xumu shouyisi, *Zhongguo xumuye tongji 1949–1989*, 21.

³⁰⁵ Liu Yangshen 刘扬深, “Mofan nü siyangyuan Shao Henglan” 模范女饲养员邵恒兰, *Renmin Ribao* 人民日报 (June 28, 1959).

每当有空，邵都会跑到山里为动物们收割各种植物。邵把它们晒干，然后储存起来作为青贮饲料。这堆植物代表着她和她的动物们能够度过又一个严冬的希望。³⁰⁵

Year	大型牲畜数量	
	总数	役畜数量
1955	87,750,000	55,710,000
1956	87,730,000	54,740,000
1957	83,820,000	53,680,000
1958	77,680,000	49,920,000
1959	79,120,000	46,600,000
1960	73,360,000	41,240,000
1961	69,490,000	38,180,000
1962	70,200,000	40,180,000
1963	75,050,000	40,330,000
1964	79,430,000	41,520,000
1965	84,210,000	43,220,000
1966	87,400,000	N/A
1967	89,820,000	N/A
1968	91,790,000	N/A
1969	92,280,000	N/A
1970	94,360,000	49,350,000
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Table 3. The 总计 umber of large livestock (bovines、马科动物和 camels) raised in C中国, 1955年至今 1977. Source: Zhonghua renming gongheguo nongyebu xumu shouyisi, *Zhongguo xumuye tongji 1949–1989*, 21.

³⁰⁵ Liu Yangshen 刘扬深, “Mofan nü siyangyuan Shao Henglan” 模范女饲养员邵恒兰, *Renmin Ribao* 人民日报 (1959年6月28日)。

According to an unpublished document produced for the animal husbandry work meeting in Tongxin County, Ningxia, Wang Chuanhe, a production team caretaker in Linyi County, Shanxi, was running out of fodder in the Famine years. Like Shao, Wang picked local plants for the oxen. But the oxen still lost weight and production team leaders had no solution. Wang decided to plunder his own nearly bare kitchen. In the dead of night, he made about 25 kg of corn noodles to feed the animals without his wife knowing.³⁰⁶ Once again, collective animals represented the common good in more-than-people’s communes, whereas only caring about oneself and one’s family was a sign of, in Lǚ Pin’s words, “the small cage of individualism.”

The end of the Famine also brought about changes in central politics as Mao retreated from the front lines. Liu Shaoqi and Deng Xiaoping then enforced the “readjustment” program in the first half of the 1960s. Implicitly admitting the impossibility of incorporating every economic activity and actor into collectivist and concentrated planning, the CCP now toned down the intensity of collectivism and introduced material incentives to grassroots collectives.³⁰⁷ More importantly, the core of post-Great Leap Forward reforms to animal policy was to expand the private sector in tandem with keeping the collective economy alive. Liu signaled that the CCP would selectively tolerate private ownership, reinforcing individual household animal caretaking.³⁰⁸ It is particularly

³⁰⁶ Wang Chuanhe 王传河, “Yixin weile geming: Shanxi Linyi Chengxi gongshe Haoyi dadui siyangyuan Wang Chuanhe zishu” 一心为了革命: 山西临猗城西公社好义大队饲养员王传河自述 (October 21, 1965), Tongxin xian xumu gongzuo huiyi wenjian 同心县畜牧工作会议文件, Unpublished material.

³⁰⁷ For example, in 1964, caretakers’ salary and bonus were specified. Those who took better care of fecund mother cows had a higher reward than their mediocre colleagues. Also, written contracts between collectives and caretakers were more commonly drawn up, stipulating that although animals were still collectively owned, it was caretakers’ individual responsibility to manage the animals. Kai xian renmin weiyuanhui 开县人民委员会, *Muniu siyangyuan gongzuozheng* 母牛饲养员工作证 (N.p., 1964).

³⁰⁸ The new slogan of “developing collective pig raising and ensuring the simultaneous lifting of the public and individual” (发展集体养猪, 保证公私并举) reflected this policy change. Shanxi sheng nongyeting xumu shouyiju 山西省农业厅畜牧兽医局, ed. *Zhude siliao shi rongyi jiejuede: Jiejue yangzhu siliao dianxing jingyan xuanji*. 猪的饲料是容易解决的: 解决养猪饲料典型经验选集 (Taiyuan: Shanxi sheng nongyeting xumu shouyiju, 1966), 5.

根据宁夏同心县畜牧工作会议的一份未发表文件，山西临猗县生产队饲养员王传河在饥荒年代饲料告罄。和邵一样，王也为牛采摘当地植物。但牛仍在掉膘，生产队领导束手无策。王决定搜刮自己几乎空无一物的厨房。在深夜，他背着妻子做了大约25公斤玉米面来喂牲畜。³⁰⁶ 再次，集体动物代表了超人民公社的公共利益，而只关心自己和家人则是，用吕频的话说，“个人主义的小牢笼”。

饥荒的结束也带来了中央政治的变化，毛泽东从前线撤退。刘少奇和邓小平随后在1960年代上半年推行了“调整”计划。中共含蓄地承认将所有经济活动和参与者纳入集体主义和集中计划是不可能的，因此现在降低了集体主义的强度，并向基层集体引入了物质激励。³⁰⁷ 更重要的是，大跃进后动物政策改革的核心是在保持集体经济活力的同时扩大私营部门。刘少奇表示，中共将有选择地容忍私有制，加强个体家庭的动物饲养。³⁰⁸ 这尤其

³⁰⁶ Wang Chuanhe 王传河, “Yixin weile geming: Shanxi Linyi Chengxi gongshe Haoyi dadui siyangyuan Wang Chuanhe zishu” 一心为了革命: 山西临猗城西公社好义大队饲养员王传河自述 (October 21, 1965), Tongxin xian xumu gongzuo huiyi wenjian 同心县畜牧工作会议文件, Unpublished material.

³⁰⁷ For example, in 1964, caretakers’ salary and bonus were specified. Those who took better care of fecund mother cows had a higher reward than their mediocre colleagues. Also, written contracts between collectives and caretakers were more commonly drawn up, stipulating that although animals were still collectively owned, it was caretakers’ individual responsibility to manage the animals. Kai xian renmin weiyuanhui 开县人民委员会, *Muniu siyangyuan gongzuozheng* 母牛饲养员工作证 (N.p., 1964).

³⁰⁸ “发展集体养猪, 保证公私并举”的新口号反映了这一政策变化。山西省农业厅畜牧兽医局编：《猪的饲料是容易解决的：解决养猪饲料典型经验选集》（太原：山西省农业厅畜牧兽医局，1966），第5页。

worth noting that in these years, peasants were allowed to own not only pigs but also draft animals.³⁰⁹ The purpose of the revolutionary social experiment changed from pushing ahead with a collective animal economy to finding a new balance between the collective and the private.

Not surprisingly, the readjustment policies faced a backlash from Mao and his loyal followers who initiated the Socialist Education Movement (1963–1966) and the Great Proletarian Cultural Revolution (1966–1976). They would give Liu Shaoqi and his assistants the fatal label of being a “capitalist roader.” Opposing such “counterrevolutionary revisionism,” Mao and advocates of the Cultural Revolution reprioritized communal and public interests, symbolized by the quantity and quality of collectively owned animals, over private approaches to material life.

The experimental coexistence of collective and individual animal keeping, nevertheless, fundamentally remained throughout the rest of the Maoist years and even outlived the Chairman, who died in 1976. Neither disappeared until the post-Mao leadership disbanded all the grassroots cooperatives between 1978 and 1985.³¹⁰ Some domestic animals, especially pigs, kept by individual villagers, not in collective pens, were still privately owned, taken care of, and disposed of. Within collective economic networks, animals as shared property were owned by collectives, but managed by designated individual caretakers.

³⁰⁹ Zhonggong zhongyang 中共中央 and Guowuyuan 国务院, “Guanyu fazhan dashengchu de jixiang guiding” 关于发展大牲畜的几项规定 (November 22, 1962), in *Xumu gongzuo wenjian xuanbian*, ed. Shaanxi sheng xumu shouyi zongzhan (N.p.: Shaanxi sheng xumu shouyi zongzhan, 1974), 90–92.

³¹⁰ The first three years of the Cultural Revolution (1966–1969) might be an exception. It seems not uncommon that some grassroots caretakers were dragged into the mass violence and extreme chaos of the time. For example, when the order was restored in 1970, Lai Tianwu, a caretaker from Yanjin County, Yunnan, testified that some of his predecessors abandoned the collective animals in the fall of 1967. Lai Tianwu 来天武, “Wei geming gandang yibeizi siyangyuan” 为革命甘当一辈子饲养员 (October 1970), Zhaotong zhuanqu di er ci huoxue huoyong Mao Zedong sixiang jiji fenzi daibiao huiyi cailiao 昭通专区第二次活学活用毛泽东思想积极分子代表会议材料, Unpublished material.

worth noting that in these years, peasants were allowed to own not only pigs but also draft animals.³⁰⁹ The purpose of the revolutionary social experiment changed from pushing ahead with a collective animal economy to finding a new balance between the collective and the private.

不出所料，调整政策遭到了毛泽东及其忠实追随者的强烈反对，他们发起了社会主义教育运动（1963-1966）和无产阶级文化大革命（1966-1976）。他们给刘少奇及其助手贴上了“走资派”的致命标签。为了反对这种“反革命修正主义”，毛泽东和文化大革命的倡导者重新将集体和公共利益放在首位，这体现在集体所有动物的数量和质量上，而非物质生活的私人方式。

然而，集体和个体养殖的实验性共存，在毛泽东时代的剩余岁月里基本保持不变，甚至比1976年去世的毛主席活得更久。直到毛泽东去世后的领导层在1978年至1985年间解散了所有基层合作社，这两种方式才消失。³¹⁰ 一些由个体村民饲养的家畜，尤其是猪，并非在集体猪圈中饲养，仍然是私人所有、私人照料和私人处置的。在集体经济网络中，动物作为共享财产归集体所有，但由指定的个体饲养员管理。

³⁰⁹ 中共中央和国务院, “关于发展大牲畜的几项规定”（1962年11月22日），载陕西省畜牧兽医总站编：《畜牧工作文件选编》（无出版地：陕西省畜牧兽医总站，1974年），第90-92页。

³¹⁰ 文化大革命的头三年（1966-1969）可能是一个例外。一些基层饲养员被卷入当时的群众暴力和极端混乱中，这似乎并不少见。例如，1970年秩序恢复后，云南盐津县的饲养员赖天武作证说，他的一些前任在1967年秋天遗弃了集体动物。赖天武, “为革命甘当一辈子饲养员”（1970年10月），《昭通专区第二次活学活用毛泽东思想积极分子代表会议材料》，未出版材料。

Against this backdrop, the state might suspect that the readjustment period could spoil local caretakers, giving them a taste of the private and neglecting the value of commitment to collectivism.

What was it like to be expected to leave one’s own animals behind and put the collective’s animals first? Why and how did one should cherish animals that one did not own? While maintaining the dual system of collective and private animal caretaking, the state now needed to strengthen propaganda that the collectivist ethics of animal care must remain essential. Therefore, collective caretakers needed to be more than another type of agricultural worker or animal lover. They were at the forefront of the state’s effort to continue the social experiment of collectivism in the wake of the failure of the Great Leap Forward and the arguably “capitalistic” readjustment.

An unpublished document produced at the inception of the Cultural Revolution describes a model caretaker, Cheng Jinqun. Cheng avidly read Mao’s texts during his military service. Among the most powerful lines that Cheng remembered was that “only seeking one’s own interests (自私自利), passivity and slowdown, embezzlement and corruption, careerism, and so on are the most despicable, whereas pursuing greater publicness and self-dedication (大公无私), initiative and endeavor, overcoming self and serving common good (克己奉公), and the spirit of giving oneself over to hard working are truly respectable.”³¹¹ Imprinting the ethics of “overcoming self and serving common good” on his soul, Cheng dreamed of being a “good” collective caretaker.³¹²

³¹¹ The text of Mao cited by Cheng is available: Mao Zedong 毛泽东, *Zhongguo gongchandang zai minzu zhanzheng zhong de diwei* 中国共产党在民族战争中的地位 (Beijing: Renmin chubanshe, 1952 [1938]).

³¹² Cheng Jinqun 程金全, “Yixin wei geming, danghao siyangyuan” 一心为革命, 当好饲养员 (December 1966), 工指第三次后勤工作会议经验介绍 Gongzhi di san ci houqin gongzuo huiyi jingyan jieshao zhiwu, Unpublished material.

在此背景下，国家可能怀疑调整期会宠坏当地饲养员，让他们尝到私有的甜头，从而忽视了对集体主义承诺的价值。人们被期望抛弃自己的动物，将集体的动物放在首位，这是一种怎样的体验？为什么以及如何才能珍惜不属于自己的动物？在维持集体和私人动物饲养双重制度的同时，国家现在需要加强宣传，即动物饲养的集体主义道德必须保持其核心地位。因此，集体饲养员不仅仅是另一种类型的农业工人或动物爱好者。在大跃进失败和“资本主义”调整之后，他们站在国家努力继续集体主义社会实验的最前沿。

一份在文化大革命初期编写的未发表文件描述了一位模范饲养员程金泉。程在服兵役期间如饥似渴地阅读毛泽东的著作。程记忆中最有力量的句子之一是：“只有谋求一己私利（自私自利）、消极怠工、贪污腐化、投机钻营等等，才是最可鄙的；而追求大公无私（大公无私）、积极进取、克己奉公（克己奉公）和艰苦奋斗的精神，才是真正值得尊敬的。”³¹¹ 程金泉将“克己奉公”的道德观念铭刻在灵魂深处，梦想成为一名“好”的集体饲养员。³¹²

³¹¹ 程引用的毛泽东著作如下：毛泽东，《中国共产党在民族战争中的地位》（北京：人民出版社，1952 [1938]）。

³¹² 程金全，《一心为革命，当好饲养员》（1966年12月），《工指第三次后勤工作会议经验介绍之五》，未出版资料。

Perhaps a more complicated voice of caretakers at the height of promoting collectivism was that of Ma Zhi'an from Wenxi County, Shanxi. He seemed to have had a great reputation as a model caretaker in his commune; he was therefore entitled to present his experience at a meeting of activist Mao-readers. According to Ma's autographic notes, his fellow peasants praised him as an excellent caretaker who already "broke the private and stood for the public" (破私立公). But even such a "red" caretaker like Ma had difficulty devoting himself to taking care of the animals owned by the cooperative. He admitted struggling with the question of "whether for the public or for the individual." It was the "fundamental issue regarding the deepest parts of individuals' soul." Struggling with this moral dilemma, Ma came to think that if he just picked a side, his life would be simpler. Ma gave up keeping his own animals at home and worked beside collective oxen night and day.³¹³

Putting his caretaking job ahead of his individual and family affairs made his mother and wife resentful. When he stopped by his home to pick up supplies to take care of a sick ox, his wife told him that their daughter was sick too and that he needed to stay home. Ma was confused. Should he nurse his daughter or go back to the byre to treat the ox? He might have thought that his daughter was being taken care of by her mother; the ox had no one. Ma left his ill daughter behind and ran back to the bovine patient, believing that he had made the right decision, even though it was painful. Ma treated the ox with some traditional remedies without needing a veterinarian's help. The ox recovered. His daughter got better slowly. After many years of struggling, Ma Zhi'an concluded that "to me, the course of caretaking collective animals is no other than the course of erasing the

³¹³ Ma Zhi'an 马治安, "Ting Mao zhuxi de hua wei geming weihao shengkou" 听毛主席的话为革命喂好牲口 (1967), Shanxi sheng Wenxi xian xuexi Mao zhuxi zhuzuo jiji fenzi daibiao dahui dianxing cailiao 山西省闻喜县学习毛主席著作积极分子代表大会典型材料, Unpublished material.

在集体主义盛行时期，文水县马治安的饲养员声音或许更为复杂。他似乎在公社中享有盛誉，是一名模范饲养员；因此，他有资格在毛泽东思想积极分子会议上介绍自己的经验。根据马治安的亲笔笔记，他的同乡称赞他是一位已经“破私立公”的优秀饲养员（破私立公）。但即使是马治安这样一位“红色”饲养员，也很难全身心投入到合作社的牲畜饲养工作中。他承认自己一直在纠结“为公还是为私”的问题。这是“关乎个人灵魂最深处的根本问题”。在与这种道德困境作斗争时，马治安开始认为，如果他只选择一边，他的生活就会更简单。马治安放弃了在家饲养自己的牲畜，日夜与集体耕牛一起劳作。³¹³

他把照料工作放在个人和家庭事务之前，这让他的母亲和妻子心生怨恨。当他顺道回家取用品去照料一头生病的牛时，他的妻子告诉他，他们的女儿也病了，他需要待在家里。马感到困惑。他应该照顾女儿还是回到牛棚治疗牛？他可能认为女儿有母亲照顾；而牛却无人照料。马把生病的女儿留在家里，跑回了生病的牛身边，他相信自己做出了正确的决定，尽管这很痛苦。马用一些传统疗法治疗了牛，没有寻求兽医的帮助。牛康复了。他的女儿慢慢好转。经过多年的挣扎，马志安总结道：“对我来说，照料集体动物的过程无非是抹去

³¹³ 马治安，“听毛主席的话为革命喂好牲口”（1967），山西省闻喜县学习毛主席著作积极分子代表大会典型材料，未出版资料。

letter of *si* (私, the individual or private) in my brain and establishing that of *gong* (公, the common or public) by relying on Maoist Thought.”³¹⁴

Pushing ahead with the social experiment of collectivism in the mid- and late-1960s, Mao and his followers once again eagerly circulated a didactic message that animals in more-than-people’s communes symbolized the collective and public good which might otherwise could remain intangible and secondary to grassroots caretakers. But some propagandistic materials hint that it was by no means natural or easy for caretakers to come to terms with this ideological demand to prioritize collective interests over individual gains. The next section focuses on this subtle crack in model caretakers’ autobiographic materials and in “bad” caretakers’ self-criticism. In so doing, I suggest that not every local caretaker internalized the state’s vision of more-than-people’s commune based on the selfless care of collective animals.

Outliers in More-than-People’s Communes

Propagandistic materials on animal care produced in the mid- and late-1960s often spotlighted outliers. “Bad” caretakers disliked or devalued animal caretaking. They were portrayed as success-driven individuals who wanted to have a job which was of higher social status than that of animal caretakers. They did not love collective animals because they had not yet overcome their personal selfishness. The stories of these outliers were presented as a lesson to the local population. But the necessity of discussing these undesirable caretakers can be seen as circumstantial evidence that there were certainly many people like them—those who did not accept the collectivist ethics.

³¹⁴ Ma Zhi’an, “Ting Mao zhuxi de hua wei geming weihao shengkou.”

将我脑海中私（私，个人或私人的）的字母抹去，并依靠毛泽东思想确立公（公，共同或公共的）的地位。”³¹⁴

在20世纪60年代中后期，毛泽东及其追随者大力推行集体主义的社会实验，再次积极传播一种说教式的信息，即超人公社中的动物象征着集体和公共利益，否则这些利益可能对基层饲养员来说仍然是无形和次要的。但一些宣传材料暗示，饲养员要接受这种将集体利益置于个人利益之上的意识形态要求绝非自然或易事。下一节将重点探讨模范饲养员自传材料和“不良”饲养员自我批评中这种微妙的裂痕。通过这样做，我认为并非每个地方饲养员都内化了国家基于对集体动物的无私照料而形成的超人公社愿景。

超人公社中的异类

20世纪60年代中后期制作的动物饲养宣传材料经常聚焦于“异类”。“坏”饲养员不喜欢或贬低动物饲养工作。他们被描绘成追求成功的人，渴望从事比动物饲养员社会地位更高的工作。他们不爱集体动物，因为他们尚未克服个人自私。这些“异类”的故事被当作给当地居民的教训。但讨论这些不受欢迎的饲养员的必要性，可以被视为旁证，表明肯定有许多像他们一样的人——那些不接受集体主义道德的人。

³¹⁴ 马治安，“听毛主席的话为革命喂好牲口。”

According to an unpublished self-criticism circulated in Likou Commune in Jia County, Henan Province, the peasants of the Second Production Team of Xiaozanzhuang Brigade under the commune frequently derided a collective caretaker Zan Po as a “work point maniac” (工分迷) who was seriously compromised by “capitalist ideas.” The people knew that he “was not concerned about and did not love animals,” even beating and mistreating them. Zan only cared about his own “small plot of land” (自留地). Zan was also indifferent to the animals’ diets. Regardless of what the animals ate, they would evacuate their bowels, and he would get extra work points for collecting the manure. It was only a matter of time before Zan’s poor reputation with his neighbors reached the local cadres’ ears.³¹⁵

Zan Po was eventually required to join the Chairman Mao’s Texts Reading Group organized by the brigade leaders in 1965. After reading Mao’s “In Memory of Norman Bethune,” Zan seemed inspired by the Canadian martyr (1890–1939), a surgeon who supported the CCP during the Sino-Japanese War and died of blood poisoning while operating on Communist soldiers in 1939. Mao praised Dr. Bethune as someone who “never benefits himself and exclusively benefits others” (毫不利己, 专门利人). It is unclear if the reading group changed Zan’s thoughts and emotions. But he openly pledged that “I will not beat animals again and take animals’ fodder” in front of his fellow commune members.³¹⁶

³¹⁵ Zan Po 咎坡, “Bu xuexi Mao zhuxi zhuzuo jiushi zuida de mangben” 不学习毛主席著作就是最大的忘本 (November 22, 1966), Henan sheng Jia xian Likou gongshe Xiaozanzhuang dadui di er shengchandui siyangyuan Zan Po fayan 河南省郑县李口公社小咎庄大队第二生产队饲养员咎坡发言, Zhonggong Jia xian weiyuanhui xuexi Mao zhuxi zhuzuo jiji fenzi daibiao huiyi fayan cailiao 中共郑县委员会学习毛主席著作积极分子代表会议发言材料, Unpublished material.

³¹⁶ Zan Po, “Bu xuexi Mao zhuxi zhuzuo jiushi zuida de mangben.”

根据河南省郑县李口公社流传的一份未发表自我批评，社里小张庄大队第二生产队的农民经常嘲笑集体饲养员赞坡是“工分迷”（工分迷），他受到“资本主义思想”的严重腐蚀。人们知道他“不关心也不爱护牲畜”，甚至殴打和虐待它们。赞坡只关心自己的“自留地”（自留地）。赞坡对牲畜的饮食也漠不关心。无论牲畜吃什么，它们都会排泄，而他会因为收集粪便而获得额外的工分。赞坡在邻居中的坏名声传到当地干部耳中只是时间问题。³¹⁵

赞坡最终被要求参加大队领导在1965年组织的毛主席著作学习小组。在阅读了毛泽东的《纪念白求恩》后，赞坡似乎受到了这位加拿大烈士（1890-1939）的启发。白求恩是一位外科医生，在中日战争期间支持中国共产党，并于1939年在为共产党员士兵手术时死于败血症。毛泽东赞扬白求恩医生是“毫不利己，专门利人”（毫不利己，专门利人）的人。目前尚不清楚学习小组是否改变了赞坡的思想和情感。但他当着社员的面公开承诺“我不再打牲畜，也不会再拿牲畜的饲料了”。³¹⁶

³¹⁵ Zan Po 咎坡, “Bu xuexi Mao zhuxi zhuzuo jiushi zuida de mangben” 不学习毛主席著作就是最大的忘本 (November 22, 1966), Henan sheng Jia xian Likou gongshe Xiaozanzhuang dadui di er shengchandui siyangyuan Zan Po fayan 河南省郑县李口公社小咎庄大队第二生产队饲养员咎坡发言, Zhonggong Jia xian weiyuanhui xuexi Mao zhuxi zhuzuo jiji fenzi daibiao huiyi fayan cailiao 中共郑县委员会学习毛主席著作积极分子代表会议发言材料, Unpublished material.

³¹⁶ 赞坡, “不学习毛主席著作就是最大的盲本。”

Zhang Yongting presented a similar case. Zhang was a commune caretaker from Wenxi County, Shanxi. In 1965, Zhang was dismayed to read a cover story in *Shanxi Daily* on the “model caretaker” Wang Chuanhe.³¹⁷ Both started animal caretaking in the same year in the mid-1950s. Zhang’s commune was very close to Wang’s in Linyi County. However, their reputations could not have been more different. Wang was admired throughout the province; Zhang’s incompetence was well known to his fellow commune members as the animals under his care were “undersized” and “feeble.”³¹⁸ Such a divergence led Zhang to “self-criticize:

What makes me feel the most ashamed was that while Wang Chuanhe never benefits himself and exclusively benefits others for the sake of the revolution with a sincere heart, I self-indulgently took 10 kg of fodder for [collective animals] to my home [to feed the privately owned animals]. No one knows about this as I have never told anyone. Today, compared to Wang, I think I’m despicable. [...] From now on, I will never cause damage to collectives, act against the revolution, and disgrace [revolutionary] poor peasants.³¹⁹

Similar self-criticisms were frequent between the mid-1960s and mid-1970s. In many cases, they were forced by either local cadres or fellow peasants. It is hard to tell how trustworthy the information in these records is. Nevertheless, we could speculate on the reason that the state circulated such stories. The Communists might have focused as much on external behavior as it was on internal psychology. As the cases of Zan Po and Zhang Yongting suggest, self-criticizers were expected to meet the standard set by model caretakers like Wang. The process of in-commune performance evaluation, comparison, and self-criticism, and the Maoist ideological norms of

³¹⁷ Zhonggong Shanxi shengwei 中共山西省委, “Guanyu xuexi he xuanchuan Wang Chuanhe tongzhi de jue ding” 关于学习和宣传王传河同志的决定, *Shanxi Ribao* 山西日报 (June 4, 1965).

³¹⁸ Zhang Yongting 张永亭, “Juexin dang Wang Chuanhe shi de siyangyuan” 决心当王传河式的饲养员, *Renmin Ribao* 人民日报 (August 18, 1965).

³¹⁹ Zhang Yongting, “Juexin dang Wang Chuanhe shi de siyangyuan.”

张永庭也提出了一个类似的案例。张是山西闻喜县的一名公社饲养员。1965年，张永庭读到《山西日报》上关于“模范饲养员”王传河的封面报道时，感到非常沮丧。³¹⁷ 两人都是在20世纪50年代中期开始从事牲畜饲养工作的。张所在的公社与临猗县王所在的公社非常近。然而，他们的声誉却截然不同。王在全省都备受推崇；而张的无能则在公社社员中广为人知，因为他饲养的牲畜“体型过小”且“虚弱”。³¹⁸这种差异导致张“自我批评”道：

最让我感到羞愧的是，王传河从不为自己谋利，而是为了革命一心一意地为他人谋利，而我却自作主张地将10公斤饲料从[集体牲畜] 那里拿回家[喂养私人牲畜]。没有人知道这件事，因为我从未告诉过任何人。今天，与王相比，我觉得自己很卑鄙。[……] 从今以后，我绝不会再损害集体利益，绝不会再做出反革命的事情，绝不会再给[革命] 贫农丢脸。³¹⁹

类似的自我批评在20世纪60年代中期到70年代中期频繁出现。在许多情况下，它们是由地方干部或农民同伴强迫进行的。这些记录中的信息有多大可信度很难说。然而，我们可以推测国家传播这些故事的原因。共产党人可能既关注外部行为，也关注内部心理。正如赞坡和张永庭的案例所示，自我批评者被期望达到像王这样的模范看护人设定的标准。社内绩效评估、比较和自我批评的过程，以及毛泽东思想的意识形态规范

³¹⁷ 中共山西省委, 《关于学习和宣传王传河同志的决定》, 山西日报 (1965年6月4日)。

³¹⁸ 张永亭, “决心当王传河式的饲养员”, 《人民日报》 (1965年8月18日)。

³¹⁹ 张永亭, “决心当王传河式的饲养员”。

prioritizing the collective over the individual, were expected to be co-constituted and mutually reinforcing.³²⁰

Ideology may become stronger when it is practices at home. Li Zongxiu was a seasoned caretaker who started taking care of collectively grown animals in suburban Beijing in 1960 when he was 14 years old. He had difficulty focusing on his job and finding it rewarding. He “hated the muckiness and was sick of exhaustion” of working with animals. Li was sloppy in the pigsty.

Whenever possible, he went back home to loaf. In 1962, his mother finally lost her patience and started admonishing her son, saying that “your dad passed away when you were one year old.”³²¹ She continued:

I brought up you and your five siblings by myself and took you all to the liberation. You went to school and are now working at the collective animal farm. What you eat and what you wear have all changed. If it were the [pre-revolutionary] old society, what would sturdy [yet poor] young men like you have done? You would have meaninglessly worked for other [rich] families and meaninglessly emptied your rice bowl. You would have been worried that one day the [employer] families would not need you anymore. Don’t forget where you came from!³²²

Since then, according to this archival document, Li became grateful for the opportunity to manage the collective’s precious property, the animals, and serve the public good. Perhaps, Li’s destitute family never got the sense of what private ownership was before the 1949 communist revolution. It might not be always the case that one needs to experience private ownership in order to develop a sense of communal ownership. But it seems that members of the younger generation,

³²⁰ On self-criticism, also see Dittmer, Lowell. “The Structural Evolution of ‘Criticism and Self-Criticism.’” *The China Quarterly* 56.10 (December 1973): 708–729.

³²¹ Li Zongxiu 李宗秀, “Wei geming yanghao zhu: Siyangyuan Li Zongxiu” 为革命养好猪: 北京市海淀区九王坟畜牧场饲养员李宗秀 (April 1966), Beijing Municipal Archives, 002-021-00689-00137.

³²² Li Zongxiu, “Wei geming yanghao zhu: Siyangyuan Li Zongxiu.”

集体优先于个人，两者应共同构成并相互促进。³²⁰

当意识形态在家中得到实践时，它可能会变得更强大。李宗秀是一位经验丰富的饲养员，1960年他14岁时开始在北京郊区照看集体饲养的动物。他很难专注于自己的工作，也觉得这份工作没有意义。他“讨厌与动物打交道的肮脏和疲惫”。李在猪圈里工作马虎。只要有可能，他就回家偷懒。1962年，他母亲终于失去了耐心，开始训斥儿子，说“你一岁的时候你爸就去世了。”³²¹她接着说：

我一个人把你们六个兄弟姐妹拉扯大，把你们都带到了解放。你上了学，现在在集体动物农场工作。你吃什么穿什么都变了。如果是[革命前的] 旧社会，像你这样健壮[却贫穷的] 年轻人会做什么？你会毫无意义地为其他[富裕的] 家庭工作，毫无意义地吃光你的饭碗。你会担心有一天[雇主] 家庭不再需要你。不要忘记你来自哪里！³²²

从那时起，根据这份档案文件，李开始感激有机会管理集体的宝贵财产——牲畜，并为公共利益服务。也许，李的贫困家庭在1949年共产主义革命之前从未体会过什么是私有制。一个人不一定非要经历私有制才能产生集体所有权意识。但似乎年轻一代的成员，

³²⁰ 关于自我批评，另见Dittmer, Lowell。 “‘批评与自我批评’的结构演变。”《中国季刊》56.10（1973年12月）：708-729。

³²¹ 李宗秀，“为革命养好猪：饲养员李宗秀”，北京市海淀区九王坟畜牧场饲养员李宗秀（1966年4月），北京市档案馆，002-021-00689-00137。

³²² 李宗秀，“为革命养老猪：饲养员李宗秀。”

like Li, could not naturally dedicate themselves to the ideal of collective ownership and responsibility. It was Li’s mother who educated the troubled young man on what working with commune animals would mean to his family. This could have been the exact point that the state wanted to make.³²³

In the midst of the Cultural Revolution, stories of caretakers who underwent moral transformations still needed to be circulated. Animals were central figures in these narratives as catalysts of acceptance of the collectivist ideology. According to an unpublished document created by the Revolutionary Committee of Shixing County, Guangdong Province for a meeting of Maoist activists, Qiu Nanying learned about animal care while in the army. When he heard the words “raising ducks and geese” come out of his superior’s mouth, he felt like “half of his heart got frozen.” Since he had attended an upper middle school before joining the PLA in March 1966, Qiu believed he was eligible to be a technician. His face turned red when comrades mockingly called him a “duck commander” (鸭司令). One day, his parents wrote him to ask what he did in the camp. Qiu equivocated: “It is a military secret. I cannot let you know. Anyway, it is about revolution!” He had no choice but to come to terms with the job. His halfhearted attitude was soon detected by the regiment leaders and he was advised to “intensively” read Mao. Qiu’s autographic narrative suggested that raising animals could be a way of “serving the People more completely and thoroughly.” Then, all of a sudden, ducks and geese started to “look adorable” to him. Two years

³²³ Li Zongxiu, “Wei geming yanghao zhu: Siyangyuan Li Zongxiu.”

像李一样，无法自然而然地投身于集体所有制和责任制的理想。正是李的母亲教育了这个陷入困境的年轻人，让他明白与公社动物一起工作对他的家庭意味着什么。这可能正是国家想要表达的观点。³²³

在文化大革命期间，看护者经历道德转变的故事仍需传播。动物是这些叙事中的核心角色，它们是接受集体主义意识形态的催化剂。根据广东省始兴县革命委员会为毛主义积极分子会议编写的一份未发表文件，邱南英在部队期间学会了动物饲养。当他听到上级说出“养鸭养鹅”时，他感到“半边心都冻住了”。由于他在1966年3月参军前曾就读于高中，邱认为自己有资格成为一名技术员。当同志们嘲讽地称他为“鸭司令”（鸭司令）时，他涨红了脸。有一天，他的父母写信问他在营地做什么。邱含糊其辞地说：“这是军事秘密。我不能告诉你们。反正，这是关于革命的！”他别无选择，只能接受这份工作。他敷衍了事的态度很快被团领导发现，并被建议“深入”阅读毛主席著作。邱的自述表明，饲养动物可以是一种“更彻底、更全面地为人民服务”的方式。然后，突然之间，鸭子和鹅开始在他看来“变得可爱起来”。两年后

³²³ 李宗秀，“为革命养好猪：饲养员李宗秀。”

later, Qiu found himself shouting out “Li-li lai-lai” and smiling as a flock of birds came running toward him.³²⁴

Animal Care and Feminism

The association of the ideology of animal care in Maoist China with local caretakers and peasants had a collectivist mindset. However, there seemed to be an additional layer: gender. Since care itself was regarded as a feminine duty in China just as it was in many other parts of the world, animal caretaking, especially pig raising, tended to be thought as a job for females.³²⁵ As such, some propaganda materials were intended for at women in the countryside. In the process, if necessary, the state tried to change conventional gender roles for the sake of its collectivist priorities.³²⁶

Certain gender biases shaped the discourse on who was eligible to be an animal caretaker. Many male caretakers felt embarrassed to have been sent to a collective pigpen. Xiong Yihe was one such man. When he first started to take care of pigs for his collective in 1956, Xiong frequently

后来，邱发现自己大喊“哩哩啦啦”，并微笑着，一群鸟儿向他跑来。³²⁴

动物护理与女权主义

毛泽东时代中国动物护理意识形态与当地护理人员和农民的结合具有集体主义心态。然而，似乎还有一层额外的因素：性别。由于护理本身在中国被视为女性的职责，就像在世界许多其他地方一样，动物护理，特别是养猪，往往被认为是女性的工作。³²⁵ 因此，一些宣传材料是针对农村妇女的。在此过程中，如有必要，国家为了集体主义优先事项，试图改变传统的性别角色。³²⁶

某些性别偏见影响了关于谁有资格成为动物饲养员的讨论。许多男性饲养员对于被派往集体猪圈感到尴尬。熊贻和就是其中之一。1956年，当他第一次开始为集体养猪时，熊经常

³²⁴ Qiu, Nanying 邱南英, “Zai Mao Zedong sixiangde buyuxia zijue jieshou zaijiaoyu” 在毛泽东思想的哺育下自觉接受再教育 (February 20, 1969), Guangzhou junqu di si jie xuexi Mao Zedong sixiang jiji fenzi daibiao dahui fayan cailiao 广州军区第四届学习毛泽东思想积极分子代表大会发言材料, Unpublished material.

³²⁵ In terms of the gender dynamics of animal-related jobs, it could be helpful here to clarify that driving a mule cart or having oxen work in the field were not part of caretakers’ work. They were “animal servicers” (使役员) business. Unlike caretakers, servicers were mostly male.

³²⁶ On the rich account of plural ways in which the Chinese revolution was gendered as well as and the category of “rural woman” was made during the collective period, see Gail Hershatter, *The Gender of Memory: Rural Women and China’s Collective Past* (Berkeley: University of California Press, 2011). On the state feminists within the PRC apparatus, see Zheng Wang, *Finding Women in the State: A Socialist Feminist Revolution in the People’s Republic of China, 1949–1964* (Oakland: University of California Press, 2017). On the Maoist “scientific experiment movement” and rural gender relations, see Sigrid Schmalzer, *Red Revolution, Green Revolution: Scientific Farming in Socialist China* (Chicago: University of Chicago Press, 2016), 119–123.

³²⁴ 邱南英，“在毛泽东思想的哺育下自觉接受再教育”（1969年2月20日），《广州军区第四届学习毛泽东思想积极分子代表大会发言材料》，未出版资料。

³²⁵ 就动物相关工作的性别动态而言，在此澄清一点可能会有所帮助：赶骡车或让牛在地里干活不属于饲养员的工作。那是“使役员”的活儿。与饲养员不同，使役员大多是男性。

³²⁶ 关于中国革命在集体化时期如何被赋予性别以及“农村妇女”这一范畴如何形成的丰富论述，请参见盖尔·赫沙特（Gail Hershatter）的《记忆的性别：农村妇女与中国的集体化过去》（伯克利：加州大学出版社，2011）。关于中华人民共和国体制内的国家女权主义者，请参见王政（Zheng Wang）的《在国家中寻找女性：中华人民共和国的社会主义女权主义革命，1949-1964》（奥克兰：加州大学出版社，2017）。关于毛泽东时代的“科学实验运动”和农村性别关系，请参见西格丽德·施马尔策（Sigrid Schmalzer）的《红色革命，绿色革命：社会主义中国的科学农业》（芝加哥：芝加哥大学出版社，2016），第119-123页。

complained that his career would be ruined because “raising pigs is what gossipy old ladies do.”³²⁷

However, as the following cases show, it is noteworthy that mainstream discourse on animal caretaking as feminine labor tended to exclude young or adolescent women. It might be physically difficult for many girls to manage strong animals. Cleaning their excrement was also too dirty.

Animal caretaking, which was not seen as socially promising, must have not been on the list of dream jobs for anyone’s daughters.

In hope of changing such views, the earliest materials to encourage females to become model caretakers in the mid-1950s appealed to their maternal instincts. They contained the message that every single birth of a baby animal could be a miraculous moment for caretakers. It is said that a model caretaker Hua Yinfeng never forgot the “ineffable joy” of feeling the warmth of newborn piglets. She carefully removed the thin mucous membrane, cut the umbilical cord, and slowly put the squirming piglets in a wooden basket.³²⁸ Xu Xuelin and Zhang Jinmei, two hard-working female caretakers from a collective near Shanghai, also wanted to enjoy the “indescribable happiness” every so often. Two seasoned caretakers knew that one of the sows would give birth very soon as it had

³²⁷ Xiong Yihe, “Wo shi zenyang zuodao si nian bu si zhu.”

³²⁸ Zhang Zheng, “Yige nongchang nü siyangyuan de gongxian.”

抱怨说他的事业会毁掉，因为“养猪是爱嚼舌根的老太太做的事。”³²⁷然而，如下面的案例所示，值得注意的是，关于动物饲养作为女性劳动的 H A RV ARD U N IV ERS ITY 主流论述往往将年轻或青春期女性排除在外。对于许多女孩来说，管理强壮的动物可能在体力上很困难。清理它们的排泄物也太脏了。动物饲养，在社会上不被看好，肯定不在任何人的女儿的理想工作之列。

为了改变这种观点，20 世纪 50 年代中期鼓励女性成为模范饲养员的最早材料诉诸于她们的母性本能。它们传达的信息是，每一个小动物的诞生对饲养员来说都可能是一个奇迹般的时刻。据说，模范饲养员华银凤从未忘记感受新生仔猪温暖的“难以言喻的喜悦”。她小心翼翼地去除薄薄的粘膜，剪断脐带，然后慢慢地将蠕动的仔猪放入木篮中。³²⁸来自上海附近一个集体农场的两位勤劳的女性饲养员徐雪林和张金梅，也想时不时地享受“难以形容的幸福”。两位经验丰富的饲养员知道其中一头母猪很快就会分娩，因为它已经

³²⁷ 熊益和，“我是怎样做到四年不死猪。”

³²⁸ 张正，“一个农场女饲养员的贡献。”

been about 115 days, the average swine gestation period, since it was mated with a boar. So Xu and Zhang spent a few nights in the pigsty (Figure 10). Before long, they met the new piglets.³²⁹

自与公猪交配以来，已经过去了大约115天，这是猪的平均妊娠期。于是徐和张在猪圈里度过了几个夜晚（图10）。不久，他们就见到了新出生的小猪。³²⁹



Figure 10. Xu Xuelin and Zhang Jinmei standing by a pregnant sow in the pigsty. Source: Chen Xigen 陈锡根, *Nü siyangyuan Xu Xuelin* 女饲养员徐雪林 (Shanghai: Shanghai renmin chubanshe. 1956), 9.

³²⁹ Chen Xigen, *Nü siyangyuan Xu Xuelin*, 7–9.



图10. 徐雪林和张金梅站在猪圈里一头怀孕的母猪旁。来源：陈锡根，《女饲养员徐雪林》（上海：上海人民出版社，1956），第9页。

³²⁹ 陈锡根，《女饲养员徐雪林》，第7-9页。

Some didactic narratives were bolder, juxtaposing feminist self-realization with collectivist commitment. A teenage girl, Zhao Aiyi from Kuandian County, Liaoning, was encouraged by local cadres to participate in a three-month animal husbandry technical training session in 1956. Zhao wished to learn how to drive a tractor and, therefore, had never thought of becoming an animal caretaker. But after listening to the inspirational speech of a model caretaker Li Shulan during the session, she came to realize that by taking good care of animals, she could live up to the expectations of “the Party, collective leaders, and fellow villagers.” Zhao said that “I must follow and study her example and will be the next Li Shulan.”³³⁰

Reading the life story of Han Meimei, another female model caretaker, Zhao Wenjun had, for the first time, a positive impression of the job. But a more decisive moment came a few years later in 1957 when her cooperative in Beizhen County, Liaoning established a new collective pigsty. “Watching 50 newly acquired boars passing by [her] front door,” the 17-year-old girl “could not take [her] eyes off the lustrous pigs and came to deeply adore them.” That night she ran to the local cadre’s house and applied to be a swine caretaker. She was disappointed when the leader refused to let her work in the pigsty as he found the task “too mucky and arduous” for a young girl. Zhao, who believed in “thorough liberation of women” (妇女彻底解放), refused to accept this. She tried to persuade the leader, shouting that “there is no fundamental difference between men and women in terms of production, and if men can do something, why on earth women cannot do the same!?” She

³³⁰ Zhao Ai’yi 赵爱义, “Yangzhu guniang” 养猪姑娘, in *Hongse qingnian siyangyuan*, ed. Gongqingtuan Liaoning sheng wei bangongshi, 19–20.

一些说教式叙事更大胆，将女性主义的自我实现与集体主义承诺并置。来自辽宁宽甸县的少女赵爱义在1956年被当地干部鼓励参加为期三个月的畜牧技术培训班。赵爱义希望学习如何驾驶拖拉机，因此从未想过要成为一名动物饲养员。但在培训期间听了模范饲养员李淑兰的励志演讲后，她意识到通过悉心照料动物，她可以不辜负“党、集体领导和乡亲们”的期望。赵爱义说：“我一定要学习她的榜样，成为下一个李淑兰。”

330

赵文君在阅读另一位女性模范饲养员韩美美的生平故事时，第一次对这份工作产生了积极的印象。但一个更具决定性的时刻发生在几年后的1957年，当时她在辽宁北镇县的合作社建立了一个新的集体猪圈。“看着50头新买的公猪从[她]家门前经过，”这位17岁的女孩“眼睛离不开那些油光发亮的猪，并深深地爱上了它们。”那天晚上，她跑到当地干部家，申请成为一名养猪员。当领导拒绝让她在猪圈工作时，她感到很失望，因为他认为这项任务对一个年轻女孩来说“太脏太辛苦”。赵文君相信“妇女彻底解放”，拒绝接受这一点。她试图说服领导，喊道：“男女在生产上没有根本区别，如果男人能做的事情，女人为什么就不能做呢？！”她

³³⁰ 赵爱义 赵爱义, “养猪姑娘” 养猪姑娘, 载《红色青年饲养员》，共青团辽宁省委办公室编，第19-20页。

was refused five more times. But Zhao was relentless and ended up becoming the first female animal caretaker in her cooperative.³³¹

Du Xiuhua from the Hedong swinery of Hanzhong, Shaanxi was another feminist caretaker who fought against the gendered division of labor, what she deemed an “outdated custom.” On a spring day in April 1959, according to a *People’s Daily* article, the mating season came to more than 160 sows in Du’s swinery. She and other female caretakers were about to carry out the “breeding work” (配种工作) when a few of her colleagues stopped her, whispering that “there are too many people [watching]. Why don’t we wait for a while until they are gone?” She refused, and began working immediately, saying “Do it now! Is there anything disgraceful [in our work]?”³³²

All the efforts of both state propagandists and feminist caretakers who might have actually existed, however, could not entirely overcome the negative perception of women looking after collective animals. Young women who wanted to care for collectives’ animals were still struggling even during the radical Cultural Revolution. Yang Jianhui was a return youth and a member of the Communist Youth League (CYL) in Lushan County, Henan.³³³ Yang shared her experience at the local CYL meeting in 1978, which was recorded as an unpublished document. When she discussed

³³¹ Zhao Wenjun 赵文君, “Hongse siyangyuan” 红色饲养员, in *Hongse qingnian siyangyuan*, ed. Gongqingtuan Liaoning sheng wei bangongshi, 4–6.

³³² “Du Xiuhua yaozuo yibeizi siyangyuan” 杜秀华要作一辈子饲养员, *Renmin Ribao* 人民日报 (January 4, 1960). Although having young women participate in animal breeding was an effective way to subvert or attack traditional gender roles, it is doubtful whether it was part of caretakers’ regular work. There were “boar men” who kept boars and brought them around villages to provide mating services. Schmalzer, *Red Revolution, Green Revolution*, 153. According to primary documents on breeding, these “boar men” were mostly widowers or disabled old men and officially called “private breeding households” (民营配种户) who worked under the auspices of “public breeding stations” (国营配种站). Another category of “collective breeders” (集体配种员) later emerged, and animal mating and improvement of local breed became their job. See Chapter Five of this dissertation.

³³³ For more comprehensive account for the educated youth in rural Maoist China, see Schmalzer, *Red Revolution, Green Revolution*, Chapter 7. Also see Emily Honig and Xiaojian Zhao, *Across the Great Divide: The Sent-Down Youth Movement in Mao’s China, 1968–1980* (Cambridge: Cambridge University Press, 2019).

又被拒绝了五次。但赵毫不气馁，最终成为她所在合作社的第一位女性动物饲养员。³³¹

来自陕西汉中河东养猪场的杜秀华是另一位与性别分工（她认为这是一种“过时习俗”）作斗争的女权主义饲养员。根据《人民日报》的一篇文章，1959年4月的一个春天，杜的养猪场里160多头母猪进入了发情期。她和其他女饲养员正准备进行“配种工作”（配种工作），这时她的一些同事拦住了她，小声说：“有太多人[在看]。我们为什么不等到他们走了再做？”她拒绝了，立刻开始工作，说：“现在就做！我们的工作有什么可耻的吗[?]”³³²

然而，国家宣传人员和可能真实存在过的女权主义饲养员所做的一切努力，都未能完全克服人们对女性照料集体动物的负面看法。即使在激进的文化大革命期间，那些想照料集体动物的年轻女性仍然举步维艰。杨建辉是河南省鲁山县的一名回乡青年，也是共产主义青年团（CYL）的成员。³³³杨在1978年当地共青团会议上分享了她的经历，这段经历被记录为一份未发表的文件。当她谈到

³³¹ 赵文君 Zhao Wenjun, “红色饲养员” Hongse siyangyuan, 载于《红色青年饲养员》，共青团辽宁省委办公室编，4-6页。

³³² “杜秀华要作一辈子饲养员” Du Xiuhua yaozuo yibeizi siyangyuan, 《人民日报》 Renmin Ribao (1960年1月4日)。尽管让年轻女性参与动物饲养是颠覆或攻击传统性别角色的有效方式，但它是否是饲养员日常工作的一部分值得怀疑。有“公猪人”饲养公猪并将其带到村庄提供配种服务。Schmalzer, *Red Revolution, Green Revolution*, 153页。根据关于育种的原始文献，这些“公猪人”大多是鳏夫或残疾老人，官方称之为“民营配种户”，他们在“国营配种站”的主持下工作。后来又出现了另一类“集体配种员”，动物配种和地方品种改良成为他们的工作。参见本论文第五章。

³³³ 有关毛泽东时代中国农村知识青年更全面的描述，请参见 Schmalzer, *Red Revolution, Green Revolution*, 第7章。另请参见 Emily Honig 和 Xiaojian Zhao, *Across the Great Divide: The Sent-Down Youth Movement in Mao’s China, 1968–1980* (Cambridge: Cambridge University Press, 2019)。

being a caretaker with her mother in 1975, the older woman admonished her daughter: “I am not preventing you from feeding pigs. But look, other girls in our neighborhood were working at science experiment stations or studying at schools. They are no better than you. Then why of all jobs you are going to work yourself to death in a piggery?” Yang hesitated. But she soon thought of the renowned model worker and revolutionary martyr Lei Feng. She wanted to stick to her belief that raising pigs was what young girls like herself could do. However physically demanding the work was, she could persevere with an indomitable will like Lei Feng’s. As of 1978 when the Cultural Revolution ended, she was still working with pigs, and her fellow peasants exalted her as a “pig raising maniac” (养猪迷).³³⁴

If the collectivist ideology of animal care was not universally accepted among the people, neither was the propagandistic combination of feminism and animal care. I interviewed two former female caretakers who worked and still live in Bobai County, Guangxi where the hog industry has been huge. Notwithstanding all the ideological work, some young mothers seemed to have become caretakers for more practical reasons. Mrs. Ling previously worked at a supply and marketing cooperative (供销社), which was the institution for the distribution of food and daily necessities in an agricultural collective. But right after giving birth to her eldest son in 1957, Ling spontaneously asked the local leaders to transfer her to a collective pigsty which had just been built. She thought that the position would allow her to be with her son during work hours. As long as she fed the pigs

³³⁴ Yang Jianhui 杨建慧, “Zhi zai yangzhuchang, nao huai da mubiao: Malou gongshe Laojiang dadui yangzhuchang siyangyuan Yang Jianhui” 志在养猪场, 脑怀大目标: 马楼公社老将大队养猪场饲养员杨建慧 (May 1978), Gongqingtuan Lushan xian di qi ci daibiao dahui fayan cailiao 共青团鲁山县第七次代表大会发言材料, Unpublished material.

1975年，她和母亲一起做饲养员，老妇人告诫女儿：“我不是不让你喂猪。但是你看，我们社区的其他女孩都在科学实验站工作或在学校学习。她们不比你强。那么，为什么在所有工作中，你却要在养猪场累死呢？”杨犹豫了。但她很快想到了著名的模范工人、革命烈士雷锋。她想坚持自己的信念，即养猪是像她这样的年轻女孩可以做的事情。无论工作多么艰苦，她都能像雷锋一样以不屈不挠的意志坚持下去。截至1978年文化大革命结束时，她仍在养猪，她的农民同伴们称赞她为“养猪迷”（养猪迷）。³³⁴

如果说集体主义的动物护理意识形态并未被民众普遍接受，那么女权主义与动物护理的宣传性结合也同样如此。我采访了两名前女性饲养员，她们曾在广西博白县工作，现在仍居住在那里，当地的生猪产业规模庞大。尽管进行了所有的思想工作，一些年轻的母亲似乎是出于更实际的原因而成为饲养员的。凌女士以前在供销社工作，供销社是农业集体中分配食物和日用品的机构。但在1957年生下大儿子后，凌女士主动要求当地领导将她调到刚建好的集体猪圈。她认为这个职位可以让她在工作时间陪伴儿子。只要她喂猪

³³⁴ 杨建慧, “志在养猪场, 脑怀大目标: 马楼公社老将大队养猪场饲养员杨建慧” (1978年5月), 共青团鲁山县第七次代表大会发言材料, 未出版材料。

three times a day and did some routine tasks, she could look after her son in the pigsty without being interrupted by other people.³³⁵

This was why Mrs. Hong, who came from the same production team as Mrs. Ling, likewise volunteered to be a caretaker. She had brought up her three children in the pigsty, working as a caretaker from 1974 to 1981. Yet beyond the advantage of being able to parent, Hong found that animal caretaking was financially rewarding for a woman who never went to school. The work points from pig raising itself were not very high, and the workload was heavy. But unlike other agricultural jobs, caretakers could earn work points every day, weekdays or weekends, rain or shine, as animals under the canopy of the pen needed to be cared for every single day. Thanks to the work points that caring for collective animals gave her, as well as some cash that private pigs brought in, Hong could raise her son who was born in 1974 and two daughters born in 1977 and 1979. Mrs. Hong still keeps pigs. After she and I wrapped up our interview, she went off to give her pigs a shower. “Pigs are sensible” (懂事), she said. “They love to be clean. If I was sometimes late to help them take a bath, they put themselves close to me as soon as they saw me. They wanted me to quickly give them a shower.”³³⁶

Conclusion

Establishing collectivism, as both an economic system and moral code, was at the core of the social experiment attempted by Mao Zedong and the CCP. Regardless of how strongly they believed that

³³⁵ Author’s interview with former caretaker Mrs. Ling in Bobai County, Guangxi, June 2021.

³³⁶ Author’s interview with former caretaker Mrs. Hong in Bobai County, Guangxi, June 2021.

一天三次，并做一些日常工作，她就可以在猪圈里照看儿子，而不会被其他人打扰。³³⁵

这就是为什么与凌女士来自同一生产队的洪女士也自愿成为一名饲养员。她于1974年至1981年期间在猪圈里养大了她的三个孩子，担任饲养员。然而，除了能够养育孩子的好处之外，洪女士发现对于一个从未上过学的女性来说，动物饲养在经济上是有回报的。养猪本身的工作量很大，但工分并不高。但与其他农业工作不同的是，饲养员每天都可以获得工分，无论是工作日还是周末，无论是晴天还是雨天，因为猪圈里的动物每天都需要照料。多亏了照料集体动物所获得的工分，以及私人养猪带来的一些现金，洪女士得以抚养她1974年出生的儿子和1977年和1979年出生的两个女儿。洪女士现在仍然养猪。我和她结束采访后，她就去给她的猪洗澡了。“猪很懂事，”她说，“它们喜欢干净。如果我有时迟了帮它们洗澡，它们一看到我就会靠近我。它们想让我快点给它们洗澡。”³³⁶

结论

建立集体主义，作为一种经济制度和道德准则，是毛泽东和中国共产党所尝试的社会实验的核心。无论他们多么坚信

³³⁵ 作者于2021年6月在广西博白县采访了前看护人凌女士。

³³⁶ 作者于2021年6月在广西博白县采访了前看护人洪女士。

the experiment was bound to succeed,³³⁷ they knew that they had to transform local people’s thinking and behavior. Thousands of pieces of propaganda were therefore necessary. Among them were stories of model caretakers, or “livestock maniacs,” who selflessly took care of collectively owned animals. As we have seen, animal care was an indispensable part of the state’s ideological work, as socialist, collectivized communities were envisioned from the outset as more-than-people’s communes: The revolutionary people and their invaluable common assets, domestic animals, would be flourish together in the community of multispecies care that would make China a wealthy socialist nation.

The didactic stories on devoted caretakers, however, ironically reveal that intended targets of this grassroots ideology were never a tabula rasa. Being aware that their everyday subsistence was closely connected to the presence of diverse animals, local collective members had many other ways of living with animals that were not subdued to the norms of collectivism. The socialist state regarded some of their practices as shortsighted, self-concerned, and even careless. But this solid reality—another meaning of more-than-people’s commune—had existed long before Maoism and continued long after. If so, with this case study of Maoist animal care, we may want to highlight the futility of all the ideological works yet again.

I wonder if a more productive way of dealing with the past of Maoist China, at least in terms of animal care, is not dismissing it, but raising a question for ourselves.³³⁸ If we can choose to accept some aspects of the collectivist ethics of animal care, how much would we be willing to do so? It is

³³⁷ Rui Kunze and Marc Andre Matten, *Knowledge Production in Mao-era China: Learning from the Masses* (Lanham: Lexington Books, 2021), xix–xxiii. Kunze and Matten call this belief in the eventual success of socialist experiments, both scientific and social, which is based on Mao’s epistemology of practice, “experiment without contingency.”

³³⁸ Here I am inspired by Aminda Smith, “Foreword: The Maoism of PRC History,” *Positions: Asia Cultures Critique* 29.4 (November 2021): 659–674.

实验注定会成功，³³⁷他们知道必须改变当地人的思想和行为。因此，数千份宣传材料是必不可少的。其中包括模范饲养员或“牲畜狂人”的故事，他们无私地照料集体所有的动物。正如我们所见，动物饲养是国家意识形态工作中不可或缺的一部分，因为社会主义集体化社区从一开始就被设想为超越人类的公社：革命人民及其宝贵的共同财产——家畜，将在多物种关怀的社区中共同繁荣，使中国成为一个富裕的社会主义国家。

然而，这些关于忠诚饲养员的教诲故事，却讽刺地揭示了这种基层意识形态的目标受众从来都不是一张白纸。当地集体成员意识到他们的日常生计与各种动物的存在密切相关，他们有许多其他与动物共存的方式，这些方式并不受集体主义规范的约束。社会主义国家认为他们的一些做法是短视的、自私的，甚至是粗心的。但这种坚实的现实——超越人类公社的另一层含义——早在毛泽东思想之前就已存在，并在此之后长期延续。如果是这样，通过这个关于毛泽东时代动物饲养的案例研究，我们可能想再次强调所有意识形态工作的徒劳。

我不知道，处理毛泽东时代中国历史的一种更有效的方式，至少在动物护理方面，不是否定它，而是向我们自己提出一个问题。³³⁸ 如果我们选择接受集体主义动物护理伦理的某些方面，我们愿意在多大程度上这样做？它是

³³⁷ Rui Kunze and Marc Andre Matten, *Knowledge Production in Mao-era China: Learning from the Masses* (Lanham: Lexington Books, 2021), xix–xxiii. Kunze和Matten将这种基于毛泽东实践认识论的对社会主义实验（包括科学和社会实验）最终成功的信念称为“无偶然性实验”。

³³⁸ 在此，我受到 Aminda Smith 的启发，其著作《前言：中华人民共和国历史中的毛泽东主义》（“Foreword: The Maoism of PRC History”，*Positions: Asia Cultures Critique* 29.4 (2021 年 11 月): 659–674）中有所阐述。

likely that the political and material situations many of us find ourselves in are different from those in the Maoist period. We may no longer depend on animals for plowing, transportation, or the supply of fertilizers. Many animals other than canine and feline companions have been out of sight from our everyday life. Nonetheless, it seems our only more-than-people’s commune, Earth, requires us to reflect on the ways in which human beings relate to nonhumans.³³⁹ In that case, by acknowledging that care always comes at a cost,³⁴⁰ we may be able to find some usefulness in the central message of the Maoist ideology of animal care: Restrain the interests and convenience of self and put the common good in which even nonhuman others are looked after first.

³³⁹ Donna Haraway, *Staying with the Trouble: Making Kin in the Chthulucene* (Durham: Duke University Press, 2016); Erle C Ellis, *Anthropocene: A Very Short Introduction* (Oxford: Oxford University Press, 2018); Kohei Saito 斎藤幸平, *Hitoshinsei no ‘Shibonron’* 人新世の「資本論」 (Tokyo: Shueisha, 2020).

³⁴⁰ Janet M. Davis, *The Gospel of Kindness: Animal Welfare and the Making of Modern America* (Oxford: Oxford University Press, 2016); Juno Salazar Parreñas, *Decolonizing Extinction: The Work of Care in Orangutan Rehabilitation* (Durham: Duke University Press, 2018); Nicole Elizabeth Barnes, *Intimate Communities: Wartime Healthcare and the Birth of Modern China, 1937–1945* (Oakland: University of California Press, 2018).

我们许多人所处的政治和物质环境可能与毛泽东时代不同。我们可能不再依赖动物进行耕作、运输或提供肥料。除了犬类和猫科动物伴侣，许多其他动物已从我们的日常生活中消失。尽管如此，我们唯一一个超越人类的公社——地球，似乎要求我们反思人类与非人类的关系。³³⁹在这种情况下，通过承认关怀总是需要付出代价的，³⁴⁰我们或许能从毛泽东动物关怀思想的核心信息中找到一些用处：克制个人利益和便利，将包括非人类在内的共同利益放在首位。

³³⁹ Donna Haraway, 《与麻烦共存：在触手世中结亲》（达勒姆：杜克大学出版社，2016）；Erle C Ellis, 《人类世：极简介绍》（牛津：牛津大学出版社，2018）；Kohei Saito 斎藤幸平, 《人新世的“资本论”》人新世の「資本論」（东京：集英社，2020）。

³⁴⁰ Janet M. Davis, 《善良的福音：动物福利与现代美国的形成》（Oxford: Oxford University Press, 2016）；Juno Salazar Parreñas, 《去殖民化灭绝：猩猩康复中的关怀工作》（Durham: Duke University Press, 2018）；Nicole Elizabeth Barnes, 《亲密社区：战时医疗与现代中国的诞生，1937-1945》（Oakland: University of California Press, 2018）。

Chapter Five:
Animal Eugenics

Introduction

Is eugenics anthropocentric? Francis Galton defined “eugenics” as “the study of all agencies under human control which can improve or impair the racial quality of future generations, either mentally or physically.”³⁴¹ If we follow this definition, eugenics does sound like an endeavor that is particularly applicable to human races. Therefore, we may say that both eugenic means and subjects of control are human-centric. However, I would call attention to the little-known fact that Galton himself used the term “eugenics” for the first time in 1883 to refer to “science of improving stock, which is by no means confined to questions of judicious mating.”³⁴² If so, nonhuman animals, too, are part of the history of eugenics. Of course, we should not reduce the rich discourse on the nature and workings of eugenics to Galton alone.³⁴³ But here we may agree that the term “animal eugenics” is not necessarily wrong, and this chapter is based on the premise that animal breeding, or any deliberate efforts to increase and improve animal populations, is eugenic.³⁴⁴

³⁴¹ Leslie Francis ed., *The Oxford Handbook of Reproductive Ethics* (Oxford: Oxford University Press, 2017), p. 146.

³⁴² Francis Galton, *Inquiries Into Human Faculty and Its Development* (London: Macmillan, 1883), p. 25.

³⁴³ On the history of eugenics, see Daniel J. Kevles, *In the Name of Eugenics: Genetics and the Uses of Human Heredity* (Cambridge, Mass.: Harvard University Press, 1985); Mark B. Adams ed., *The Wellborn Science: Eugenics in Germany, France, Brazil, and Russia* (Oxford: Oxford University Press, 1990); Yuehtsen Juliette Chung, *Struggle for National Survival: Eugenics in Sino-Japanese Contexts, 1896–1945* (London: Routledge, 2002); Alexandra Minna Stern, *Eugenic Nation: Faults and Frontiers of Better Breeding in Modern America* (Berkeley: University of California Press, 2005); Nancy Leys Stepan, “*The Hour of Eugenics*”: *Race, Gender, and Nation in Latin America* (Ithaca, NY: Cornell University Press, 2015).

³⁴⁴ In a similar vein, Amir Zelinger calls for thinking the question of human “race” in tandem with that of animal “breed,” given that the animal and the human have been intertwined and capitalized on each other. On the one hand, Zelinger stresses that “animals are a banal metaphor in the service of racial animosity by humans against humans.” On the other, when one sees individual animals in terms of breed in the same way that individual humans are subsumed to racial categories, “it [the concept of breed] reduces them to superficial beings judged by their conformation to their breed type rather than seeing them as singular creature with unique characteristics.” The framing of this chapter is

第五章：
动物优生学

导言

优生学是以人类为中心的吗？弗朗西斯·高尔顿将“优生学”定义为“研究人类控制下所有能够改善或损害未来世代种族（无论是精神上还是身体上）质量的机构”。³⁴¹ 如果我们遵循这个定义，优生学听起来确实是一项特别适用于人类种族的努力。因此，我们可以说优生学的手段和控制对象都是以人类为中心的。然而，我想提请大家注意一个鲜为人知的事实，高尔顿本人在1883年首次使用“优生学”一词时，指的是“改善畜群的科学，这绝不局限于审慎交配的问题”。³⁴² 如果是这样，非人类动物也是优生学历史的一部分。当然，我们不应将关于优生学性质和运作的丰富论述仅仅归结为高尔顿一人。³⁴³ 但在这里我们可以同意，“动物优生学”这个术语不一定是错误的，本章基于这样一个前提：动物育种，或任何旨在增加和改善动物种群的蓄意努力，都是优生学的。³⁴⁴

³⁴¹ Leslie Francis 编著，《牛津生殖伦理学手册》（牛津：牛津大学出版社，2017 年），第 146 页。

³⁴² 弗朗西斯·高尔顿，《人类能力及其发展研究》（伦敦：麦克米伦出版社，1883 年），第 25 页。

³⁴³ On the history of eugenics, see Daniel J. Kevles, *In the Name of Eugenics: Genetics and the Uses of Human Heredity* (Cambridge, Mass.: Harvard University Press, 1985); Mark B. Adams ed., *The Wellborn Science: Eugenics in Germany, France, Brazil, and Russia* (Oxford: Oxford University Press, 1990); Yuehtsen Juliette Chung, *Struggle for National Survival: Eugenics in Sino-Japanese Contexts, 1896–1945* (London: Routledge, 2002); Alexandra Minna Stern, *Eugenic Nation: Faults and Frontiers of Better Breeding in Modern America* (Berkeley: University of California Press, 2005); Nancy Leys Stepan, “*The Hour of Eugenics*”: *Race, Gender, and Nation in Latin America* (Ithaca, NY: Cornell University Press, 2015).

³⁴⁴ 同样，Amir Zelinger 呼吁在思考人类“种族”问题时，同时思考动物“品种”问题，因为动物和人类相互交织并相互利用。一方面，Zelinger 强调“动物是人类对人类种族敌意的平庸隐喻”。另一方面，当人们以对待种族类别的方式看待个体动物的品种时，“它[品种的概念] 将它们简化为根据其品种类型的符合性来判断的肤浅存在，而不是将其视为具有独特特征的独特生物。”本章的框架是

More-than-people’s communes in Maoist China engaged in animal breeding. Working with state-run breeding farms and establishing their own breeding stations, local collectives became places where diverse actors—state cadres, state and commune breeders, private breeding households (or “boar men”), and peasants—propagated and improved the genetic stock of pigs, cattle, horses, and donkeys (Figure 11). However, these historical actors had different views on what constituted “superior” livestock, the treatment of “inferior” animals, and how the value produced by animals needed to be distributed. This chapter charts the tensions between the state and local collectives over these issues and shows that how the eugenic idea and practices of animal breeding were an essential part of the story of human-animal relations in more-than-people’s communes.



Figure 11. Local breeders in Yanheng Cattle Breeding Station, Sichuan. December 1952. Source: Author’s Collection.

undoubtedly indebted to Zelinger’s problematics. Amir Zelinger, “Race and Animal-Breeding: A Hybridized Historiography,” *History and Theory* 58.3 (September 2019): 364, 374.

毛泽东时代的“超人民公社”从事动物育种。通过与国营育种农场合作并建立自己的育种站，地方集体成为各种参与者——国家干部、国家和公社育种员、私人育种户（或“养猪人”）和农民——繁殖和改良猪、牛、马和驴的遗传种群的场所（图11）。然而，这些历史参与者对什么是“优良”牲畜、“劣等”动物的处理以及动物产生的价值如何分配持有不同观点。本章阐述了国家和地方集体在这些问题上的紧张关系，并表明动物育种的优生学思想和实践是“超人民公社”人与动物关系故事的重要组成部分。



图 11。1952 年 12 月，四川盐亭牛场当地饲养员。来源：作者收藏。

无疑得益于泽林格的问题。阿米尔·泽林格（Amir Zelinger），《种族与动物育种：混合史学》（Race and Animal-Breeding: A Hybridized Historiography），*History and Theory* 58.3（2019年9月）：364, 374。

Terms, Technologies, and Assumptions

Before diving into the practices of animal breeding, the terminology and the underlying assumptions shared by historical actors need to be explained. What was called in China the “breeding work” (配种工作) included any activities designed to increase the quantity or improve the quality of livestock. In principle, for the rapid quantitative development, humans could mate (配种) males with as many female animals as possible, thereby propagating (繁殖) the maximum number of offspring in a short period of time.³⁴⁵ The efforts for the improvement of livestock breed (畜种改良), although often carried out with the encouragement of mass mating and propagating, was predicated on a distinct logic: the selection of “superior” breeds and the destruction of “inferior” ones. Maximizing the number of “inferior” animals was, at least from the perspective of the state, pointless. In this vein, we may have no reason to shy away from describing breeding in China as a eugenic endeavor.

Facilitating mating was a more accessible practice than improving a breed. It did require knowledge, though. One had to possess some biological information about reproduction—such as since what age a piglet acquires reproductive capacity and what the estrous cycle of a mare is (Table 4). Those who were in charge of this work learned either by experience or from textbooks how to recognize when animals were aroused or pregnant without hormone testing or ultrasound,³⁴⁶ not to

³⁴⁵ The Chinese term “*peizhong*” (配种) therefore means both mating in a narrow sense and breeding more generally.

³⁴⁶ The Chinese breeders in the Maoist period relied upon observable signals. For example, according to a 1960 manual, these are four ways to determine whether or not a sow has entered estrus: 1) the sow becomes nervous and edgy; 2) the sow’s appetite decreases; 3) the sow’s genitals become congested, inflamed, and swollen. Then fluid or mucus drains from the vagina; 4) when a person massages the sow’s midsection with both hands, the sow’s ears become erect with pleasure. Fu Mianye 富绵业, *Zhu de peizhong yu fenmian* 猪的配种与分娩 (Changchun: Jilin renmin chubanshe, 1960), 2.

术语、技术和假设

在深入探讨动物育种实践之前，需要解释历史参与者共享的术语和基本假设。在中国被称为“配种工作”的活动包括旨在增加牲畜数量或提高牲畜质量的任何活动。原则上，为了快速实现数量发展，人类可以使雄性与尽可能多的雌性动物交配（配种），从而在短时间内繁殖（繁殖）最大数量的后代。³⁴⁵ 畜种改良的努力，尽管通常在鼓励大规模交配和繁殖的情况下进行，但其前提是基于一独特的逻辑：选择“优良”品种，淘汰“劣质”品种。至少从国家的角度来看，最大限度地增加“劣质”动物的数量是毫无意义的。从这个角度来看，我们没有理由回避将中国的育种描述为一项优生学事业。

促进交配比改良品种更容易实现。但这确实需要知识。人们必须掌握一些关于繁殖的生物学信息——例如，小猪从多大年龄开始具备繁殖能力，以及母马的发情周期是怎样的（表4）。负责这项工作的人通过经验或教科书学习如何识别动物何时发情或怀孕，而无需进行激素测试或超声波检查，³⁴⁶更不用说

³⁴⁵ 因此，中文术语“配种”既指狭义上的交配，也指更广义上的育种。

³⁴⁶ 毛泽东时期的中国育种者依赖可观察的信号。例如，根据1960年的一本手册，有四种方法可以判断母猪是否进入发情期：1）母猪变得紧张不安；2）母猪食欲下降；3）母猪生殖器充血、发炎、肿胀。然后有液体或粘液从阴道流出；4）当人双手按摩母猪中段时，母猪的耳朵会因愉悦而竖起。富绵业，《猪的配种与分娩》（长春：吉林人民出版社，1960），第2页。

mention pointers for increasing the number of successful pregnancies. With this knowledge, Chinese breeders and peasants in the Maoist era carried out natural mating and artificial insemination.

	Horses	Cattle	Sheep	Pigs
Progenitive age	3–4 years	2 years	1.5 years	1 year
Duration of estrus	4–11 days	1–3 days	1.5–3 days	2–4 days
Postpartum recovery	2–10 days	21–28 days	4–6 months	5–6 weeks
Best timing for mating	The second day of estrus	The second half of estrus	12 hours after the start of estrus	2 days after the start of estrus
Pregnancy period	330–340 days	285 days	150 days	112–120 days
Labor	30–60 minutes	3 hours	3 hours	1–6 hours
Weaning period	After 6 months	After 3 months	After 2–4 months	After 1–2 months
Estrous cycle	17–25 days	17–23 days	16–19 days	20–22 days
Lifespan	25–40 years	15–20 years	10–12 years	15 years

Table 4. The knowledge about the reproduction of major livestock. Source: Shanxi sheng renmin zhengfu nongyeting 山西省人民政府农业厅, *Peizhong hangjia xunlian jiaocai* 配种行家训练教材 (Taiyuan: Shanxi sheng renmin zhengfu nongyeting, 1951), 8–9.

Artificial insemination was promoted by the state from the early 1950s as a more efficient and advanced technology to replace natural mating.³⁴⁷ In theory, as long as there is one high quality male animal, breeders can impregnate all female animals in an area, with the expectation that the females will give birth to high quality offspring. It took less time to administer sperm to a female animal’s body than to arrange for two animals to mate.³⁴⁸ However, extracting sperm from a male, chemically processing and storing it, and then injecting it into a female’s cervix (Figure 12) were

³⁴⁷ Shanxi sheng renmin zhengfu nongyeting 山西省人民政府农业厅, *Peizhong hangjia xunlian jiaocai* 配种行家训练教材 (Taiyuan: Shanxi sheng renmin zhengfu nongyeting, 1951), 14–17.

³⁴⁸ Zhonggong Pei xianwei bangongshi 中共沛县委办公室, “Zhangzhai gongshe muchu peizhong fanzhi jingyan: Rengong cuqing rengong shoujing” 张砦公社母畜配种繁殖经验: 人工催情人工受精 (July 17, 1959), Jiangsu sheng Pei xian neibu ziliao 江苏省沛县内部资料, Unpublished material.

提及增加成功妊娠数量的要点。有了这些知识，毛泽东时代的中国育种者和农民进行了自然交配和人工授精。

	马	牛	羊	Pigs
Progenitive age	3-4年	2年	1.5 年	1 年
持续时间 of 发情期	4–11 天	1–3 天	1.5–3 天	2–4 天
产后 恢复	2–10 天	21–28 天	4–6个月	5–6周
最佳时机 for 交配	发情期 第二天	发情期后半段 发情期	发情开始后12小时 发情开始	发情开始后2天 e 发情开始
妊娠期 cy period	330–340 天	285 天	150 天	112–120 天
劳动	30–60 分钟	3 小时	3 小时	1–6 小时
断奶 g period	6 个月后	3个月后	2-4个月后	1-2个月后 s
Estrous cycle	17-25天	17-23天	16-19天	20–22 天
寿命	25–40 年	15–20 年	10–12 年	15 年

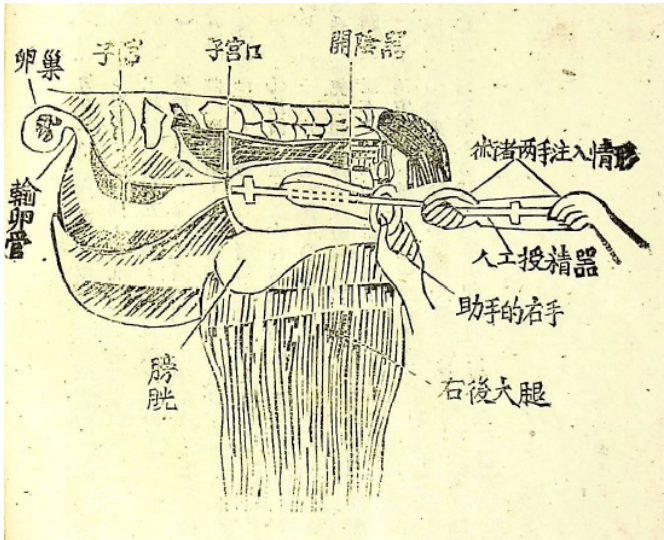
Table 4. The knowledge about the reproduction of major livestock. Source: Shanxi sheng renmin zhengfu nongyeting 山西省人民政府农业厅, *Peizhong hangjia xunlian jiaocai* 配种行家训练教材 (Taiyuan: Shanxi sheng renmin zhengfu nongyeting, 1951), 8–9.

从20世纪50年代初开始，国家推广人工授精，认为这是一种更高效、更先进的技术，可以取代自然交配。³⁴⁷理论上，只要有一只高质量的雄性动物，饲养员就可以使一个地区的所有雌性动物受孕，并期望这些雌性动物能产下高质量的后代。将精子注入雌性动物体内比安排两只动物交配花费的时间更少。³⁴⁸然而，从雄性动物体内提取精子，对其进行化学处理和储存，然后将其注入雌性动物的子宫颈（图12）是

³⁴⁷山西省人民政府农业厅，《配种行家训练教材》（太原：山西省人民政府农业厅，1951），第14-17页。

³⁴⁸中共沛县委办公室，“张砦公社母畜配种繁殖经验：人工催情人工受精”（1959年7月17日），江苏省沛县内部资料，未出版材料。

tasks too difficult for most people at the grassroots. Although there is evidence that artificial insemination was more often carried out for horses than for other livestock from the mid-1950s,³⁴⁹ it did not become popular before the late 1970s. As of November 1955, there was only one livestock artificial insemination station (家畜人工授精站) in all of Guangxi Province.³⁵⁰ According to a survey report conducted in a prefecture of Shaanxi Province, one of leading provinces in animal husbandry and livestock production, from May to November 1973, there were 750,000 pigs, 230,000 oxen, and 100,000 sheep, almost all of which were the result of natural mating, not artificial insemination.³⁵¹



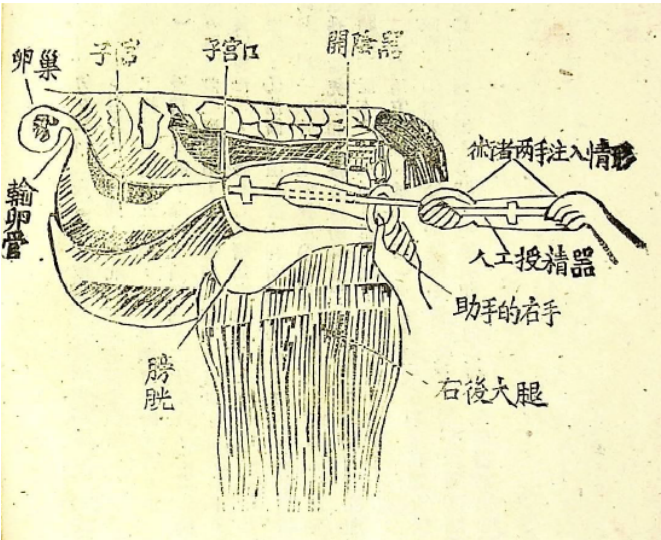
³⁴⁹ For example, see Nongyebu xumu shouyi zongju 农业部畜牧兽医总局, “Shanshi zhongmachang 1953 nian peizhong gongzuo zhuyao jingyan jiaoxun” 山市种马场 1953 年配种工作主要经验教训 (March 15, 1954), Guangxi Zhuang Autonomous Region Archives (hereafter GARA), X163-001-0392-0004.

³⁵⁰ Guangxi sheng nongyeting 广西省农业厅, “Guangxi sheng 1955 nian zhongdian shengchu peizhong gongzuo huiyi baogao” 广西省 1955 年重点牲畜配种站工作会议报告 (November 14, 1955), GARA, X163-002-0138-0079. In 1955, Guangxi was still a province. It was restructured to Guangxi Tong Autonomous Region in 1958 and then Guangxi Zhuang Autonomous Region in 1965.

³⁵¹ Shaanxi sheng Shangluo diqu xumu shouyi gongzuozhan 陕西省商洛地区畜牧兽医工作站, *Shangluo diqu niu yang pucha ziliao huibian* 商洛地区牛羊普查资料汇编 (N.p: Shaanxi sheng Shangluo diqu xumu shouyi gongzuozhan, 1974), 3 and 25.

对于大多数基层民众来说，这些任务太难了。尽管有证据表明，从20世纪50年代中期开始，人工授精在马匹上的应用比在其他牲畜上更常见，³⁴⁹但在20世纪70年代末之前，它并未普及。截至1955年11月，整个广西省只有一个家畜人工授精站（家畜人工授精站）。³⁵⁰根据对畜牧业大省陕西省某地区进行的一项调查报告，从1973年5月到11月，该地区有75万头猪、23万头牛和10万只羊，几乎所有这些都是自然交配的结果，而非人工授精。

³⁵¹



³⁴⁹例如，参见农业部畜牧兽医总局，“山市种马场1953年配种工作主要经验教训”（1954年3月15日），广西壮族自治区档案馆（以下简称GARA），X163-001-0392-0004。

³⁵⁰ Guangxi sheng nongyeting 广西省农业厅, “Guangxi sheng 1955 nian zhongdian shengchu peizhong gongzuo huiyi baogao” 广西省 1955 年重点牲畜配种站工作会议报告 (November 14, 1955), GARA, X163-002-0138-0079. In 1955, Guangxi was still a province. It was restructured to Guangxi Tong Autonomous Region in 1958 and then Guangxi Zhuang Autonomous Region in 1965.

³⁵¹ Shaanxi sheng Shangluo diqu xumu shouyi gongzuozhan 陕西省商洛地区畜牧兽医工作站, *Shangluo diqu niu yang pucha ziliao huibian* 商洛地区牛羊普查资料汇编 (N.p: Shaanxi sheng Shangluo diqu xumu shouyi gongzuozhan, 1974), 3 and 25.

Figure 12. Visual explanation on how to conduct artificial insemination. Source: Shanxi sheng renmin zhengfu nongyeting 山西省人民政府农业厅, *Peizhong hangjia xunlian jiaocai* 配种行家训练教材 (Taiyuan: Shanxi sheng renmin zhengfu nongyeting, 1951), 16.

Let us move from the question of quantity to that of quality. Throughout the Maoist period, two methods were used to improve livestock. The first one was “selecting good breeding animals and selectively mating on the ground” (当地选种选配). It was taken when rural communities could not introduce superior males (优良种畜) from the outside. Local peasant-breeders selected a few healthy male animals which were well-built, known to be easily fatten, or had fine, even-colored coats. Sometimes the selection criteria were determined through “public evaluation and comparison” (群众评比) led by local villagers.³⁵² By encouraging the mating between selected males and local female livestock and preventing non-selected males from breeding by castrating or, in limited cases, culling them, peasants and breeders sought to improve the quality of local livestock population. This method was often touted as in line with the Maoist emphasis on local self-reliance (自力更生).³⁵³

The second method, “hybrid improvement” (杂交改良), referred to the breed betterment through the hybridization of model-bred males and native females. So long as the model males were taken good care of, this method was more reliable than the on-the-ground selection. However, hybrid improvement worked in a more top-down way, as the development, dissemination, and quality control of the preferred breeds—such as White Yorkshire hogs and Xinjing fine-wool

³⁵² Guangxi sheng nongyeting 广西省农业厅, “Guangxi sheng 1954 nian jiachu bayou jihua dagang cao’an” 广西省1954年家畜保育计划大纲草案 (1954), GARA, X163-001-0394-0083.

³⁵³ In the similar vein, the slogan of “self-breeding and self-raising” (自繁自养) gained currency in the realm of animal husbandry and veterinary medicine. “Nuli fanzhi dashengchu” 努力繁殖大牲畜, *Renmin Ribao* 人民日报 (May 19, 1963).

图12。人工授精操作的图解。来源：山西省人民政府农业厅，《配种行家训练教材》（太原：山西省人民政府农业厅，1951），第16页。

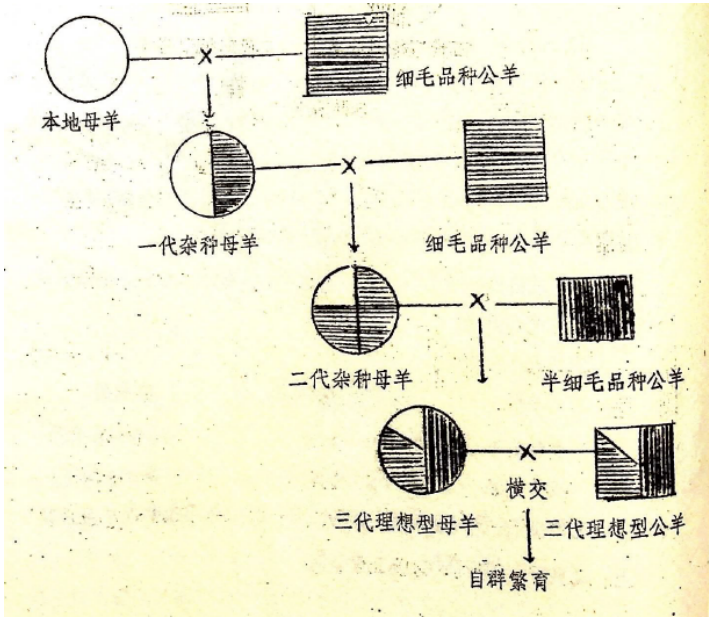
我们从数量问题转向质量问题。在整个毛泽东时代，有两种方法用于改良牲畜。第一种是“当地选种选配”（当地选种选配）。当农村社区无法从外部引进优良雄性牲畜（优良种畜）时，就采用这种方法。当地农民育种者会选择一些健康、体格健壮、易于育肥或毛色均匀的雄性牲畜。有时，选择标准是通过当地村民主导的“群众评比”（群众评比）来确定的。³⁵²通过鼓励选定的雄性牲畜与当地雌性牲畜交配，并通过阉割或在少数情况下淘汰非选定雄性牲畜来阻止其繁殖，农民和育种者力求提高当地牲畜种群的质量。这种方法常被宣传为符合毛泽东时代强调的自力更生（自力更生）原则。³⁵³

第二种方法是“杂交改良”，指的是通过模式培育的雄性与本地雌性杂交来改良品种。只要模式雄性得到良好照料，这种方法就比实地选择更可靠。然而，杂交改良是一种更自上而下的方式，因为优良品种（如白色约克夏猪和新疆细毛羊）的开发、传播和质量控制

³⁵² 广西省农业厅，《广西省1954年家畜保育计划大纲草案》（1954），GARA, X163-001-0394-0083。

³⁵³ 同样，“自繁自养”的口号在畜牧兽医领域也流行起来。“努力繁殖大牲畜”，《人民日报》（1963年5月19日）。

sheep—were spearheaded by state-owned breeding stations (国营配种站).³⁵⁴ As collectivization intensified after the late 1950s and the ideal of mass science became salient, the central and provincial governments urged state-owned breeding stations to extend knowledge and “assign missions to collectives and let them proceed on their own” (交社自办).³⁵⁵ The subtle mismatch between the top-down technology and the stress on bottom-up efforts for self-reliance was not addressed until the 1970s. The ultimate goal of hybrid improvement was then set as achieving the status of “self-reproduction within the local herds” (自群繁殖). That being said, if a collective once becomes able to reproduce sufficient numbers of model-bred males and three generations of hybrid offspring, it would no longer need to bring in model-bred males from outside (Figure 13).³⁵⁶

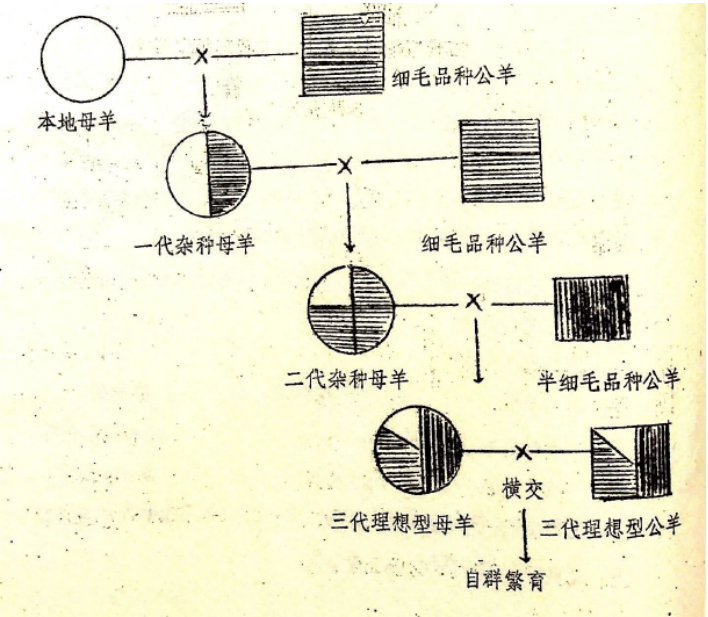


³⁵⁴ Institutions called “state-run breeding livestock farms” (国营种畜场) or “state-run livestock farms” (国营畜牧场) played similar roles across rural China.

³⁵⁵ Zhonghua renmin gongheguo nongyebu 中华人民共和国农业部, “Guanyu quanguo mianyang gailiang zuotanhui de baogao” 关于全国绵羊改良座谈会的报告 (circa June 1957), in *Xumu gongzuo wenjian xuanbian* 畜牧工作文件选编, ed. Shaanxi sheng xumu shouyi zongzhan 陕西省畜牧兽医总站 (N.p., 1974), 130.

³⁵⁶ Shanxi nongxueyuan xumu shouyixi yangyang keyan xiaozu 山西农学院畜牧兽医系养羊科研小组, “Zouchu xiaomen dagao mianyang yuzhong gongzuo” 走出校门大搞绵羊育种工作 (March 1973), Unpublished material, 4–8.

绵羊) 的推广，都是由国营配种站率先开展的。³⁵⁴ 20世纪50年代后期，随着集体化深入和大众科学理念的兴起，中央和省级政府敦促国营配种站推广知识，“交社自办”。³⁵⁵ 这种自上而下的技术与自下而上自力更生努力之间的微妙不匹配，直到20世纪70年代才得到解决。杂交改良的最终目标是实现“自群繁殖”。也就是说，如果一个集体一旦能够繁殖足够数量的模式培育雄性动物和三代杂交后代，它就不再需要从外部引入模式培育雄性动物（图13）。³⁵⁶



³⁵⁴ 机构被称为“国营种畜场”或“国营畜牧场”在中国农村地区扮演着类似的角色。

³⁵⁵ 中华人民共和国农业部, “关于全国绵羊改良座谈会的报告” (约1957年6月), 载陕西省畜牧兽医总站编《畜牧工作文件选编》(出版地不详, 1974年), 第130页。

³⁵⁶ 山西农学院畜牧兽医系养羊科研小组, “走出校门大搞绵羊育种工作” (1973年3月), 未出版资料, 第4-8页。

Figure 13. The roadmap to “self-reproduction within the local herds.” Source: Shanxi nongxueyuan xumu shouyixi yangyang keyan xiaozu 山西农学院畜牧兽医系养羊科研小组, “Zouchu xiaomen dagao mianyang yuzhong gongzuo” 走出校门大搞绵羊育种工作 (March 1973), Unpublished material, 4.

Although the state was eager to push forward with animal breeding, people at the grassroots did not necessarily share the state’s enthusiasm. Local villagers understood that having more oxen should be conducive to the increase of agricultural production. But they feared that if their village had too many draft animals, it could be nearly impossible to retain enough fodder to feed them.³⁵⁷ Some middle and poor peasants, especially in the pre- or earlier collectivization stage in the mid-1950s, also worried about the possibility of being recategorized as landlords or rich peasants, lethal stigmas under Maoism, as a result of breeding too many animals. Moreover, some local Communist cadres did not necessarily welcome the mission of propagating as many livestock as possible. It was simply too overwhelming and too difficult.³⁵⁸

The notion of breed improvement was even more problematic from the perspective of some local peasants. It was because the state forced them to castrate “faulty male animals” (不良公畜) and “strictly slaughter livestock which contracted animal infectious diseases.”³⁵⁹ In the face of the arrival of well-bred males, villages were often also required to desex native male livestock.³⁶⁰ After

³⁵⁷ Guangxi sheng nongyeting 广西省农林厅, “Guanyu xiezhu Wutang chu peizhongzhan gongzuo de baogao” 关于协助五塘畜配种站工作的报告 (November 8, 1953), GARA, X163-001-0246-0053.

³⁵⁸ “Hunan sheng 1953 nian jiachu bayou qingkuang ji 1954 nian gongzuo yijian deng cailiao” 湖南省 1953 年家畜保育情况及 1954 年工作意见等材料 (1954), GARA, X163-001-0391-0070.

³⁵⁹ Guangxi xumu shiyanchang 广西畜牧试验场, “Guanyu dui quanguo shengchu peizhong yewu zuotanhui de jidian yijian” 关于对全国牲畜配种业务座谈会的几点意见 (February 19, 1953), GARA, X163-001-0245-0068.

³⁶⁰ Guangxi xumu shiyanchang 广西畜牧试验场, “Huangniu rengong shoujing gongzuo de jiancha baogao” 黄牛人工授精工作的检查报告 (1954), GARA, X163-001-0392-0006.

图 13。“地方畜群内部自我繁殖”的路线图。资料来源：山西农学院畜牧兽医系养羊科研小组, 《走出校门大搞绵羊育种工作》（1973 年 3 月），未出版材料，第 4 页。

尽管国家急于推进畜牧业发展，但基层民众不一定认同国家的热情。当地村民明白，拥有更多牛只应有助于农业生产的增加。但他们担心，如果村里役畜过多，几乎不可能保留足够的饲料来喂养它们。³⁵⁷一些中贫农，尤其是在 20 世纪 50 年代中期集体化之前或早期阶段，也担心由于饲养过多牲畜而被重新划分为地主或富农，这在毛泽东时代是致命的耻辱。此外，一些地方共产党员干部不一定欢迎尽可能多地繁殖牲畜的任务。这简直太令人难以承受，也太困难了。³⁵⁸

从一些当地农民的角度来看，品种改良的概念问题更多。这是因为国家强迫他们阉割“不良公畜”并“严格屠宰感染动物传染病的牲畜”。³⁵⁹ 面对优良品种公畜的到来，村庄通常还被要求对本地雄性牲畜进行去势。³⁶⁰ 之后

³⁵⁷ 广西省农林厅, 《关于协助五塘畜配种站工作的报告》（1953年11月8日），GARA, X163-001-0246-0053。

³⁵⁸ 《湖南省1953年家畜保育情况及1954年工作意见等材料》（1954），GARA, X163-001-0391-0070。

³⁵⁹ 广西畜牧试验场, “关于对全国牲畜配种业务座谈会的几点意见”（1953年2月19日），GARA, X163-001-0245-0068。

³⁶⁰ 广西畜牧试验场, “黄牛人工授精工作的检查报告”（1954），GARA, X163-001-0392-0006。

the completion of collectivization in the late 1950s, the central and provincial governments extended its reach to local collectives and urged them to “immediately cull what needed to be culled” to avoid “a waste of labor and fodder.”³⁶¹ The higher-level cadres and planners were not attentive to the fact that different actors could have different views on the eugenic measures.

The dividing lines between, on the one hand, the “superior” and “well-bred” and, on the other, the “inferior” and “faulty” were indeed ambiguous and controversial in rural Maoist China. Some peasants in Lingui County, Guangxi, for example, preferred medium-sized to gigantic bulls given the local ecological conditions—the prevalence of narrow wet paddies and the shortage of grassland. The state-run breeding stations in the county, nonetheless, planned to breed large cattle.³⁶² Perhaps for a similar reason, people in “very many parts” of Guangxi bred small pigs in 1955 despite the presence of improved breeds. Still, the provincial Ministry of Agriculture deemed the people’s preference a “faulty practice” (不良习惯) that would bring about the “degeneration” of swine breed.³⁶³

The central and provincial governments thus realized in the mid-1950s that they would need to strengthen propaganda and education to “explain [to the masses] the advantages of culling inferior breeds (劣种) and improving the quality of breeds.”³⁶⁴ However, the “faulty practices” of

³⁶¹ “Gansu sheng jianli gengchu fanzhi jidi de juti jihua cao’an” 甘肃省建立耕畜繁殖基地的具体计划草案 (N.d.), in *Xumu shouyi jingyan 4: Gengchu de peizhong fanzhi* 畜牧兽医经验 4: 耕畜的配种繁殖, ed. Gansu sheng xumuting 甘肃省畜牧厅 (Lanzhou: Gansu renmin chubanshe, 1958), 12–15.

³⁶² Guangxi xumu shiyanchang, “Huangniu rengong shoujing gongzuo de jiancha baogao.”

³⁶³ Guangxi sheng nongyeting 广西省农业厅, “Lin tingzhang zai Guangxi sheng 1955 nian shengchu peizhong jianyizhan gongzuo huiyi shang de baogao” 林厅长在广西省 1955 年牲畜配种检疫站工作会议上的报告 (1955), GARA, X163-002-0138-0059.

³⁶⁴ Guangxi sheng nongyeting, “Guanyu xiezhu Wutang chu peizhongzhan gongzuo de baogao.”

在1950年代末集体化完成后，中央和省级政府将其影响力扩展到地方集体，并敦促他们“立即淘汰需要淘汰的”，以避免“浪费劳动力和饲料”。³⁶¹ 高级干部和规划者没有注意到不同行为者可能对优生措施有不同的看法。

在毛泽东时代的中国农村，“优等”和“良种”与“劣等”和“有缺陷”之间的界限确实模糊且充满争议。例如，广西临桂县的一些农民更喜欢中等体型的公牛，而不是巨型公牛，因为当地的生态条件——狭窄的湿稻田和草地短缺。然而，该县的国营育种站却计划培育大型牛。³⁶²也许出于类似的原因，尽管有改良品种，1955年广西“许多地方”的人们还是饲养了小型猪。尽管如此，省农业厅仍将人们的偏好视为一种“不良习惯”（不良习惯），认为这会导致猪种“退化”。³⁶³

因此，中央和地方政府在1950年代中期意识到，他们需要加强宣传和教育，以“向[群众]解释淘汰劣种和提高品种质量的优势”。³⁶⁴ 然而，“错误的做法”

³⁶¹ “甘肃省建立耕畜繁殖基地的具体计划草案” (N.d.)，载于甘肃省畜牧厅编《畜牧兽医经验 4: 耕畜的配种繁殖》（兰州：甘肃人民出版社，1958），第12-15页。

³⁶² 广西畜牧实验场，“黄牛人工授精工作的检查报告”。

³⁶³ 广西省农业厅，“林厅长在广西省 1955 年牲畜配种检疫站工作会议上的报告”（1955），GARA, X163-002-0138-0059。

³⁶⁴ 广西省农业厅，“关于协助武塘畜配种站工作的报告”。

peasants and breeders at the grassroots never died out. The state’s anxiety about the degeneration of livestock breed seemed not to have been dispelled until the 1970s. The leaders of Shaanxi Province were alarmed in 1973 that some superior breeds that they had diligently propagated for many years had become seriously degenerated. “Breeding of new breeds has been too slow. The heredity is still unstable. The systematic management of bringing in and reproducing the superiors, both from other parts of China and from foreign countries, has not been satisfactory either. All these phenomena have led to the decrease of [livestock] productivity.”³⁶⁵ The sections to follow will explore these tensions over animal breeding between the state and local communities.

Institutional Change

We may be able to call those who worked to increase the quantity and quality of livestock breeders. But in the early 1950s, the Communist state had not yet systematically organized them, so the occupational category of “breeder” was not clearly defined. There were certainly people who had the requisite knowledge and experience, but the scope of their work was not limited to breeding. They were also herders, caretakers, veterinarians, and entrepreneurs. Wang Zemin was such a person. He was born in 1915 to a middle peasant family in a village in Yongji County, Jilin Province. Since childhood, Wang took care of his family’s animals. In 1951, he ran a carriage business with a dozen horses and mules he owned. Wang knew how to breed horses, how to prevent equine venereal

³⁶⁵ “Shaanxi sheng chuqin pinzhong gailiang gongzuo zuotanhui jiyao” 陕西省 畜禽品种改良工作座谈会纪要 (March 25, 1973), in *Xumu gongzuo wenjian xuanbian*, ed. Shaanxi sheng xumu shouyi zongzhan, 30–31.

基层农民和饲养员从未消失。国家对牲畜品种退化的担忧似乎直到 20 世纪 70 年代才消除。1973 年，陕西省领导人惊恐地发现，他们多年来辛勤培育的一些优良品种已经严重退化。“新品种的培育速度太慢。遗传性状仍然不稳定。从中国其他地区 and 国外引进和繁殖优良品种的系统管理也不尽如人意。所有这些现象都导致了牲畜生产力的下降。”³⁶⁵ 接下来的章节将探讨国家与地方社区之间在动物育种方面的这些紧张关系。

制度变革

我们或许可以称那些致力于提高牲畜数量和质量的人为饲养员。但在20世纪50年代初，共产党国家尚未系统地组织他们，因此“饲养员”这一职业类别没有明确的定义。当然，有些人具备必要的知识和经验，但他们的工作范围不限于饲养。他们还是牧民、看护员、兽医和企业家。王泽民就是这样一个人。他于1915年出生在吉林省永吉县一个村庄的中农家庭。王从小就照料家里的牲畜。1951年，他用自己拥有的十几匹马和骡子经营马车生意。王懂得如何繁殖马匹，如何预防马匹性病。

³⁶⁵ “陕西省畜禽品种改良工作座谈会纪要”（1973年3月25日），载陕西省畜牧兽医总站编《畜牧工作文件选编》，第 30-31页。

diseases, and how to treat common equine diseases with both “native remedies” (土方) and “Western” disinfectants.³⁶⁶

In this earlier period, the “model workers” in animal husbandry were those who worked hard and “rebuilt the family fortunes and got rich” (发家致富). Lin Sangma was a typical model shepherd-breeder-peasant. In 1951, she was 23 years old and living in Chakhar League in Inner Mongolia. Before the revolution, Lin looked after sheep, goats, and cows for wealthy animal owners. After the Communists took power in Inner Mongolia in 1945, leveraging the political changes and her own knowledge of animal breeding, Lin boldly propagated her own livestock herds and came to own 203 sheep, 55 cows, 10 horses, 8 camels, and 220 goats in 1950.³⁶⁷

Similarly, Zhong Dui was a former Tibetan monk who saw the Communist revolution in Inner Mongolia as an opportunity to grow wealthy. His herds increased from 50 animals in 1945 to over 1,800 in 1947. The local Communist cadres made him a “model” and encouraged him to share his breeding skills. Zhong thus organized community-wide collective breeding programs that brought all the fertile male and female livestock to a designated location for breeding.³⁶⁸

The state’s control over animal breeding tightened after the central government convened a national livestock breeding workshop in early 1953. The workshop was held to facilitate the building of state-run breeding stations (国营配种站) across the country. The state stations would be in charge of developing superior livestock breeds; disseminating advanced breeding technologies such

³⁶⁶ “Xumu mofan Wang Zemin” 畜牧模范王泽民, in Shen Yancheng 沈延成, *Miqiulin xueshuo zai xumujie de yingyong* 米邱林学说在畜牧界的应用 (Nanjing: Xumu shouyi tushu chubanshe, 1951), 86–87.

³⁶⁷ “Xumuye mofan Lin Sangma” 畜牧业模范林桑玛, in Shen Yancheng, *Miqiulin xueshuo zai xumujie de yingyong*, 89–90.

³⁶⁸ “Xumuye shengchan mofan Zhong Dui” 畜牧业生产模范忠对, in Shen Yancheng, *Miqiulin xueshuo zai xumujie de yingyong*, 87–88.

疾病，以及如何用“土方”（土方）和“西式”消毒剂治疗常见的马匹疾病。³⁶⁶

在早期，畜牧业的“模范工人”是那些努力工作并“发家致富”（发家致富）的人。林桑玛就是一位典型的模范牧羊人兼农民。1951年，她23岁，居住在内蒙古察哈尔盟。革命前，林为富有的牲畜主人照看绵羊、山羊和奶牛。1945年共产党在内蒙古掌权后，林利用政治变革和她自己的动物饲养知识，大胆地繁殖自己的牲畜群，到1950年拥有了203只绵羊、55头奶牛、10匹马、8峰骆驼和220只山羊。³⁶⁷

同样，钟堆是一位前藏族僧侣，他将内蒙古的共产主义革命视为致富的机会。他的牲畜从1945年的50头增加到1947年的1800多头。当地的共产党员干部将他树立为“模范”，并鼓励他分享自己的饲养技能。因此，钟堆组织了社区范围的集体饲养计划，将所有可繁殖的雄性和雌性牲畜带到指定地点进行繁殖。³⁶⁸

1953年初，中央政府召开全国畜牧业生产会议后，国家对畜牧业的控制进一步加强。此次会议旨在推动全国范围内国营配种站的建设。这些国家配种站将负责培育优良畜种，推广先进的育种技术，例如

³⁶⁶ 沈延成，《畜牧模范王泽民》，载《米邱林学说在畜牧界的应用》（南京：畜牧兽医图书出版社，1951），第86-87页。³⁶⁷ 沈延成，《畜牧业模范林桑玛》，载《米邱林学说在畜牧界的应用》，第89-90页。

³⁶⁸ 沈延成，《畜牧业生产模范忠对》，载《米邱林学说在畜牧界的应用》，第87-88页。

as artificial insemination; encouraging peasants in nearby villages to take good care of fertile female animals; and supervising private breeding households (民营配种户) on the ground.³⁶⁹ In carrying out this central decision, provincial governments rushed to establish foothold breeding stations (重点配种站). Hunan Province, for instance, opened 15 provincial stations by the end of 1953. The provincial government urged these newly established stations to “accept the Soviet advanced experiences” and mobilize private breeding households and peasants, thereby “implementing breed selection and propagation and improving the quality of livestock.”³⁷⁰

It is worth noting that a new social category of “private breeding households” came into being at this time. As the state breeding stations started to emerge in 1953, animal breeding became a distinct subfield under the auspices of animal husbandry and veterinary work (畜牧兽医工作). This process progressed in tandem with specifying the non-state counterparts that engaged in animal breeding. Although some of the versatile herdsman-breeders-peasants in previous years came to specialize in equine or bovine breeding, the state paid more attention to the larger number of private breeding households, whom Sigrid Schmalzer called “boar men.”³⁷¹ They were “in general lonely widowers or disabled elders.” From the perspective of the state, the boars and other male animals raised by these elderly men were by no means healthy or superior, and the men’s skills were “backward.” However, these people had played an indispensable role in livestock breeding in village communities. Hence, one of the major tasks of the state breeding stations was to help them

³⁶⁹ Guangxi xumu shiyanchang, “Guanyu dui quanguo shengchu peizhong yewu zuotanhui de jidian yijian.”

³⁷⁰ “Hunan sheng 1953 nian jiachu bayou qingkuang ji 1954 nian gongzuo yijian deng cailiao.”

³⁷¹ Sigrid Schmalzer, *Red Revolution, Green Revolution: Scientific Farming in Socialist China* (Chicago: University of Chicago Press, 2016), 124–125 and 153.

如人工授精；鼓励附近村庄的农民照料好可生育的雌性动物；以及监督当地的民营配种户（民营配种户）。³⁶⁹ 在执行这项中央决定时，各省纷纷建立重点配种站（重点配种站）。例如，湖南省在1953年底前开设了15个省级配种站。省政府敦促这些新成立的配种站“接受苏联先进经验”，动员民营配种户和农民，从而“实施品种选择和繁殖，提高牲畜质量”。³⁷⁰

值得注意的是，“民营配种户”这一新的社会类别在此时应运而生。随着国家配种站于1953年开始出现，动物育种成为畜牧兽医工作下的一个独立分支领域。这一过程与明确从事动物育种的非国家对应方同步进行。尽管前些年一些多才多艺的牧民-育种者-农民开始专门从事马或牛的育种，但国家更关注数量更多的民营配种户，西格丽德·施马尔泽（Sigrid Schmalzer）称他们为“公猪人”。³⁷¹ 他们“通常是孤独的鳏夫或残疾老人”。从国家的角度来看，这些老年人饲养的公猪和其他雄性动物绝不健康或优良，他们的技能也“落后”。然而，这些人在乡村社区的牲畜育种中发挥了不可或缺的作用。因此，国家配种站的主要任务之一就是帮助他们

³⁶⁹ 广西畜牧实验场，“关于对全国牲畜配种业务座谈会的几点意见。”

³⁷⁰ “湖南省1953年家畜罢优情况及1954年工作意见等材料。”

³⁷¹ Sigrid Schmalzer, *Red Revolution, Green Revolution: Scientific Farming in Socialist China* (芝加哥：芝加哥大学出版社，2016)，124–125 和 153。

eliminate “inferior” beasts; awaken their class consciousness; improve their technical capability; and elevate their social status.³⁷²

This institutionalization of animal breeding with a focus on state breeding stations was consolidated over the next few years. By November 1955, Guangxi Province had a vertical structure of breeding stations: one provincial-level artificial insemination station; six prefecture-level foothold breeding stations; and 44 county-level breeding stations. Each prefectural station was supposed to manage six bulls and four boars of superior breeds. At least four bulls and two boars needed to be raised in a county station and were mated with local female livestock.³⁷³ In this manner, a new animal breeding network featuring state stations as primary and private breeding households as secondary nodes took root in rural China in the mid-1950s.

In the late 1950s, however, this system underwent another transformation as full-fledged collectivization interlocked with animal breeding. The state now expected grassroots collectives to foster their own breeding centers that were administratively distinct from the state-owned breeding stations. As we have seen, the central government tried to “assign missions” of hybrid improvement “to collectives and let them proceed on their own” as early as 1957.³⁷⁴ But this does not mean that the state attempted to hand over all the tasks that state breeding stations were expected to do to collectives. More precisely, the state’s goal was to make collectives quantitative “propagation bases” (繁殖基地), letting state stations continue to lead the qualitative improvement of breed. As such, in 1958, Gansu Provincial Government ordered 35 counties to encourage subordinate communes,

³⁷² Guangxi xumu shiyanchang, “Guanyu dui quanguo shengchu peizhong yewu zuotanhui de jidian yijian.”

³⁷³ Guangxi sheng nongyeting, “Guangxi sheng 1955 nian zhongdian shengchu peizhong gongzuo huiyi baogao.”

³⁷⁴ Zhonghua renmin gongheguo nongyebu, “Guanyu quanguo mianyang gailiang zuotanhui de baogao,” 130.

淘汰“劣等”牲畜；唤醒它们的阶级意识；提高它们的技术能力；提升它们的社会地位。

372

这种以国家育种站为重点的动物育种制度在接下来的几年中得到巩固。到1955年11月，广西省建立了垂直的育种站体系：一个省级人工授精站；六个地级育种站；以及44个县级育种站。每个地级站应管理六头公牛和四头优良品种的公猪。每个县级站至少需要饲养四头公牛和两头公猪，并与当地的母畜进行交配。³⁷³通过这种方式，一个以国家育种站为主、私人育种户为辅的新型动物育种网络于20世纪50年代中期在中国农村扎根。

然而，在20世纪50年代后期，随着全面集体化与动物育种的相互交织，这一体系又经历了一次转型。国家现在期望基层集体建立自己的育种中心，这些中心在行政上与国有育种站有所区别。正如我们所见，早在1957年，中央政府就试图“将杂交改良的任务‘分配给集体，让它们自行开展’”。³⁷⁴但这并不意味着国家试图将国有育种站应承担的所有任务都移交给集体。更确切地说，国家的目标是使集体成为数量上的“繁殖基地”，让国有育种站继续主导品种的质量改良。因此，1958年，甘肃省政府命令35个县鼓励下属公社，

³⁷² 广西畜牧实验场，“关于对全国牲畜配种业务座谈会的几点意见。”³⁷³ 广西省农业厅，“广西省1955年重点牲畜配种工作会议报告。”³⁷⁴ 中华人民共和国农业部，“关于全国绵羊改良座谈会的报告”，130。

brigades, and teams to build their own breeding stations and mobilize them in rapidly increasing the number of draft animals.³⁷⁵

Against this backdrop, leaders of communes, brigades, and teams came to nominate some local residents as “collective breeders” (集体配种员). Just as Chapter Four showed suitable persons were found to be collective caretakers, those who had either relevant knowledge or political enthusiasm became breeders. Diverse short-training sessions for novice breeders were also widely available. It might have been possible that the boar men and private breeding households who handled other male animals were incorporated into this new rank of collective breeders or the majority of these elderly men might have died in the 1950s and early 1960s. In either event, the term “private breeding households” rarely appears in the documents produced in these years and after.³⁷⁶

The collective breeders—also known as “breeding handlers” (配种行家), “breeding technicians” (配种技术人员), or “well-bred male animal caretakers” (种公畜饲养员)—were the workers in collective breeding stations who looked after male animals, either locally selected or superior ones received from the state breeding stations, and facilitated mating the males with local female herds. Their job requirements also included taking a regular livestock census; acquainting themselves with the information about the estrus of local female animals; and promoting the state’s policies within their collectives.³⁷⁷ In addition, these breeders were expected to collaborate with

³⁷⁵ “Gansu sheng jianli gengchu fanzhi jidi de juti jihua cao’an,” 12–13 and 20.

³⁷⁶ According to the online database of the *People’s Daily* 人民日报, articles which include the term “private breeding households” were found only in the period between 1952 and 1956. We will further look into these boar men in the next section.

³⁷⁷ Minle xian renmin weiyuanhui 民乐县人民委员会, “Chenguang nongyeshe tigao muchu shoutailü he youchu chenghuolü de jingyan” 晨光农业社提高母畜受胎率和幼畜成活率的经验 (N.d.), in *Xumu shouyi jingyan 4: Gengchu de peizhong fanzhi*, ed. Gansu sheng xumuting, 26–27.

大队和生产队建立自己的配种站，并动员它们迅速增加役畜数量。³⁷⁵

在此背景下，公社、大队和生产队的领导开始提名一些当地居民为“集体配种员”（集体配种员）。正如第四章所示，找到了合适的人选担任集体饲养员，那些具有相关知识或政治热情的人成为了配种员。针对新手配种员的各种短期培训班也广泛开设。处理其他雄性动物的猪倌和私人养殖户可能被纳入了集体配种员这一新行列，或者这些老年人中的大多数可能在20世纪50年代和60年代初已经去世。无论如何，“私人养殖户”一词在这些年及以后产生的文件中很少出现。³⁷⁶

集体配种员——也称为“配种行家”、“配种技术人员”或“种公畜饲养员”——是集体配种站的工作人员，他们负责照料雄性动物，无论是本地选育的还是从国家配种站接收的优良品种，并促进雄性动物与当地雌性畜群的交配。他们的工作要求还包括定期进行牲畜普查；熟悉当地雌性动物的发情信息；以及在集体内部宣传国家政策。³⁷⁷ 此外，这些配种员还需与

³⁷⁵ “甘肃省建立耕畜繁殖基地的具体计划草案”，12-13 和 20。

³⁷⁶ 根据《人民日报》的在线数据库，包含“私人养殖户”一词的文章仅在1952年至1956年期间出现。我们将在下一节中进一步探讨这些“猪倌”。

³⁷⁷ 民乐县人民委员会民乐县人民委员会，“晨光农业社提高母畜受胎率和幼畜成活率的经验” 晨光农业社提高母畜受胎率和幼畜成活率的经验（无日期），载于《畜牧兽医经验4：耕畜的配种繁殖》，甘肃省畜牧厅编，26-27页。

veterinarians, caretakers, animal servicers (使役员), and animal disease prevention workers (防疫员) to maintain the general health of livestock in more-than-people’s communes.³⁷⁸

Thanks to the presence and labor of such collective breeders, Zhejiang Provincial Government reported in 1959 that the province had already fulfilled the goals of “self-breeding and self-raising” (自繁自养), “local self-propagation” (就地繁殖), and “local self-supply [of livestock]” (就地供应).³⁷⁹ In the same year, according to a report written by the Communist Party Yunnan Chengjiang County Committee, the county operated 31 commune-level breeding stations with about 700 full-time breeders.³⁸⁰ By 1962, breeding stations or breeding posts (配种点) in communes, brigades, and teams were widespread across China’s vast countryside.³⁸¹ In other words, the late 1950s and the early 1960s saw the animal breeding system based on the division of labor between the state-owned and collective breeding stations. The former focused on the qualitative improvement of livestock breed, whereas the latter took bigger initiatives in the quantitative increase of domestic animals.

³⁷⁸ “Gansu sheng jianli gengchu fanzhi jidi de juti jihua cao’an,” 15; Shaanxi sheng Weinan xian renmin weiyuanhui 陕西省渭南县人民委员会, “Weinan xian de shengchu baohu he fanzhi gongzuo” 渭南县的牲畜保护和繁殖工作 (N.d.), in *Xumu shouyi jingyan 4: Gengchu de peizhong fanzhi*, ed. Gansu sheng xumuting, 41.

³⁷⁹ Zhejiang sheng renmin weiyuanhui 浙江省人民委员会, “Guanyu zengyang muzhu he zhujin muzhu peizhong gongzuo de zhishi” 关于增养母猪和抓紧母猪配种工作的指示 (April 5, 1959), Unpublished material.

³⁸⁰ Zhonggong Chengjiang xianwei 中共澄江县委, “Dashengchu manhuai fengchan jingyan” 大牲畜满怀丰产经验 (N.d.), in *Shengchu fanzhi peizhong: Yunnan sheng 1959 nian nongye xianjin jingyan xumu zhi liu* 牲畜繁殖配种: 云南省 1959 年农业先进经验畜牧之六, ed. Yunnan sheng 1959 nian nongye shehui zhuyi jianshe xianjin danwei shengchanzhe dai biao dahui 云南省 1959 年农业社会主义建设先进单位生产者代表大会 (Kunming: Yunnan renmin chubanshe, 1960), 12–13.

³⁸¹ Yuncheng xian renmin weiyuanhui 运城县人民委员会, “Guanyu zhaokai peizhong renyuan huiyi de tongzhi” 关于召开配种人员会议的通知 (March 1, 1962), Unpublished material.

兽医、饲养员、动物服务人员（使役员）和动物疾病预防工作者（防疫员），以维持人民公社以上牲畜的整体健康。³⁷⁸

得益于这些集体饲养员的存在和劳动，浙江省政府在1959年报告称，该省已实现“自繁自养”、“就地繁殖”和“就地供应[牲畜]”的目标。³⁷⁹同年，根据中共云南澄江县委的一份报告，该县运营着31个公社级畜牧站，拥有约700名全职饲养员。³⁸⁰到1962年，公社、大队和生产队的畜牧站或配种点已遍布中国广阔的农村地区。³⁸¹换句话说，1950年代末和1960年代初，动物繁育系统是建立在国营和集体繁育站之间分工的基础上的。前者侧重于牲畜品种的质量改进，而后者则在家畜数量的增加方面采取了更大的举措。

³⁷⁸ “Gansu sheng jianli gengchu fanzhi jidi de juti jihua cao’an,” 15; Shaanxi sheng Weinan xian renmin weiyuanhui 陕西省渭南县人民委员会, “Weinan xian de shengchu baohu he fanzhi gongzuo” 渭南县的牲畜保护和繁殖工作 (N.d.), in *Xumu shouyi jingyan 4: Gengchu de peizhong fanzhi*, ed. Gansu sheng xumuting, 41.

³⁷⁹ 浙江省人民委员会, “关于增养母猪和抓紧母猪配种工作的指示”（1959年4月5日），未出版材料。

³⁸⁰ Zhonggong Chengjiang xianwei 中共澄江县委, “Dashengchu manhuai fengchan jingyan” 大牲畜满怀丰产经验 (N.d.), in *Shengchu fanzhi peizhong: Yunnan sheng 1959 nian nongye xianjin jingyan xumu zhi liu* 牲畜繁殖配种: 云南省 1959 年农业先进经验畜牧之六, ed. Yunnan sheng 1959 nian nongye shehui zhuyi jianshe xianjin danwei shengchanzhe dai biao dahui 云南省 1959 年农业社会主义建设先进单位生产者代表大会 (Kunming: Yunnan renmin chubanshe, 1960), 12–13.

³⁸¹ 运城县人民委员会, “关于召开配种人员会议的通知”（1962年3月1日），未出版材料。

During the devastating Great Leap Famine (1959–1961), many local animal breeding institutions temporarily halted their activities. To collective communities, it made no sense to increase the number of livestock in the midst of a severe food shortage. Breeders and peasants hence castrated or spayed highbred male and local female animals. The last thing they wanted to do was to slaughter local livestock; in the long term, humans could not survive without their animals.³⁸² After the Famine and with the inception of the Readjustment Policy led by Liu Shaoqi in the first half of the 1960s, the central government exhorted county governments to resume local animal breeding by offering economic incentives to breeders of male animals and caretakers of females to mate their animals.³⁸³ In addition, soft-pedaling the Maoist ideal of communal self-reliance, an editorial article in the *People's Daily* of May 19, 1963 argued that though collectives would have to cling to the “self-breeding and self-raising,” if they lacked superior male animals and other necessary resources, they could rely on assistance from neighboring collectives.³⁸⁴

These changes in policies notwithstanding, the state’s priorities remained unchanged. The same *People's Daily* article urged local collectives to normalize breeding stations and organize collective breeders and other local veterinary workers for the maximization of both the quantitative increase and the qualitative improvement of livestock. The article stressed that collective breeders had to grasp the health conditions of the local herds of female livestock of breeding age.³⁸⁵ Beijing’s

³⁸² Nanchang xian Jiangxiang qu gongsuo 南昌县蒋巷区公所, “Zhuanfa xian renmin weiyuanhui guanyu xunsu huifu yu jianli gengniu peizhongzhan de tongzhi” 转发县人民委员会关于迅速恢复与建立耕牛配种站的通知 (May 1963), Unpublished material.

³⁸³ Nanchang xian Jiangxiang qu gongsuo, “Zhuanfa xian renmin weiyuanhui guanyu xunsu huifu yu jianli gengniu peizhongzhan de tongzhi.”

³⁸⁴ “Nuli fanzhi dashengchu.”

³⁸⁵ “Nuli fanzhi dashengchu.”

在毁灭性的大跃进饥荒（1959-1961）期间，许多地方动物育种机构暂时停止了活动。对于集体社区来说，在严重的粮食短缺中增加牲畜数量毫无意义。因此，饲养员和农民对高繁殖雄性动物和本地雌性动物进行了阉割或绝育。他们最不想做的事情就是屠宰本地牲畜；从长远来看，人类离不开他们的动物。³⁸²饥荒过后，随着刘少奇在20世纪60年代上半叶推行调整政策，中央政府鼓励县政府通过向雄性动物饲养者和雌性动物看护者提供经济激励，让他们进行动物交配，从而恢复地方动物育种。³⁸³此外，1963年5月19日《人民日报》的一篇社论文章淡化了毛泽东关于集体自力更生的理想，认为尽管集体必须坚持“自繁自养”，但如果他们缺乏优质雄性动物和其他必要资源，他们可以依靠邻近集体的帮助。³⁸⁴

尽管政策有所调整，但国家优先事项保持不变。同一篇《人民日报》文章敦促地方集体将畜牧站规范化，并组织集体饲养员和其他地方兽医，以最大限度地提高牲畜的数量和质量。文章强调，集体饲养员必须掌握当地适龄母畜群的健康状况。³⁸⁵北京的

³⁸²南昌县蒋巷区公所， “转发县人民委员会关于迅速恢复与建立耕牛配种站的通知”（1963年5月）， 未出版材料。

³⁸³南昌县蒋巷区公所， “转发县人民委员会关于迅速恢复与建立耕牛配种站的通知”。

³⁸⁴ “Nuli fanzhi dashengchu.”

³⁸⁵ “Nuli fanzhi dashengchu.”

emphasis on controlling female nonhuman bodies passed down through the administrative hierarchy to provincial governments, county governments, and sub-county collectives. In 1964, the Communist Party Committee of Wanrong County, Shanxi Province ordered every production team within the county to retain a maternal herd (母畜群) of at least five extremely fertile female animals.³⁸⁶ Collective breeding stations invariably needed to function as propagation bases even when the Readjustment Policy unfolded in the first half of the 1960s.

The Cultural Revolution (1966–1976) once again foregrounded communal self-reliance. It criticized the “expert line” (专家路线) and the idea of “counting on outside aid” (依靠外援) that allegedly harmed animal breeding in the Readjustment period. More specifically, local breeding stations and breeders were now required to break down the practices of “relying on the state for [breed] improvement, on experts for technologies, and on heaven for raising [livestock].” Instead, by strengthening collective breeding stations’ initiatives in the improvement, collectives had to achieve the milestone of “self-reproduction within the local herds.”³⁸⁷

But this did not mean that the collective stations could replace the state stations. The National Breeding Farms Workshop held in December 1973 clarified the relationship between the two lines of breeding stations. Even though state breeding stations could easily be associated with the expert line, the workshop declared that to consolidate the “mass line” in animal breeding, the state stations would have to dedicate themselves to supporting local collective stations with a focus

³⁸⁶ Shanxi sheng Wanrong xianwei 山西省万荣县委, “Zuochu jinyibu fazhan dashengchu de shi xiang jue ding” 作出进一步发展大牲畜的十项决定 (N.d.), in *Xumu shouyi gongzuo cankao ziliao* 畜牧兽医工作参考资料, ed. Gansu sheng xumuting 甘肃省畜牧厅 (Lanzhou: Gansu sheng xumuting, 1964), 132–135.

³⁸⁷ Shaanxi sheng xumu shouyi zongzhan 陕西省畜牧兽医总站, “Guanyu mianyang gailiang zuotanhui qingkuang de baogao” 关于绵羊改良座谈会情况的报告 (June 1, 1971), in *Xumu gongzuo wenjian xuanbian*, ed. Shaanxi sheng xumu shouyi zongzhan, 148–151.

对控制雌性非人动物的重视通过行政层级下达到省级政府、县级政府和县以下集体。1964年，山西省万荣县共产党委员会命令县内每个生产队至少保留五头极度肥沃的雌性动物作为母畜群。³⁸⁶ 即使在20世纪60年代上半叶的调整政策时期，集体繁殖站也必然需要作为繁殖基地发挥作用。

文化大革命（1966-1976）再次强调了集体自力更生。它批判了“专家路线”（专家路线）和“依靠外援”（依靠外援）的观念，认为这些观念在调整时期损害了动物育种。更具体地说，地方育种站和育种者现在被要求打破“依靠国家进行[品种]改良，依靠专家提供技术，以及依靠上天饲养[牲畜]”的做法。相反，通过加强集体育种站在改良方面的举措，集体必须实现“地方畜群内部自我繁殖”的里程碑。³⁸⁷

但这并不意味着集体站可以取代国家站。1973年12月举行的全国育种农场研讨会阐明了这两类育种站之间的关系。尽管国家育种站很容易与专家路线联系起来，但研讨会宣布，为了巩固畜牧业的“群众路线”，国家站必须致力于支持地方集体站，重点是

³⁸⁶ 山西省万荣县委, “作出进一步发展大牲畜的十项决定” (N.d.), 载甘肃省畜牧厅编《畜牧兽医工作参考资料》（兰州：甘肃省畜牧厅，1964），第132-135页。

³⁸⁷ 陕西省畜牧兽医总站, “关于绵羊改良座谈会情况的报告”（1971年6月1日），载陕西省畜牧兽医总站编《畜牧工作文件选编》，第148–151页。

on the extension of the best male animals as well as artificial insemination technologies.³⁸⁸ The state and collective breeding stations coexisted as long as the former could conform with the collective-centered mass line of animal breeding.³⁸⁹ The official discourses hence prescribed that animal breeding policies during the Cultural Revolution period would be predicated upon the “mass-based breeding system that took people’s communes as foundation, state-run breeding farms as backbone, and scientific research and educational institutions as advisor.”³⁹⁰

Black and White Pigs

Based on the conceptual background and institutional change outlined in the previous sections, this and following sections explore the diverse and specific relationalities among different animals, state breeders, and collective breeders, and peasants. We start with the most prominent nonhuman animal: pigs. In the early 1950s before agricultural collectivization, the presence of private breeding households, or “boar men,” drew the central government’s attention. When there was a lack of male animals for breeding, the state established new state breeding stations, on the one hand, and “more actively helped” the private breeding households “refine their performance,” on the other. The role of these breeding households was emphasized in terms of swine breeding. The central and

³⁸⁸ “Quanguo zhongchuchang zuotanhui jiyao” 全国种畜场座谈会纪要 (December 24, 1973), in *Xumu gongzuo wenjian xuanbian*, ed. Shaanxi sheng xumu shouyi zongzhan, 21–25.

³⁸⁹ If we can associate the state breeding lines with *yang* science and the collective breeding stations with *tu* science, the strange symbiosis between the two reflected the Maoist calls for “harnessing tu and yang together in productive partnership.” On the discussion on the *tu/yang* binary, see Schmalzer, *Red Revolution, Green Revolution*, 34–38.

³⁹⁰ “Guanyu quanguo ban ximao yang yuzhong xiezuo huiyi de baogao” 关于全国半细毛羊育种协作会议的报告 (August 29, 1975), in *Quanguo ban ximao yang yuzhong xiezuo huiyi ziliao xuanbian* 全国半细毛羊育种协作会议资料选编, ed. Qinghai sheng xumu shouyi yanjiusuo 青海省畜牧兽医研究所 and Dongbei diqu mianyang yuzhong weiyuanhui 东北地区绵羊育种委员会 (N.p: Qinghai sheng xumu shouyi yanjiusuo and Dongbei diqu mianyang yuzhong weiyuanhui, 1975), 1–2.

推广优良雄性动物以及人工授精技术。³⁸⁸ 只要国家和集体育种站能与以集体为中心的畜牧业群众路线保持一致，两者就可以共存。³⁸⁹ 因此，官方论述规定，文化大革命期间的畜牧业政策将以“以人民公社为基础、国营种畜场为骨干、科研教育机构为顾问的群众性畜牧业体系”为前提。³⁹⁰

黑白猪

基于前几节概述的概念背景和制度变革，本节及以下各节探讨了不同动物、国家育种者、集体育种者和农民之间多样而具体的关系。我们从最突出的非人类动物——猪开始。在20世纪50年代初农业集体化之前，私人养殖户或“公猪人”的存在引起了中央政府的关注。当缺乏用于繁殖的雄性动物时，国家一方面建立了新的国家育种站，另一方面“更积极地帮助”私人养殖户“改进其表现”。这些养殖户在生猪养殖方面的作用得到了强调。中央和

³⁸⁸ “全国种畜场座谈会纪要”（1973年12月24日），载陕西省畜牧兽医总站编《畜牧工作文件选编》，第21—25页。

³⁸⁹ 如果我们将国家育种系与“洋”科学联系起来，将集体育种站与“土”科学联系起来，那么两者之间奇怪的共生关系反映了毛泽东提出的“土洋结合，共同生产”的号召。关于“土/洋”二元论的讨论，参见施马尔泽（Schmalzer），《红色革命，绿色革命》，第34—38页。

³⁹⁰ “关于全国半细毛羊育种协作会议的报告”（1975年8月29日），载青海省畜牧兽医研究所和东北地区绵羊育种委员会编《全国半细毛羊育种协作会议资料选编》（无出版地：青海省畜牧兽医研究所和东北地区绵羊育种委员会，1975），第1—2页。

provincial governments therefore aggressively reached out to these breeding households to reform and organize them.³⁹¹

The leaders of Neijiang County, Sichuan Province articulated how they dealt with such private breeders in 1953. There were 231 private breeding households in the county. Holding 285 boars, these households provided mating services to peasants who raised sows. Eighty-five percent of the households were disabled men. Since these private breeders carried 4-to-12-month-old male pigs and desexed them before the animals grew too large or stubborn for them to handle, the local cadres and state breeders in the state-run breeding stations regarded the boar men’s boars as too small, weak, and generally inferior. Moreover, there was a “superstitious” belief among the peasants in the county that working as a private pig breeder would render a person sterile or infertile. The social status of these breeders was very low, and they had long suffered from “self-depreciation” (自卑心理). They stayed in the job only because they could find no other way to make a living.³⁹²

To uplift and mobilize the private boar breeders in the country’s animal breeding mission, the local cadres and state breeders organized them into “breeding groups” (配种小组). In the spring of 1953, the cadres and state breeders asked the private breeders to select better male animals from what they had and pushed the other peasants to raise more sows. From April to August, the private breeders received intensive training and then were assigned to several sub-districts to tour and provide mating services to the locals. In this process, each private breeder’s physical disability was taken into consideration. The county government also standardized the breeding service fee to

³⁹¹ “Xin Zhongguo xumu shiye gaikuang” 新中国畜牧事业概况, *Renmin Ribao* 人民日报 (October 29, 1952).

³⁹² Nongyebu xumu shouyi zongju 农业部畜牧兽医总局, “Guanyu jieshao Sichuan Neijiang zhuanqu zhongzhuchang zuzhi zhongzhuhu tuijian xuanzhong xuanpei jingyan de tongzhi” 关于介绍四川内江专区种猪场组织种猪户推荐选种选配经验的通知 (February 2, 1954), GARA, X163-001-0392-0001.

因此，各省政府积极主动地与这些养殖户联系，对其进行改革和组织。³⁹¹

四川省内江县的领导阐述了他们如何在1953年处理这些私人养殖户。该县有231户私人养殖户。这些养殖户拥有285头公猪，为饲养母猪的农民提供配种服务。其中85%的养殖户是残疾人。由于这些私人养殖户饲养4到12个月大的公猪，并在它们长得太大或太难驯服之前对其进行去势，因此当地干部和国营养殖站的国有养殖员认为这些公猪太小、太弱，总体上劣质。此外，该县农民中流传着一种“迷信”观念，认为从事私人养猪业会使人绝育或不育。这些养殖户的社会地位很低，长期以来一直遭受“自卑心理”的困扰（自卑心理）。他们之所以继续从事这项工作，只是因为他们找不到其他谋生方式。

³⁹²

为了提升和动员全国的私人种猪饲养者参与国家动物育种任务，地方干部和国家饲养员将他们组织成“配种小组”。1953年春天，干部和国家饲养员要求私人饲养者从他们现有的动物中挑选出更好的雄性动物，并推动其他农民饲养更多的母猪。从4月到8月，私人饲养者接受了强化培训，然后被分配到几个分区进行巡回服务，为当地人提供配种服务。在此过程中，每个私人饲养者的身体残疾都得到了考虑。县政府还规范了配种服务费，规定为

³⁹¹ “新中国畜牧事业概况”，《人民日报》（1952年10月29日）。

³⁹² 农业部畜牧兽医总局，“关于介绍四川内江专区种猪场组织种猪户推荐选种选配经验的通知”（1954年2月2日），GARA, X163-001-0392-0001。

prevent some peasants from intimidating the private breeders into paying no fees. The private breeders, according to the county government’s document, were satisfied. Hu Songyun, a private breeder, was happy that “unlike the past when I was not paid, I have no longer such problems. Moreover, I can earn technologies!” Another private breeder, Hu Songlin, recovered his fortune by supporting the state’s policies. He was eventually able to marry and have a daughter. He said that he was the counterexample of the superstition that private breeders would become impotent or sterile. Private breeders like them in Neijiang County propagated 64,294 piglets in that year alone.³⁹³

In the mid-1950s, the honeymoon between the state and the private breeders came to an end as the state initiated the improvement of swine breeds along with agricultural collectivization. The central and provincial governments now hoped to propagate the highbred Yorkshire hogs in nascent cooperatives. The Yorkshire pig, or “Large White,” was a celebrated British breed that was known around the world for its suitability for the industrial production of pork. Preferring these “superior” pigs, Communist cadres and state breeders came to devalue the small black native pigs bred by private breeders. However, local villagers did not readily conform to the state’s preferences. It was “very strange” for villagers in some parts of Guangxi, for example, “to extend the Yorkshires as they were a completely new kind.” They also “found raising purebred pigs burdensome.” The peasants were afraid that they would have to get some special salt and bone meal for the Yorkshire pigs. Large White pigs were unappealing because of their thick skin and the toughness of their meat. Local cadres, nevertheless, tried to recruit some young peasants who supported collectivization and had a passion for animal husbandry as cooperative breeders to facilitate the breeding of the

³⁹³ Nongyebu xumu shouyi zongju, “Guanyu jieshao Sichuan Neijiang zhuanqu zhongzhuchang zuzhi zhongzhuhu tuijian xuanzhong xuanpei jingyan de tongzhi.”

防止一些农民恐吓私人饲养员，使其不支付费用。根据县政府的文件，私人饲养员对此表示满意。私人饲养员胡松云高兴地说：“不像过去我拿不到报酬，我现在再也没有这样的问题了。而且，我还能学到技术！”另一位私人饲养员胡松林通过支持国家政策恢复了财富。他最终得以结婚并有了一个女儿。他说，他是私人饲养员会变得阳痿或不育这种迷信的反例。内江县像他们这样的私人饲养员仅在那一年就繁殖了64294头仔猪。

393

20世纪50年代中期，随着农业集体化以及国家开始改良生猪品种，国家与私人养殖户之间的“蜜月期”宣告结束。中央和省级政府现在希望在新兴的合作社中推广约克夏猪。约克夏猪，又称“大白猪”，是一种著名的英国猪种，以其适合工业化生产猪肉而闻名于世。共产主义干部和国家育种员偏爱这些“优良”猪种，开始贬低私人养殖户饲养的本地小型黑猪。然而，当地村民并不轻易顺从国家的偏好。例如，广西一些地区的村民觉得“推广约克夏猪非常奇怪，因为它们是全新的品种”。他们还“觉得饲养纯种猪很麻烦”。农民们担心他们必须为约克夏猪准备一些特殊的盐和骨粉。大白猪因其皮厚肉韧而不受欢迎。尽管如此，当地干部还是试图招募一些支持集体化并对畜牧业充满热情的年轻农民作为合作社饲养员，以促进

³⁹³ 农业部畜牧兽医总局，“关于介绍四川内江专区种猪场组织种猪户推荐选种选配经验的通知。”

Yorkshires. Few volunteered. There was a widespread idea that only widowers or people without a family would raise boars. The state’s effort to disseminate the “superior” pigs was adrift.³⁹⁴

Local cadres from other parts of Guangxi, however, faced a different situation. Local people, according to their 1955 report, were initially unfamiliar with the White Larges. But soon after, the foreign pigs became popular, and were even “idolized.” The unexpected result of this phenomenon was that the private breeders could no longer eke out a living by breeding native black boars; the villagers did not want the small black pigs. To solve with this problem, local cadres educated the peasants on the need to produce superior-bred pigs and cull inferior ones. At the same time, local cadres and state breeders cynically tried to convince them that the native pigs were just as good as the Yorkshires.³⁹⁵ The Guangxi cadres blamed themselves for the resulting confusion, saying that “when we encouraged the masses to select breeds and expected them to accomplish the animal breeding work, the direction of the work was not specific enough.”³⁹⁶

In November 1956, the state clearly expressed its positive stance toward the Yorkshire pigs. Local peasants still resisted. Some county-level governments including Pinghe County, Fujian Province unconditionally prohibited peasants from spaying sows with reproductive capacity. Such a “coercive and peremptory method” (强迫命令的做法) met “opposition from the masses.” For example, a poor widow who could not afford more piglets wanted her sow to be sterilized but had no way of doing so. The sow soon became aroused. Yet the widow refused to mate her. The sow

³⁹⁴ Guangxi sheng nonglinting 广西省农林厅, “Nandan xian tuiguang yuekexia gongzhu de jiancha baogao” 南丹县推广约克夏公猪的检查报告 (1954), GARA, X163-001-0392-0052.

³⁹⁵ Guangxi sheng nongyeting, “Guangxi sheng 1955 nian zhongdian shengchu peizhong gongzuo huiyi baogao.”

³⁹⁶ Guangxi sheng nongyeting 广西省农业厅, “Guangxi sheng 1955 nian zhongdian shengchu peizhongzhan gongzuo huiyi zongjie ji jinhou gongzuo juti cuoshi” 广西省 1955 年重点牲畜配种站工作会议总结及今后工作具体措施 (November 21, 1955), GARA, X163-002-0138-0001.

约克夏猪。很少有人自愿饲养。人们普遍认为只有鳏夫或没有家庭的人才会饲养公猪。国家推广 “优良” 猪种的努力陷入困境。³⁹⁴

然而，广西其他地区的当地干部却面临着不同的情况。根据他们1955年的报告，当地人最初不熟悉大白猪。但很快，这种外国猪变得流行起来，甚至被 “偶像化”。这种现象意想不到的结果是，私人饲养者无法再通过饲养本地黑公猪来维持生计；村民们不想要小黑猪。为了解决这个问题，当地干部教育农民需要生产优良品种的猪，淘汰劣质猪。与此同时，当地干部和国家饲养员却玩世不恭地试图说服他们，本地猪和约克夏猪一样好。³⁹⁵ 广西干部将由此造成的混乱归咎于自己，称 “当我们鼓励群众选择品种并期望他们完成动物育种工作时，工作方向不够明确。”³⁹⁶

1956年11月，国家明确表示对约克夏猪持积极态度。但地方农民仍然抵制。包括福建省平和县在内的一些县级政府无条件禁止农民为具有生育能力的母猪绝育。这种 “强迫命令的做法” 遭到了 “群众的反对”。例如，一位贫穷的寡妇买不起更多的小猪，想让她母猪绝育，但却无计可施。母猪很快发情。然而寡妇拒绝让它交配。这头母猪

³⁹⁴ 广西省农林厅， 《南丹县推广约克夏公猪的检查报告》（1954）， GARA, X163-001-0392-0052。

³⁹⁵ 广西省农业厅， 《广西省1955年重点牲畜配种工作会议报告》。

³⁹⁶ 广西省农业厅， “广西省1955年重点牲畜配种站工作会议总结及今后工作具体措施” （1955年11月21日）， GARA, X163-002-0138-0001。

eventually “started running around the sty in a frenzy and screaming all day.” The widow complained that “I have never heard of the ban of pig sterilization in all of history.”³⁹⁷

Similarly, peasants in Jinhua Prefecture, Zhejiang Province accused the state breeding station in the prefecture of stealing from them and spreading contagious swine diseases to the lower cooperatives. The state stations placed a high price on Yorkshires. Although state breeders argued that raising and rehoming purebred Yorkshires took extra resources and labor and that therefore the price was reasonable, the peasants refused to pay more than twice as much for a White Large than for an ordinary hog. And when a swine disease broke out in Jinhua in 1956, its origin was traced to the state breeding station. The peasants used this accident as a reason to refuse to purchase Yorkshires.³⁹⁸

After the Great Leap Forward, the state intensified its requisition of local communes’ pigs to squeeze every last bit of surplus value from the countryside.³⁹⁹ As pork became a significant export after the late 1950s, Bobai County, Guangxi which was not far from Hong Kong was under pressure to breed the Yorkshire-based hybrid pigs for export. At the expense of maintaining and increasing the use value that other domestic animals could give to the collectives, the collectives in Bobai input manpower and resources to raise and send the hybrid hogs to the international trade port where the

³⁹⁷ “Dapo fazhan shengchu shengchan de zhang’ai” 打破发展生猪生产的障碍, *Renmin Ribao* 人民日报 (November 18, 1956).

³⁹⁸ “Dapo fazhan shengchu shengchan de zhang’ai.”

³⁹⁹ On the question of the “primitive socialist accumulation” in Maoist China and the roles of rural women’s labor in the process, see Jacob Eyferth, “State Socialism and the Rural Household: How Women’s Handloom Weaving (and Pig-Raising, Firewood-Gathering, Food-Scavenging) Subsidized Chinese Accumulation,” *International Review of Social History* (February 2022): 1–19, <https://doi.org/10.1017/S0020859021000717>. I am indebted to Eyferth’s insight that rural China’s capacity of producing value was by no means limitless and that, therefore, surplus value that the state wanted to extract from rural China and use value that local collectives hoped to retain at the grassroots were in zero-sum relations. Also see Jason W. Moore, *Capitalism in the Web of Life: Ecology and the Accumulation of Capital* (New York: Verso, 2015).

最终“开始在猪圈里疯狂地跑来跑去，整天尖叫。”寡妇抱怨说：“我从未听说过历史上禁止给猪绝育。”³⁹⁷

同样，浙江省金华府的农民指责该府的国营育种站偷窃他们，并向低级合作社传播传染性猪病。国营站对约克夏猪要价很高。尽管国营育种员辩称，饲养和安置纯种约克夏猪需要额外的资源和劳动力，因此价格合理，但农民拒绝为一头白色大猪支付比普通猪高出两倍以上价格。1956年金华爆发猪瘟时，其源头被追溯到国营育种站。农民以此事故为由拒绝购买约克夏猪。³⁹⁸

大跃进之后，国家加大了对地方公社生猪的征用，以榨取农村的每一分剩余价值。³⁹⁹ 20世纪50年代后期，猪肉成为重要的出口产品，离香港不远的广西博白县面临着饲养约克夏杂交猪出口的压力。为了维持和增加其他家畜能给集体带来的使用价值，博白县的集体投入人力和资源饲养杂交猪并将其送往国际贸易港口，在那里

³⁹⁷ “打破发展生猪生产的障碍”，《人民日报》（1956年11月18日）。

³⁹⁸ “打破发展生猪生产的障碍。”

³⁹⁹ 关于毛泽东时代中国的“原始社会主义积累”问题以及农村妇女劳动在此过程中的作用，参见 Jacob Eyferth, “国家社会主义与农村家庭：妇女手工织布（以及养猪、拾柴、觅食）如何补贴中国积累”，《国际社会史评论》（2022年2月）：1-19页，<https://doi.org/10.1017/S0020859021000717>。我非常感谢 Eyferth 的真知灼见，即中国农村创造价值的能力绝非无限，因此，国家希望从中国农村榨取的剩余价值与地方集体希望在基层保留的使用价值之间存在零和关系。另见 Jason W. Moore, 《生命之网中的资本主义：生态与资本积累》（纽约：Verso, 2015）。

animals were slaughtered and loaded onto freighters as meat. In 1964, Bobai County exported 41,662 live pigs, while locally consuming 30,957. Among these pigs were native and Large Whites. In 1965, 70 percent of the local pigs were native; only 30 percent were Yorkshire-native hybrid hogs. The central and provincial governments, however, ordered the county to desex or cull native pigs and make the entire swine population hybrid by 1966. The price of meat from the Bobai natives was only a tenth of that of other well-bred or hybrid hogs in Hong Kong dollars. Nonetheless, collective breeders and peasants were reluctant to increase the population of hybrid pigs. The Bobai locals agreed that the hybrids were temperamentally much more vicious than the natives. The hybrids were also known to be picky eaters. Grassroots breeders therefore had difficulty fattening the hybrid hogs in time. This bottom-up disinclination forced the county government to use a carrot: offering 50 kilograms of chemical fertilizer for each hybrid pig and 35 for each native one to collectives which were enthusiastic about meeting the requisition quota.⁴⁰⁰

Cattle Must Plow a Field

Local peasants and the state did not seem to have arrived at a consensus on cattle breeding either. The Guangxi Provincial Government tested its livestock breeding policies in a few places from 1952 to 1954. In 1953, Gonghe, You’ai, and Yingguang townships were designated as testing grounds for the breeding of a superior breed of shorthorn cattle. But the peasants there were skeptical. They were concerned that as highbred bulls were exempted from field labor, the draft animals would lose their “shoulders” (肩膀). Then the offspring of these incapable animals, the peasants thought,

⁴⁰⁰ “Zhuanfa xian waimaozhan guanyu shengzhu gailiang yijian de baogao” 转发县外贸站关于生猪改良意见的报告 (March 3, 1965), Guangxi Bobai County Archives (hereafter GBCA), 1-2-034-29.

动物被屠宰并作为肉类装载到货船上。1964年，博白县出口了41662头活猪，本地消费了30957头。这些猪包括本地猪和大白猪。1965年，本地猪占当地猪的70%；只有30%是约克夏本地杂交猪。然而，中央和省级政府命令该县在1966年前对本地猪进行去势或淘汰，使整个生猪种群杂交化。博白本地猪肉的价格仅为香港其他优良品种或杂交猪肉价格的十分之一。尽管如此，集体养殖户和农民不愿增加杂交猪的数量。博白当地人认为，杂交猪的性情比本地猪凶猛得多。杂交猪也以挑食而闻名。因此，基层养殖户很难及时育肥杂交猪。这种自下而上的不情愿迫使县政府采取奖励措施：对积极完成征用配额的集体，每头杂交猪提供50公斤化肥，每头本地猪提供35公斤化肥。⁴⁰⁰

牛必须耕地

Local peasants and the state did not seem to have arrived at a consensus on cattle breeding either. The Guangxi Provincial Government tested its livestock breeding policies in a few places from 1952 to 1954. In 1953, Gonghe, You’ai, and Yingguang townships were designated as testing grounds for the breeding of a superior breed of shorthorn cattle. But the peasants there were skeptical. They were concerned that as highbred bulls were exempted from field labor, the draft animals would lose their “shoulders” (肩膀). Then the offspring of these incapable animals, the peasants thought,

⁴⁰⁰ “转发县外贸站关于生猪改良意见的报告” (1965年3月3日), 广西博白县档案馆（以下简称GBCA），1-2-034-29。

would not be able to plow a field. Moreover, these townships were chronically short of grassland.

The peasants rarely welcomed animals who “are too bulky and eat much” (体大食多). More

fundamentally, they rejected the state’s idea that villagers had to prioritize breeding over putting

oxen to work when there was a conflict between the two. The peasants cared about oxen because

they could pull a plow.⁴⁰¹

Lingui County was a foothold livestock county (畜牧重点县) in Guangxi Province. The county government designated seven townships under its jurisdiction as bovine artificial insemination areas. The townships tested the artificial insemination method on 101 cows for the first time from October to December 1952. The results were disappointing; only 25 of the cows conceived. The local cadres focused on increasing the number of healthy cows in the next year and resumed the experiment in 1954 with 528 bovine mothers. The target fertilization rate was 70 percent. But the two-year preparation fell short. The actual fertilization rate in 1954 was only 34.8 percent. The peasants made jokes about these failed experiments, saying that the “mysterious” method using a “potion” (药水)—i.e. artificially collected semen—would not impregnate cows.

They also thought that it made no sense to lead cows to faraway experimental stations for artificial insemination that would not work. What was even harder to accept, the peasants reported, was the directive to cull inferior local bulls. The cadres regarded the reproductive ability of the animals as an unnecessary variable affecting the experiment.⁴⁰²

To correct this problem, the local cadres who were in charge of artificial insemination held scientific education sessions. They demonstrated to the peasants how to extract sperm from superior

⁴⁰¹ Guangxi sheng nongyeting, “Guanyu xiezhu Wutang chu peizhongzhan gongzuo de baogao.”

⁴⁰² Guangxi xumu shiyanchang, “Huangniu rengong shoujing gongzuo de jiancha baogao.”

无法耕地。此外，这些乡镇长期缺乏草地。农民们很少欢迎那些“体大食多”的动物（体大食多）。更根本的是，他们拒绝了国家提出的当耕牛繁殖和耕作发生冲突时，村民必须优先考虑繁殖而非耕作的观点。农民们关心耕牛是因为它们能拉犁。⁴⁰¹

临桂县是广西壮族自治区的一个畜牧重点县（畜牧重点县）。县政府将其管辖下的七个乡镇指定为牛人工授精区。1952年10月至12月，这些乡镇首次对101头母牛进行了人工授精试验。结果令人失望；只有25头母牛受孕。当地干部在次年致力于增加健康母牛的数量，并于1954年恢复了试验，使用了528头母牛。目标受精率为70%。但两年的准备工作未能达到预期。1954年的实际受精率仅为34.8%。农民们拿这些失败的试验开玩笑，说使用“药水”（药水）——即人工收集的精液——的“神秘”方法不会使母牛怀孕。他们还认为，将母牛带到遥远的实验站进行无效的人工授精毫无意义。农民们报告说，更难以接受的是，上级指示要淘汰劣质的本地公牛。干部们认为动物的繁殖能力是影响实验的不必要变量。⁴⁰²

为了纠正这个问题，负责人工授精的当地干部举办了科学教育课程。他们向农民演示了如何从优良的

⁴⁰¹ 广西省农业厅，“关于协助武塘畜配种站工作的报告。”

⁴⁰² 广西畜牧实验场，“黄牛人工授精工作的检查报告。”

bulls and inject it into cows. They borrowed microscopes from the authorities and let the local people look at motile sperm in the collected semen. The cadres wanted the peasants to understand that there was nothing magical about the procedure. They admitted, however, that to “completely address the masses’ mistrust,” they would need to increase the fertilization rate of artificial insemination. They problematized the situation that a few cadres transferred from the prefecture government took seats in the township-level experiment stations to learn about and spread the new technology. They emphasized the necessity of training more specialized breeders and artificial insemination technicians.⁴⁰³ Despite these efforts, livestock artificial insemination remained rare until the end of the Maoist era.

In addition to the dissent over the assumptions and methodologies of the state’s animal breeding policies, local peasants were particularly sensitive to the mating schedule of their bovines. The local leaders of Yulin, a prefecture under Guangxi, stated in the spring of 1960 that as the result of the expansion of intensive cultivation (精耕细作), the increase of arable land, and the high demand for short-distance transportation, the shortage of animal labor became more conspicuous. Given that draft animals would remain the prime source of power in agrarian production until full mechanization, the cadres continued, every commune, brigade, and team should take advantage of the golden chance for breeding cattle that the spring brought—a warm climate, enough water, and plenty of grass. But this encouragement of “spring mating” (春配) was at odds with the peasants’ preference for “autumn mating” (秋配). Once cows were pregnant, they needed to be protected and carefully monitored until the calves were born. This meant that the cows could not work as much as

⁴⁰³ Guangxi xumu shiyanchang, “Huangniu rengong shoujing gongzuo de jiancha baogao.”

公牛的精液注入母牛体内。他们向有关部门借来显微镜，让当地人观察收集到的精液中能活动的精子。干部们希望农民明白，这个过程没有什么神奇之处。然而，他们承认，要“彻底解决群众的不信任”，就需要提高人工授精的受精率。他们指出，一些从地区政府调来的干部在乡镇实验站任职，学习和推广这项新技术，这使得情况变得复杂。他们强调了培训更多专业育种员和人工授精技术员的必要性。⁴⁰³ 尽管做出了这些努力，但在毛泽东时代结束之前，牲畜人工授精仍然很少见。

除了对国家畜牧政策的假设和方法存在异议之外，当地农民对牛的交配时间也特别敏感。1960年春天，广西玉林地区的地方领导指出，由于精耕细作的推广、耕地面积的增加以及对短途运输的高需求，畜力短缺变得更加突出。干部们接着说，鉴于役畜在农业生产中仍将是主要的动力来源，直到实现全面机械化，每个公社、大队和生产队都应该抓住春天带来的繁殖牛的黄金机会——温暖的气候、充足的水源和茂盛的牧草。但这种对“春配”的鼓励与农民偏爱“秋配”的习惯相悖。母牛一旦怀孕，就需要受到保护和仔细监测，直到小牛出生。这意味着母牛不能像以前那样劳作，

⁴⁰³ 广西畜牧试验场，“黄牛人工授精工作的检查报告。”

when they were not pregnant. Hence, the peasants found that spring mating irrational and unfavorable for farming.⁴⁰⁴

The bigger problem of spring mating, however, was not about cows, but calves. If cows were impregnated in spring, the calves would arrive in mid-winter, when they were least likely to survive. In contrast, autumn mating ensured that calves would be born in late summer, when their survival was more likely. The calves also could have four to six months before their first winter. In 1962, the leaders of the Communist Party Yulin Prefecture Committee changed their position to support autumn mating.⁴⁰⁵ At least when it came to the health of bovines, the peasantry had the knowledge to overrule the state.

The discord between the state and local collectives continued into the last decade of the Maoist era. A 1965 report from Zhangqing Commune’s breeding station in Hukou County, Jiangxi Province showed that the situation was analogous to that of Guangxi. Collective breeders, caretakers, and peasants all understood that having more calves than they could take care of was a “burden.” They “wanted to raise neither highbred bulls nor native cows.” They complained that “pregnant cows were not good at working.” In short, the peasants in Zhangqing Commune “did not trust the breeding work.”⁴⁰⁶ As late as 1975, the peasants in Shangluo Prefecture, Shaanxi still had

⁴⁰⁴ Zhonggong Guangxi Zhuangzu zizhiqu Yulin defang weiyuanhui 中共广西僮族自治区玉林地方委员会 and Guangxi Zhuangzu zizhiqu Yulin zhuanyuan gongshu 广西僮族自治区玉林专员公署, “Guanyu zuohao gengniu fanzhi yanghu gongzuo de lianhe tongzhi” 关于做好耕牛繁殖养护工作的联合通知 (March 12, 1960), Guangxi Yulin Municipal Archives (hereafter GYMA), 005-005-018-23.

⁴⁰⁵ Guangxi Zhuangzu zizhiqu Yulin zhuanyuan gongshu 广西僮族自治区玉林专员公署, “Guanyu renzhen zhuhao gengniu qiupei gongzuo de tongzhi” 关于认真抓好耕牛秋配工作的通知 (October 17, 1962), GYMA, 030-009-013-03.

⁴⁰⁶ Hukou xian kexue jishu xiehui 湖口县科学技术协会, “Banhao peizhongzhan gengniu fanzhi kuai: Hukou xian Zhangqing gongshe jianli gengniu peizhongzhan de jingyan he zuofa” 办好配种站耕牛繁殖快: 湖口县张青公社建立耕牛配种站的经验和作法 (April 1965), Unpublished material.

在它们没有怀孕的时候。因此，农民们认为春季交配是不合理且不利于耕作的。

404

然而，春季交配更大的问题不在于母牛，而在于牛犊。如果母牛在春天受孕，牛犊将在隆冬时节出生，那时它们的存活率最低。相比之下，秋季交配确保牛犊在夏末出生，那时它们的存活率更高。牛犊在第一个冬天到来之前也有四到六个月的生长时间。1962年，中共玉林地委领导改变立场，支持秋季交配。⁴⁰⁵至少在牛的健康问题上，农民们有知识推翻国家的决定。

国家与地方集体之间的不和一直持续到毛泽东时代的最后一个十年。1965年江西省湖口县张青公社育种站的一份报告显示，情况与广西类似。集体饲养员、看护员和农民都明白，牛犊过多而无法照料是一种“负担”。他们“既不想养良种公牛，也不想养本地母牛”。他们抱怨“怀孕的母牛不善于劳作”。简而言之，张青公社的农民“不信任育种工作”。⁴⁰⁶直到1975年，陕西商洛地区的农民仍然

⁴⁰⁴ Zhonggong Guangxi Zhuangzu zizhiqu Yulin defang weiyuanhui 中共广西僮族自治区玉林地方委员会 and Guangxi Zhuangzu zizhiqu Yulin zhuanyuan gongshu 广西僮族自治区玉林专员公署, “Guanyu zuohao gengniu fanzhi yanghu gongzuo de lianhe tongzhi” 关于做好耕牛繁殖养护工作的联合通知 (March 12, 1960), Guangxi Yulin Municipal Archives (hereafter GYMA), 005-005-018-23.

⁴⁰⁵ 广西僮族自治区玉林专员公署, 《关于认真抓好耕牛秋配工作的通知》（1962年10月17日），GYMA, 030-009-013-03。

⁴⁰⁶ Hukou xian kexue jishu xiehui 湖口县科学技术协会, “Banhao peizhongzhan gengniu fanzhi kuai: Hukou xian Zhangqing gongshe jianli gengniu peizhongzhan de jingyan he zuofa” 办好配种站耕牛繁殖快: 湖口县张青公社建立耕牛配种站的经验和作法 (April 1965), Unpublished material.

not fully adopted the notion of breed improvement. They “never selected superior breeds strictly.” The local people even did not pay attention to the differences in breed and frequently “castrated the superior, not the inferior” (阉优不阉劣).⁴⁰⁷

Reflecting on these cases, most local peasants expected cattle to plow a field or pull a wagon in more-than-people’s communes. The use value of such cattle did not need to be always increased or improved. It seems that what they wanted was to optimize the number and quality of the herds in accordance with the size of farmland and the availability of fodder. I see no reason to deny the fact that based on their needs, the masses at the grassroots had knowledge and experience to sustain bovine herds on their own. Perhaps it is now time to examine local breeding knowledge and practices more closely.

Gendered Local Knowledge of Equine Reproduction

In this section, I pay special attention to the ways in which collective breeders propagated and improved horses, donkeys, and mules in the late 1950s and the early 1960s. In 1958, state breeders in Gansu Province, one of the leading livestock provinces, had specific goals in equine breeding. On the one hand, they introduced the Karabair from Uzbekistan and Tajikistan and the Russian Don from the Soviet Union to China. These purebred horses and the hybrids between these foreign horses and natives were welcomed by the cadres and breeders in state breeding stations.⁴⁰⁸

⁴⁰⁷ Shaanxi sheng Shangluo diqu xumu shouyi gongzuozhan, *Shangluo diqu niu yang pucha ziliao huibian*, 25–26.

⁴⁰⁸ “Gansu sheng jianli gengchu liangzhong jidi juti jihua cao’an” 甘肃省建立耕畜良种基地具体计划草案 (N.d.), in *Xumu shouyi jingyan 4: Gengchu de peizhong fanzhi*, ed. Gansu sheng xumuting, 16–17.

没有完全接受品种改良的概念。他们“从不严格选择优良品种”。当地人甚至不注意品种差异，经常“阉优不阉劣”（阉优不阉劣）。⁴⁰⁷

回顾这些案例，大多数当地农民期望牛在人民公社中能耕地或拉车。这类牛的使用价值不需要总是增加或提高。他们似乎想要的是根据耕地面积和饲料供应情况来优化牛群的数量和质量。我没有理由否认这样一个事实：基于他们的需求，基层群众拥有维持牛群的知识 and 经验。也许现在是时候更仔细地审视当地的育种知识和实践了。

马科动物繁殖的性别化地方知识

在本节中，我特别关注集体饲养员在1950年代末和1960年代初繁殖和改良马、驴和骡子的方式。1958年，作为主要畜牧省份之一的甘肃省的国有饲养员在马科动物育种方面有具体目标。一方面，他们从乌兹别克斯坦和塔吉克斯坦引进了卡拉拜尔马，并从苏联引进了顿河马到中国。这些纯种马以及这些外国马与本地马的杂交后代受到国家育种站干部和饲养员的欢迎。⁴⁰⁸

⁴⁰⁷ 陕西省商洛地区畜牧兽医工作站， 《商洛地区牛羊普查资料汇编》， 25–26页。

⁴⁰⁸ “甘肃省建立耕畜良种基地具体计划草案”（无日期）， 载甘肃省畜牧厅编： 《畜牧兽医经验4： 耕畜的配种繁殖》， 16–17页。

In addition to these foreign horses, the state breeders were eager to promote superior Chinese breeds. Hequ horses bred from Gannan Tibetan Autonomous Prefecture and Chakouyi horses from Tianzhu Tibetan Autonomous County were spotlighted as these breeds were known to run very well and were therefore suitable for “both riding and dragging cargo” (挽乘兼用).

Longdong Large donkeys, native to Qingyang, Gansu, also deserved to be bred. They were “extremely patient with hard working and adversity” (极耐劳苦). Taking advantage of the accelerated formation of people’s communes in that year, the state breeders wanted to train a large number of commune breeders and charge them with breeding the three remarkable native breeds. As usual, short-term intensive trainings became a venue for collective breeders recruited from the peasantry to learn breeding technologies and share their prior knowledge and experience.⁴⁰⁹

By 1958, there were 40 commune breeders in Cheli District, Jingning County, Gansu. Three communes in the district co-organized a training session in which 32 were newly trained. Among these breeders, Zhang Wanfu was the most skilled. Zhang was especially adept at mating male horses and female donkeys to produce hinnies.⁴¹⁰ Zhang and other commune breeders often encountered male horses who “were physically weak and had not had sex for long.” These horses did not “feel comfortable with female donkeys.” Local peasants, perhaps projected their ideas about human sexuality onto these nonhuman animals, called them “trash” (废物) and did not expect them to bear offspring. Zhang, however, posited that breeders were able to help the males by “disciplining” them. Under the auspices of the district head, the breeder provided a kind of sex

⁴⁰⁹ “Gansu sheng jianli gengchu liangzhong jidi juti jihua cao’an,” 16 and 20.

⁴¹⁰ The offspring of a male horse and a female donkey is called hinny (驴骡), while that of a male donkey and a female horse is mule (骡 or 骡子).

除了这些外来马匹，国家育种者还热衷于推广优良的中国本土品种。来自甘南藏族自治州的河曲马和天祝藏族自治县的岔口驿马备受关注，因为这些品种以善跑而闻名，因此适合“挽乘兼用”。产自甘肃庆阳的陇东大驴也值得繁育。它们“极耐劳苦”。国家育种者利用当年人民公社加速组建的有利时机，希望培训大量公社育种员，并责成他们繁育这三个卓越的本土品种。像往常一样，短期强化培训成为从农民中招募的集体育种员学习育种技术和分享他们先前知识和经验的场所。⁴⁰⁹

到1958年，甘肃省静宁县车里区有40名公社育种员。该区三个公社联合举办了一次培训班，其中32人是新培训的。在这些育种员中，张万福技术最娴熟。张尤其擅长让公马和母驴交配以产出骡子。⁴¹⁰ 张和其他公社育种员经常遇到“身体虚弱，长期没有性行为”的公马。这些马对母驴“感觉不舒服”。当地农民，也许是将他们对人类性行为的观念投射到这些非人类动物身上，称它们为“废物”，并且不指望它们能生育后代。然而，张认为育种员可以通过“管教”它们来帮助这些公马。在区长的支持下，这位育种员提供了一种性

⁴⁰⁹ “Gansu sheng jianli gengchu liangzhong jidi juti jihua cao’an,” 16 and 20.

⁴¹⁰ 公马和母驴的后代称为驴骡，而公驴和母马的后代称为骡子。

education—details are not recorded—to one stallion for about a month, eventually revitalizing the horse. Sixty-seven mules and hinnies were born in Cheli in that year.⁴¹¹

Li Wanming and Wu Xiufeng were seasoned collective breeders from Fangshan County, Hebei. They, too, selected and bred local horses. But rather than seeing horses based on their breed, Li and Wu seemed to value the individuality of the equines. They believed that breeders “first must figure out whether a stallion has a good or bad character.” Once they picked gorgeous yet docile horses, it would now be humans’ turn to be nice: Li and Wu treated the animals in a “gentle and affable way.” The relationship between breeders and horses, Li and Wu continued, should not be “rough and violent.” If breeders whipped their horses, the horses would be frightened and would inevitably become “self-defensive,” biting and kicking the humans. If breeders and horses reached such a situation, the breeders would never be able to guide the horses to mate as planned. In other words, without a bond between the human and the horse, breeding work would not be possible at all.⁴¹²

In Changjiaping Commune, Huichuan County, Gansu, two breeders were in charge of the breeding. They looked after two stallions, a male donkey, and a male yak. The animals were all locally selected and did not belong to any notable breeds. The two breeders somehow shared an idea that the native male horses in their commune had short penises, but the native female horses had deep vaginas. Hence, when they mated the horses, they tried to find a slope so that the female would be lower than the male. This idea made the breeders prefer to ask local caretakers to bring female

⁴¹¹ Jingning xian renmin weiyuanhui 静宁县人民委员会, “Jingning xian Cheli qu fanzhi lüluo jingyan” 静宁县车李区繁殖驴骡经验 (N.d.), in *Xumu shouyi jingyan 4: Gengchu de peizhong fanzhi*, ed. Gansu sheng xumuting, 20–23.

⁴¹² Hebei sheng Fangshan xian renmin weiyuanhui 河北省房山县人民委员会, “Fangshan xian peizhongyuan dui pei lüluo jishu de ji dian tihui” 房山县配种员对配驴骡技术的几点体会 (N.d.), in *Xumu shouyi jingyan 4: Gengchu de peizhong fanzhi*, ed. Gansu sheng xumuting, 58–60.

教育——细节未被记录——对一匹种马进行了一个月的教育，最终使这匹马恢复了活力。那一年，车里出生了六十七头骡子和馱。411

李万明和吴秀峰是河北房山县经验丰富的集体饲养员。他们也选择并饲养当地的马匹。但李和吴似乎不是根据马的品种来看待马匹，而是重视马匹的个性。他们认为饲养员“首先必须弄清楚一匹种马的性格是好是坏”。一旦他们选择了漂亮而温顺的马匹，接下来就轮到人类友善了：李和吴“温柔和蔼”地对待动物。李和吴接着说，饲养员和马匹之间的关系不应该是“粗暴和暴力的”。如果饲养员鞭打他们的马匹，马匹会受到惊吓，并不可避免地变得“自卫”，咬人和踢人。如果饲养员和马匹达到这种境地，饲养员将永远无法按照计划引导马匹交配。换句话说，如果人与马之间没有建立联系，繁殖工作根本不可能进行。412

在甘肃省徽川县昌吉坪公社，有两名饲养员负责育种。他们照料着两匹种马、一头公驴和一头公牦牛。这些动物都是当地选育的，不属于任何著名品种。这两名饲养员不知何故都认为他们公社的本地公马阴茎短，而本地母马阴道深。因此，他们在马匹交配时，会尽量找一个斜坡，让母马低于公马。这种想法使得饲养员更倾向于让当地饲养员带来母马

⁴¹¹ 静宁县人民委员会, 《静宁县车李区繁殖驴骡经验》（不详），载甘肃省畜牧厅编：《畜牧兽医经验4：耕畜的配种繁殖》，第20-23页。

⁴¹² Hebei sheng Fangshan xian renmin weiyuanhui 河北省房山县人民委员会, “Fangshan xian peizhongyuan dui pei lüluo jishu de ji dian tihui” 房山县配种员对配驴骡技术的几点体会 (N.d.), in *Xumu shouyi jingyan 4: Gengchu de peizhong fanzhi*, ed. Gansu sheng xumuting, 58–60.

animals from their brigades and teams to the commune breeding station where there were many slopes; the breeders rarely went down to lower collectives as they did not want to be bothered to find slopes. Caretakers—most of whom were female—therefore found animal breeding onerous.⁴¹³

But they seemed to have no better alternative. They might not have wanted or been able to learn breeding skills as breeding was largely regarded as male work.

Let us focus more on the gendered nature of the local knowledge regarding equine breeding. If the abovementioned breeders reflected certain cultural norms about stallions, Tian Guangyi, a collective breeder from Shangqiu County, Henan Province, had similar views about mares. Tian was certain that “mares whose frame is big, waist is short, belly is big, and hip is short are good at getting pregnant.” If the “vulva is big and the pelvis is short, the mares may give birth to foals more easily.” He continued: “Once a mare is rutting, it turns delirious and does not eat well. When a stallion meets its eyes, the mare’s vagina would expand. And they would breathe hard and hit on the stallion.” The model breeder’s “technical” remarks were not supported by any scientific evidence other than “his experience.” Tian then introduced his “day-after-day mating,” “alternating-days mating,” “blindfolded mating,” “human assistive mating,” and the like. What is worth emphasizing here is that explaining when one may want to use a certain skill, Tian discussed only the different conditions of fillies and mares—“how old a mare is; whether a mare’s character is docile or not; and how strong a mare’s sexual desire is”—as if males were not a source of contingency.⁴¹⁴

⁴¹³ Huichuan xian renmin weiyuanhui 会川县人民委员会, “Changjiaping xiang shengchu peizhongzhan peizhong jingyan” 常家坪乡牲畜配种站配种经验, in *Xumu shouyi jingyan 4: Gengchu de peizhong fanzhi*, ed. Gansu sheng xumuting, 32–34.

⁴¹⁴ Henan sheng Shangqiu xian renmin weiyuanhui 河南省商邱县人民委员会, “Tian Guangyi fanzhi luluo de jingyan” 田广义繁殖驴骡的经验, in *Xumu shouyi jingyan 4: Gengchu de peizhong fanzhi*, ed. Gansu sheng xumuting, 89–91.

将牲畜从各自的生产队和生产组带到公社的畜牧站，那里有很多斜坡；饲养员很少下到较低的集体单位，因为他们不想费心去寻找斜坡。因此，饲养员——其中大多数是女性——觉得动物繁殖很麻烦。⁴¹³但她们似乎没有更好的选择。她们可能不想或不能学习繁殖技能，因为繁殖在很大程度上被认为是男性工作。

让我们更多地关注马匹繁殖方面地方知识的性别化性质。如果说上述饲养员反映了某些关于种马的文化规范，那么河南省商丘县的集体饲养员田广义对母马也有类似的看法。田广义确信“体格大、腰短、肚子大、臀部短的母马善于怀孕”。如果“外阴大、骨盆短，母马可能更容易产下马驹”。他接着说：“一旦母马发情，它就会变得神志不清，吃不好。当种马与它目光相遇时，母马的阴道会扩张。它们会大口喘气并撞击种马。”这位模范饲养员的“技术”言论除了“他的经验”外，没有任何科学证据支持。田广义随后介绍了他的“日日交配”、“隔日交配”、“蒙眼交配”、“人工辅助交配”等。这里值得强调的是，在解释何时可以使用某种技能时，田广义只讨论了小母马和母马的不同情况——“母马的年龄；母马的性格是否温顺；以及母马的性欲有多强”——仿佛雄性不是偶然性的来源。⁴¹⁴

⁴¹³ 会川县人民委员会, “常家坪乡牲畜配种站配种经验”, 载《畜牧兽医经验4: 耕畜的配种繁殖》, 甘肃省畜牧厅编, 第32–34页。

⁴¹⁴ 河南省商邱县人民委员会, “田广义繁殖驴骡的经验”, 载《畜牧兽医经验4: 耕畜的配种繁殖》, 甘肃省畜牧厅编, 第89–91页。

In 1960, Gongbaolaga Commune, Taibus Banner, Inner Mongolia established a breeding station. The commune leaders recruited six young herdsmen as commune breeders and sent them to an upper-level institution to be trained. After reading some books and participating in field practices, they “became quickly accustomed to the artificial insemination technology in a short period of time.” The six young breeders now exclusively looked after a purebred superior Soviet stallion and 104 mares. But perhaps because of their lack of expertise, artificial insemination did not work well. Facing the disappointment of local cadres and fellow herdsmen, the breeders blamed the low success rate of artificial insemination on the mares. They assumed that some hypersexual mares still needed to see some stallions, even though the actual mating would never take place. The breeders eventually brought three “stimulating male horses” (试情公马) to the station.⁴¹⁵

In many other places, similarly, mother horses were pointed out as potential obstacle to the successful breeding work. Breeders in Xinjiang, for example, were dissatisfied with the result of equine artificial insemination. The fertilization rate in their station was only 50 percent. Once again, the mares were blamed. The breeders stated that “mares who are not sexually active” were “vile mares” (恶癖的母马). Such mares therefore should be “strictly monitored” to prevent breeders from failing to notice estrous signs which were difficult to see.⁴¹⁶

⁴¹⁵ “Taipusi qi Gongbao laga gongshe qunma rengong shoujing shoutailü dadao 96.8% de jingyan” 太仆寺旗贡宝拉嘎公社群马人工受精受胎率达到 96.8%的经验, in *Dachu peizhong fanzhi gailiang jingyan huibian* 大畜配种繁殖改良经验汇编, ed. Xi lin guo le meng dang’anguan 锡林郭勒盟档案馆 (N.p.: Xi lin guo le meng dang’anguan, 1960), 38–40.

⁴¹⁶ Xinjiang shengchan jianshe bingtuan nong si shi zhaoguanchu zhongmachang 新疆生产建设兵团农四师昭管处种马场, “Mapi peizhong ji fangzhi liuchan de jingyan” 马匹配种及防止流产的经验 (N.d.), in *Guoying nongchang xumu shouyi cankao ziliao 2* 国营农场畜牧兽医参考资料 2, ed. Nongkenbu nongye shengchanju 农垦部农业生产局 (N.p.: Nongkenbu nongye shengchanju, 1963), 6–8.

1960年，内蒙古太仆寺旗贡宝拉格公社成立了一个育种站。公社领导招募了六名年轻牧民作为公社育种员，并派他们到上级机构接受培训。在阅读了一些书籍并参加了实地实践后，他们“在短时间内迅速掌握了人工授精技术”。这六名年轻的育种员现在专门照料一匹纯种的苏联优良种马和104匹母马。但也许是由于他们缺乏专业知识，人工授精效果不佳。面对当地干部和牧民的失望，育种员将人工授精成功率低归咎于母马。他们认为，一些性欲旺盛的母马仍然需要看到一些公马，尽管实际的交配永远不会发生。育种员最终将三匹“试情公马”（试情公马）带到了育种站。⁴¹⁵

在许多其他地方，同样地，母马被认为是成功育种工作的潜在障碍。例如，新疆的育种员对马匹人工授精的结果不满意。他们育种站的受精率只有50%。母马再次受到指责。育种员声称“性不活跃的母马”是“恶癖的母马”（恶癖的母马）。因此，应“严格监控”此类母马，以防止育种员未能发现难以察觉的发情迹象。⁴¹⁶

⁴¹⁵ “太仆寺旗贡宝拉嘎公社群马人工受精受胎率达到96.8%的经验”，载《大畜配种繁殖改良经验汇编》，锡林郭勒盟档案馆编（无出版地：锡林郭勒盟档案馆，1960），第38-40页。

⁴¹⁶ 新疆生产建设兵团农四师昭管处种马场，“马匹配种及防止流产的经验”（无日期），载《国营农场畜牧兽医参考资料2》，农垦部农业生产局编（无出版地：农垦部农业生产局，1963），第6-8页。

It is likely that the gender biases about sexual difference in rural China were projected onto animals. I lacked documentary evidence to investigate the questions of what those ideas about human sexual differences were and how they were linked to the discourse on and practice of horses, donkeys, mules, and hinnies. Nonetheless, what is clear is that human females were largely excluded from the animal breeding work. I have not found a single document that mentioned female breeders. As seen in Chapter Four, female caretakers who took care of female livestock occasionally participated in breeding. But their presence when animals mated always caused controversy. This gendered foundation of the knowledge of animal breeding persisted until the end of the Maoist period.

Conclusion

The quantitative increase and qualitative improvement of livestock populations were constant priorities of the Maoist state. However, diverse local actors—private breeding households, collective breeders, and peasants—did not always agree “the more the merrier.” They considered the economic and environmental conditions and rather sought to optimize the necessary number of domestic animals within such limitations. In terms of the question of quality, it was also obvious that the state and non-state actors held different views. Although the state framed the breeding work as systematic, advanced, and scientific, local collectives did not readily abandon their preconceptions, preferences, and practices that the state often denounced as “faulty” and “superstitious.” They also presented different criteria of what was “superior” or “inferior.” These discords in animal breeding reflected a clash of interest between the state and grassroots communities: the former sought to maximize the extraction of surplus value from the latter with limited material support, while the latter had to communally uphold the use value derived from animals.

中国农村对性别差异的偏见很可能投射到了动物身上。我缺乏文献证据来调查关于人类性别差异的观念是什么，以及它们如何与关于马、驴、骡子和馱的论述和实践联系起来。尽管如此，显而易见的是，女性在很大程度上被排除在动物育种工作之外。我没有找到任何一份提及女性育种员的文件。正如第四章所见，照料雌性牲畜的女性饲养员偶尔会参与育种。但她们在动物交配时的出现总是引起争议。这种基于性别的动物育种知识基础一直持续到毛泽东时代结束。

结论

牲畜数量的增加和质量的提高是毛泽东时代国家持续的优先事项。然而，不同的地方参与者——私人繁殖户、集体繁殖者和农民——并不总是认同“越多越好”。他们考虑经济和环境条件，而是在这些限制内寻求优化家畜的必要数量。在质量问题上，国家和非国家参与者显然也持有不同观点。尽管国家将繁殖工作框定为系统、先进和科学的，但地方集体并没有轻易放弃国家经常谴责为“错误”和“迷信”的先入之见、偏好和实践。他们还提出了关于什么是“优等”或“劣等”的不同标准。动物繁殖中的这些不和反映了国家与基层社区之间的利益冲突：前者试图在有限的物质支持下最大限度地从后者那里榨取剩余价值，而后者则必须共同维护动物的使用价值。

It is not easy, however, simply to set a dichotomy between the eugenic state and non-eugenic peasants. Both the state agents and members of collectives attempted to eugenically control the nonhuman populations, although their assumptions and measures could be different. Animal eugenics, in every sense, existed in Maoist China. In addition, we do not have to romanticize the indigenous knowledge of local breeders. It was based on gendered epistemologies and communal structures. All these unappealing practices were part of the ways in which more-than-people's communes maintained their multispecies well-being.

A tentative insight that we might be able to excavate from the experience of animal breeding in Maoist China, is the possibility of denaturalizing the concept of “breed.” If some collective breeders and peasants prioritized maintaining the use value over producing surplus exchange value, and if they succeeded in securing enough use value from local animals, either “superior” or “inferior”; “foreign” or “native,” might they have had no reason to share the state's obsessive attention to the category of “breed” itself? If one had not adopted the concept of breed per se, the entire schemes for the qualitative increase and quantitative improvement of livestock could be seen as unnecessary. One could have just appreciated the use value of the individual animals. When that was not enough, one could produce more offspring or introduce foreign stock. Nonetheless, one might have been able to affirm the existence of every animal as it was and just live with them in a more-than-people's commune.

然而，简单地在优生国家和非优生农民之间划清界限并不容易。国家代理人和集体成员都试图对非人类种群进行优生控制，尽管他们的假设和措施可能有所不同。动物优生学，从任何意义上说，都存在于毛泽东时代的中国。此外，我们不必美化当地饲养者的本土知识。它基于性别认识论和公共结构。所有这些令人不快的做法都是“超人公社”维持其多物种福祉的方式的一部分。

我们或许可以从毛泽东时代动物育种的经验中挖掘出一个初步的见解，那就是“品种”概念去自然化的可能性。如果一些集体饲养者和农民优先考虑维持使用价值而不是生产剩余交换价值，并且如果他们成功地从本地动物（无论是“优等”还是“劣等”；“外来”还是“本地”）那里获得了足够的使用价值，那么他们可能就没有理由像国家那样痴迷于“品种”这个范畴本身了。如果一个人没有采纳“品种”这个概念本身，那么所有旨在提高牲畜质量和数量的计划都可能被视为不必要的。人们可能只会欣赏个体动物的使用价值。当这还不够时，人们可以生产更多的后代或引入外来品种。尽管如此，人们或许能够肯定每一种动物的存在，并与它们在“超人公社”中共同生活。

Epilogue

In June 2021, I was scavenging for archival sources in Guangxi Zhuang Autonomous Region. Departing from Nanning, the capital of the provincial region, and going southeast, I made an unplanned stop at several local archives. These visits to archives turned out to be unexpectedly productive. Some prefectural or county-level archives in China were bonanzas. But what I was most looking forward to was an interview with a former veterinarian who was living in a remote village of 30-40 households that I will call Tower Ground in Bobai County, Guangxi.

Perhaps I was following the same route to the village as Mao-era state cadres. From the provincial heartland, Nanning, to the prefectural center, Yulin, to the main areas in Bobai, and then to the village, I traveled 180 miles of bumpy mountain roads. Back in the 1960s or 1970s the trip must have been even more difficult. If so, the Communist state must have had quite a long reach to have found this secluded community.

The interview had been arranged by my Chinese colleague, Professor Li Yufeng, and his student, “Baobao,” from Nanning Normal University. Baobao was born in Tower Ground. His grandmother told him that one of their neighbors had worked as a veterinarian in the past. Yufeng and Baobao kindly accompanied me as tour guides and interpreters, knowing that I did not speak Hakka, the language of this village community, and its inhabitants did not speak mandarin Chinese.

Tower Ground, like many other communities in Bobai, had been hog-producing collectives in the Maoist period. Walking around the village, I could still see cement pigsties. Although Baobao’s parents work in Shenzhen, the rest of the family still raise hundreds of pigs in the village. The pigs are no longer grown for communal purposes; now they are sold to megacities like Guangzhou,

结语

2021年6月，我在广西壮族自治区搜寻档案资料。我从省会南宁出发，一路向东南，意外地在几个地方档案馆停留。这些档案馆之行出乎意料地富有成效。中国的一些地市级或县级档案馆简直是宝库。但我最期待的是采访一位前兽医，他住在一个由30-40户人家组成的偏远村庄，我称之为广西博白县的塔地村。

也许我走的路线和毛泽东时代的国家干部去村里的路线一样。从省中心南宁到地级市玉林，再到博白的主要地区，然后到村里，我走了180英里的崎岖山路。回到20世纪60年代或70年代，那次旅行一定更加艰难。如果是这样，共产党国家一定有相当长的影响力才能找到这个僻静的社区。

这次采访是由我的中国同事、南宁师范大学的李玉峰教授和他的学生“宝宝”安排的。宝宝出生在塔地。他的奶奶告诉他，他们的一位邻居以前是兽医。玉峰和宝宝很友善地陪我做向导和翻译，因为他们知道我不会说客家话（这个村庄社区的语言），而当地居民也不会说普通话。

塔地，像博白县的许多其他社区一样，在毛泽东时代曾是生猪生产集体。在村子里走动时，我仍然能看到水泥猪圈。尽管宝宝的父母在深圳工作，但家里其他人仍然在村子里养了几百头猪。这些猪不再是为了集体目的而饲养；现在它们被卖到广州这样的大城市，

Shenzhen, and Hong Kong to generate income for the family. At some point after 2000, the family could finally afford to build some three-story houses. After briefly enjoying the air-conditioning in Baobao’s place, I and Baobao went to the home of the former veterinarian.

A Village Veterinarian in the Post-Mao era

The 86-year-old veterinarian, “Mr. Chen,” greeted us. We were surprised to learn that he had not started working as a veterinarian until after 1980. As I had expected to hear about the historical experience of a collective veterinarian in the Maoist era, I was disappointed. But soon after the interview began, I realized that his life story could inform me of legacies of Maoist more-than-people’s communes in the post-Maoist era. Unless otherwise noted, this section is based on my interviews with Mr. Chen on June 26 and 27, 2021. I asked questions in mandarin Chinese, and Mr. Chen answered in Hakka. Baobao translated our conversations.

Mr. Chen was born in 1935 to a poor peasant family in Tower Ground. The wars and revolution delayed his education. In 1954, he graduated from middle school in “Tiger Pond Township.” Tiger Pond later became a people’s commune. Tower Ground became a production team. Between Tiger Pond and Tower Ground, there was a production brigade called “Eternal Happiness.” Mr. Chen was lucky to enter an agricultural school located at the center of Bobai County. He studied horticultural sciences there and graduated in 1960. After working at a state-run horticultural station for the next few years, the state sent him back to his hometown. Working around Tiger Pond, Eternal Happiness, and some other nearby collectives for the rest of the Maoist period, he held several jobs, including forest protector (护林员) and work point recorder (记分员). Not surprisingly, agricultural labor was common.

深圳和香港打工，为家里挣钱。2000年后的某个时候，这家人终于有能力盖起几栋三层楼的房子。在宝包家短暂享受了空调后，我和宝包去了前兽医的家。

后毛泽东时代的一位乡村兽医

86岁的兽医“陈先生”接待了我们。我们惊讶地得知，他直到1980年后才开始从事兽医工作。我本以为会听到毛泽东时代集体兽医的历史经验，所以有些失望。但采访开始后不久，我意识到他的生平故事可以让我了解毛泽东时代人民公社在后毛泽东时代的遗产。除非另有说明，本节内容基于我于2021年6月26日和27日对陈先生的采访。我用普通话提问，陈先生用客家话回答。宝包翻译了我们的对话。

陈先生于1935年出生在塔地的一个贫农家庭。战争和革命耽误了他的学业。1954年，他从“虎塘乡”中学毕业。虎塘后来成为一个人民公社。塔地成为一个生产队。在虎塘和塔地之间，有一个名为“永乐”的生产大队。陈先生很幸运地进入了博白县中心的一所农业学校。他在那里学习园艺科学，并于1960年毕业。在接下来的几年里，他在一个国营园艺站工作，之后国家将他送回了家乡。在毛泽东时代剩下的时间里，他在虎塘、永乐和附近其他一些集体单位工作，担任过多个职务，包括护林员和记分员。不出所料，农业劳动很常见。

The end of the Mao era arrived in Mr. Chen’s village in 1979-80, when the local collectives were gradually disbanded and the collective animal farms were abolished. The communally owned livestock was sold to local markets or individual peasants. The work points and food stamps were all gone. This post-Maoist transition made it difficult for Mr. Chen’s family to earn a livelihood. While continuing to farm, he decided to practice veterinary medicine as a sideline in 1980. But this decision was affected by his Mao-era experience.

When Mr. Chen studied horticulture at the Bobai Agricultural School in the late 1950s, one of his close classmates, “Mr. Sun,” majored in veterinary medicine. After working at a state-run breeding farm for a few years in the early 1960s, Mr. Sun was transferred to the veterinary station of a commune close to Mr. Chen’s. Since then, Mr. Sun occasionally visited Tower Ground to see his old friend and gather medicinal herbs. One day, Mr. Sun came across and treated an ill ox in Tower Ground. The villagers were surprised. They thought that bovine diseases were much more difficult to treat than swine disorders because the only brigade veterinarian available, nicknamed “Cyclops Vet,” had always trouble treating sick cattle. Hence, whenever Mr. Sun visited Mr. Chen’s place, the people of Tower Ground Team and Eternal Happiness Brigade eagerly called in Mr. Sun to see their oxen and cows. In 1980, Mr. Chen remembered that the villagers needed a cattle doctor. He spent two months at Mr. Sun’s house and intensively learned veterinary medicine from him.

Cyclops Vet, the official brigade veterinarian, continued to provide veterinary services as a private practitioner in the early post-Mao period. Villagers knew that he could treat pigs but not cattle. If Cyclops Vet had been trained as a collective or barefoot doctor in a less privileged institution or through apprenticeship, as explained in Chapter Three, he could have been trained to specialize in swine, reflecting the state’s prioritization of pigs over other livestock. As Cyclops Vet passed away a few years ago before my visit, I was unable to confirm my hypothesis. In any case,

毛泽东时代于1979-80年间在陈先生的村庄结束，当时当地的集体农场逐渐解散，集体动物农场被废除。公有牲畜被出售给当地市场或个体农民。工分和粮票都消失了。这种后毛泽东时代的转型使陈先生一家难以谋生。在继续务农的同时，他决定在1980年兼职从事兽医工作。但这个决定受到了他毛泽东时代经历的影响。

20世纪50年代末，陈先生在博白农业学校学习园艺时，他的一位亲密同学“孙先生”主修兽医学。20世纪60年代初，孙先生在一家国营养殖场工作了几年后，被调到离陈先生家不远的一个公社的兽医站。从那时起，孙先生偶尔会到塔地看望他的老朋友并采集草药。有一天，孙先生在塔地偶然发现并治疗了一头生病的牛。村民们很惊讶。他们认为牛的疾病比猪的疾病更难治疗，因为当时唯一的生产队兽医，绰号“独眼兽医”，在治疗病牛方面总是遇到困难。因此，每当孙先生到陈先生家时，塔地队和永乐大队的村民们都急切地请孙先生给他们的牛看病。1980年，陈先生想起村民们需要一位牛医生。他在孙先生家住了两个月，向他 intensively 学习了兽医学。

独眼兽医，这位公社的官方兽医，在毛泽东时代结束后初期继续作为私人执业者提供兽医服务。村民们知道他能治疗猪，但不能治疗牛。如果独眼兽医是在一个不那么优越的机构或通过学徒制（如第三章所述）接受集体或赤脚医生培训的，他本可以被培养成专门治疗猪的兽医，这反映了国家对猪的重视程度高于其他牲畜。由于独眼兽医在我到访前几年去世，我无法证实我的假设。无论如何，

under Mr. Sun’s tutelage, Mr. Chen aimed at this niche market: healing cattle. In December 1980, he cured his first ailing ox. In his words, “the ox made me famous.” In these years, not unlike the Maoist era, the trust of fellow villagers still mattered. An animal could confirm a veterinarian’s authority.

For the next three decades, Mr. Chen practiced internal veterinary medicine in Tower Ground and villages within a 15-kilometer radius. Most of his patients were cattle and pigs with diarrhea, fever, respiratory trouble, and similar symptoms. In addition to Mr. Sun’s teaching on bovine medicine, Mr. Chen taught himself about swine medicine by reading books on TCVM. (He did not remember the titles.) I suspect that he might have read some of the TCVM monographs co-authored by veterinary elites such as Gao Guojing, Cui Diseng, and Jiang Cisheng, mentioned in Chapter Two. Such different channels of his learning of bovine and swine remedies appeared to have led to some therapeutic differences: Mr. Chen preferred using local herbs to treat cattle disorders, but used what he called “Western drugs” such as penicillin, streptomycin, and cephalosporin to treat pigs.

Mr. Chen did not consider his expertise to be scientific, although he proudly claimed to have correctly diagnosed about 90 percent of all the clinical cases he encountered in his decades-long practice. “It was no science. Everything was based on experience, and then I made a decision.” Experience, according to him, was the sum of clinical information that he could get from observing animals’ diet and behavior and taking their temperature. He gave me an example. “If you see an ox or cow sleeping facing forward and if their legs droop straight, that means the animal may have some problem with its heart. In that case, there would be no medicine in the whole world that could cure it.” I asked if he had felt pity for such animal patients. “No,” he answered. “Pigs can be ill; Cattle can be ill; humans, too, can be ill. That is a law of nature.” I then said, “Nonetheless, when at

在孙先生的指导下，陈先生瞄准了这个小众市场：治疗牛。1980年12月，他治愈了他的第一头病牛。用他的话说，“这头牛让我出了名。”在那些年里，与毛泽东时代一样，村民的信任仍然很重要。一头动物可以证实兽医的权威性。

在接下来的三十年里，陈先生在塔基及其15公里半径范围内的村庄从事兽医内科工作。他的大多数病人是患有腹泻、发烧、呼吸困难和类似症状的牛和猪。除了孙先生关于牛病学的教导，陈先生还通过阅读中兽医书籍自学了猪病学（他记不起书名了）。我怀疑他可能读过第二章中提到的由高国经、崔涤生、蒋次生等兽医精英合著的一些中兽医专著。他学习牛和猪治疗方法的不同途径似乎导致了一些治疗上的差异：陈先生更喜欢使用当地草药治疗牛病，但使用他所谓的“西药”，如青霉素、链霉素和头孢菌素来治疗猪。

陈先生认为自己的专业知识并非科学，尽管他自豪地声称在他几十年的行医生涯中，他所遇到的临床病例中约有90%都得到了正确诊断。“那不是科学。一切都基于经验，然后我做出决定。”他认为，经验是他通过观察动物的饮食和行为以及测量体温所能获得的临床信息的总和。他给我举了一个例子。“如果你看到一头牛睡觉时头朝前，腿直直地垂着，那说明这头动物的心脏可能有问题。在这种情况下，全世界都没有药能治好它。”我问他是否对这样的动物病人感到怜悯。“不，”他回答道，“猪会生病；牛会生病；人也会生病。这是自然规律。”我接着说，“尽管如此，当在

least some animals were ill, you were there to help them.” Mr. Chen softly smiled. “I was just an amateur [animal] doctor” (野医).

Mr. Chen recalled that among epizootic and zoonotic diseases, swine fever was the most common. Of course, he meant classical swine fever (CSF), not African swine fever (ASF) which has no vaccine or cure. But his memories regarding swine fever were less impressive than I expected. When there were outbreaks of swine fever after the 1980s, the state swiftly stepped in and took charge of the situation. The state distributed vaccines to villages, instead of expecting the villagers to take care of themselves. All that Mr. Chen needed to do was to give animals a shot. He did not seem to know much about disease control. With the demise of the “Comprehensive Prevention and Treatment” scheme of the Maoist era, discussed in Chapter One, the communal initiatives in animal disease control might be gone too.

After the interview, Baobao invited me to take a look at his family’s hog farm. I saw both Yorkshires and black natives. As explained in Chapter Five, animal eugenics was a part of more-than-people’s communes. I was glad to see that the black native pigs had survived the combination of eugenic breeding practices and capitalist economy. Outside the pigpens, some cattle, chickens, and dogs roamed freely on the roads. Their sounds and smells reminded me that Tower Ground is still a multispecies community.

But it was no longer a more-than-people’s commune in the absence of collectively shared animals, their caretakers, and breeders. Animal doctors such as Mr. Chen were part of the legacy of collective and barefoot veterinarians in the Maoist era. However, I could not get a sense of any communal subsistence during the interview. When I asked my last question, he answered, “When

至少有些动物生病了，你在那里帮助它们。”陈先生温柔地笑了笑。“我只是个业余的[兽医]（野医）。”

陈先生回忆说，在动物流行病和人畜共患病中，猪瘟是最常见的。当然，他指的是传统猪瘟（CSF），而不是没有疫苗或治愈方法的非洲猪瘟（ASF）。但他对猪瘟的记忆并没有我预期的那么深刻。20世纪80年代后，当猪瘟爆发时，国家迅速介入并控制了局势。国家向村庄分发疫苗，而不是指望村民自力更生。陈先生所需要做的就是给动物注射。他似乎对疾病控制知之甚少。随着第一章中讨论的毛泽东时代“综合防治”计划的消亡，动物疾病控制方面的社区举措可能也随之消失了。

采访结束后，宝宝邀请我参观他家的养猪场。我看到了约克夏猪和本地黑猪。正如第五章所解释的，动物优生学是超越人民公社的一部分。我很高兴看到本地黑猪在优生育种实践和资本主义经济的结合下幸存下来。猪圈外，一些牛、鸡和狗在路上自由漫步。它们的声音和气味提醒我，塔基仍然是一个多物种社区。

但由于没有集体共享的动物、它们的饲养员和育种者，它不再是一个超越人民公社的存在。像陈先生这样的兽医是毛泽东时代集体兽医和赤脚兽医遗产的一部分。然而，在采访中我没有感受到任何公共生计。当我问最后一个问题时，他回答说：“当

was I happiest as a veterinarian? Haha, well, I don't know. Maybe when I made a little money and supplemented the fortune of my family?"

Recognizing without Celebrating More-Than-People's Communes

More-than-people's communes once existed. Yet it seems that they rarely remain—like a footprint of the Buddha. If the more-than-people's commune is only a vestige of the abortive Maoist revolution, why should we care?

Transspecies diseases—COVID-19 is only one of the most recent examples—draw attention to the notion of One Health as a possible solution. But if the causes of this latest public health crisis are rooted in deeper, planetary problems of the Anthropocene, our understanding of One Health should not be limited to biomedical languages and practices. In other words, the framework of One Health needs to be more inclusive of non-biosecurity-centered and non-Western human-nonhuman relations. The history of Maoist China's more-than-people's communes in this sense can diversify One Health by foregrounding a unique set of contexts making a deep connection between livestock and livelihood of a collectivized community salient.

Such contexts in which more-than-people's communes were situated were both socialist and Chinese—in other words, Maoist. The communes functioned as centers of multispecies subsistence and survival, uniting human and nonhuman individuals and filling in for the state bureaucracy. The communes denaturalized private ownership and the meritocratic stratification of socio-intellectual prerogative. The communes did not always revere arguably universal modern sciences and technocratic rationality. The communes instead revisited their folk knowledge and experience which

我当兽医的时候最开心吗？哈哈，我不知道。也许是当我赚了点钱，为我的家庭增添了财富的时候吧？”

承认而非颂扬“超人公社”

“超人公社”曾经存在。然而，它们似乎很少留存下来——就像佛陀的足迹。如果“超人公社”只是夭折的毛泽东革命的遗迹，我们为什么要关心呢？

跨物种疾病——COVID-19只是最近的例子之一——引起了人们对“同一健康”概念的关注，认为它可能是一种解决方案。但是，如果这场最新公共卫生危机的根源在于人类世更深层次的全球性问题，那么我们对“同一健康”的理解就不应局限于生物医学语言和实践。换句话说，“同一健康”的框架需要更具包容性，涵盖非以生物安全为中心和非西方的“人-非人”关系。从这个意义上说，毛泽东时代中国“超人公社”的历史可以通过突出独特的背景来丰富“同一健康”的内涵，这些背景使得牲畜与集体化社区生计之间的深层联系变得突出。

多于人的公社所处的语境既是社会主义的，也是中国的——换言之，是毛泽东思想的。公社是多物种生存和繁衍的中心，它将人类和非人类个体联合起来，并取代了国家官僚机构。公社使私有制和以绩效为基础的社会智力特权分层失去了其自然属性。公社并不总是崇尚普遍的现代科学和技术官僚理性。相反，公社重新审视了它们的民间知识和经验，这些知识和经验

featured acupuncture and medicinal herbs. The communes, either willingly or reluctantly, embraced the insecurity of seeking alternatives to Modernity.

The possibilities of more-than-people’s communes were, at the same time, their limitations. As a product of the revolution, the experimental nature of more-than-people’s communes was never settled. Maoist China’s search for something beyond Modernity now fuels post-Maoist, fully capitalistic China’s nationalism and essentialism. China may be de-Westernizing but not necessarily decolonizing the world, as Walter Mignolo has noted.⁴¹⁷

The most problematic of more-than-people’s communes was found in the ways in which their human actors engaged with their nonhuman animals. More-than-people’s communes were unmistakably anthropocentric. I am not talking about whether veterinary workers and peasants were affectionate to animals or squarely exploitative. By anthropocentrism, I observe that humans were the end, whereas nonhumans were the means, not vice versa. Healing and taking care of animals was subordinated to placing animals in service to humans. The category of breed was less frequently questioned in the country where the concepts of ethnic minority and race could cause fierce conflicts.⁴¹⁸

Putting all these possibilities and limits together, this dissertation does not romanticize or fetishize the more-than-people’s communes. But anti-communism, anti-Chinese racism, and the worship of Modernity may not be helpful either. Rather, I do suggest a way of recognizing the

⁴¹⁷ Walter D. Mignolo, “Coloniality and Globalization: A Decolonial Take,” *Globalizations* 18.5 (July 2021): 720–737.

⁴¹⁸ For example, see Ben Hillman and Gray Tuttle, ed., *Ethnic Conflict and Protest in Tibet and Xinjiang: Unrest in China’s West* (New York: Columbia University Press, 2016); Xiaoyuan Liu, *To the End of Revolution: The Chinese Communist Party and Tibet, 1949-1959* (New York: Columbia University Press, 2020).

以针灸和草药为特色。公社，无论是自愿还是不情愿，都接受了寻求现代性替代方案的不确定性。

超人公社的可能性，同时也是它们的局限性。作为革命的产物，超人公社的实验性质从未确定。毛泽东时代中国对超越现代性的探索，如今助长了后毛泽东时代、完全资本主义的中国的民族主义和本质主义。正如沃尔特·米尼奥洛所指出的，中国可能正在去西方化，但未必正在使世界去殖民化。⁴¹⁷

超人公社最成问题之处在于其人类行为者与非人类动物互动的方式。超人公社无疑是以人类为中心的。我不是在讨论兽医工作者和农民是否对动物有感情，或者是否完全剥削动物。我所说的以人类为中心，是指人类是目的，而非人类是手段，反之则不然。治疗和照顾动物是次要的，将动物置于为人类服务的地位。在少数民族和种族概念可能引起激烈冲突的国家，品种类别很少受到质疑。⁴¹⁸

综合所有这些可能性和局限性，本论文并未将超人公社浪漫化或神化。但反共主义、反华种族主义和对现代性的崇拜也可能无济于事。相反，我确实提出了一种承认

⁴¹⁷ Walter D. Mignolo, “Coloniality and Globalization: A Decolonial Take” , *Globalizations* 18.5 (2021年7月)：720–737。

⁴¹⁸ 例如，参见 Ben Hillman 和 Gray Tuttle，编著，*《Ethnic Conflict and Protest in Tibet and Xinjiang: Unrest in China’s West》*（纽约：哥伦比亚大学出版社，2016）；Xiaoyuan Liu，*《To the End of Revolution: The Chinese Communist Party and Tibet, 1949-1959》*（纽约：哥伦比亚大学出版社，2020）。

communes without celebrating them: Keeping our critical distance from more-than-people’s commune in the past, present-day China, and the Westernized modern world simultaneously.

How could subnational communities and nation-states or their coalitions form a better way to protect the health of humans, nonhumans, and the planet?⁴¹⁹ Do we always want to prioritize exclusive possession over sharing of knowledge and animate and inanimate things? Can we continue the intellectual experiments with elitism, egalitarianism, meritocracy, and anti-intellectualism in a less violent way? Even though we are not equating humans with nonhumans, would not there be hitherto unthinkable ways to soften the binary of humans as the end and nonhumans as the means? These are the questions raised by the more-than-people’s communes in Maoist China. In envisioning fresh visions for and modes of dwelling on Earth as a more-than-people’s commune, I hope that these lessons will be learned.

⁴¹⁹ This question is ultimately connected to the “question of ‘political community’ [...] beyond the archetype of the nation-state,” in Christopher Lee’s words. See Christopher J. Lee, ed., *Making a World After Empire: The Bandung Moment and Its Political Afterlives* (Athens: Ohio University Press, 2010), 22.

公社，但不歌颂它们：同时与过去、当代中国和西方化的现代世界中的超人公社保持批判性距离。

次国家社区和民族国家或其联盟如何才能形成更好的方式来

保护人类、非人类和地球的健康？⁴¹⁹ 我们是否总是想优先考虑知识和有生命及无生命事物的独占所有权，而不是共享？我们能否以一种不那么暴力的方式继续进行精英主义、平等主义、英才教育和反智主义的智力实验？即使我们不将人类等同于非人类，难道就没有迄今为止无法想象的方式来软化人类是目的而非人类是手段的二元对立吗？这些是毛泽东时代超人公社提出的问题。在构想地球作为超人公社的全新愿景和居住模式时，我希望这些教训能够被吸取。

⁴¹⁹ 这个问题最终与克里斯托弗·李（Christopher Lee）所说的“超越民族国家原型”的“‘政治共同体’问题[...]”相关。参见 Christopher J. Lee, ed., *Making a World After Empire: The Bandung Moment and Its Political Afterlives* (Athens: Ohio University Press, 2010), 22。

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